Chapter 19

Integrals

Solutions

SECTION - A

Objective Type Questions (One option is correct)

[Basic Indefinite Integrals by Transformation]

1. The family of anti-derivatives of

$$f(x) = \frac{1}{ax^2 + 2\sqrt{ab}x + b + 1} \text{ is}$$

(1) $\frac{1}{2\sqrt{a}}\log|\sqrt{a}\,x + \sqrt{b}| + C$

(2) $\frac{1}{\sqrt{a}}\tan^{-1}(\sqrt{a}\,x + \sqrt{b}) + C$

(3) $\sin^{-1}(\sqrt{a}\,x + \sqrt{b}) + C$

(4) $e^{\sqrt{a}\,x + \sqrt{b}} + C$

Sol. Answer (2)

$$\int \frac{1}{(\sqrt{a}x + \sqrt{b})^2 + 1}dx = \frac{1}{\sqrt{a}}\tan^{-1}(\sqrt{a}x + b) + C$$

2. $\int\left(\sqrt[3]{x} + \frac{1}{\sqrt[3]{x}}\right)^3 dx$ equals

(1) $x^3 + 3x^2 + 3x + C$

(2) $\frac{3}{4}x^{\frac{4}{3}} + \log|\sqrt[3]{x}| + C$

(3) $\frac{x^2}{2} + \frac{9}{4}x^{\frac{4}{3}} + \frac{9}{2}x^{\frac{2}{3}} + \log|x| + C$

(4) $x + \frac{1}{x} + \sqrt{3}x + \frac{3}{\sqrt{x}} + C$

Sol. Answer (3)

$$\int\left(\sqrt[3]{x} + \frac{1}{\sqrt[3]{x}}\right)^3 dx = \int\left(x + \frac{1}{x} + 3\sqrt[3]{x} + \frac{3}{\sqrt[3]{x}}\right)dx = \frac{x^2}{2} + \log|x| + \frac{9}{4}x^{4/3} + \frac{9}{2}x^{2/3} + C$$

3. $\int \frac{x^4+3x^2+1}{\sqrt{x}}dx$ equals

(1) $\frac{2}{7}x^{\frac{7}{2}}+2x^{\frac{3}{2}}+x^{-\frac{1}{2}}+C$

(2) $\frac{2}{9}x^{\frac{7}{2}}+\frac{2}{3}x^{\frac{3}{2}}+\sqrt{x}+C$

(3) $\frac{2}{5}x^{\frac{5}{2}}+\frac{2}{3}x^{\frac{3}{2}}+2x^{\frac{1}{2}}+C$

(4) $\frac{2}{9}x^{\frac{9}{2}}+\frac{6}{5}x^{\frac{5}{2}}+2x^{\frac{1}{2}}+C$

Sol. Answer (4)

$$\int \frac{x^4+3x^2+1}{\sqrt{x}}dx = \int\left(x^{\frac{7}{2}}+3x^{\frac{3}{2}}+x^{-\frac{1}{2}}\right)dx = \frac{2}{9}x^{\frac{9}{2}}+\frac{6}{5}x^{\frac{5}{2}}+2x^{\frac{1}{2}}+C$$

4. $\int(2^x-4\cos x+2)dx$ equals

(1) $2^x\log_2 e-4\sin x+2x+C$

(2) $2^x+4\sin x+2x+C$

(3) $2^x\log_e 2+4\sin x+2x+C$

(4) $e^x\log_2 e+4\sin x+2x+C$

Sol. Answer (1)

$$\int(2^x-4\cos x+2)dx$$

$$= \int 2^x dx-4\int\cos x\, dx+2\int dx$$

$$= \frac{2^x}{\log_e 2}-4(\sin x)+2x+C$$

$$= 2^x\log_2 e-4\sin x+2x+C$$

5. $\int(\tan x+\cot x)^2dx$ equals

(1) $\tan x+\cot x+C$

(2) $\sec x+\operatorname{cosec} x+C$

(3) $\tan x-\cot x+C$

(4) $\sec x-\operatorname{cosec} x+C$

Sol. Answer (3)

$$\int(\tan x+\cot x)^2dx$$

$$= \int(\tan^2 x+\cot^2 x+2)dx$$

$$= \int(\sec^2 x-1+\operatorname{cosec}^2 x-1+2)dx$$

$$= \int(\sec^2 x+\operatorname{cosec}^2 x)dx$$

$$= \int\sec^2 x\, dx+\int\operatorname{cosec}^2 x\, dx$$

$$= \tan x-\cot x+C$$

6. $\int \sec^2(3-2x)dx$ equals

(1) $\frac{1}{2}\tan(2x-3)+C$ (2) $\frac{1}{2}\sec(2x-3)+C$ (3) $\frac{1}{2}\cos(3-2x)+C$ (4) $\frac{1}{2}\tan(3-2x)+C$

Sol. Answer (1)

$\int \sec^2(3-2x)dx$ Substituting $3 - 2x = t$

we get, $-2dx = dt$

$= -\frac{1}{2}\int \sec^2 t\, dt$

$= -\frac{1}{2}\tan t + C$

$= -\frac{1}{2}\tan(3-2x)+C$

$= \frac{1}{2}\tan(2x-3)+C$

7. $\int\left(\frac{\sec x}{\sec x - \tan x}\right)dx$ equals

(1) $\sec x - \tan x + C$ (2) $\log|\sec x + \tan x| + C$

(3) $\log|\sec x - \tan x| + C$ (4) $\sec x + \tan x + C$

Sol. Answer (4)

$\int\left(\frac{\sec x}{\sec x - \tan x}\right)dx$

$= \int \frac{\sec x(\sec x + \tan x)}{\sec^2 x - \tan^2 x}dx$

$= \int \sec x(\sec x + \tan x)dx$

$= \int \sec^2 x\, dx + \int \sec x \tan x\, dx$

$= \tan x + \sec x + C$

8. If $f'(x) = \sqrt{x}$ and $f(1) = 2$ then $f(x)$ is equal to

(1) $\frac{3}{2}x^{\frac{3}{2}}$ (2) $\frac{3}{2}x^{\frac{3}{2}}+\frac{4}{3}$ (3) $\frac{2}{3}x^{\frac{3}{2}}$ (4) $\frac{2}{3}x^{\frac{3}{2}}+\frac{4}{3}$

Sol. Answer (4)

$$f'(x) = \sqrt{x} \quad \Rightarrow \quad f(x) = \int \sqrt{x}\,dx = \frac{2}{3}x^{\frac{3}{2}} + C$$

$$f(1) = 2 \quad \Rightarrow \quad 2 = \frac{2}{3} + C \quad \Rightarrow \quad C = \frac{4}{3}$$

$$\therefore \quad f(x) = \frac{2}{3}x^{\frac{3}{2}} + \frac{4}{3}$$

9. $\int f(x)dx = 2(f(x))^3 + C$, and $f(0) = 0$ then $f(x)$ is

(1) $\frac{x}{2}$ (2) $\sqrt{\frac{x}{3}}$ (3) $2\sqrt{\frac{x}{3}}$ (4) $\frac{x^2}{2}$

Sol. Answer (2)

$$\int f(x)dx = 2(f(x))^3 + C$$

Differentiating both sides with respect to x

$$f(x) = 6(f(x))^2 \cdot f'(x)$$

or $f(x) \cdot f'(x) = \frac{1}{6}$

Hence, integrating both sides,

$$\int f(x)f'(x)dx = \int \frac{1}{6}dx$$

or $\frac{1}{2}(f(x))^2 = \frac{1}{6}x + C'$

or $f(x) = \sqrt{\frac{x}{3} + C}$

$f(0) = 0 \quad \Rightarrow \quad 0 = \sqrt{C} \quad$ *i.e.*, $C = 0$

Hence, $f(x) = \sqrt{\frac{x}{3}}$

10. If $f(0) = f'(0) = 0$ and $f''(x) = \tan^2 x$, then $f(x)$ is

(1) $\log|\sec x| - \frac{1}{2}x^2$ (2) $\log \cos x + \frac{1}{2}x^2$ (3) $\log|\sec x| + \frac{1}{2}x^2$ (4) $x^4 + x^3 + 1$

Sol. Answer (1)

$$f''(x) = \tan^2 x$$

Integrating both sides w.r.t. x, we get,

$$\int f''(x)dx = \int \tan^2 x\,dx$$

or $f'(x) = \int (\sec^2 x - 1)dx$

$= \tan x - x + C$

Given, $f'(0) = 0$

Hence, $0 = \tan 0 - 0 + C$

i.e., $C = 0$

Hence, $f'(x) = \tan x - x$

Integrating both sides w.r.t. x once more, we get,

$$\int f'(x)dx = \int (\tan x - x)dx$$

or $f(x) = \log \sec x - \dfrac{x^2}{2} + C$

Given, $f(0) = 0$

Hence, $0 = \log(\sec 0) - \dfrac{0^2}{2} + C$

or $C = 0$

Hence, $f(x) = \log \sec x - \dfrac{x^2}{2}$

11. $\int \left(\cos\dfrac{x}{2} - \sin\dfrac{x}{2}\right)^2 dx$ equals

(1) $x + \cos x + C$

(2) $2\cos^2 \dfrac{x}{2} + C$

(3) $\dfrac{1}{3}\left(\cos\dfrac{x}{2} - \dfrac{x}{2}\right)^3 + C$

(4) $x - \cos x + C$

Sol. Answer (1)

$$\int \left(\cos\frac{x}{2} - \sin\frac{x}{2}\right)^2 dx$$

$$= \int \left(\cos^2\frac{x}{2} + \sin^2\frac{x}{2} - 2\sin\frac{x}{2}\cos\frac{x}{2}\right) dx$$

$$= \int (1 - \sin x)dx$$

$= x + \cos x + C$

12. $\int \frac{dx}{\sqrt{x+a}+\sqrt{x+b}}$ equals

(1) $\frac{2}{3(a-b)}\left\{(x+a)^{\frac{3}{2}}+(x+b)^{\frac{3}{2}}\right\}+C$

(2) $\frac{2}{3(a-b)}\left\{(x+a)^{\frac{3}{2}}-(x+b)^{\frac{3}{2}}\right\}+C$

(3) $3\left\{(x+a)^{\frac{3}{2}}+(x+b)^{\frac{3}{2}}\right\}+C$

(4) $(x+a)^2+(x+b)^2+C$

Sol. Answer (2)

$$\int \frac{dx}{\sqrt{x+a}+\sqrt{x+b}}$$

$$= \int \frac{\sqrt{x+a}-\sqrt{x+b}}{\{(x+a)-(x+b)\}}dx$$

$$= \frac{1}{a-b}\int(\sqrt{x+a}-\sqrt{x+b})dx$$

$$= \frac{1}{(a-b)}\cdot\frac{2}{3}\left\{(x+a)^{\frac{3}{2}}-(x+b)^{\frac{3}{2}}\right\}+C$$

$$= \frac{2}{3(a-b)}\left\{(x+a)^{\frac{3}{2}}-(x+b)^{\frac{3}{2}}\right\}+C$$

13. If $\int f(x)dx = x\cos\pi x + C$, then $f\left(\frac{1}{2}\right)$ is equal to

(1) 0 (2) π (3) $-\frac{\pi}{2}$ (4) $\frac{\pi}{2}$

Sol. Answer (3)

$$\int f(x)dx = x\cos\pi x + C$$

or $f(x) = \cos\pi x - \pi x\sin\pi x$

$$f\left(\frac{1}{2}\right) = \cos\frac{\pi}{2}-\frac{\pi}{2}\sin\frac{\pi}{2} = -\frac{\pi}{2}$$

14. $\int\sqrt{1+2\cot x(\cot x+\operatorname{cosec} x)}\,dx$ equals

(1) $2\ln\cos\frac{x}{2}+C$ (2) $2\ln\sin\frac{x}{2}+C$ (3) $\frac{1}{2}\ln\cos\frac{x}{2}+C$ (4) $\frac{1}{2}\ln\sin\frac{x}{2}+C$

Sol. Answer (2)

$$\int\sqrt{1+2\cot^2 x+2\cot x\,\text{cosec}\,x}\;dx$$

$$=\int\sqrt{\text{cosec}^2 x-\cot^2 x+2\cot^2 x+2\cot x\,\text{cosec}\,x}\;dx$$

$$=\int\sqrt{(\text{cosec}\,x+\cot x)^2}\;dx=\int|\text{cosec}\,x+\cot x|\;dx$$

$$=\int\left|\frac{1}{\sin x}+\frac{\cos x}{\sin x}\right|dx=\int\cot x/2\;dx=2\log\sin x/2+C$$

15. $\int\{1+2\tan x(\tan x+\sec x)\}^{1/2}\,dx$ is equal to

(1) $\log\left|\sec x\cdot\cot\left(\frac{x}{2}-\frac{\pi}{4}\right)\right|+C$

(2) $\log\left|\cos x\cdot\cot\left(\frac{x}{2}+\frac{\pi}{4}\right)\right|+C$

(3) $-\log\left|\sin x\cdot\cot\left(\frac{x}{2}-\frac{\pi}{4}\right)\right|+C$

(4) $\log\left|\text{cosec}\,x\cdot\cot\left(\frac{x}{2}-\frac{\pi}{4}\right)\right|+C$

Sol. Answer (1)

$$\int\{1+2\tan x\,(\tan x+\sec x)\}^{1/2}\,dx$$

$$=\int\{1+2\tan^2 x+2\tan x\sec x\}^{1/2}\,dx$$

$$=\int\{\tan^2 x+2\tan x\sec x+\sec^2 x\}^{1/2}\,dx$$

$$=\int(\sec x+\tan x)^{2\cdot\frac{1}{2}}\,dx$$

$$=\int(\sec x+\tan x)\,dx$$

$$=\left[\log|\sec x|+\log\tan\left|\frac{\pi}{4}+\frac{x}{2}\right|\right]+C$$

$$=\left[\log|\sec x|+\log\left|\cot\left(\frac{\pi}{2}-\frac{\pi}{4}-\frac{x}{2}\right)\right|\right]+C$$

$$=\left[\log|\sec x|+\log\left|\cot\left(\frac{\pi}{4}-\frac{x}{2}\right)\right|\right]+C$$

$$\log\left|\sec x\cot\left(\frac{\pi}{4}-\frac{x}{2}\right)\right|=\log\sec x\cot\left(\frac{x}{2}-\frac{\pi}{4}\right)+C$$

[Integration by Substitution]

16. If $\int \sin 3x \cos 5x \, dx = 0$ when $x = 0$, then the value of constant of integration

(1) $\frac{1}{16}$ (2) $-\frac{1}{16}$ (3) $\frac{3}{16}$ (4) $-\frac{3}{16}$

Sol. Answer (4)

$$\int \sin 3x \cos 5x \, dx$$

$$= \int \frac{1}{2}(\sin 8x + \sin(-2x))dx$$

$$= \frac{1}{2}\left\{\int \sin 8x \, dx - \int \sin 2x \, dx\right\}$$

$$= \frac{1}{2}\left\{-\frac{1}{8}\cos 8x + \frac{1}{2}\cos 2x\right\} + C$$

when, $x = 0$

$$\frac{1}{2}\left\{-\frac{1}{8}\cos 0 + \frac{1}{2}\cos 0\right\} + C = 0$$

i.e., $C = -\frac{3}{16}$

17. $\int \frac{dx}{x\left(1+(\log x)^2\right)}$ equals

(1) $\log(\tan^{-1} x) + C$ (2) $\log(\tan^{-1} x) + \tan^{-1}(\log x) + C$

(3) $\tan^{-1}(\log(\tan^{-1} x)) + C$ (4) $\tan^{-1}(\log x) + C$

Sol. Answer (4)

$$\int \frac{dx}{x\{1+(\log x)^2\}}$$ Substituting $\log x = t$

$$= \int \frac{dt}{1+t^2} = \tan^{-1} t + C$$ $\frac{1}{x}dx = dt$

$$= \tan^{-1}(\log x) + C$$

18. $\int \frac{e^{\cos^{-1} x}}{\sqrt{1-x^2}} dx$ equals

(1) $-e^{\sin^{-1} x} + C$ (2) $e^{\tan^{-1} x} + C$ (3) $-e^{\cos^{-1} x} + C$ (4) $e^{\sec^{-1} x} + C$

Sol. Answer (3)

$\int \frac{e^{\cos^{-1} x}}{\sqrt{1-x^2}} dx$ Substituting $\cos^{-1} x = t$

$= -\int e^t dt$ we get, $-\frac{1}{\sqrt{1-x^2}} dx = dt$

$= -e^t + C$

$= -e^{\cos^{-1} x} + C$

19. $\int \frac{x^2\, dx}{(a+bx)^2}$ equals

(1) $\frac{1}{b^2}\left[x + \frac{2a}{b}\log|(a+bx)| - \frac{a^2}{b}\frac{1}{(a+bx)}\right] + C$ (2) $\frac{1}{b^2}\left[x - \frac{2a}{b}\log|(a+bx)| + \frac{a^2}{b}\frac{1}{(a+bx)}\right] + C$

(3) $\frac{1}{b^2}\left[x + \frac{a}{b} - \frac{2a}{b}\log|(a+bx)| - \frac{a^2}{b}\frac{1}{(a+bx)}\right] + C$ (4) $\frac{1}{b^2}\left[x + \frac{a}{b} + \frac{2a}{b}\log|(a+bx)| + \frac{a^2}{b}\frac{1}{(a+bx)}\right] + C$

Sol. Answer (3)

$\int \frac{x^2 dx}{(a+bx)^2}$ Substituting $a + bx = t$

we get, $b\, dx = dt$

also, $x = \frac{t-a}{b}$

$= \int \left(\frac{t-a}{b}\right)^2 \cdot \frac{1}{t^2}\frac{dt}{b}$

$= \frac{1}{b^3}\int \left(\frac{t^2 - 2at + a^2}{t^2}\right) dt$

$= \frac{1}{b^3}\int \left(1 - \frac{2a}{t} + \frac{a^2}{t^2}\right) dt$

$= \frac{1}{b^3}\left\{t - 2a\log|t| - \frac{a^2}{t}\right\} + C$

$$= \frac{1}{b^3}\left\{a+bx-2a\log|(a+bx)|-\frac{a^2}{a+bx}\right\}+C$$

$$= \frac{1}{b^2}\left\{x+\frac{a}{b}-\frac{2a}{b}\log|(a+bx)|-\frac{a^2}{b}\cdot\frac{1}{(a+bx)}\right\}+C$$

20. $\int \frac{a^{\sqrt{x}}}{\sqrt{x}}dx$ equals

(1) $2a^{\sqrt{x}}\log_e a+C$ (2) $2a^{\sqrt{x}}\log_a e+C$ (3) $2a^{\sqrt{x}}\log_{10} a+C$ (4) $2a^{\sqrt{x}}\log_a 10+C$

Sol. Answer (2)

$\int \frac{a^{\sqrt{x}}}{\sqrt{x}}dx$ Substituting $\sqrt{x}=t$

$$\frac{1}{2}\cdot\frac{1}{\sqrt{x}}dx=dt$$

$$= 2\int a^{\sqrt{x}}\left(\frac{1}{2\sqrt{x}}dx\right)$$

$$= 2\int a^t\,dt = \frac{2a^t}{\log_e a}+C$$

$$= 2a^{\sqrt{x}}\log_a e+C$$

21. $\int x\frac{e^{\tan^{-1}x^2}}{x^4+1}dx$ equals

(1) $e^{x^2}\tan^{-1}x^2+C$ (2) $\frac{1}{2}e^{\tan^{-1}x^2}+C$ (3) $\log(\tan^{-1}x^2)+C$ (4) $\frac{x^2}{2}\log(x^2)+C$

Sol. Answer (2)

$\int \frac{xe^{\tan^{-1}x^2}}{x^4+1}dx$ Substituting $\tan^{-1}x^2 = t$

$$= \frac{1}{2}\int e^{\tan^{-1}x^2}\cdot\frac{2x}{1+(x^2)^2}dx \qquad \frac{2x}{1+x^4}dx=dt$$

$$= \frac{1}{2}\int e^t\,dt = \frac{1}{2}e^t+C$$

$$= \frac{1}{2}e^{\tan^{-1}x^2}+C$$

22. If $f(x)=\int\frac{(x^2+\sin^2 x)}{1+x^2}\sec^2 x\,dx$ and $f(0) = 0$, then $f(1)$ is equal to

(1) $1-\frac{\pi}{4}$ (2) $\frac{\pi}{4}-1$ (3) $\tan 1-\frac{\pi}{4}$ (4) $\frac{\pi}{4}$

Sol. Answer (3)

$$f(x) = \int\frac{(x^2+\sin^2 x)}{1+x^2}\sec^2 x\,dx$$

$$= \int\left\{\frac{x^2+(1-\cos^2 x)}{1+x^2}\right\}\sec^2 x\,dx$$

$$= \int\left\{\sec^2 x-\frac{1}{1+x^2}\right\}dx\,]$$

$$= \tan x-\tan^{-1}x+C$$

$\because$ $f(0) = 0$

$\therefore$ $C = 0$

$\Rightarrow$ $f(x) = \tan x - \tan^{-1} x$

Hence, $f(1) = \tan 1 - \tan^{-1} 1$

$$= \tan 1-\frac{\pi}{4}$$

23. If $\int f(x)dx = f(x)$, then $\int\{f(x)\}^2 dx$ is equal to

(1) $\frac{1}{2}\{f(x)\}^2+C$ (2) $\{f(x)\}^2+C$ (3) $\frac{1}{3}\{f(x)\}^2+C$ (4) $\frac{1}{5}\{f(x)\}^2+C$

Sol. Answer (1)

$$\int f(x)\,dx = f(x)$$

$$\Rightarrow \frac{d}{dx}[f(x)] = f(x)$$

$$\Rightarrow \int\frac{1}{f(x)}d\{f(x)\} = \int dx$$

$$\Rightarrow \log|f(x)| = x+\log C$$

$$\Rightarrow f(x) = C_1 e^x$$

$$\therefore \int\{f(x)\}^2 dx = \int C_1^2 e^{2x}dx = \frac{C_1^2 e^{2x}}{2}+C = \frac{1}{2}\{f(x)\}^2+C$$

24. $\int \frac{nx^{n-1} + m^x \log m}{x^n + m^x} dx$ equals

(1) $m^x - x^n + C$ (2) $m^x + x^n + C$ (3) $(m^x - x^n)^{-1} + C$ (4) $\log(m^x + x^n) + C$

Sol. Answer (4)

Let $t = x^n + m^x$

then, $dt = (nx^{n-1} + m^x \log m)\, dx$

Hence, $\int \frac{nx^{n-1} + m^x \log m}{x^n + m^x} dx$

$= \int \frac{dt}{t} = \log|t| + C$

$= \log|x^n + m^x| + C$

25. Let $\int x^6 \sin(5x^7) dx = \frac{k}{5}\cos(5x^7) + C_1,\ x \neq 0,$ and $\int \frac{dx}{7x+14} = m \log|7x+14| + C_2$ then $k + m$ equals

(1) 0 (2) $\frac{1}{2}$ (3) $-\frac{1}{2}$ (4) 3

Sol. Answer (1)

$\int x^6 \sin(5x^7) dx$

$= \frac{1}{35}\int 35x^6 \sin(5x^7) dx$

$= \frac{1}{35}\int \sin(5x^7)\, d(5x^7)$

$= -\frac{1}{35}\cos(5x^7) + C_1$

i.e., $\frac{k}{5} = -\frac{1}{35}$

or $k = -\frac{1}{7}$

$\int \frac{dx}{7x+14} = \frac{1}{7}\log|7x+14| + C_2$

Hence, $m = \frac{1}{7}$

i.e., $k + m = -\frac{1}{7} + \frac{1}{7} = 0$

26. If $\int f(x)dx = F(x)$, $f(x)$ is a continuous function, then $\int \frac{f(x)}{F(x)}dx$ equals

(1) $\log_e|f(x)| + C$ (2) $\log_e|F(x)| + C$ (3) $F(x) + C$ (4) $(f(x))^2 + C$

Sol. Answer (2)

$$\int f(x)dx = F(x)$$

Hence, $\frac{d}{dx}(F(x)) = f(x)$

$$\therefore \quad \int \frac{f(x)}{F(x)}dx = \int \frac{d(F(x))}{F(x)}$$

$$= \log_e|F(x)| + C$$

27. $\int \frac{\sin 2x}{2\cos^2 x + 3\sin^2 x}dx$ equals

(1) $\log(2 + \sin x) + C$ (2) $\log(2 + \cos^2 x) + C$

(3) $\log(2 + \sin^2 x) + C$ (4) $\log(2 + \cos x) + C$

Sol. Answer (3)

$$\int \frac{\sin 2x}{2\cos^2 x + 3\sin^2 x}dx$$

$$= \int \frac{2\sin x \cos x}{2 - 2\sin^2 x + 3\sin^2 x}dx$$

$$= 2\int \frac{\sin x \cos x}{2 + \sin^2 x}dx$$

Substituting $2 + \sin^2 x = t$

we get $2\sin x \cos x\, dx = dt$

$$= \int \frac{dt}{t} = \log|t| + C$$

$$= \log|2 + \sin^2 x| + C$$

28. $\int \frac{x\,dx}{\sqrt{x^4 - 1}}$ equals to

(1) $\frac{1}{2}\log|x^2 - \sqrt{x^4 - 1}| + C$ (2) $\frac{1}{2}\log|x^2 + \sqrt{x^4 - 1}| + C$

(3) $-\frac{1}{2}\log|x^2 + \sqrt{x^4 + 1}| + C$ (4) $\log|x^4 + 1| + C$

Sol. Answer (2)

$$\int \frac{x\,dx}{\sqrt{x^4-1}} = \int \frac{dx}{x\sqrt{1-\frac{1}{x^4}}}$$

Substituting $\frac{1}{x^2} = \sin\theta$

we get, $-2x^{-3}\,dx = \cos\theta\,d\theta$

i.e., $dx = -\frac{1}{2}\operatorname{cosec}^{\frac{3}{2}}\theta\cos\theta\,d\theta$

$x^2 = \operatorname{cosec}\theta$

$$= -\frac{1}{2}\int \frac{\sqrt{\sin\theta}\operatorname{cosec}^{\frac{3}{2}}\theta\cos\theta\,d\theta}{\sqrt{1-\sin^2\theta}}$$

$$= -\frac{1}{2}\int \operatorname{cosec}\theta\,d\theta$$

$$= -\frac{1}{2}\log|\operatorname{cosec}\theta - \cot\theta| + C$$

$$= -\frac{1}{2}\log\left|\frac{\operatorname{cosec}^2\theta - \cot^2\theta}{\operatorname{cosec}\theta + \cot\theta}\right| + C$$

$$= \frac{1}{2}\log|\operatorname{cosec}\theta + \cot\theta| + C$$

$$= \frac{1}{2}\log|x^2 + \sqrt{x^4-1}| + C$$

Alternately :

$$\int \frac{\frac{1}{x^5}}{\frac{1}{x^4}\sqrt{1-\frac{1}{x^4}}}\,dx$$

Let $1 - \frac{1}{x^4} = t^2$

$\Rightarrow \frac{4}{x^5}dx = 2t\,dt$

$$\int \frac{1}{2}\frac{t\,dt}{(1-t^2)t} = \frac{1}{2\times 2}\log\left|\frac{1+t}{1-t}\right| + C = \frac{1}{4}\log\left|\frac{1+\sqrt{1-\frac{1}{x^4}}}{1-\sqrt{1-\frac{1}{x^4}}}\right| + C = \frac{1}{4}\log\left|\frac{x^2+\sqrt{x^4-1}}{x^2-\sqrt{x^4-1}}\right| + C$$

$$= \frac{1}{4}\log\left|\frac{(x^2+\sqrt{x^4-1})^2}{x^4-x^4+1}\right| + C = \frac{1}{2}\log\left|x^2+\sqrt{x^4-1}\right| + C$$

29. $\int \sec x^{\circ}\, dx$ equals to

(1) $\log|\sec x^{\circ}+\tan x^{\circ}|+C$

(2) $\frac{\pi}{180^{\circ}}\log\tan\left(\frac{\pi}{4}+\frac{x}{2}\right)+C$

(3) $\frac{180^{\circ}}{\pi}\log\tan\left(\frac{\pi}{4}+\frac{x}{3}\right)+C$

(4) $\frac{180^{\circ}}{\pi}\log\tan\left(\frac{\pi}{4}+\frac{\pi x}{360^{\circ}}\right)+C$

Sol. Answer (4)

$$\int \sec x^{\circ}\, dx = \int \sec\frac{\pi x}{180^{\circ}}\, dx \qquad \text{Substituting } \frac{\pi x}{180^{\circ}} = t$$

$$= \frac{180^{\circ}}{\pi}\int \sec t\, dt \qquad \text{we get, } \frac{\pi}{180^{\circ}}dx = dt$$

$$\int \sec x^{\circ}\, dx = \frac{180^{\circ}}{\pi}\log|\sec t+\tan t|+C$$

$$= \frac{180^{\circ}}{\pi}\log\left|\frac{1+\sin t}{\cos t}\right|+C$$

$$= \frac{180^{\circ}}{\pi}\log\left|\frac{\cos^2\frac{t}{2}+\sin^2\frac{t}{2}+2\sin\frac{t}{2}\cos\frac{t}{2}}{\cos^2\frac{t}{2}-\sin^2\frac{t}{2}}\right|+C$$

$$= \frac{180^{\circ}}{\pi}\log\left|\frac{\left(\cos\frac{t}{2}+\sin\frac{t}{2}\right)^2}{\left(\cos\frac{t}{2}-\sin\frac{t}{2}\right)\left(\cos\frac{t}{2}+\sin\frac{t}{2}\right)}\right|+C$$

$$= \frac{180^{\circ}}{\pi}\log\left|\frac{\cos\frac{t}{2}+\sin\frac{t}{2}}{\cos\frac{t}{2}-\sin\frac{t}{2}}\right|+C$$

$$= \frac{180^{\circ}}{\pi}\log\left|\frac{1+\tan\frac{t}{2}}{1-\tan\frac{t}{2}}\right|+C$$

$$= \frac{180^{\circ}}{\pi}\log\left|\tan\left(\frac{\pi}{4}+\frac{t}{2}\right)\right|+C$$

$$= \frac{180^{\circ}}{\pi}\log\left|\tan\left(\frac{\pi}{4}+\frac{\pi x}{360^{\circ}}\right)\right|+C$$

30. $\int \frac{x+1}{x(1+xe^x)^2}dx$ equals

(1) $\frac{1}{1+xe^x}+C$

(2) $\log\left(\frac{xe^x}{1+xe^x}\right)+C$

(3) $\frac{1}{1+xe^x}+\log\left(\frac{xe^x}{1+xe^x}\right)+C$

(4) $x(e^x + 1) + C$

Sol. Answer (3)

Put $(1 + xe^x) = t$

we get, $(e^x + xe^x)\,dx = dt$

i.e., $\frac{(1+x)dx}{x} = \frac{dt}{(t-1)}$

$$\int \frac{x+1}{x(1+xe^x)^2}dx = \int \frac{dt}{t^2(t-1)}$$

$$= \int\left(-\frac{1}{t^2}-\frac{1}{t}+\frac{1}{t-1}\right)dt$$

$$= \frac{1}{1+xe^x}+\log\left|\frac{xe^x}{1+xe^x}\right|+C$$

31. $\int \frac{x^2+1}{x^4+1}dx$ equals

(1) $\frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{x^2-1}{\sqrt{2}x}\right)+C$

(2) $\frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{1-x^2}{\sqrt{2}x}\right)+C$

(3) $\frac{1}{2}\tan^{-1}\left(\frac{x^2-1}{2x}\right)+C$

(4) $\frac{1}{2}\tan^{-1}\left(\frac{1-x^2}{2x}\right)+C$

Sol. Answer (1)

Substituting $x^2 = t$

we get, $2x\,dx = dt$

Also $x = \sqrt{t}$

Hence, $\int \frac{x^2+1}{x^4+1}dx$

$$= \int \frac{1+\frac{1}{x^2}}{x^2+\frac{1}{x^2}}dx$$

$$= \int \frac{1+\frac{1}{x^2}}{\left(x-\frac{1}{x}\right)^2+2}dx$$

$$= \frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{x-\frac{1}{x}}{\sqrt{2}}\right)+C$$

$$= \frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{x^2-1}{\sqrt{2}x}\right)+C$$

32. If $\int \frac{e^x-1}{e^x+1}dx = f(x)+C$, then $f(x)$ is equal to

(1) $2 \log |e^x + 1|$

(2) $\log |e^{2x} - 1|$

(3) $2 \log (e^x + 1) - x$

(4) $\log (e^{2x} + 1)$

Sol. Answer (3)

$$\int \frac{e^x-1}{e^x+1}dx$$

$$= \int\left(\frac{2e^x}{e^x+1}-1\right)dx$$

$$= 2\int \frac{e^x dx}{e^x+1} - \int dx$$

$$= 2\log(e^x+1) - x + C$$

Hence, $f(x) = 2\log(e^x+1) - x$

33. $\int \frac{dx}{e^x+e^{-x}}$ equals

(1) $\tan^{-1}(e^{-x})+C$

(2) $\tan^{-1}(e^x)+C$

(3) $\log|e^x - e^{-x}|+C$

(4) $\log(e^x+e^{-x})+C$

Sol. Answer (2)

$$\int \frac{dx}{e^x+e^{-x}} = \int \frac{e^x dx}{e^{2x}+1}$$

$$= \int \frac{e^x dx}{1+(e^x)^2}$$

$$= \tan^{-1}(e^x)+C$$

34. $\int \cos\sqrt{x}\,dx$ equals

(1) $2\left[\sqrt{x}\sin\sqrt{x}+\cos\sqrt{x}\right]+C$

(2) $2\left[\sqrt{x}\sin\sqrt{x}-\cos\sqrt{x}\right]+C$

(3) $2\left[\cos\sqrt{x}-\sqrt{x}\sin\sqrt{x}\right]+C$

(4) $-2\left[\sqrt{x}\sin\sqrt{x}+\cos\sqrt{x}\right]+C$

Sol. Answer (1)

Put $\sqrt{x}=t \Rightarrow \frac{1}{2\sqrt{x}}dx = dt \Rightarrow dx = 2t\cdot dt$

we get $\int \cos\sqrt{x}\,dx = \int 2t\cdot\cos t\,dt$

$= 2\left\{t\sin t - \int \sin t\,dt\right\}$

$= 2t\sin t + 2\cos t$

$= 2\{\sqrt{x}\sin\sqrt{x}+\cos\sqrt{x}\}+C$

35. $\int \frac{dx}{\cos x+\sin x}$ is equal to

(1) $\frac{1}{\sqrt{2}}\log\tan\left(\frac{x}{2}+\frac{\pi}{8}\right)+C$

(2) $\frac{1}{\sqrt{2}}\log\left[\operatorname{cosec}\left(\frac{x}{2}+\frac{\pi}{8}\right)+\cot\left(\frac{x}{2}+\frac{\pi}{8}\right)\right]+C$

(3) $-\frac{1}{\sqrt{2}}\log\left[\operatorname{cosec}\left(\frac{x}{2}+\frac{\pi}{8}\right)-\cot\left(\frac{x}{2}+\frac{\pi}{8}\right)\right]+C$

(4) $-\frac{1}{\sqrt{2}}\log\left[\operatorname{cosec}\left(\frac{x}{2}-\frac{\pi}{8}\right)+\cot\left(\frac{x}{2}-\frac{\pi}{8}\right)\right]+C$

Sol. Answer (1)

$$\int\frac{dx}{\cos x+\sin x} = \frac{1}{\sqrt{2}}\int\frac{dx}{\frac{1}{\sqrt{2}}\cos x+\frac{1}{\sqrt{2}}\sin x}$$

$$=\frac{1}{\sqrt{2}}\int\frac{dx}{\sin\frac{\pi}{4}\cos x+\cos\frac{\pi}{4}\sin x} = \frac{1}{\sqrt{2}}\int\frac{dx}{\sin\left(x+\frac{\pi}{4}\right)}$$

$$=\frac{1}{\sqrt{2}}\int\operatorname{cosec}\left(x+\frac{\pi}{4}\right)dx = \frac{1}{\sqrt{2}}\log\tan\left(\frac{x}{2}+\frac{\pi}{8}\right)+C$$

36. $\int\frac{dx}{\sqrt{2-3x-x^2}}$ equals

(1) $\tan^{-1}\left(\frac{2x+3}{\sqrt{17}}\right)+C$

(2) $\sec^{-1}\left(\frac{2x+3}{\sqrt{17}}\right)+C$

(3) $\sin^{-1}\left(\frac{2x+3}{\sqrt{17}}\right)+C$

(4) $\cos^{-1}\left(\frac{2x+3}{\sqrt{17}}\right)+C$

Sol. Answer (3)

$$\int \frac{dx}{\sqrt{2-3x-x^2}}$$

$$= \int \frac{dx}{\sqrt{2-\left(x^2+3x+\frac{9}{4}\right)+\frac{9}{4}}}$$

$$= \int \frac{dx}{\sqrt{\left(\sqrt{\frac{17}{4}}\right)^2-\left(x+\frac{3}{2}\right)^2}}$$

$$= \sin^{-1}\left(\frac{x+\frac{3}{2}}{\sqrt{\frac{17}{4}}}\right)+C$$

$$= \sin^{-1}\left(\frac{2x+3}{\sqrt{17}}\right)+C$$

37. $\int \frac{(x+x^3)^{1/3}}{x^4}\,dx$ is equal to

(1) $\frac{3}{8}\left(\frac{1}{x^2}-1\right)^{4/3}+C$

(2) $-\frac{3}{8}\left[\left(1+\frac{1}{x^2}\right)^{4/3}\right]+C$

(3) $\frac{1}{8}\left(1+\frac{1}{x^2}\right)^{4/3}+C$

(4) $\frac{1}{8}\left(\frac{1}{x^2}-1\right)^{4/3}+C$

Sol. Answer (2)

$$\int \frac{(x+x^3)^{1/3}}{x^4}\,dx = \int \frac{\left(1+\frac{1}{x^2}\right)^{1/3} dx}{x^3} \quad \text{Put } 1+\frac{1}{x^2}=t \text{ then } \frac{-2}{x^3}\,dx = dt$$

$$= \frac{-1}{2}\int t^{1/3}\,dt = \frac{-1}{2}\left[\frac{t^{4/3}}{4/3}\right]+C = \frac{-3}{8}t^{4/3}+C = \frac{-3}{8}\left[\left(1+\frac{1}{x^2}\right)^{4/3}\right]+C$$

38. Primitive of $\frac{(3x^4-1)}{(x^4+x+1)^2}$ w.r.t. x is

(1) $\frac{x}{x^4+x+1}+C$

(2) $\frac{-x}{x^4+x+1}+C$

(3) $\frac{x+1}{x^4+x+1}+C$

(4) $\frac{-x-1}{x^4+x+1}+C$

Sol. Answer (2)

$$\int\frac{3x^4-1}{(x^4+x+1)^2}\,dx \qquad \frac{d}{dx}\left(\frac{x}{x^4+x+1}\right)=\frac{-3x^4+1}{(x^4+x+1)^2}$$

$$\int\frac{d}{dx}\frac{x\,dx}{(x^4+x+1)^2}=\int\frac{-3x^4+1}{(x^4+x+1)^2}\,dx$$

$$\int\frac{3x^4-1}{(x^4+x+1)^2}\,dx=\frac{-x}{x^4+x+1}+C$$

39. $\displaystyle\int\frac{(2x+1)}{(x^2+4x+1)^{\frac{3}{2}}}dx$

(1) $\dfrac{x^3}{(x^2+4x+1)^{1/2}}+C$ (2) $\dfrac{x}{(x^2+4x+1)^{1/2}}+C$

(3) $\dfrac{x^2}{(x^2+4x+1)^{1/2}}+C$ (4) $\dfrac{1}{(x^2+4x+1)^{1/2}}+C$

Sol. Answer (2)

$$\int\frac{(2x+1)}{(x^2+4x+1)^{3/2}}\,dx$$

Since, $\displaystyle\frac{d}{dx}\left(\frac{x}{(x^2+4x+1)^{1/2}}\right)=\frac{(x^2+4x+1)^{1/2}\cdot 1-\frac{x}{2}\cdot\frac{(2x+4)}{(x^2+4x+1)^{1/2}}}{(x^2+4x+1)}$

$$=\frac{(x^2+4x+1)-(2x^2+4x)\cdot\frac{1}{2}}{(x^2+4x+1)\sqrt{x^2+4x+1}}$$

$$=\frac{(x^2+4x+1)-(x^2+2x)}{(x^2+4x+1)^{3/2}}$$

$$=\frac{2x+1}{(x^2+4x+1)^{3/2}}$$

Therefore, $\displaystyle\int\frac{2x+1}{(x^2+4x+1)^{3/2}}dx=\frac{x}{(x^2+4x+1)^{1/2}}+C$

40. $\displaystyle\int 7^{7^{7^x}}\cdot 7^{7^x}\cdot 7^x\,dx=$

(1) $\dfrac{7^{7^{7^x}}}{(\log 7)^3}+C$ (2) $\dfrac{7^{7^{7^x}}}{(\log 7)^2}+C$ (3) $7^{7^{7^x}}\cdot(\log 7)^3+C$ (4) $7^{7^{7^x}}$

Sol. Answer (1)

$$\int 7^{7^{7^x}} \cdot 7^{7^x} \cdot 7^x \, dx$$

Put $7^{7^{7^x}} = t$, then $7^{7^{7^x}} \cdot 7^{7^x} \cdot 7^x (\log 7)^3 \, dx = dt$

$$\int \frac{dt}{(\log 7)^3} = \frac{t}{(\log 7)^3} = \frac{7^{7^{7^x}}}{(\log 7)^3} + C$$

41. If $\int f(x)dx = f(x)$, then $\int \{f(x)\}^2 \cdot x\, dx$ is equal to

(1) $\frac{x\{f(x)\}^2}{2} - \frac{\{f(x)\}^2}{4} + C$

(2) $x^2 \frac{\{f(x)\}^2}{2} - \frac{\{f(x)\}^4}{4} + C$

(3) $xf(x) - \frac{\{f(x)\}^2}{4} + C$

(4) $xf(x) + C$

Sol. Answer (1)

If $\int f(x)\, dx = f(x)$,

Put $f(x) = e^x$ $\quad\therefore\quad f(x) = f'(x)$

$f'(x) = e^x = f(x)$ $\quad\Rightarrow\quad \int \frac{f'(x)}{f(x)} dx = \int 1 \cdot dx$

then $\int \{f(x)\}^2 \cdot x \cdot dx =$

$\int (e^x)^2 \cdot x\, dx = \int e^{2x} \cdot x\, dx$ $\quad\Rightarrow\quad \log_e f(x) = x$

$= x\frac{e^{2x}}{2} - \int \frac{e^{2x}}{2} dx$ $\quad\Rightarrow\quad f(x) = e^x$

integration by parts

$$= \frac{xe^{2x}}{2} - \frac{e^{2x}}{4} = \frac{x\{f(x)\}^2}{2} - \frac{\{f(x)\}^2}{4} + C$$

42. $\int \frac{1-x^7}{x(1+x^7)} dx$ equals

(1) $\ln x + \frac{2}{7}\ln(1+x^7) + C$

(2) $\ln x - \frac{2}{7}\ln(1-x^7) + C$

(3) $\ln x - \frac{2}{7}\ln(1+x^7) + C$

(4) $\ln x + \frac{2}{7}\ln(1-x^7) + C$

Sol. Answer (3)

$$\int \frac{1-x^7}{x(1+x^7)}\,dx = -\int \frac{x^7+1-2}{(x^7+1)\,x}\,dx$$

$$= -\int \frac{1}{x}\,dx + 2\int \frac{x^6}{x^7}\,\frac{dx}{(x^7+1)} \quad \text{Let } x^7 = t,\ 7x^6\,dx = dt$$

$$= -\int \frac{1}{x}\,dx + \frac{2}{7}\int\left(\frac{1}{t} - \frac{1}{t+1}\right)dt$$

$$= -\log x + \frac{2}{7}\log\frac{x^7}{1+x^7} = -\log x + \frac{2}{7}\log x^7 - \frac{2}{7}\log(1+x^7)$$

$$= +\log x - \frac{2}{7}\log(1+x^7) + C$$

43. $\int \sin x\, d(\cos x)$ is equal to

(1) $\frac{1}{4}\sin 2x + \frac{x}{2} + c$ (2) $\frac{1}{4}\sin 2x - \frac{x}{2} + c$ (3) $2\sin 2x + c$ (4) $\sin x + \cos x + c$

Sol. Answer (2)

Consider $I = \int \sin x\, d(\cos x)$

Put $\cos x = t$ so that $d(\cos x) = dt$

And, $\sin x = \sqrt{1-t^2}$

$$\therefore\ I = \int\sqrt{1-t^2}\,dt = \frac{1}{2}\sqrt{1-t^2} + \frac{1}{2}\sin^{-1}t + c$$

$$= \frac{\cos x}{2}\cdot\sin x + \frac{1}{2}\sin^{-1}\cdot(\cos x) + c$$

$$= \frac{\sin 2x}{4} + \frac{1}{2}\sin^{-1}(\sin c\frac{1}{2} - x) + c$$

$$= \frac{\sin 2x}{4} - \frac{x}{2} + c$$

44. $\int \frac{dx}{x\cdot(\ln x)(\ln(\ln x))............\underbrace{(\ln \ln \ln ..... \ln x)}_{20\text{ times}}}$ is equal to

(1) $\underbrace{(\log\ \log\ \log........x)}_{20\text{ times}} + c$

(2) $\underbrace{(\log\ \log\ \log........x)}_{19\text{ times}} + c$

(3) $\underbrace{(\log\ \log\ \log........x)}_{21\text{ times}} + c$

(4) $\underbrace{(\log\ \log\ \log........x)}_{22\text{ times}} + c$

Sol. Answer (3)

Put $\ln\ln\underbrace{\ln.....}_{20\text{ times}}x = t$

$$\frac{1}{x}\cdot\frac{1}{\ln x}\cdot\frac{1}{(\ln\ln x)...\ln\underbrace{\ln\cdot\ln}_{19\text{ times}}...x} = dt$$

So $I = \int\frac{dt}{t} = \ln t = (\ln\underbrace{\ln\ln\ln.....x}_{21\text{ times}}) + c$

45. $\int\frac{x\sin x^2 e^{\sec x^2}}{\cos^2 x^2}dx =$

(1) $\frac{1}{2}e^{\sec x^2}+C$ (2) $\frac{1}{2}e^{\sin x^2}+C$ (3) $\frac{1}{2}\sin x^2\cdot e^{\cos^2 x^2}+C$ (4) None of these

Sol. Answer (1)

$$\int\frac{x\sin x^2 e^{\sec x^2}}{\cos^2 x^2}dx = \int x\cdot\left(\frac{\sin x^2}{\cos x^2}\right)\cdot\left(\frac{1}{\cos x^2}\right)e^{\sec x^2}dx$$

$$= \int x\cdot\tan x^2\cdot\sec x^2 e^{\sec x^2}dx$$

Put $\sec x^2 = t$

$2x\sec x^2\tan x^2\, dx = dt$

$$= \frac{1}{2}\int e^t\,dt = \frac{1}{2}e^t = \frac{1}{2}e^{\sec x^2}+C$$

46. $\int x.2^{\ln(x^2+1)}dx$ equals

(1) $\frac{2^{\ln(x^2+1)}}{2(\ln 2+1)}+C$ (2) $\frac{(x^2+1)2^{\ln(x^2+1)}}{2(\ln 2+1)}+C$ (3) $\frac{(x^2+1)^{\ln 2}}{2(\ln 2+1)}+C$ (4) $\frac{(x^2+1)^{\ln 2}}{2(\ln 2+1)}+C$

Sol. Answer (2)

$$\int x\cdot 2^{\ln(x^2+1)}dx = \frac{1}{2}\int 2^{\log y}\cdot dy = \frac{1}{2}\int 2^z\cdot e^z\,dz$$

$$= \frac{1}{2}\int(2e)^z\,dz$$

$$= \frac{1}{2}\frac{(2e)^z}{(\log 2+1)}$$

$x^2+1 = y$

$2x\,dx = dy$

$= \dfrac{2^{\log y} \cdot y}{2(\log 2 + 1)}$ $\qquad \log y = z$, $\dfrac{1}{y}dy = dz$

$= \dfrac{2^{\log(x^2+1)} \cdot (x^2+1)}{2(\log 2+1)} + C$ $\qquad dy = e^z\, dz$

47. $\displaystyle\int \frac{\ln(6x^2)}{x}dx$ equals

(1) $\dfrac{1}{8}(\ln 6x^2)^3 + C$ (2) $\dfrac{1}{4}(\ln(6x^2))^2 + C$ (3) $\dfrac{1}{2}(\ln(6x^2)) + C$ (4) $\dfrac{1}{16}(\ln(6x^2)) + C$

Sol. Answer (2)

$$\int \frac{\ln(6x^2)}{x}dx = \int \frac{x\ln(6x^2)}{x^2}dx$$

Let $6x^2 = z$

$12x\, dx = dz$

$$= \frac{1}{2} \cdot \frac{(\log z)^2}{2} + C = \frac{1}{4}(\log(6x^2))^2 + C$$

48. $\displaystyle\int \frac{d(x^2+1)}{\sqrt{x^2+2}}$ is equal to

(1) $2\sqrt{x^2+2}+c$ (2) $\sqrt{x^2+2}+c$ (3) $x\sqrt{x^2+2}+c$ (4) $x^2\sqrt{x-1}+c$

Sol. Answer (1)

We know $d(x^2+1) = 2x \cdot dx$

$$I = \int \frac{d(x^2+1)}{x^2+1} = \int \frac{2x}{\sqrt{x^2+1}}dx = 2\sqrt{x^2+1}+c$$

[Integration by Partial Fraction]

49. $\displaystyle\int \frac{dx}{x(1+4x^3+3x^6)}$ equals

(1) $\dfrac{1}{3}\log|x^3| + \dfrac{1}{6}\log|1+x^3| + C$ (2) $\dfrac{1}{3}\log|x^3| + \dfrac{1}{3}\log|1+x^3| + C$

(3) $\dfrac{1}{3}\log|x^3| + \dfrac{1}{6}\log|1+x^3| - \dfrac{1}{2}\log|1+3x^3| + C$ (4) $\log|x^3+1| + C$

Sol. Answer (3)

$$\int \frac{dx}{x(1+4x^3+3x^6)}$$

Substituting $x^3 = t$

we get, $3x^2\, dx = dt$

$$= \frac{1}{3}\int \frac{dt}{t(1+4t+3t^2)}$$

$$\frac{1}{t(1+t)(1+3t)} = \frac{A}{t}+\frac{B}{1+t}+\frac{C}{1+3t}$$

$$= \frac{1}{3}\int\left\{\frac{1}{t}+\frac{1}{2}\frac{1}{1+t}-\frac{9}{2}\frac{1}{(1+3t)}\right\}dt$$

$$= \frac{A(1+t)(1+3t)+Bt(1+3t)+Ct(1+t)}{t(1+t)(1+3t)}$$

$$= \frac{1}{3}\log|t|+\frac{1}{6}\log|1+t|-\frac{1}{2}\log|1+3t|+K$$

Hence, $\left.\begin{array}{l}3A+3B+C=0\\4A+B+C=0\\A=1\end{array}\right\} \Rightarrow A=1, B=\frac{1}{2}, C=-\frac{9}{2}$

$$= \frac{1}{3}\log|x^3|+\frac{1}{6}\log|1+x^3|-\frac{1}{2}\log|1+3x^3|+K$$

50. $\int \frac{x^2+1}{(x-1)^2(x+3)}dx$ equals

(1) $e^{x-1}+\frac{1}{4}\cdot\frac{1}{(x-1)}+\tan^{-1}(x+3)+\log x+C$

(2) $\frac{3}{8}\log|x-1|-\frac{1}{2}\frac{1}{(x-1)}+C$

(3) $\frac{3}{8}\log|x-1|-\frac{1}{2}\cdot\frac{1}{(x-1)}+\frac{5}{8}\log|x+3|+C$

(4) $\frac{1}{3}\tan^{-1}(x+1)+\log|x+3|+C$

Sol. Answer (3)

$$\int \frac{x^2+1}{(x-1)^2(x+3)}dx$$

$$\frac{x^2+1}{(x-1)^2(x+3)} = \frac{Ax+B}{(x-1)^2}+\frac{C}{(x+3)}$$

$$= \int\left(\frac{\frac{3}{8}x+\frac{1}{8}}{(x-1)^2}+\frac{\frac{5}{8}}{x+3}\right)dx$$

$$= \frac{Ax(x+3)+B(x+3)+C(x^2-2x+1)}{(x-1)^2(x+3)}$$

$$= \frac{3}{8}\int\frac{x+\frac{1}{3}}{(x-1)^2}dx+\frac{5}{8}\int\frac{dx}{x+3}$$

$$= \frac{(A+C)x^2+(3A+B-2C)x+3B+C}{(x-1)^2(x+3)}$$

$$= \frac{3}{8}\left\{\int\frac{x-1}{(x-1)^2}dx+\int\frac{\frac{4}{3}}{(x-1)^2}dx\right\}+\frac{5}{8}\int\frac{dx}{x+3}$$

Hence, $\left.\begin{array}{l}A+C=1\\3A+B-2C=0\\3B+C=1\end{array}\right\} \Rightarrow A=\frac{3}{8}, B=\frac{1}{8}, C=\frac{5}{8}$

$$= \frac{3}{8}\left\{\int \frac{dx}{x-1} + \frac{4}{3}\int \frac{dx}{(x-1)^2}\right\} + \frac{5}{8}\int \frac{dx}{x+3}$$

$$= \frac{3}{8}\left\{\log|x-1| - \frac{4}{3}\frac{1}{(x-1)}\right\} + \frac{5}{8}\log|x+3| + C$$

51. If $\int \frac{2x+3}{(x-1)(x^2+1)}dx = \log_e\{(x-1)^{\frac{5}{2}}(x^2+1)^a\} - \frac{1}{2}\tan^{-1}x + C$, $x > 1$ where C is any arbitrary constant, then the value of 'a' is

(1) $\frac{5}{4}$ (2) $-\frac{5}{3}$ (3) $-\frac{5}{6}$ (4) $-\frac{5}{4}$

Sol. Answer (4)

$$\int \frac{2x+3}{(x-1)(x^2+1)}dx$$

$$= \int \frac{5\,dx}{2(x-1)} + \int \frac{-\left(\frac{5}{2}x + \frac{1}{2}\right)}{x^2+1}dx$$

$$= \frac{5}{2}\log(x-1) - \frac{5}{4}\log(1+x^2) - \frac{1}{2}\tan^{-1}x + C$$

$$= \log(x-1)^{\frac{5}{2}}(1+x^2)^{-\frac{5}{4}} - \frac{1}{2}\tan^{-1}x + C$$

On comparing $a = -\frac{5}{4}$

52. If $\int \frac{x^4+1}{x(x^2+1)^2}dx = A\ln|x| + \frac{B}{1+x^2} + C$ then

(1) $A = 1, B = -1$ (2) $A = -1, B = 1$ (3) $A = B = 1$ (4) $A = B = -1$

Sol. Answer (3)

$$\int \frac{x^4+1}{x(x^2+1)^2}\,dx = A\log|x| + \frac{B}{1+x^2} + C$$

$$= \int \frac{(x^4+1)x}{x^2(x^2+1)^2}\,dx \qquad \text{Let } x^2+1 = y,\ 2x\,dx = dy$$

$$= \frac{1}{2}\int \frac{(y-1)^2+1}{(y-1)y^2}\,dy = \frac{1}{2}\int \frac{y^2-2y+2}{(y-1)y^2}\,dy$$

$$= \frac{y^2-2y+2}{(y-1)y^2} = \frac{A}{y}+\frac{B}{y^2}+\frac{C}{y-1} = \frac{-2}{y^2}+\frac{1}{y-1}$$

$$= \frac{1}{2}\int\frac{y^2-2y+2}{(y-1)y^2}\,dy = \frac{1}{2}\int\left(\frac{-2}{y^2}+\frac{1}{y-1}\right)dy = \frac{1}{2}\left(\frac{2}{y}+\log(y-1)\right)+C$$

$$=\frac{1}{2}\left(\frac{2}{x^2+1}\right)+\frac{1}{2}\log\left|x^2+1-1\right|+C$$

$$=\frac{1}{x^2+1}+\log|x|+C$$

$A = 1$, $B = 1$

53. If $\int\frac{2x^2+3}{(x^2-1)(x^2+4)}dx = a\ln\left(\frac{x-1}{x+1}\right)+b\tan^{-1}\frac{x}{2}+c$, then values of a and b are respectively.

(1) $\frac{1}{2}, \frac{1}{3}$ (2) 1, 1 (3) $\frac{1}{2}, \frac{1}{2}$ (4) $\frac{1}{3}, \frac{1}{3}$

Sol. Answer (3)

$$\int\frac{2x^2+3}{(x^2-1)(x^2+4)}\,dx = \int\frac{dx}{x^2-1}+\int\frac{dx}{x^2+4} \qquad \text{[By partial fraction]}$$

$$= \frac{1}{2}\ln\left(\frac{x-1}{x+2}\right)+\frac{1}{2}\tan^{-1}\frac{x}{2}+c$$

Clearly, $a = \frac{1}{2}$, $b = \frac{1}{2}$

[Integration by Parts]

54. $\int e^{3x}\sin 2x\,dx$ equals

(1) $\frac{e^{3x}}{13}(3\cos 2x+2\sin 2x)+C$

(2) $\frac{e^{3x}}{13}(3\sin 2x-2\cos 2x)+C$

(3) $\frac{e^{3x}}{13}(-3\sin 2x-2\cos 2x)+C$

(4) $\frac{e^{3x}}{13}(3\sin 2x+2\cos 2x)+C$

Sol. Answer (2)

$$I = \int e^{3x}\cdot\sin 2x\,dx$$

$$= \sin 2x\cdot\frac{e^{3x}}{3}-\int 2\cos 2x\cdot\frac{e^{3x}}{3}dx$$

$$= \frac{1}{3}e^{3x}\sin 2x-\frac{2}{3}\int e^{3x}\cos 2x\,dx$$

$$= \frac{1}{3}e^{3x}\sin 2x - \frac{2}{3}\left\{\cos 2x \cdot \frac{e^{3x}}{3} - \int(-2\sin 2x)\cdot\frac{e^{3x}}{3}dx\right\}$$

$$= \frac{1}{3}e^{3x}\sin 2x - \frac{2}{9}e^{3x}\cos 2x - \frac{4}{9}I$$

i.e., $\frac{13}{9}I = \frac{e^{3x}}{9}(3\sin 2x - 2\cos 2x) + C'$

i.e., $I = \frac{e^{3x}}{13}(3\sin 2x - 2\cos 2x) + C$

55. If $\int f(x)dx = g(x)$, then $\int f^{-1}(x)dx$ is equal to

(1) $g^{-1}(x)+C$

(2) $xf^{-1}(x) - g(f^{-1}(x)) + C$

(3) $xf^{-1}(x) - g^{-1}(x) + C$

(4) $f^{-1}(x) + C$

Sol. Answer (2)

$\int f(x)dx = g(x)$

$$\int f^{-1}(x)\cdot 1\,dx = f^{-1}(x)\int dx - \int\left\{\frac{d}{dx}f^{-1}(x)\right\}\left(\int dx\right)dx = xf^{-1}(x) - \int x\frac{d}{dx}f^{-1}(x)\,dx = xf^{-1}(x) - \int xd\{f^{-1}(x)\}$$

Let $f^{-1}(x) = t \Rightarrow x = f(t)$

we get, $d\{f^{-1}(x)\} = dt$

Hence $xf^{-1}(x) - \int f(t)\cdot dt = xf^{-1}(x) - g(t) = xf^{-1}(x) - g\{f^{-1}(x)\} + C$

56. $\int \tan^{-1}\left(\sqrt{\frac{1-x}{1+x}}\right)dx$ equals

(1) $\frac{1}{2}(x\cos^{-1}x - \sqrt{1-x^2}) + C$

(2) $\frac{1}{2}(x\cos^{-1}x + \sqrt{1-x^2}) + C$

(3) $\frac{1}{2}(x\sin^{-1}x - \sqrt{1-x^2}) + C$

(4) $\frac{1}{2}(x\sin^{-1}x + \sqrt{1-x^2}) + C$

Sol. Answer (1)

$$\int \tan^{-1}\left(\sqrt{\frac{1-x}{1+x}}\right)dx$$

$$= \frac{1}{2}\int \cos^{-1}x\,dx$$

$$= \frac{1}{2}\int 1\cdot\cos^{-1}x\,dx$$

$$= \frac{1}{2}\left\{x\cos^{-1}x - \int x\cdot\frac{-1}{\sqrt{1-x^2}}dx\right\}$$

$$= \frac{1}{2}\left\{x\cos^{-1}x - \sqrt{1-x^2}\right\} + C$$

57. $\int e^{\sqrt{x}}\,dx$ is equal to

(1) $e^{\sqrt{x}} + C$

(2) $\frac{1}{2}e^{\sqrt{x}} + C$

(3) $2(\sqrt{x}-1)e^{\sqrt{x}} + C$

(4) $2(\sqrt{x}+1)e^{\sqrt{x}} + C$

Sol. Answer (3)

Substituting $x = t^2$

we get, $dx = 2t\,dt$

$$\therefore \quad \int e^{\sqrt{x}}dx = \int 2t\,e^t\,dt$$

$$= 2\left\{t\,e^t - \int e^t\,dt\right\}$$

$$= 2\{t\,e^t - e^t\} + C$$

$$= 2e^t(t-1) + C$$

$$= 2e^{\sqrt{x}}(\sqrt{x}-1) + C$$

58. $\int(\log x)^2 dx$ equals

(1) $x(\log x)^2 - 2x\log x - 2x + C$

(2) $x(\log x)^2 - 2x\log x - x + C$

(3) $x(\log x)^2 - 2x\log x + 2x + C$

(4) $x(\log x)^2 - 2x\log x + x + C$

Sol. Answer (3)

$$\int(\log x)^2 dx$$

$$= (\log x)^2\int dx - \int\left[\left\{\frac{d}{dx}(\log x)^2\right\}\left(\int dx\right)\right]dx$$

$$= x(\log x)^2 - \int 2\log x\,dx$$

$$= x(\log x)^2 - 2\{x\log x - x\} + C$$

$$= x(\log x)^2 - 2x\log x + 2x + C$$

59. $\int\{f(x)g''(x)-f''(x)g(x)\}dx$ equals

(1) $\frac{f(x)}{g(x)}+C$ (2) $f(x)g'(x)-f'(x)g(x)+C$

(3) $f(x)g'(x)+f'(x)g(x)+C$ (4) $f(x)\cdot g(x)+C$

Sol. Answer (2)

$\int\{f(x)g''(x)-f''(x)g(x)\}dx$

$= \int f(x)g''(x)dx-\int f''(x)g(x)dx$

$= \{f(x)\cdot\int g''(x)dx-\int f'(x)(\int g''(x)dx)dx\} -\{g(x)\int f''(x)dx-\int g'(x)(\int f''(x)dx)dx\}$

$= \{f(x)g'(x)-\int f'(x)g'(x)dx\} -\{g(x)f'(x)-\int g'(x)f'(x)dx\}$

$= f(x)g'(x)-g(x)f'(x)+C$

60. $\int(1-x^2)\log x\, dx$ equals

(1) $\left(x-\frac{x^3}{3}\right)\log x-\left(x-\frac{x^3}{9}\right)+C$ (2) $\left(x-\frac{x^3}{3}\right)\log x+\left(x-\frac{x^3}{9}\right)+C$

(3) $\left(x+\frac{x^3}{3}\right)\log x+\left(x+\frac{x^3}{9}\right)+C$ (4) $x\log x+C$

Sol. Answer (1)

$\int(1-x^2)\log x\, dx$

$= \int\log x\, dx-\int x^2\log x\, dx$

$= x\log x-x-\left\{\log x\int x^2dx-\int\left(\frac{1}{x}\int x^2dx\right)dx\right\}$

$= x\log x-x-\frac{x^3}{3}\log x+\frac{x^3}{9}+C$

$= \left(x-\frac{x^3}{3}\right)\log x-\left(x-\frac{x^3}{9}\right)+C$

61. $\int(x^6+7x^5+6x^4+5x^3+4x^2+3x+1)e^x\, dx$ equals

(1) $\sum_{i=0}^{6}x^ie^x+C$ (2) $\sum_{i=1}^{7}x^ie^x+C$ (3) $\sum_{i=1}^{6}x^ie^x+C$ (4) x^ie^x+C

Sol. Answer (3)

$$\int(x^6+7x^5+6x^4+5x^3+4x^2+3x+1)e^x dx$$

$$= \int e^x(x^6+6x^5)dx+\int e^x(x^5+5x^4)dx+\int e^x(x^4+4x^3)dx+\int e^x(x^3+3x^2)dx+\int e^x(x^2+2x)dx+\int e^x(x+1)dx$$

$$= x^6e^x+x^5e^x+x^4e^x+x^3e^x+x^2e^x+xe^x+C$$

$$= e^x(x^6+x^5+x^4+x^3+x^2+x)+C$$

$$= \sum_{i=1}^{6} x^i e^x + C$$

62. Evaluate $\int \cot^{-1}(x^2+x+1)\,dx$.

(1) $x\cot^{-1}(x^2+x+1)+C$

(2) $\frac{1}{2}\log\left(\frac{1+x^2}{2+x^2+2x}\right)+\tan^{-1}(x+1)+C$

(3) $\tan^{-1}(x+1)+C$

(4) $x\cot^{-1}(x^2+x+1)+\frac{1}{2}\log\left(\frac{1+x^2}{2+x^2+2x}\right)+\tan^{-1}(x+1)+C$

Sol. Answer (4)

$$= \int \tan^{-1}\left(\frac{1}{x^2+x+1}\right)dx$$

$$= \int \tan^{-1}\left\{\frac{(x+1)-x}{1+x(x+1)}\right\}dx$$

$$= \int\left\{\tan^{-1}(x+1)-\tan^{-1}(x)\right\}dx$$

$$= \int \tan^{-1}(x+1)\,dx-\int \tan^{-1}(x)\,dx$$

Applying Integration by parts, we get

$$= \left\{\tan^{-1}(x+1)\right\}x-\int x.\frac{1}{1+(x+1)^2}dx-\left\{\left(\tan^{-1}x\right)x\int x.\frac{1}{1+x^2}\,dx\right\}$$

$$= x\left\{\tan^{-1}(x+1)-\tan^{-1}x\right\}+\int\frac{x}{1+x^2}.dx-\int\frac{x}{1+(1+x)^2}\,dx$$

$$I = x\left\{\tan^{-1}(x+1) - \tan^{-1}(x)\right\} + \frac{1}{2}\log\left(1+x^2\right) - I_1$$

Let $I_1 = \int \frac{x}{1+(1+x)^2}\,dx;$ put $1+x=t$

$$\Rightarrow \quad I = x\left\{\tan^{-1}(x+1) - \tan^{-1}(x)\right\} + \frac{1}{2}\log\left(1+x^2\right) - \frac{1}{2}\log(1+(1+x)^2) + \tan^{-1}(1+x) + C$$

63. $\int \frac{1}{x}\ln\left(\frac{x}{e^x}\right)dx$ equals

(1) $\frac{1}{2}e^x - \ln x + C$ (2) $\frac{1}{2}\ln x - e^x + C$ (3) $\frac{1}{2}\ln^2 x - x + C$ (4) $\frac{e^x}{2x} + C$

Sol. Answer (3)

$$\int \frac{1}{x}\ln\left(\frac{x}{e^x}\right)dx = \int \frac{1}{x}\left(\log x - \log e^x\right)dx$$

$$= \int\left(\frac{\log x}{x}\,dx - 1\right)dx$$

$$= \frac{(\log x)^2}{2} - x + C$$

64. $\int e^{\tan^{-1}x}\left(\frac{1+x+x^2}{1+x^2}\right)dx =$

(1) $xe^{\tan^{-1}x} + C$ (2) $x^2 e^{\tan^{-1}x} + C$ (3) $\frac{1}{x}e^{\tan^{-1}x} + C$ (4) $\frac{(1+x^2)}{2}e^{\tan^{-1}x} + C$

Sol. Answer (1)

$\int e^{\tan^{-1}x}\left(\frac{1+x+x^2}{1+x^2}\right)dx$ Put $\tan^{-1}x = t$

$= \int e^t \cdot (1+\tan t + \tan^2 t)\cdot dt$ $\left(\frac{1}{1+x^2}\right)dx = dt$

$$= \int e^t(\tan t + \sec^2 t)\,dt$$

$$= e^t \tan t + C$$

$$= x\,e^{\tan^{-1}x} + C$$

65. $\int \frac{1}{\log_x e} dx$ equals

(1) $\log \log_x e + C$ (2) $\frac{1}{(\log_x e)^2} + C$ (3) $x \log_e \left(\frac{x}{e}\right) + C$ (4) $e^x + C$

Sol. Answer (3)

$$\int \frac{1}{\log_x e} dx = \int \log_e x \, dx$$

$$= \log_e x \int dx - \int \left(\frac{1}{x} . \int dx\right) dx$$

$$= x \log_e x - x + C$$

$$= x \log_e \left(\frac{x}{e}\right) + C$$

66. $\int \frac{\tan^{-1} x - \cot^{-1} x}{\tan^{-1} x + \cot^{-1} x} dx$ equals

(1) $\frac{4x}{\pi} \tan^{-1} x + \frac{2}{\pi} \ln(1 + x^2) - x + C$ (2) $\frac{4x}{\pi} \tan^{-1} x - \frac{2}{\pi} \ln(1 + x^2) + x + C$

(3) $\frac{4x}{\pi} \tan^{-1} x + \frac{2}{\pi} \ln(1 + x^2) + x + C$ (4) $\frac{4x}{\pi} \tan^{-1} x - \frac{2}{\pi} \ln(1 + x^2) - x + C$

Sol. Answer (4)

$$\int \frac{\tan^{-1} x - \cot^{-1} x}{\tan^{-1} x + \cot^{-1} x} dx = \frac{2}{\pi} \int \tan^{-1} x - \left(\frac{\pi}{2} - \tan^{-1} x\right) dx = \frac{2}{\pi} \int \left(2 \tan^{-1} x - \frac{\pi}{2}\right) dx = \frac{4}{\pi} \int \tan^{-1} x \, dx - x + C$$

Let $\tan^{-1} x = \theta$

$x = \tan\theta$

$dx = \sec^2 \theta \, d\theta$

$$\int \tan^{-1} x \, dx = \int \theta \cdot \sec^2 \theta \, d\theta = \theta \tan\theta - \int \tan\theta \, d\theta$$

$$= \theta \tan\theta - \log \sec\theta + C$$

$$= \theta \tan\theta - \frac{1}{2} \log (1 + \tan^2 \theta) + C$$

$$= \frac{4}{\pi}\left[x \tan^{-1} x - \frac{1}{2} \log (1 + x^2)\right] - x + C$$

67. $\int \sin 2x \cdot \log_e \cos x \, dx$ is equal to

(1) $\left(\frac{1}{2}+\log_e \cos x\right)\cdot \cos^2 x + C$

(2) $\cos^2 x \cdot \log_e \cos x + C$

(3) $\left(\frac{1}{2}-\log_e \cos x\right)\cdot \cos^2 x + C$

(4) $(1 - \log_e \cos x) + C$

Sol. Answer (3)

$\int \sin 2x \log_e \cos x \, dx$ Put $\log_e \cos x = t$

$= \int 2 \sin x \cos x \log_e \cos x \, dx$ $\cos x = e^t$

$= -\int 2e^t \cdot t \cdot e^t \, dt = -2\int e^{2t} \, t \, dt$ $-\sin x \, dx = e^t \, dt$

Integrating by parts

$$= -2\left[t\cdot\frac{e^{2t}}{2} - \int 1\cdot\frac{e^{2t}}{2}\,dt\right]$$

$$= -2\left[\frac{te^{2t}}{2} - \frac{e^{2t}}{4}\right] = -2e^{2t}\left(\frac{t}{2}-\frac{1}{4}\right)$$

$$= -2\cos^2 x\cdot\left(\frac{\log_e \cos x}{2}-\frac{1}{4}\right)$$

$$= \frac{1}{2}\cos^2 x - (\log_e \cos x)(\cos^2 x) + C$$

$$= \left(\frac{1}{2}-\log_e \cos x\right)\cos^2 x + C$$

68. $\int \frac{\cot^{-1}(e^x)}{e^x}dx$ equals

(1) $\frac{1}{2}\ln(e^{2x}+1) - \frac{\cot^{-1}(e^x)}{e^x} + x + C$

(2) $\frac{1}{2}\ln(e^{2x}+1) + \frac{\cot^{-1}(e^x)}{e^x} + x + C$

(3) $\frac{1}{2}\ln(e^{2x}+1) - \frac{\cot^{-1}(e^x)}{e^x} - x + C$

(4) $\frac{1}{2}\ln(e^{x}+1) + \frac{\cot^{-1}(e^x)}{e^x} - x + C$

Sol. Answer (3)

$$\int \frac{\cot^{-1}(e^x)\,dx}{e^x} = \cot^{-1}(e^x)\int e^{-x}\,dx - \int \frac{dx}{1+(e^x)^2}$$

Let $e^x = t$

$e^x\,dx = dt$

$$= \frac{-\cot^{-1}e^x}{e^x} - \int \frac{1}{t}\,\frac{dt}{(1+t^2)} \quad \left(\because\ \frac{1}{t(1+t^2)} = \frac{1}{t} - \frac{1}{2}\left(\frac{2t}{1+t^2}\right)\right)$$

$$= \frac{-\cot^{-1}e^x}{e^x} - \log t + \frac{1}{2}\log(1+t^2) + C = \frac{1}{2}\log(1+e^{2x}) - \log e^x - \frac{\cot^{-1}(e^x)}{e^x} + C$$

69. $\int e^{\tan\theta}(\sec\theta - \sin\theta)d\theta$ equals

(1) $-e^{\tan\theta}\sin\theta + C$ (2) $e^{\tan\theta}\sin\theta + C$ (3) $e^{\tan\theta}\sec\theta + C$ (4) $e^{\tan\theta}\cos\theta + C$

Sol. Answer (4)

$$\int e^{\tan\theta}(\sec\theta - \sin\theta)\,d\theta$$

$$\int e^{\tan\theta}\left(\frac{\cos\theta}{\cos^2\theta} - \sin\theta\right)d\theta$$

$$\int \left(e^{\tan\theta}(\cos\theta\sec^2\theta) - e^{\tan\theta}\cdot\sin\theta\right)d\theta$$

$$\int \frac{d}{d\theta}(e^{\tan\theta}\cdot\cos\theta)\,d\theta = e^{\tan\theta}\cdot\cos\theta + C$$

70. $\int x^n \log x\,dx$ equals

(1) $\frac{x^{n+1}}{n+1}\left\{\log x + \frac{1}{n+1}\right\} + C$ (2) $\frac{x^{n+1}}{n+1}\left\{\log x + \frac{2}{n+1}\right\} + C$

(3) $\frac{x^{n+1}}{n+1}\left\{2\log x - \frac{1}{n+1}\right\} + C$ (4) $\frac{x^{n+1}}{n+1}\left\{\log x - \frac{1}{n+1}\right\} + C$

Sol. Answer (4)

$$\int x^n \log x\,dx$$

$$= \log x\cdot\int x^n dx - \int \frac{d}{dx}(\log x)\left(\int x^n dx\right)dx$$

$$= \frac{x^{n+1}}{n+1}\log x - \int \frac{1}{x}\cdot\frac{x^{n+1}}{n+1}dx$$

$$= \frac{x^{n+1}}{n+1}\log x - \frac{1}{n+1}\int x^n\, dx$$

$$= \frac{x^{n+1}}{n+1}\log x - \frac{x^{n+1}}{(n+1)^2}$$

$$= \frac{x^{n+1}}{n+1}\left\{\log x - \frac{1}{n+1}\right\} + C$$

71. If $\int e^{3x}\cos 4x dx = e^{3x}(A\sin 4x + B\cos 4x) + C$ then

(1) $4A = 3B$ (2) $2A = 3B$ (3) $3A = 4B$ (4) $3A = 2B$

Sol. Answer (3)

$$\int e^{3x}\cos 4x\, dx = \frac{e^{3x}}{25}(3\cos 4x + 4\sin 4x) + C$$

because, $\int e^{ax}\cos bx\, dx = \frac{e^{ax}}{a^2+b^2}(a\cos bx + b\sin bx) + C$ $= e^{3x}\left(\frac{3}{25}\cos 4x + \frac{4}{25}\sin 4x\right) + C$

$$\int e^{3x}\cos 4x\, dx = e^{3x}(A\sin 4x + B\cos 4x) + C$$

$$A = \frac{4}{25},\ B = \frac{3}{25}$$

$$\frac{A}{B} = \frac{4}{3} \Rightarrow 4B = 3A$$

[Miscellaneous Problems on Indefinite Integrals]

72. $\int \frac{x^2-1}{x\sqrt{x^4+3x^2+1}}dx$ equals

(1) $\log\left|\frac{x^2+1+\sqrt{x^4+3x^2+1}}{x}\right| + C$

(2) $\log\left|\frac{x+1+\sqrt{x^2+3x+1}}{x}\right| + C$

(3) $\log\left|\frac{x^2+1+\sqrt{x^2+1}}{x^2}\right| + C$

(4) $\tan^{-1}x + C$

Sol. Answer (1)

$$\int \frac{x^2-1}{x\sqrt{x^4+3x^2+1}}dx$$

$$= \int \frac{x^2 - 1}{x^2\sqrt{x^2 + \frac{1}{x^2} + 3}} dx$$

$$= \int \frac{\left(1 - \frac{1}{x^2}\right) dx}{\sqrt{\left(x + \frac{1}{x}\right)^2 + 1}}$$

$$= \int \frac{dt}{\sqrt{t^2 + 1}}$$

Substituting $t = x + \frac{1}{x}$

we get, $dt = \left(1 - \frac{1}{x^2}\right) dx$

$$= \log\left|\frac{x^2 + 1 + \sqrt{x^4 + 3x^2 + 1}}{x}\right| + C$$

73. If $\int \frac{dx}{\sqrt[3]{\sin^{11} x \cdot \cos x}} = -\frac{a\,(b + c\tan^2 x)}{d\tan^2 x(\tan x)^{2/3}} + k$, then $a + b + c + d$ equal to

(1) 8 (2) 16 (3) 24 (4) 32

Sol. Answer (2)

Put $\tan x = t$ so that $\frac{dx}{\cos^2 x} = dt$

Given integrand becomes

$$I = \int \frac{1 + t^2}{t^{11/3}} dt$$

$$= \int (t^{-11/3} + t^{-5/3})\, dt$$

$$= \frac{-3}{8} \cdot t^{-8/3} \frac{-3}{2} \cdot t^{-2/3} + k$$

$$= \frac{-3}{8} \frac{(1 + 4\tan^2 x)}{\tan^2 x \cdot (\tan x)^{3/2} + k}$$

Clearly $a + b + c + d = 16$

74. $\int\{1+\tan x\cdot\tan(x+A)\}dx = \cot A\cdot\ln|f(x)|+c$, then $f(x)$ equal to

(1) $\dfrac{\sec(x+A)}{\sec x}$ (2) $\dfrac{\text{cosec }(x+A)}{\text{cosec }(x-A)}$ (3) $\dfrac{\text{cosec }(x-A)}{\text{cosec }x}$ (4) $\dfrac{\sec(x-A)}{\sec(x)}$

Sol. Answer (1)

$$I = \int\{1+\tan x\tan(x+A)\}\,dx$$

$$= \int\left\{1+\frac{\sin x}{\cos x}\cdot\frac{\sin(x+A)}{\cos(x+A)}\right\}dx$$

$$= \int\frac{\cos x\cdot\cos(x+A)+\sin(x+A)\cdot\sin x}{\cos x\cdot\cos(x+A)}\,dx$$

$$= \int\frac{\cos A}{\cos x\cdot\cos(x+A)}\,dx$$

$$= \cot A\cdot\int\frac{\sin(x+A-x)}{\cos x\cdot\cos(x+A)}\,dx = \cot A\cdot\int(\tan(x+A)-\tan x)$$

$$= \cot A\cdot[\ln\sec(x+A)-\ln(\sec x)]+c$$

$$= \cot A\cdot\ln\left[\frac{\sec(x+A)}{\sec x}\right]+c$$

75. The value of $\int\frac{x^2-1}{(x^2+1)\sqrt{x^4+1}}dx$ equal to $\frac{1}{\sqrt{2}}\sec^{-1}(f(x))+c$, then $f(x)$ is

(1) $\dfrac{x^2-1}{2x}$ (2) $\dfrac{x^2-1}{\sqrt{2}x}$ (3) $\dfrac{x^2+1}{2x}$ (4) $\dfrac{x^2+1}{\sqrt{2}x}$

Sol. Answer (4)

$$I = \int\frac{x^2-1}{(x^2+1)\sqrt{(x^4+1)}} = \int\frac{1-\frac{1}{x^2}}{\left(x^2+\frac{1}{x^2}\right)}dx = \int\frac{1-\frac{1}{x^2}}{\left(x+\frac{1}{x}\right)\sqrt{\left(x+\frac{1}{x}\right)^2-2}}$$

Put $x+\frac{1}{x}=t$ so that $\left(1-\frac{1}{x^2}\right)dx = dt$

$$I = \int\frac{dt}{t\sqrt{t^2-2}} = \frac{1}{\sqrt{2}}\sec^{-1}\left(\frac{t}{\sqrt{2}}\right)+c = \frac{1}{\sqrt{2}}\sec^{-1}\left(\frac{x^2+1}{\sqrt{2}x}\right)+c$$

Clearly $f(x) = \dfrac{x^2+1}{\sqrt{2}x}$

76. If $\int f(x)\cos x\,dx = \frac{1}{2}f^2(x)+c$, then $f(x)$ can be

(1) x (2) 1 (3) $\cos x$ (4) $\sin x$

Sol. Answer (4)

$$\int f(x)\cdot\cos x\,dx = \frac{1}{2}f^2(x)+c$$

Differentiating, we get

$$f(x)\cdot\cos x = \frac{1}{2}\cdot 2 - f(x)\cdot f'(x)$$

$$f'(x) = \cos x$$

$$f(x) = \sin x + c$$

77. If $\int\frac{(x^2-1)\,dx}{(x^4+3x^2+1)\tan^{-1}\left(x+\frac{1}{x}\right)} = \ln\cdot(f(x))+c$, then $f(x)$ equal to

(1) $\cot^{-1}\left(\frac{x^2+1}{x}\right)$ (2) $\tan^{-1}\left(\frac{x^2+1}{x}\right)$ (3) $\sin^{-1}\left(\frac{x^2+1}{x}\right)$ (4) $\cos^{-1}\left(\frac{x^2+1}{x}\right)$

Sol. Answer (2)

We have

$$I = \int\frac{(x^2-1)\,dx}{(x^4+3x^2+1)\cdot\tan^{-1}\left(x+\frac{1}{x}\right)} = \int\frac{\left(1-\frac{1}{x^2}\right)dx}{\left(x^2+\frac{1}{x^2}+3\right)\tan^{-1}\left(x+\frac{1}{x}\right)} = \int\frac{\left(1-\frac{1}{x^2}\right)dx}{\left(\left(x+\frac{1}{x}\right)^2+1\right)\tan^{-1}\left(x+\frac{1}{x}\right)}$$

Put $x+\frac{1}{x} = t$ so that $\left(1-\frac{1}{x^2}\right)dx = dt$

$$I = \int\frac{dt}{(t^2+1)\cdot\tan^{-1}t} = \ln|\tan^{-1}t| + c$$

$$= \ln\left|\tan^{-1}\left(\frac{x^2+1}{x}\right)\right| + c$$

78. $\int\left\{\frac{(\log x-1)}{1+(\log x)^2}\right\}^2 dx$ is equal to

(1) $\frac{x}{(\log x)^2+1}+C$ (2) $\frac{xe^x}{1+x^2}+C$ (3) $\frac{x}{x^2+1}+C$ (4) $\frac{\log x}{(\log x)^2+1}+C$

Sol. Answer (1)

$$\int\left\{\frac{(\log x-1)}{1+(\log x)^2}\right\}^2 dx$$

Consider $f(x)=\dfrac{x}{(\log x)^2+1}$ by cross check method

$$f'(x)=\frac{(1-\log x)^2}{(1+(\log x)^2)^2}$$

$$\int\left\{\frac{1-\log x}{1+(\log x)^2}\right\}^2 dx=\int f'(x)\,dx=f(x)+C=\frac{x}{(\log x)^2+1}+C$$

Hence option (1) is correct answer and we check the other choices by the similar argument.

79. $\int[\sin\log x+\cos\log x]\,dx$ is equal to

(1) sin (log x) + cos(log x) + C (2) x sin(log x) + C

(3) x cos(log x) + C (4) x tan x + C

Sol. Answer (2)

$\int(\sin(\log x)+\cos(\log x)]\,dx$ Put $\log_e x=t$

$=\int e^t(\sin t+\cos t)\,dt$ $x=e^t$, then

$=e^t\cdot\sin t+C$ $dx=e^t\,dt$

$=x\sin(\log_e x)+C$

80. $\displaystyle\int\frac{1}{|(2x-7)|\sqrt{x^2-7x+12}}\,dx=$

(1) 2 $\sec^{-1}(2x-7)+C$ (2) $\sec^{-1}(2x-7)+C$

(3) $\dfrac{1}{2}\sec^{-1}(2x-7)+C$ (4) $\sec^{-1}(2x+7)+C$

Sol. Answer (2)

$$\int\frac{dx}{|2x-7|\sqrt{x^2-7x+12}}=2\int\frac{dx}{|2x-7|\sqrt{(2x-7)^2-1}}$$

Put $2x-7=t$, then $2dx=dt$

$$\int\frac{dt}{t\sqrt{t^2-1}}=\sec^{-1}t=\sec^{-1}(2x-7)+C$$

81. If $\int f(x)dx = F(x)$, then $\int x^3 f(x^2)dx$ is equal to

(1) $\frac{1}{2}x^2F(x^2) - \int xF(x^2)dx$

(2) $\frac{1}{2}\left(x^2F(x^2) - \int F(x^2)dx\right)$

(3) $\frac{1}{2}\left(x^2F(x) - \frac{1}{2}\int F(x^2)dx\right)$

(4) $\frac{1}{2}\left(x^2F(x) - \frac{3}{2}\int F(x^2)dx\right)$

Sol. Answer (1)

If $\int f(x)\,dx = F(x)$ then $\int x^3 f(x^2)\,dx = \int x \cdot x^2 f(x^2)\,dx$

Put $x^2 = t$; then $2x\,dx = dt$

$$= \frac{1}{2}\int t\,f(t)\,dt$$

$$= \frac{1}{2}\left[t \cdot \int f(t)\,dt - \int\left(\int f(t)\,dt\right)dt\right] \quad [f(x)\,dx = F(x)]$$

$$= \frac{1}{2}\left[t \cdot F(t) - \int F(t)\,dt\right]$$

$$= \frac{1}{2}\left[x^2 F(x^2) - \int F(x^2)\,dx\right]$$

82. $\int \frac{\ln|x|}{x\sqrt{1+\ln|x|}}dx$ equals

(1) $\frac{2}{3}\sqrt{1+\ln|x|}(\ln|x|-2)+C$

(2) $\frac{2}{3}\sqrt{1+\ln|x|}(\ln|x|+2)+C$

(3) $\frac{1}{3}\sqrt{1+\ln|x|}(\ln|x|-2)+C$

(4) $2\sqrt{1+\ln|x|}(\ln|x|+2)+C$

Sol. Answer (1)

$$\int \frac{\log|x|\,dx}{x\sqrt{1+\log|x|}}$$

Let $1+\log|x| = t^2$ $\qquad |x| = x$

$\frac{1}{|x|}dx = 2t\,dt$ $\qquad$ if $x > 0$

$\frac{1}{x}dx = 2t\,dt$

$$= \int \frac{(t^2-1)\,2t\,dt}{t}$$

$$= 2\int (t^2-1)\,dt = \frac{2t^3}{3} - 2t + C = 2\left[\frac{(1+\log x)^{3/2}}{3} - (1+\log x)^{1/2}\right] + C$$

$$= 2\sqrt{1+\log x}\left(\frac{1+\log x}{3} - 1\right) + C = \frac{2}{3}\sqrt{1+\log x}\,(\log|x| - 2) + C$$

83. $\int x\frac{\ln(x+\sqrt{1+x^2})}{\sqrt{1+x^2}}dx$ equals

(1) $\sqrt{1+x^2}\ln(x+\sqrt{1+x^2}) - x + C$

(2) $\frac{x}{2}\ln^2(x+\sqrt{1+x^2}) - \frac{x}{\sqrt{1+x^2}} + C$

(3) $\frac{x}{2}\ln^2(x+\sqrt{1+x^2}) + \frac{x}{\sqrt{1+x^2}} + C$

(4) $\sqrt{1+x^2}\ln(x+\sqrt{1+x^2}) + x - C$

Sol. Answer (1)

$$\int \frac{x\ln\left(x+\sqrt{1+x^2}\right)dx}{\sqrt{1+x^2}}$$

Let $\ln\left(x+\sqrt{1+x^2}\right) = t \quad \Rightarrow \quad x+\sqrt{1+x^2} = e^t$

$\frac{1}{\sqrt{1+x^2}}dx = dt \qquad -x+\sqrt{1+x^2} = e^{-t}$

$$x = \frac{1}{2}[e^t - e^{-t}]$$

$$\frac{1}{2}\int (e^t - e^{-t})\cdot t\cdot dt = \frac{1}{2}\left[(te^t - e^t) - (-te^{-t} - e^{-t})\right]$$

$$= \frac{1}{2}\left[(te^t - e^t) + te^{-t} + e^{-t}\right]$$

$$= \frac{1}{2}\left[t(e^t + e^{-t}) - (e^t - e^{-t})\right]$$

$$= \left(\ln\left(x+\sqrt{1+x^2}\right)\right)\left(\sqrt{1+x^2}\right) - x + C$$

$$= \sqrt{1+x^2}\,\ln\left(x+\sqrt{1+x^2}\right) - x + C$$

84. If $\int \frac{x \cdot e^x}{\sqrt{1+e^x}} dx = f(x)\sqrt{1+e^x} - 2\ln(g(x)) + c$, then value of $f(x)$ and $g(x)$ is

(1) $f(x) = 2x+4,\ g(x) = \frac{\sqrt{1+e^x}+1}{\sqrt{e^x-1}-1}$

(2) $f(x) = 2x-4,\ g(x) = \frac{\sqrt{e^x+1}-1}{\sqrt{e^x+1}+1}$

(3) $f(x) = 3x-2,\ g(x) = \frac{\sqrt{e^x-1}+1}{\sqrt{e^x-1}-1}$

(4) $f(x) = 2x-4,\ g(x) = \frac{\sqrt{e^x-1}+1}{\sqrt{e^x+1}-1}$

Sol. Answer (2)

Let $I = \int \frac{x \cdot e^x}{\sqrt{1+e^x}} dx$

Put $1 + e^x = t^2$ so that $e^x\, dx = 2t\, dt$ and $x = \ln(t^2 - 1)$

So $I = \int \frac{\ln(t^2-1)}{t} 2t\, dt$

$= 2\int \ln(t^2-1)\, dt$

$= 2\left[t \cdot \ln(t^2-1) - 2\int \frac{t^2}{t^2-1} dt\right]$ (Integration by parts)

$= 2\left[t \ln(t^2-1) - 2t - \ln\left(\frac{t-1}{t+1}\right)\right] + C$

$= 2x\sqrt{1+e^x} - 4\sqrt{1+e^x} - 2\log\left(\frac{\sqrt{1+e^x}-1}{\sqrt{1+e^x}+1}\right) + C$

Clear $f(x) = (2x - 4)$

$g(x) = \frac{\sqrt{1+e^x}-1}{\sqrt{e^x+1}+1}$

85. If $f(x) = \frac{x+2}{2x+3}$, then $\int \left(\frac{f(x)}{x^2}\right)^{1/2} dx$ is equal to $\frac{1}{\sqrt{2}} g\left(\frac{1+\sqrt{2f(x)}}{1-\sqrt{2f(x)}}\right) - \sqrt{\frac{2}{3}} h\left(\frac{\sqrt{3f(x)}+\sqrt{2}}{\sqrt{3f(x)}-\sqrt{2}}\right) + c$, where

(1) $g(x) = \tan^{-1} x,\ h(x) = \ln|x|$

(2) $g(x) = \ln x,\ h(x) = \tan^{-1} x$

(3) $g(x) = h(x) = \tan^{-1} x$

(4) $g(x) = h(x) = \ln|x|$

Sol. Answer (4)

Let $f(x) = y^2 = \frac{x+2}{2x+3}$

$\Rightarrow \quad x = \dfrac{3y^2-2}{1-2y^2}$

$\Rightarrow \quad dx = -\dfrac{2y}{(1-2y^2)2}dy$

Now, consider $I = \displaystyle\int \frac{(f(x))^{1/2}}{x}dx$

Put $f(x) = y^2$ so that $dx = -\dfrac{2y}{(1-2y^2)2}\cdot dy$

$I = -\displaystyle\int \frac{y(1-2y^2)}{(3y^2-2)} \times \left(-\frac{2y}{(1-2y^2)^2}\right)\cdot dy$

$= \displaystyle\int \frac{-2y^2}{(3y^2-2)(1-2y^2)}dy$

$= -2\displaystyle\int\left(\frac{1}{2y^2-1} - \frac{2}{3y^2-2}\right)dy$

$= \dfrac{1}{2}\text{by}\left|\dfrac{1+\sqrt{2}y}{1-\sqrt{2}y}\right| - \sqrt{\dfrac{2}{3}}\text{ by}\left|\dfrac{\sqrt{3}y+\sqrt{2}}{\sqrt{3}y-\sqrt{2}}\right| + c$

Clearly $f(x) = g(x) = \ln|x|$

86. $\displaystyle\int\sqrt{\sec x - 1}\,dx$ equal to

(1) $-\log\left[2\cos x - 1 + 2\sqrt{\sin^2 x - \sin x}\right] + c$

(2) $+\log\left[2\cos x + 1 + 2\sqrt{\cos^2 x - \cos x}\right] + c$

(3) $-\log\left[2\cos x - 1 + 2\sqrt{\cos^2 x - \cos x}\right] + c$

(4) $-\log\left[2\cos x + 1 + 2\sqrt{\cos^2 x + \cos x}\right] + c$

Sol. Answer (4)

$I = \displaystyle\int\sqrt{\sec x - 1}\,dx$

$= \displaystyle\int\sqrt{\frac{1-\cos x}{\cos x}}\,dx = \sqrt{2}\int\frac{\sin x/2}{\sqrt{2\cos^2 x/2 \; - 1}}$

$= -2\displaystyle\int\frac{dt}{t^2-1} \quad \left[\begin{array}{l}\text{Put } \sqrt{2}\cos\dfrac{x}{2} = t \text{ so that} \\ \qquad \sqrt{2}\sin\dfrac{x}{2}\cdot dx = -2\,dt\end{array}\right]$

$= -2 \cdot \ln\left|t + \sqrt{t^2 - 1}\right| + c$

$I = -2 \cdot \cos h^{-1}(t) + c = -2 \cos h^{-1}\left(\sqrt{2}\cos\frac{x}{2}\right) + c$

But we knew that $\cos h^{-1}(x) = \sin h^{-1}\sqrt{x^2} - 1$

$I = -2 \sin h^{-1}(\sqrt{\cos x}) + c$

87. If $\int \frac{(x+2)\,dx}{(x^2+3x+3)\sqrt{(x+1)}} = \frac{2}{\sqrt{3}}\left[\tan^{-1}\left(\frac{f(x)+1}{\sqrt{3}}\right) + \tan^{-1}\left(\frac{f(x)-1}{\sqrt{3}}\right)\right] + c$ then $f(x)$ is equal to

(1) $2\sqrt{x-1}$ (2) $3\sqrt{x+1}$ (3) $3\sqrt{x-1}$ (4) $2\sqrt{x+1}$

Sol. Answer (4)

Consider $I = \int \frac{(x+2)dx}{(x^2+3x+3)\sqrt{x+1}}$

Put $x + 1 = t^2$ so that $dx = 2tdt$

$\therefore \quad I = 2\int \frac{(t^2+1)\cdot t\,dt}{[(t^2-1)^2 + 3(t^2-1) + 3]\cdot t}$

$= 2\int \frac{(t^2+1)dt}{t^4+t^2+1} = 2\int \frac{t^2+1}{(t^2-t+1)(t^2+t+1)}$

$= \int\left(\frac{1}{t^2+t+1} + \frac{1}{t^2-t+1}\right)dt$

$= \int \frac{dt}{(t+1/2)^2 + (1/2\sqrt{3})^2} + \int \frac{dt}{(t-1/2)^2 + \left(\frac{\sqrt{3}}{2}\right)^2}$

$= \frac{2}{\sqrt{3}}\left[\tan^{-1}\left(\frac{2t+1}{\sqrt{3}}\right) + \tan^{-1}\left(\frac{2t-1}{\sqrt{3}}\right)\right] + C$

$= \frac{2}{\sqrt{3}}\left[\tan^{-1}\left(\frac{2\sqrt{x+1}+1}{\sqrt{3}}\right) + \tan^{-1}\left(\frac{2\sqrt{x-1}-1}{\sqrt{3}}\right)\right] + C$

Clearly $f(x) = 2\sqrt{x+1}$

88. If $\int x^{-11}(1+x^4)^{-1/2}dx = \frac{1}{ax^{10}}(1+x^4)^m + \frac{1}{bx^6}(1+x^4)^n + \frac{1}{cx^2}(1+x^4)^p + k$ where a, b, c, m, n, p are real numbers and k be arbitrary constant, then $\frac{a+b+c}{m+n+p}$ equal to

(1) -2 (2) $-\frac{1}{2}$ (3) $\frac{1}{2}$ (4) 2

Sol. Answer (1)

Consider

$$I = \int x^{-11}(1+x^4)^{-1/2}dx$$

Put $1 + x^4 = x^4 \cdot t^2 \Rightarrow x = \frac{1}{(t^2-1)^{1/4}}$

$$dx = -\frac{tdt}{2(t^2-1)^{5/4}}$$

$$I = \frac{-1}{2}\int (t^2-1)^{11/4}\left(\frac{t^2}{t^2-1}\right)^{-1/2} \cdot \frac{tdt}{(t^2-1)^{5/4}}$$

$$= \frac{-1}{2}\int (t^2-1)^2 dt$$

$$= \frac{-t^5}{10} + \frac{t^3}{3} - \frac{t}{2} + C$$

Returning to x, we get

$$I = -\frac{1}{10 \cdot x^{10}}\ (1+x^4)^{5/2} + \frac{1}{3x^6}(1+x^4)^{3/2} - \frac{1}{2x^2}\ (1+x^4)^{1/2} + C$$

Clearly $a = -10$, $b = 3$, $c = -2$, $m = \frac{5}{2}$, $n = \frac{3}{2}$, $p = \frac{1}{2}$

$a + b + c = -10 + 3 - 2 = -9$

$m + n + p = \frac{5}{2} + \frac{3}{2} + \frac{1}{2} = \frac{9}{2}$

$\therefore\ \frac{a+b+c}{m+n+p} = -2$

89. $\int \frac{\sqrt{1+x^2}}{1-x^2}\,dx = -\frac{\sqrt{2}}{2}\ln|f(x)| + g(x) + c$, then $f(x)$ and $g(x)$ are

(1) $f(x) = \frac{\sqrt{1+x^2}-\sqrt{2}x}{\sqrt{1+x^2}+\sqrt{2}x}$; $g(x) = \ln(x + \sqrt{1+x^2}\ ) + c$

(2) $f(x) = \frac{\sqrt{1+x^2}+\sqrt{2}x}{\sqrt{1+x^2}-\sqrt{2}x}$; $g(x) = \ln(x - \sqrt{1+x^2}\ ) + c$

(3) $f(x) = \frac{\sqrt{1-x^2}-\sqrt{2}x}{\sqrt{1-x^2}+\sqrt{2}x}$; $g(x) = \ln(x + \sqrt{1-x^2}\ ) + c$

(4) $f(x) = \frac{\sqrt{1+x^2}+\sqrt{2}x}{\sqrt{1+x^2}+\sqrt{2}x}$; $g(x) = \ln(\sqrt{x^2-1} + x) + c$

Sol. Answer (1)

$$\int \frac{1+x^2}{(1-x^2)\sqrt{1+x^2}}\,dx = \int \frac{2-(1-x^2)}{(1-x^2)\sqrt{1+x^2}}\,dx$$

$$= 2\int \frac{dx}{(1-x^2)\sqrt{(1+x^2)}} - \int \frac{dx}{\sqrt{1+x^2}}$$

$$= -2\int \frac{tdt}{(t^2-1)\sqrt{t^2+1}} - \ln\left|x+\sqrt{1+x^2}\right| + c \qquad \left[\text{Put } x = \frac{1}{t} \text{ in the integral}\right]$$

Again $t^2 + 1 = v^2$; $tdt = vdv$

$$I = -2\int \frac{dv}{v^2-2} - \ln\left|x+\sqrt{1+x^2}\right| + c$$

On integrating had changing into x, we get

$$I = \frac{-\sqrt{2}}{2}\ln\left|\frac{\sqrt{1+x^2}-\sqrt{2}\,x}{\sqrt{1+x^2}+\sqrt{2}x}\right| - \ln\left|x+\sqrt{1+x^2}\right| + c$$

Clearly we can find $f(x)$ and $g(x)$.

90. If $\int \frac{dx}{e^{2x}+e^{-2x}} = \frac{A}{842}\tan^{-1}e^{2x} + C$ then A equals.....

(1) 0.5 (2) 842 (3) 1684 (4) 421

Sol. Answer (4)

$$\int \frac{dx}{e^{2x}+e^{-2x}} = \int \frac{e^{2x}}{1+(e^{2x})^2}\,dx \qquad \left[\text{Let } e^{2x} = t,\ 2e^{2x}dx = dt\right]$$

$$= \frac{1}{2}\int \frac{dt}{1+t^2} = \frac{1}{2}\tan^{-1}t + C = \frac{1}{2}\tan^{-1}(e^{2x}) + C$$

$$\int \frac{dx}{e^{2x}+e^{-2x}} = \frac{A}{842}\tan^{-1}e^{2x} + C$$

$$\frac{A}{842} = \frac{1}{2} \Rightarrow A = 421$$

91. If $\int \frac{\sin x}{1+\sin x}dx = \frac{A}{798}\sec x - \tan x + x + C$ then A equals.....

(1) –798 (2) 791 (3) 798 (4) 399

Sol. Answer (3)

$$\int \frac{\sin x}{1+\sin x}\,dx = \frac{A}{798}\sec x - \tan x + x + C$$

$$\int \frac{\sin x(1-\sin x)}{\cos^2 x}\,dx = \int (\tan x \sec x - \tan^2 x)\,dx$$

$$= \int \tan x \sec x\,dx - \int (\sec^2 x - 1)\,dx = \sec x - \tan x + x + C$$

$$\frac{A}{798} = 1 \Rightarrow A = 798$$

92. Evaluate $\int \frac{a+b\sin x}{(b+a\sin x)^2}\,dx$.

(1) $\frac{1}{a\cos x + b\sin x}$ (2) $-\frac{1}{b\sec x + a\tan x}$ (3) $\frac{1}{a\cos x - b\sin x}$ (4) $\frac{1}{b\sec x + a\tan x}$

Sol. Answer (2)

$$I = \int \frac{a+b\sin x}{(b+a\sin x)^2}\,dx = \frac{b}{a}\int \frac{\frac{a^2}{b} - b + (b+a\sin x)}{(b+a\sin x)^2}\,dx = \frac{a^2-b^2}{a}\int \frac{dx}{(b+a\sin x)^2} + \frac{b}{a}\int \frac{dx}{b+a\sin x}$$

Now let, $A = \frac{\cos x}{b+a\sin x} \Rightarrow \frac{dA}{dx} = -\frac{b\sin x - a}{(b+a\sin x)^2}$

$$\Rightarrow \frac{dA}{dx} = -\frac{b}{a}\left\{\frac{a\sin x + b + \frac{a^2}{b} - b}{(b+a\sin x)^2}\right\}$$

$$\frac{dA}{dx} = -\frac{b}{a}\left\{\frac{1}{b+a\sin x} + \frac{a^2-b^2}{b(b+a\sin x)^2}\right\}$$

Integrating w.r.t x we get

$$A = -\frac{b}{a}\int \frac{dx}{(b+a\sin x)^2} - \frac{(a^2-b^2)}{a}\int \frac{dx}{(b+a\sin x)^2}$$

$$\Rightarrow \frac{(a^2-b^2)}{a}\int \frac{dx}{(b+a\sin x)^2} = -\frac{b}{a}\int \frac{dx}{b+a\sin x} - A$$

So, we get $I = -\left(\frac{\cos x}{b+a\sin x}\right) + c$

93. $\int \sqrt{x^2+2x+5}\,dx$ equals

(1) $\left(\frac{x+1}{2}\right)\sqrt{x^2+2x+5} + 2\log|x+1+\sqrt{x^2+2x+5}| + C$

(2) $\left(\frac{x+1}{2}\right)\sqrt{x^2+2x+5} + C$

(3) $\log|x+1+\sqrt{x^2+2x+5}| + C$

(4) $\left(x^2+2x+5\right)^{\frac{3}{2}} + C$

Sol. Answer (1)

$$\int \sqrt{x^2+2x+5}\,dx = \int \sqrt{(x+1)^2+2^2}\,dx$$

$$= \frac{1}{2}\left\{(x+1)\sqrt{(x+1)^2+2^2} + 2^2\log\left|(x+1)+\sqrt{(x+1)^2+2^2}\right|\right\} + C$$

$$= \frac{1}{2}(x+1)\sqrt{x^2+2x+5} + 2\log\left|(x+1)+\sqrt{x^2+2x+5}\right| + C$$

Definite Integrals

94. The value of $\int_0^{\pi/2} \frac{\sqrt{\cot x}}{\sqrt{\cot x}+\sqrt{\tan x}}\,dx$ is equal to

(1) π (2) $\frac{\pi}{2}$ (3) $\frac{\pi}{4}$ (4) $\frac{\pi}{3}$

Sol. Answer (3)

$$I = \int_0^{\pi/2} \frac{\sqrt{\cot x}}{\sqrt{\cot x}+\sqrt{\tan x}}\,dx \quad \ldots\text{(i)}$$

Again, $I = \int_0^{\pi/2} \frac{\sqrt{\tan x}}{\sqrt{\tan x}+\sqrt{\cot x}}\,dx$ …(ii) $\left\{\text{using } \int_0^a f(x)dx = \int_0^a f(a-x)dx\right\}$

Adding (i) and (ii), we get

$$2I = \int_0^{\pi/2}\left(\frac{\sqrt{\tan x}+\sqrt{\cot x}}{\sqrt{\tan x}+\sqrt{\cot x}}\right)dx = \int_0^{\pi/2} dx = \frac{\pi}{2}$$

so, $I = \frac{\pi}{4}$.

95. The value of $\int_0^{\pi/2} \frac{dx}{1+\tan^3 x}$ is equal to

(1) 0 (2) 1 (3) $\frac{\pi}{2}$ (4) $\frac{\pi}{4}$

Sol. Answer (4)

Let $I = \int_0^{\pi/2} \frac{dx}{1+\tan^3 x} = \int_0^{\pi/2} \frac{\cos^3 x}{\sin^3 x + \cos^3 x} dx$ ...(i)

Again, $I = \int_0^{\pi/2} \frac{\sin^3 x}{\cos^3 x + \sin^3 x} dx$ ...(ii) {using $\int_0^a f(x)dx = \int_0^a f(a-x)dx$ }

Adding (i) and (ii), we get

$$2I = \int_0^{\pi/2} \frac{\sin^3 x + \cos^3 x}{\sin^3 x + \cos^3 x} dx = \int_0^{\pi/2} dx = \frac{\pi}{2}$$

so, $I = \frac{\pi}{4}$.

96. The value of $\int_2^3 \frac{\sqrt{x}}{\sqrt{5-x}+\sqrt{x}} dx$ is equal to

(1) 1 (2) 0 (3) –1 (4) $\frac{1}{2}$

Sol. Answer (4)

Here $I = \int_2^3 \frac{\sqrt{x}}{\sqrt{x}+\sqrt{5-x}} dx$ ...(i)

Again, $I = \int_2^3 \frac{\sqrt{5-x}}{\sqrt{x}+\sqrt{5-x}} dx$ ...(ii) {using $\int_a^b f(x)dx = \int_a^b f(a+b-x)dx$ }

Adding (i) and (ii), we get

$$2I = \int_2^3 \frac{\sqrt{x}+\sqrt{5-x}}{\sqrt{x}+\sqrt{5-x}} dx$$

$$= \int_2^3 dx$$

$\Rightarrow \quad I = \frac{1}{2}$.

97. The value of $\int_0^a \frac{dx}{x+\sqrt{(a^2-x^2)}}$ is equal to

(1) $\frac{\pi}{4}$ (2) $\frac{\pi}{2}$ (3) π (4) $\frac{\pi}{6}$

Sol. Answer (1)

Let $x = a\sin\theta \Rightarrow dx = a\cos\theta d\theta$

so, $I = \int_0^{\pi/2} \frac{a\cos\theta d\theta}{a\sin\theta + a\cos\theta} = \int_0^{\pi/2} \frac{\cos\theta d\theta}{\sin\theta+\cos\theta}$ …(i)

again, $I = \int_0^{\pi/2} \frac{\sin\theta d\theta}{\cos\theta+\sin\theta}$ …(ii) {using

$\int_0^a f(x)dx = \int_0^a f(a-x)dx\}$

Adding (i) and (ii), we get

$$2I = \int_0^{\pi/2}\left(\frac{\sin\theta+\cos\theta}{\sin\theta+\cos\theta}\right)d\theta = \int_0^{\pi/2} d\theta = \frac{\pi}{2}$$

So, $I = \frac{\pi}{4}$.

98. The value of $\int_0^{\pi/2n} \frac{dx}{1+\tan^n(nx)}$ is equal to, ($n \in N$)

(1) 0 (2) $\frac{\pi}{4n}$ (3) $\frac{\pi}{2n}$ (4) $\frac{\pi}{2}$

Sol. Answer (2)

Let $I = \int_0^{\pi/2n} \frac{dx}{1+\tan^n(nx)} = \int_0^{\pi/2n} \frac{\cos^n(nx)}{\sin^n(nx)+\cos^n(nx)}dx$ …(i)

Again, $I = \int_0^{\pi/2n} \frac{\sin^n(nx)}{\sin^n(nx)+\cos^n(nx)}dx$ …(ii)

Adding (i) and (ii), we get

$$2I = \int_0^{\pi/2n} dx = \frac{\pi}{2n}$$

$\Rightarrow I = \frac{\pi}{4n}$.

99. The value of $\int_0^{\pi/2} \frac{a\sin x + b\cos x}{\sin x + \cos x} dx$ is equal to

(1) 0 (2) $(a+b)\frac{\pi}{2}$ (3) $a + b$ (4) $(a+b)\frac{\pi}{4}$

Sol. Answer (4)

Let $I = \int_0^{\pi/2} \frac{a\sin x + b\cos x}{\sin x + \cos x} dx$ ...(i)

Again, $I = \int_0^{\pi/2} \frac{a\cos x + b\sin x}{\cos x + \sin x} dx$ ...(ii)

Adding (i) and (ii), we get

$$2I = \int_0^{\pi/2} \frac{(a+b)(\sin x + \cos x)}{\sin x + \cos x} dx = (a+b)\int_0^{\pi/2} dx$$

$\Rightarrow$ $I = \frac{\pi}{4}(a+b)$.

100. The value of $\int_0^{\pi/2} \frac{\cos 2x}{(\sin x + \cos x)^2} dx$ is equal to

(1) 0 (2) $\frac{\pi}{4}$ (3) $\frac{\pi}{2}$ (4) $\frac{\pi}{6}$

Sol. Answer (1)

Let $I = \int_0^{\pi/2} \frac{\cos^2 x - \sin^2 x}{(\cos x + \sin x)^2} dx = \int_0^{\pi/2} \left(\frac{\cos x - \sin x}{\cos x + \sin x}\right) dx$ ...(i)

Again, $I = \int_0^{\pi/2} \left(\frac{\sin x - \cos x}{\sin x + \cos x}\right) dx$ ...(ii) {using $\int_0^a f(x)dx = \int_0^a f(a-x)dx$ }

Adding (i) and (ii), we get

$$2I = \int_0^{\pi/2} 0\, dx = 0$$

$\Rightarrow$ $I = 0$.

101. The value of $\int_{-1}^{1} x^{17} \cos^4 x\, dx$ is equal to

(1) –2 (2) –1 (3) 0 (4) 2

Sol. Answer (3)

$\int_{-1}^{1} x^{17} \cos^4 x\, dx = 0$ {using $\int_{-a}^{a} f(x)dx = 0$, where *f(x)* is odd function}

{as $f(x) = x^{17}.\cos^4 x$ is an odd function}

102. The value of $\int_{\pi/4}^{\pi/2} \operatorname{cosec}^2 x\, dx$ is equal to

(1) –1 (2) 1 (3) 0 (4) $\frac{1}{2}$

Sol. Answer (2)

$I = \int_{\pi/4}^{\pi/2} \operatorname{cosec}^2 x\, dx$

$= \{-\cot x\}_{\pi/4}^{\pi/2}$

$= -\left\{\cot\frac{\pi}{2} - \cot\frac{\pi}{4}\right\}$

= 1.

103. The value of $\int_{-1/2}^{1/2}\left([x] + \ln\left(\frac{1+x}{1-x}\right)\right)dx$ is equals to (where [.] represents the greatest integer function)

(1) $-\frac{1}{2}$ (2) 0 (3) 1 (4) $2\log\left(\frac{1}{2}\right)$

Sol. Answer (1)

Given, $I = \int_{-1/2}^{1/2}[x]\,dx + \int_{-1/2}^{1/2}\log\left(\frac{1+x}{1-x}\right)dx$

$= \int_{-1/2}^{0}(-1)\,dx + \int_{0}^{1/2} 0\,dx + 0$ {using $\int_{-a}^{a} f(x)\,dx = 0$, where *f(x)* is odd function}

$= (-1)\left\{0 - \left(-\frac{1}{2}\right)\right\}$

$= -\frac{1}{2}$

104. $f : R \to R$, $g : R \to R$ are continuous functions. The value of integral $\int_{-\pi/2}^{\pi/2}[f(x)+f(-x)][g(x)-g(-x)]\,dx$ is equal to

(1) π (2) 1 (3) –1 (4) 0

Sol. Answer (4)

$I = \int_{-\pi/2}^{\pi/2}\{f(x)+f(-x)\}\{g(x)-g(-x)\}\,dx$

= 0. {as $\int_{-a}^{a} f(x)\,dx = 0$, where *f(x)* is an odd function}

as, here *f(x)* + *f(–x)* is an even function, whereas

g(x) – *g(–x)* is an odd function.

105. The value of $\int_0^{\pi/2} \log \sin x\, dx$ is equal to

(1) $-\left(\frac{\pi}{2}\right)\log 2$ (2) $\pi \log\frac{1}{2}$ (3) $-\pi\log\frac{1}{2}$ (4) $\frac{\pi}{2}\log 2$

Sol. Answer (1)

Let $I = \int_0^{\pi/2} \log \sin x\, dx$ ...(i)

Again, $I = \int_0^{\pi/2} \log \cos x\, dx$ ...(ii) $\{$using $\int_0^a f(x)dx = \int_0^a f(a-x)dx\}$

Adding (i) and (ii), we get

$$2I = \int_0^{\pi/2} \log(\sin x \cdot \cos x)dx$$

$$= \int_0^{\pi/2} \log\left(\frac{\sin 2x}{2}\right)dx$$

$$\Rightarrow \quad 2I = \int_0^{\pi/2} \log \sin 2x\, dx - \int_0^{\pi/2} \log 2\, dx \qquad \text{...(iii)}$$

Let $I_1 = \int_0^{\pi/2} \log \sin 2x\, dx$

Now, put $2x = z \Rightarrow 2dx = dz$

so, $I_1 = \int_0^{\pi} \log \sin z\left(\frac{dz}{2}\right)$

$$= \frac{1}{2}\int_0^{\pi} \log \sin z\, dz$$

$$= \frac{1}{2} \times 2\int_0^{\pi/2} \log \sin x\, dx$$

$= \int_0^{\pi/2} \log \sin x\, dx = I$ $\{$using $\int_0^{2a} f(x)dx = 2\int_0^a f(x)dx$, where $f(2a-x) = f(x)\}$

Now, applying (iii), we get

$$2I = I - \int_0^{\pi/2} \log 2\, dx$$

$$\Rightarrow \quad I = -\frac{\pi}{2}\log 2$$

106. If $f(x) = \begin{cases} e^{\cos x}\sin x & \text{for } |x| \le 2 \\ 2 & \text{otherwise} \end{cases}$ then the value of $\int_{-2}^{3} f(x)dx$ is equal to

(1) 0 (2) 1 (3) 2 (4) 3

Sol. Answer (3)

Here, $\int_{-2}^{3} f(x)dx = \int_{-2}^{2} e^{\cos x}\sin x dx + \int_{2}^{3} 2dx$ $\left\{\begin{array}{l}\text{as, } f(x) = e\cos x.\sin x \text{ is an odd function.} \\ \text{so, } \int_{-2}^{2} e^{\cos x}\cdot\sin x\, dx = 0\end{array}\right\}$

$$= 0 + 2\{x\}_2^3 = 2$$

107. The value of $\int_0^{2/3} \frac{dx}{4+9x^2}$ is equal to

(1) $\frac{\pi}{12}$ (2) $\frac{\pi}{24}$ (3) $\frac{\pi}{4}$ (4) 0

Sol. Answer (2)

$$I = \int_0^{2/3} \frac{dx}{2^2 + (3x)^2}$$

$$= \frac{1}{3}\cdot\frac{1}{2}\left\{\tan^{-1}\left(\frac{3x}{2}\right)\right\}_0^{2/3}$$

$$= \frac{1}{6}\left(\frac{\pi}{4}\right) = \frac{\pi}{24}.$$

108. The value of $\int_{\log 1/2}^{\log 2} \sin\left\{\frac{e^x - 1}{e^x + 1}\right\} dx$ is equal to

(1) $\cos\frac{1}{3}$ (2) $\sin\frac{1}{2}$ (3) 2cos2 (4) 0

Sol. Answer (4)

$I = \int_{-\log 2}^{\log 2} \sin\left\{\frac{e^x - 1}{e^x + 1}\right\} dx = 0$ {as, $\frac{e^x - 1}{e^x + 1}$ is an odd function, therefore $\sin\left(\frac{e^x - 1}{e^x + 1}\right)$ is also an odd function}.

109. The value of $\int_0^1 \sqrt{\frac{1-x}{1+x}}\, dx$ is equal to

(1) $\left(\frac{\pi}{2} - 1\right)$ (2) $\left(\frac{\pi}{2} + 1\right)$ (3) $\frac{\pi}{2}$ (4) $(\pi + 1)$

Sol. Answer (1)

Let $x = \cos\theta \Rightarrow dx = -\sin\theta d\theta$

so, $I = \int_{\pi/2}^{0} \sqrt{\frac{1-\cos\theta}{1+\cos\theta}}(-\sin\theta)d\theta$

$$= \int_0^{\pi/2} \left(\frac{1-\cos\theta}{\sin\theta}\right)\sin\theta d\theta$$

$$= \int_0^{\pi/2} (1-\cos\theta)d\theta$$

$$= \{\theta - \sin\theta\}_0^{\pi/2}$$

$$= \frac{\pi}{2} - 1$$

110. The value of $\int_0^{\pi/2} \frac{\cos x\, dx}{1+\cos x+\sin x}$ is equal to

(1) $\frac{\pi}{4}+\log 2$ (2) $\frac{\pi}{4}+\frac{1}{2}\log 2$ (3) $\pi-\frac{1}{2}\log 2$ (4) $\frac{\pi}{4}-\frac{1}{2}\log 2$

Sol. Answer (4)

Let $I = \int_0^{\pi/2} \frac{\cos x}{1+\sin x+\cos x}dx$ ...(i)

Again, $I = \int_0^{\pi/2} \frac{\sin x}{1+\cos x+\sin x}dx$ ...(ii) $\{$using $\int_0^{a} f(x)dx = \int_0^{a} f(a-x)dx \}$

Adding (i) and (ii), we get

$$2I = \int_0^{\pi/2} \frac{\sin x+\cos x}{1+\sin x+\cos x}dx$$

$$= \int_0^{\pi/2} dx - \int_0^{\pi/2} \frac{dx}{1+\sin x+\cos x}$$

$$= \frac{\pi}{2} - \int_0^{\pi/2} \frac{\left(1+\tan^2\frac{x}{2}\right)dx}{\left(\tan^2\frac{x}{2}+1\right)+2\tan\frac{x}{2}+\left(1-\tan^2\frac{x}{2}\right)}$$

$$\Rightarrow \quad 2I = \frac{\pi}{2} - \int_0^{\pi/2} \frac{\sec^2\frac{x}{2}dx}{2\left(1+\tan\frac{x}{2}\right)} \quad \text{...(iii)}$$

Let $I_1 = \int_0^{\pi/2} \frac{\sec^2\frac{x}{2}dx}{2\left(1+\tan\frac{x}{2}\right)}$

Put $\tan\frac{x}{2} = z \Rightarrow \frac{1}{2}\sec^2\frac{x}{2}dx = dz$

so, $I_1 = \int_0^1 \frac{dz}{1+z} = \{\log(1+z)\}_0^1 = \log 2$

Using (iii), we have

$I = \frac{\pi}{4} - \frac{1}{2}\log 2$.

111. If $I_1 = \int_0^{n\pi} f(|\cos x|)dx$ and $I_2 = \int_0^{5\pi} f(|\cos x|)dx$, then

(1) $\frac{I_1}{I_2} = \frac{5}{n}$ (2) $\frac{I_1}{I_2} = \frac{n}{5}$ (3) $I_1 + I_2 = n + 5$ (4) $I_1 - I_2 = n - 5$

Sol. Answer (2)

$I_1 = \int_0^{n\pi} f(|\cos x|)dx = n\int_0^{\pi} f(|\cos x|)dx$ {as $\int_0^{nT} f(x)dx = n\int_0^{T} f(x)dx$, where T is period of the function}

$I_2 = \int_0^{5\pi} f(|\cos x|)dx = 5\int_0^{\pi} f(|\cos x|)dx$

So, $\frac{I_1}{I_2} = \frac{n}{5}$.

112. If for every integer n, $\int_n^{n+1} f(x)dx = n^2$, then the value of $\int_{-2}^{4} f(x)dx$ is equal to

(1) 16 (2) 14 (3) 19 (4) 20

Sol. Answer (3)

$\int_{-2}^{4} f(x)dx = \int_{-2}^{-1} f(x)dx + \int_{-1}^{0} f(x)dx + \int_{0}^{1} f(x)dx + \int_{1}^{2} f(x)dx + \int_{2}^{3} f(x)dx + \int_{3}^{4} f(x)dx$

$= (-2)^2 + (-1)^2 + 0^2 + 1^2 + 2^2 + 3^2 = 19.$

113. The value of $\int_{-\pi/2}^{\pi/2} \frac{dx}{e^{\sin x}+1}$ is equal to

(1) 0 (2) 1 (3) $-\frac{\pi}{2}$ (4) $\frac{\pi}{2}$

Sol. Answer (4)

Let $I = \int_{-\pi/2}^{\pi/2} \frac{dx}{e^{\sin x}+1}$ …(i)

again $I = \int_{-\pi/2}^{\pi/2} \frac{dx}{e^{-\sin x}+1}$ {using $\int_a^b f(x)dx = \int_a^b f(a+b-x)dx$}

$= \int_{-\pi/2}^{\pi/2} \frac{e^{\sin x}}{e^{\sin x}+1}dx$ …(ii)

From (i) + (ii), we get

$$2I = \int_{-\pi/2}^{\pi/2} dx = \pi$$

$$\Rightarrow \quad I = \frac{\pi}{2}.$$

114. The value of $\int_1^2 \frac{dx}{x(1+x^4)}$ is equal to

(1) $\frac{1}{4}\log\frac{17}{32}$ (2) $\frac{1}{4}\log\frac{17}{2}$ (3) $\log\frac{17}{2}$ (4) $\frac{1}{4}\log\frac{32}{17}$

Sol. Answer (4)

$$I = \int_1^2 \frac{x^3 dx}{x^4(1+x^4)}$$

Let $x^4 = t \Rightarrow 4x^3 dx = dt$

so, $I = \int_1^{16} \frac{1}{4} \cdot \frac{dt}{t(1+t)}$

$$= \frac{1}{4}\int_1^{16} \left(\frac{1}{t} - \frac{1}{1+t}\right) dt$$

$$= \frac{1}{4}\left\{\log\left(\frac{t}{1+t}\right)\right\}_1^{16}$$

$$= \frac{1}{4}\left\{\log\left(\frac{16}{17}\right) - \log\left(\frac{1}{2}\right)\right\}$$

$$= \frac{1}{4}\log\left(\frac{32}{17}\right).$$

115. The value of $\int_0^{\pi/2} \log\tan x\, dx$ is equal to

(1) $\frac{\pi}{2}\log_e 2$ (2) $-\frac{\pi}{2}\log_e 2$ (3) $\pi\log_e 2$ (4) 0

Sol. Answer (4)

$I = \int_0^{\pi/2} \log\tan x\, dx$ ...(i)

Again, $I = \int_0^{\pi/2} \log\cot x\, dx$ ...(ii) {using $\int_0^a f(x)dx = \int_0^a f(a-x)dx$ }

Adding (i) and (ii), we get

$$2I = \int_0^{\pi/2} (\log\tan x + \log\cot x)\,dx = \int_0^{\pi/2} 0\,dx = 0$$

$\Rightarrow \quad I = 0.$

116. The value of $\int_{-2}^{2} |1 - x^2|\,dx$ is equal to

(1) 2 (2) 4 (3) 6 (4) 8

Sol. Answer (2)

Here, $\int_{-2}^{2} |1 - x^2|\,dx$

$= 2\int_0^2 |1 - x^2|\,dx$ {using $\int_{-a}^{a} f(x)dx = 2\int_0^a f(x)dx$, where *f(x)* an even function}

$$= 2\left\{ \int_0^1 (1 - x^2)\,dx + \int_1^2 (x^2 - 1)\,dx \right\}$$

$$= 2\left[\left\{ x - \frac{x^3}{3} \right\}_0^1 + \left\{ \frac{x^3}{3} - x \right\}_1^2 \right]$$

$$= 2\left\{ \frac{2}{3} + \left(\frac{2}{3} - \left(-\frac{2}{3} \right) \right) \right\}$$

= 4.

117. The value of $\int_0^{\pi/2} |\sin x - \cos x|\,dx$ is equal to

(1) 0 (2) $2(\sqrt{2} - 1)$ (3) $\sqrt{2} - 1$ (4) $2(\sqrt{2} + 1)$

Sol. Answer (2)

$$I = \int_0^{\pi/2} |\sin x - \cos x|\,dx$$

$$= \int_0^{\pi/4} (\cos x - \sin x)\,dx + \int_{\pi/4}^{\pi/2} (\sin x - \cos x)\,dx$$

$$= \{\sin x + \cos x\}_0^{\pi/4} - \{\sin x + \cos x\}_{\pi/4}^{\pi/2}$$

$$= \left(\sqrt{2} - 1\right) - \left(1 - \sqrt{2}\right)$$

$$= 2\left(\sqrt{2} - 1\right)$$

118. The value of $\int_0^{\pi}|\cos x|dx$ is equal to

(1) π (2) 0 (3) 2 (4) 1

Sol. Answer (3)

$$I = \int_0^{\pi}|\cos x|dx$$

$$= \int_0^{\pi/2}\cos x\,dx + \int_{\pi/2}^{\pi}(-\cos x)dx$$

$$= \{\sin x\}_0^{\pi/2} + \{-\sin x\}_{\pi/2}^{\pi}$$

$= 1 + 1 = 2.$

119. The value of the integral $I = \int_0^1 x(1-x)^n dx$ is equal to

(1) $\frac{1}{n+1}$ (2) $\frac{1}{n+2}$ (3) $\frac{1}{n+1}-\frac{1}{n+2}$ (4) $\frac{1}{n+1}+\frac{1}{n+2}$

Sol. Answer (3)

$$I = \int_0^1 x(1-x)^n\,dx$$

$$= \int_0^1 x^n(1-x)dx \qquad \left\{\text{using } \int_0^a f(x)dx = \int_0^a f(a-x)dx\right\}$$

$$\Rightarrow \quad I = \int_0^1 (x^n - x^{n+1})dx$$

$$= \left\{\frac{x^{n+1}}{n+1}-\frac{x^{n+2}}{n+2}\right\}_0^1$$

$$= \frac{1}{n+1}-\frac{1}{n+2}.$$

120. The value of $\int_1^5(|x-3|+|1-x|)dx$ is equal to

(1) 10 (2) $\frac{5}{6}$ (3) 21 (4) 12

Sol. Answer (4)

$$I = \int_1^5|x-3|dx + \int_1^5|1-x|dx$$

$$= \int_1^3(3-x)dx + \int_3^5(x-3)dx + \int_1^5(x-1)dx$$

$$= \left\{3x - \frac{x^2}{2}\right\}_1^3 + \left\{\frac{x^2}{2} - 3x\right\}_3^5 + \left\{\frac{x^2}{2} - x\right\}_1^5$$

$$= \left(\frac{9}{2} - \frac{5}{2}\right) + \left(\frac{9}{2} - \frac{5}{2}\right) + \left(\frac{15}{2} + \frac{1}{2}\right)$$

$= 12.$

121. To find the numerical value of $\int_{-2}^{2}(px^2 + qx + s)\,dx$, it is necessary to know the value(s) of constant(s)

(1) p (2) q (3) s (4) p and s

Sol. Answer (4)

Here $I = \int_{-2}^{2} px^2\,dx + \int_{-2}^{2} qx\,dx + \int_{-2}^{2} s\,dx$

$$= 2p\int_0^2 x^2\,dx + 0 + 2\int_0^2 s\,dx$$

$$= \frac{2p}{3}\left\{x^3\right\}_0^2 + 2s\left\{x\right\}_0^2$$

$$= \frac{16p}{3} + 4s$$

i.e., depends on p and s.

122. The value of $\int_{e^{-1}}^{e^2}\left|\frac{\log_e x}{x}\right| dx$ is equal to

(1) $\frac{3}{2}$ (2) $\frac{5}{2}$ (3) 3 (4) 5

Sol. Answer (2)

$$\int_{1/e}^{1}\frac{-\log x}{x}dx + \int_{1}^{e}\frac{\log x}{x}dx$$

123. If the value of $\lim_{n\to\infty}\left\{\frac{1}{n+1} + \frac{1}{n+2} + ... + \frac{1}{6n}\right\}$ is 'K' then find value of (K – $\log_e 6$).

(1) 0 (2) 1 (3) 2 (4) –1

Sol. Answer (1)

$$\lim_{n\to\infty}\left\{\frac{1}{n+1} + \frac{1}{n+2} + ... + \frac{1}{6n}\right\} = \sum_{r=1}^{5n}\frac{1}{n+r} = \sum_{r=1}^{5n}\left[\frac{1}{n}\left\{\frac{1}{1+(r/n)}\right\}\right] \quad \left\{\because \text{ put } \frac{r}{n} = x\right\}$$

$$\Rightarrow K = \lim_{n\to\infty} S_n = \int_0^5 \frac{dx}{1+x} = \lfloor \ln(1+x) \rfloor_0^5 = \ln 6$$

$$\therefore \quad K - \ln 6 = \ln 6 - \ln 6 = 0$$

124. If [ ] and { } represents greatest integer function and fractional function then $\int_{-1}^{1}([x]+\{x\}+|x|)\,dx$ equals

(1) 0 (2) 1 (3) –1 (4) $\frac{1}{2}$

Sol. Answer (2)

We know, $[x] + \{x\} = x$

$$\therefore \quad I=\int_{-1}^{1}(x+|x|)dx=\int_{-1}^{1}x\,dx+\int_{-1}^{0}(-x)dx+\int_{0}^{1}x\,dx$$

$$\therefore \quad I=\left[\frac{x^2}{2}\right]_{-1}^{1}-\left[\frac{x^2}{2}\right]_{-1}^{0}+\left[\frac{x^2}{2}\right]_{0}^{1}=1$$

125. Value of $\int_{1}^{9}(x-2)(x-4)(x-6)(x-8)(x-5)dx$ is

(1) 0 (2) 1 (3) –1 (4) $\frac{1}{2}$

Sol. Answer (1)

Let $x - 5 = t$

$$\Rightarrow \quad I=\int_{1}^{9}t(t-1)(t+1)(t-3)(t+3)dt=\int_{1}^{9}t(t^2-1)(t^2-9)dt$$

$$\therefore \quad I=\int_{1}^{9}(t^5-10t^3+9t)dt=\left[\frac{t^6}{6}\frac{-5}{2}t^4+\frac{9}{2}t^2\right]_{1}^{9}=0$$

Alternate method :

$$I=\int_{1}^{9}(x-2)(x-4)(x-6)(x-8)(x-5)dx \;=\int_{1}^{9}(8-x)(4-x)(2-x)(2-x)(5-x)dx \;=-I$$

$\Rightarrow I = 0$

126. $\int_{1}^{\infty}(e^{x+1}+e^{3-x})^{-1}dx$ is equal to

(1) $\frac{\pi}{4e^2}$ (2) $\frac{\pi}{4e}$ (3) $\frac{\pi}{2e^2}$ (4) $\frac{\pi}{2e}$

Sol. Answer (1)

$$\int_1^{\infty}(e^{x+1}+e^{3-x})^{-1}dx = \int_1^{\infty}\frac{2}{e^{x+1}+e^{3-x}}dx$$

$$= \int_1^{\infty}\frac{1}{e^x.e^1+e^3.e^{-x}}dx = \frac{1}{e}\int\frac{e^x dx}{e^{2x}+e^2} = \frac{1}{e}\int\frac{e^x dx}{e^{2x}+e^2}$$

$$= \frac{1}{e}.\frac{1}{e}\left[\tan^{-1}\left(\frac{e^x}{e}\right)\right]_1^{\infty} = \frac{1}{e^2}\left[\tan^{-1}\left(e^{x-1}\right)\right]_1^{\infty}$$

$$= \frac{1}{e^2}(\tan^{-1}\infty-\tan^{-1}1) = \frac{1}{e^2}\left(\frac{\pi}{2}-\frac{\pi}{4}\right) = \frac{\pi}{4e^2}.$$

127. $\int_0^3\left(|x|+|x-1|+|x-2|\right)dx$ equals

(1) $\frac{9}{2}$ (2) $\frac{15}{2}$ (3) $\frac{19}{2}$ (4) $\frac{21}{2}$

Sol. Answer (3)

$$\int_0^3(|x|+|x-1|+|x-2|)dx$$

$$= \int_0^3 xdx - \int_0^1(x-1)dx + \int_1^3(x-1)dx - \int_0^2(x-2)dx + \int_2^3(x-2)dx$$

$$= \frac{19}{2}.$$

128. For a real number y, $[y]$ is greatest integer function less than or equal to y, then the value of integral $\int_{\pi/2}^{3\pi/2}[2\sin x]\,dx$ is

(1) $-\pi$ (2) 0 (3) $\frac{-\pi}{2}$ (4) $\frac{\pi}{2}$

Sol. Answer (3)

$$\int_{\pi/2}^{3\pi/2}[2\sin x]dx = \int_{\pi/2}^{5\pi/6}1.dx + \int_{5\pi/6}^{\pi}0.dx + \int_{\pi}^{7\pi/6}(-1)dx + \int_{7\pi/6}^{3\pi/2}(-2)dx$$

$$= -\frac{\pi}{2}$$

129. Let $y = (x-[x])^{[x]}$ where $[x]$ is greatest integer function, then $\int_0^3 y\,dx$ equals

(1) $\frac{5}{6}$ (2) $\frac{2}{3}$ (3) 1 (4) $\frac{11}{6}$

Sol. Answer (4)

$$y = (x-[x])^{[x]} = \{x\}^{[x]}$$

$x-[x] = \{x\}$ fraction part of x

$$\int_0^3 y dx = \int_0^3 \{x\}^{[x]} dx = \int_0^1 \{x\}^0 dx + \int_1^2 \{x\}^1 dx + \int_2^3 \{x\}^2 dx$$

$$= 1 + \int_1^2 (x-1)dx + \int_2^3 (x-2)^2 dx = \frac{11}{6}.$$

130. The expression $\dfrac{\int_0^n [x]\, dx}{\int_0^n \{x\}\, dx}$, where $[x]$ and $\{x\}$ are integral and fractional parts of x and $n \in N$ is equal to

(1) $\frac{1}{n-1}$ (2) $\frac{1}{n}$ (3) n (4) $n-1$

Sol. Answer (4)

$$\frac{\int_0^n [x]dx}{\int_0^n \{x\}dx}$$

$$\int_0^n [x]dx = \int_0^1 0.dx + \int_1^2 1dx + \int_2^3 2dx + ... + \int_{n-1}^n (n-1)dx$$

$$= 0 + 1 + 2 + 3 + \ldots\ldots\ldots + n-1 = \frac{(n-1)n}{2}$$

$$\int_0^n \{x\}dx = n\int_0^1 (x-[x])dx$$

$$= n\int_0^1 xdx - n\int_0^1 [x]dx \quad [\{x\} \text{ is a function of period 1.}]$$

$$= n\left[\frac{x^2}{2}\right]_0^1 = \frac{n}{2} \qquad \left[\int_0^{nT} f(x)dx = n\int_0^T f(x)dx\right]$$

Hence the expression $\dfrac{\int_0^n [x]dx}{\int_0^n \{x\}dx} = \dfrac{\frac{n(n-1)}{2}}{\frac{n}{2}} = n-1.$

131. Let $f(x)$ be a continuous function such that $f(a-x)+f(x)=0$ for all $x \in [0, a]$. Then the value of the integral $\int_0^a \frac{1}{1+e^{f(x)}}dx$ is equal to

(1) a (2) $\frac{a}{2}$ (3) $f(a)$ (4) $\frac{1}{2}f(a)$

Sol. Answer (2)

$f(a-x)+f(x)=0 \ \forall \ x \in [0, a]$

Let $I = \int_0^a \frac{1}{1+e^{f(x)}}dx$ … (i)

then, $I = \int_0^a \frac{1}{1+e^{f(a-x)}}dx \quad \left[\int_0^a f(x)dx = \int_0^a f(a-x)dx\right]$

$= \int_0^a \frac{1}{e^{-f(x)}+1}dx = \int_0^a \frac{e^{f(x)}}{e^{f(x)}+1}dx$ … (ii)

Adding equations (i) and (ii) we get,

$2I = \int_0^a 1.dx = a$

$I = \frac{a}{2}$.

132. Let $I_1 = \int_0^{\frac{\pi}{2}} \frac{\sin x - \cos x}{1+\sin x.\cos x}dx$; $I_2 = \int_0^{2\pi}(\cos^6 x)dx$; $I_3 = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}(\sin^2 x)dx$ and $I_4 = \int_0^1 \ln\left(\frac{1}{x}-1\right)dx$, then

(1) $I_1 = I_2 = I_3 = I_4 = 0$ (2) $I_1 = I_2 = I_3 = 0$ but $I_4 \neq 0$

(3) $I_1 = I_3 = I_4 = 0$ but $I_2 \neq 0$ (4) $I_1 = I_4 = 0$ but $I_3 \neq 0$, and $I_2 \neq 0$

Sol. Answer (4)

$I_1 = \int_0^{\pi/2} \frac{\sin x - \cos x}{1+\sin x \cos x}dx$ … (i)

$I_1 = \int_0^{\pi/2} \frac{\cos x - \sin x}{1+\sin x \cos x}dx$ … (ii) $\int_0^a f(x)dx = \int_0^a f(a-x)dx$

$2I_1 = 0 \quad \Rightarrow \quad I_1 = 0$

$I_2 = \int_0^{2\pi}\cos^6 x\,dx = 2\int_0^{\pi}\cos^6 x dx = 4\int_0^{\pi/2}\cos^6 x dx$

$= 4.\frac{5.3.1}{6.4.2}\left(\frac{\pi}{2}\right) = \frac{15}{24}\pi = \frac{5}{8}\pi$

$I_3 = \int_{-\pi/2}^{\pi/2}(\sin^2 x)dx \;=\; 2\int_0^{\pi/2}\sin^2 x dx = \frac{1}{2}.\frac{\pi}{2}.2 \;=\; \frac{\pi}{2}$

$I_4 = \int_0^1 \log\left(\frac{1}{x}-1\right)dx \;=\; \int_0^1 \log\left(\frac{1-x}{x}\right)dx$ ... (1)

$\int_0^a f(x)dx = \int_0^a f(a-x)dx$

$I_4 = \int_0^a \log\frac{(1-(1-x))}{1-x}dx = \int_0^1 \log\frac{x}{1-x}dx$

$2I_4 = 0$

$I_4 = 0$

$I_1 = I_4 = 0\,,\; I_2 \neq 0\,,\; I_3 \neq 0\,.$

133. Let f and g be continuous functions in [0, a] such that $f(x) = f(a-x)$ and $g(x) + g(a-x) = 4$ then $\int_0^a f(x).g(x)dx$ equals $\left(\int_0^a f(x)dx = K\right)$

(1) $\frac{K}{2}$ (2) $2K$ (3) K (4) $4K$

Sol. Answer (2)

$f(x) = f(a-x)$ $\qquad$ $\int_0^a f(x)dx = k$ (given)

$g(x) + g(a-x) = 4$

$I = \int_0^a f(x)g(x)dx = \int_0^a f(a-x)g(a-x)dx$

$= \int_0^a f(x)g(a-x)dx \;=\; \int_0^a f(x)(4-g(x))dx$

$\int_0^a f(x)g(x)dx = 4\int_0^a f(x)dx - \int_0^a f(x)g(x)dx$

$2\int_0^a f(x).g(x)dx = 4\int_0^a f(x)dx$

$\int_0^a f(x)g(x)dx = 2\int_0^a f(x)dx = 2k$

134. $\int_{-2}^{0}\{x^3+3x^2+3x+3+(x+1)\cos(x+1)\}dx =$

(1) –4 (2) 0 (3) 4 (4) 6

Sol. Answer (3)

$$\int_{-2}^{0}\{x^3+3x^2+3x+3+(x+1)\cos(x+1)\}dx$$

$$= \int_{-2}^{0}(x+1)^3dx+\int_{-2}^{0}2dx+\int_{-2}^{0}(x+1)\cos(x+1)dx$$

Put $x + 1 = t$

Then $dx = dt$, $x = -2 \Rightarrow t = -1$, $x = 0 \Rightarrow t = -1$

$$\int_{-1}^{1}t^3dt+[2x]_{-2}^{0}+\int_{-1}^{1}t\cos t\,dt$$

But t^3 and $t \cos t$ are odd function of t

$= -2(-2) = 4$ square units. $\int_{-a}^{a}f(x)dx=0$ if $f(-x)=-f(x)$.

135. Let $f(x)=\dfrac{e^x+1}{e^x-1}$ and $\int_0^1\dfrac{x^3(e^x+1)}{(e^x-1)}dx=\alpha$. Then, $\int_{-1}^{1}t^3f(t)\,dt$ is equal to

(1) 0 (2) α (3) 2α (4) α^2

Sol. Answer (3)

We have, $f(x)=\dfrac{e^x+1}{e^x-1}$ and $\int_0^1x^3\left(\dfrac{e^x+1}{e^x-1}\right)dx=\alpha$

Now, $\int_0^1x^3f(x)dx=\int_0^1t^3f(t)dt$

$$\int_0^af(x)dx=\int_0^af(t)dt \qquad \left[\int_a^bf(x)dx=\int_a^bf(t)dt\right]$$

$$f(t)=\frac{e^t+1}{e^t-1}$$

$$f(-t)=\frac{e^{-t}+1}{e^{-t}-1}=\frac{1+e^t}{1-e^t}=-\left(\frac{e^t+1}{e^t-1}\right)$$

$$f(-t)=-f(t)$$

$\Rightarrow$ $f(t)$ is an odd function $\Rightarrow$ $t^3f(t)$ is an even function.

Hence $\int_{-1}^{1} t^3f(t)dt = 2\int_0^1 t^3f(t)dt = 2\int_0^1 x^3f(x)dx = 2\alpha$

136. $\int_{-1/2}^{1/2}\left[\sin^{-1}(3x-4x^3)-\cos^{-1}(4x^3-3x)\right]dx$ is equal to

(1) 0 (2) $-\frac{\pi}{2}$ (3) $\frac{7\pi}{2}$ (4) $\frac{\pi}{2}$

Sol. Answer (2)

$$\int_{-1/2}^{1/2}[\sin^{-1}(3x-4x^3)-\cos^{-1}(4x^3-3x)]dx$$

$$= \int_{-1/2}^{1/2}\left[\sin^{-1}(3x-4x^3)-\left(\frac{\pi}{2}-\sin^{-1}(4x^3-3x)\right)\right]dx$$

$$= \int_{-1/2}^{1/2}\left[\sin^{-1}(3x-4x^3)-\left(\frac{\pi}{2}+\sin^{-1}(3x-4x^3)\right)\right]dx \quad \sin^{-1}(-x) = -\sin^{-1}x$$

$$= \int_{-1/2}^{1/2}-\frac{\pi}{2}dx = -\frac{\pi}{2}[x]_{-1/2}^{1/2} = -\frac{\pi}{2}.$$

137. Let $f(x) = \int\limits_0^x \frac{\sin^{100}t}{\sin^{100}t+\cos^{100}t}\,dt$, then $f(2\pi) =$

(1) $2f\left(\frac{\pi}{2}\right)$ (2) $4f\left(\frac{\pi}{2}\right)$ (3) $f\left(\frac{\pi}{2}\right)$ (4) $3f\left(\frac{\pi}{2}\right)$

Sol. Answer (2)

$f(x) = \int_0^x \frac{\sin^{100}t dt}{\sin^{100}t+\cos^{100}t}$, then $f(2\pi)$

$$f(2\pi) = \int_0^{2\pi}\frac{\sin^{100}t\,dt}{\sin^{100}t+\cos^{100}t} \quad \ldots \text{(i)}$$

$$= 4\int_0^{\pi/2}\frac{\sin^{100}tdt}{\sin^{100}t+\cos^{100}t}$$

$$= 4f\left(\frac{\pi}{2}\right), \quad \int_0^{2a}f(x)dx = 2\int_0^a f(x).dx \text{, if } f(2a-x) = f(x)$$

138. $\int_0^{\pi}(x.\sin^2 x.\cos x)dx$ is equal to

(1) 0 (2) $\frac{2}{9}$ (3) $-\frac{2}{9}$ (4) $-\frac{4}{9}$

Sol. Answer (4)

$$\int_0^{\pi}(x\sin^2 x\cos x)dx = \left[\frac{x\sin^3 x}{3}\right]_0^{\pi} - \frac{1}{3}\int_0^{\pi}\sin^3 x dx$$

$$= -\frac{1}{12}\int_0^{\pi}(3\sin x - \sin 3x)dx, \text{ as } \sin^3 x = \frac{3\sin x - \sin 3x}{4}$$

$$= \left[\frac{x\sin^3 x}{3} - \frac{1}{12}\left[-3\cos x + \frac{\cos 3x}{3}\right]\right]_0^{\pi} = -\frac{4}{9}$$

139. The value of $\int\limits_0^{1000} e^{x-[x]}\, dx$, where [ ] is G.I.F., is

(1) $\frac{e^{1000}-1}{1000}$ (2) $\frac{e^{1000}-1}{e-1}$ (3) 1000 (e – 1) (4) $\frac{e-1}{1000}$

Sol. Answer (3)

$$\int_0^{1000} e^{x-[x]}dx = \int_0^{1000} e^{\{x\}}dx$$

$= 1000\int_0^1 e^{\{x\}}dx$ [$\because$ $e^{\{x\}}$ is a periodic function having period]

$= 1000\int_0^1 e^x dx$ [$\because$ $\{x\} = x \in [0, 1)$]

$= 1000[e^x]_0^1 = 1000\,(e - e^0) = 1000\,(e - 1)$

140. $\int_{19}^{37}(\{x\}^2 + 3\sin 2\pi x)dx$, where $\{x\}$ is fractional part of x, equals

(1) 0 (2) 6 (3) 9 (4) 7

Sol. Answer (2)

We have, $\int_{19}^{37}(\{x\}^2 + 3\sin 2\pi x)dx = \int_{19}^{37}\{x\}^2 dx + 3\int_{19}^{37}\sin 2\pi x dx$

We know that

$\{x\} = x - [x]$

and $\int_{aT}^{bT} f(x)dx = (b-a)\int_0^T f(x)dx$

where $f(x+T) = f(x)$, T is the fundamental period of f.

$\{x\}$ is a function of period 1

$$= (37-19)\int_0^1 \{x\}^2 dx + 3\left[-\frac{\cos 2\pi x}{2\pi}\right]_{19}^{37} = 18\int_0^1 x^2 dx = 6$$

141. If $I_n = \int_0^{\pi/4} \tan^n x\, dx$, then $\frac{1}{I_2+I_4}, \frac{1}{I_3+I_5}, \frac{1}{I_4+I_6}, ...$ form

(1) An A.P. (2) A G.P. (3) A H.P. (4) None of these

Sol. Answer (1)

$$I_n = \int_0^{\pi/4} \tan^n x dx$$

$$I_{n-2} = \int_0^{\pi/4} \tan^{n-2} x dx$$

$$I_n + I_{n-2} = \int_0^{\pi/4} \tan^n x dx + \int_0^{\pi/4} \tan^{n-2} x dx$$

$$= \int_0^{\pi/4} \tan^{n-2} x(\sec^2 x - 1)dx + \int_0^{\pi/4} \tan^{n-2} x dx$$

$$= \int_0^{\pi/4} \tan^{n-2} x.\sec^2 x dx - \int_0^{\pi/4} \tan^{n-2} x dx + \int_0^{\pi/4} \tan^{n-2} x dx$$

$$I_n + I_{n-2} = \left[\frac{\tan^{n-1} x}{n-1}\right]_0^{\pi/4} = \frac{1}{n-1}$$

$$\therefore \quad \frac{1}{I_2+I_4} = 3, \ \frac{1}{I_3+I_5} = 4, \ \frac{1}{I_4+I_6} = 5$$

Hence $\frac{1}{I_2+I_4}, \frac{1}{I_3+I_5}, \frac{1}{I_4+I_6}$........ form an A.P. with common difference '1'.

142. $\lim\limits_{n\to\infty} \sum_{r=1}^{4n} \frac{\sqrt{n}}{\sqrt{r}(3\sqrt{r}+4\sqrt{n})^2}$ is equal to

(1) $\frac{1}{35}$ (2) $\frac{1}{14}$ (3) $\frac{1}{10}$ (4) $\frac{1}{5}$

Sol. Answer (3)

$$\lim_{n\to\infty}\sum_{r=1}^{4n}\frac{\sqrt{n}}{\sqrt{r}(3\sqrt{r}+4\sqrt{n})^2}$$

$$= \lim_{n\to\infty}\sum_{r=1}^{4n}\frac{1}{\sqrt{r/n}}\left(3\sqrt{\frac{r}{n}}+4\right)^2.\frac{1}{n}$$

$$= \int_0^4\frac{dx}{\sqrt{x}(3\sqrt{x}+4)^2}$$. Put $\sqrt{x}=t$ so that $\frac{1}{2\sqrt{x}}dx = dt$.

$$= 2\int_0^2\frac{dt}{(3t+4)^2} = \left[-\frac{2}{(3t+4)}.\frac{1}{3}\right]_0^2 = -\frac{2}{3}\left[\frac{1}{3t+4}\right]_0^2$$

$$= -\frac{2}{3}\left[\frac{1}{10}-\frac{1}{4}\right] = \frac{1}{10}$$

143. $\lim_{k\to 0}\frac{1}{k}\int_0^k(1+\sin 2x)^{\frac{1}{x}}dx$

(1) 2 (2) 1 (3) e^2 (4) Does not exists

Sol. Answer (3)

$$\lim_{k\to 0}\frac{1}{k}\int_0^k(1+\sin 2x)^{1/x}dx$$

Applying L. Hospital Rule

$$\lim_{k\to 0}(1+\sin 2k)^{1/k} = \lim_{k\to 0}\left[(1+\sin 2k)^{\frac{1}{\sin 2k}}\right]^{\frac{\sin 2k}{k}} = e^2$$

144. Let $f(x)$ be a function satisfying $f'(x) = f(x)$, $f(0) = 1$ and g be a function satisfying $f(x) + g(x) = x^2$ then $\int_0^1 f(x)g(x)dx$ equals

(1) $e-\frac{e^2}{2}-\frac{5}{2}$ (2) $e-e^2-3$ (3) $\frac{1}{2}(e-3)$ (4) $e-\frac{1}{2}e^2-\frac{3}{2}$

Sol. Answer (4)

$f'(x) = f(x), f(0) = 1,\ f(x)+g(x) = x^2$ (given)

$f(x) = e^x$

$g(x) = x^2 - f(x) = x^2 - e^x$

$$\int_0^1 f(x)g(x)dx = \int_0^1 e^x(x^2 - e^x)dx$$

$$= \left|x^2e^x\right|_0^1 - \int_0^1 e^x.2xdx - \int_0^1 e^{2x}dx$$

$$= \left[x^2e^x - 2xe^x + 2e^x - \frac{e^{2x}}{2}\right]_0^1 = e - \frac{e^2}{2} - \frac{3}{2}.$$

145. Suppose that $F(x)$ is an antiderivative of $f(x) = \frac{\sin x}{x}$, $x > 0$ then $\int_1^3 \frac{\sin 2x}{x}dx$ can be expressed as

(1) $F(6) - F(2)$ (2) $\frac{1}{2}[F(6) - F(2)]$ (3) $\frac{1}{2}[F(3) - F(1)]$ (4) $2[F(6) - F(2)]$

Sol. Answer (1)

$F(x) = \int f(x)dx = \int \frac{\sin x}{x}dx$ $x > 0$

$\int_1^3 \frac{\sin 2x}{x}dx$ put $2x = t$, $dx = \frac{dt}{2}$

$\frac{1}{2}\int_2^6 \frac{\sin t}{t/2}.dt = \int_2^6 \frac{\sin t}{t}dt = \int_2^6 \frac{\sin x}{x}dx$

$[F(x)]_2^6 = F(6) - F(2)$.

146. If $f(x) = \begin{cases} \sqrt{1-x} & ; \ 0 \le x \le 1 \\ (7x-6)^{-1} & ; \ 1 < x \le 2 \end{cases}$ then $\int_0^2 f(x)dx$ equals

(1) $\frac{2}{3} - \frac{1}{7}\log 8$ (2) $\frac{2}{3} + \frac{1}{7}\log 2$ (3) $\frac{1}{42}$ (4) $\frac{2}{3} + \frac{1}{7}\log 8$

Sol. Answer (4)

We have, $f(x) = \begin{cases} \sqrt{1-x} & ; \ 0 \le x \le 1 \\ (7x-6)^{-1} & ; \ 1 < x \le 2 \end{cases}$ then $\int_0^2 f(x)dx$

$$= \int_0^2 f(x)dx = \int_0^1 f(x)dx + \int_1^2 f(x)dx = \int_0^1 \sqrt{1-x}\,dx + \int_1^2 (7x-6)^{-1}dx$$

$$= \frac{-2}{3}\left[(1-x)^{3/2}\right]_0^1 + \frac{1}{7}\left[\log(7x-6)\right]_1^2 = \frac{2}{3} + \frac{1}{7}\log 8 = \frac{2}{3} + \frac{3}{7}\log 2$$

147. If $f:R\rightarrow R, f(x)=x+\sin x$, then value of $\int_0^{\pi}\left(f^{-1}(x)\right)dx$ equals

(1) $\frac{\pi^2}{2}-2$ (2) π^2 (3) $\frac{\pi^2}{2}+2$ (4) $\frac{\pi^2}{2}$

Sol. Answer (1)

Let $I=\int_0^x f^{-1}(x)dx$; put $\begin{matrix} x=f(t) \\ dx=f'(t)dt \end{matrix}$

$$=\int_{f^{-1}(0)}^{f^{-1}(x)} \underset{I}{t}.\underset{II}{f'(t)}dt$$

Using by parts,

$$\therefore \quad I=\left\lfloor tf(t)\right\rfloor_{f^{-1}(0)}^{f^{-1}(x)}-\int_{f^{-1}(0)}^{f^{-1}(x)} f(t)dt$$

$$\therefore \quad I=\left\lfloor t\{t+\sin t\}\right\rfloor_{f^{-1}(0)}^{f^{-1}(x)}-\int_{f^{-1}(0)}^{f^{-1}(x)} f(t)dt$$

$$\therefore \quad \int_0^{\pi} f^{-1}(x)dx=\left\lfloor t^2+t\sin t\right\rfloor_0^{\pi}-\int_0^{\pi}(t+\sin t)dt$$

$$=\pi^2-\left[\frac{t^2}{2}-\cos t\right]_0^{\pi}$$

$$=\pi^2-\left\{\frac{\pi^2}{2}+1-(0-1)\right\}$$

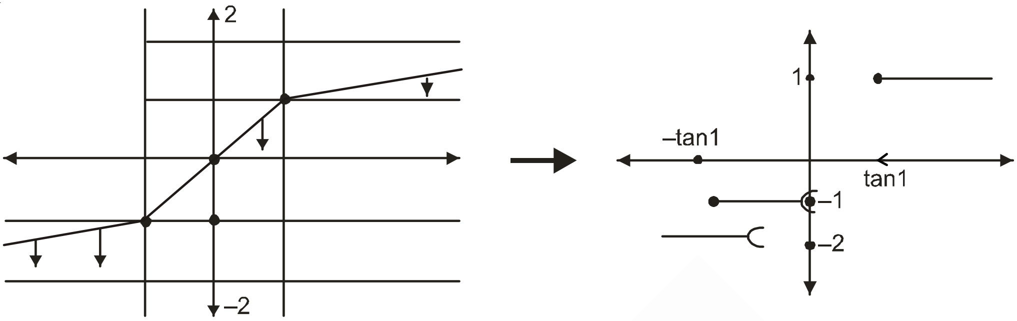
$$=\frac{\pi^2}{2}-2$$

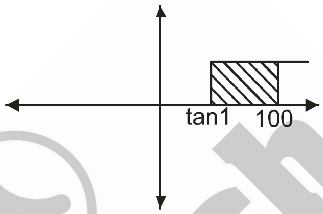
148. The value of $\int_0^{100}\left[\tan^{-1}x\right]dx$ is, (where [.] denotes greatest integer function)

(1) $\tan 1-\tan 100$ (2) $\frac{\pi}{2}-\tan 1$ (3) $100-\tan 1$ (4) $\tan 1-\frac{\pi}{2}$

Sol. Answer (3)

Let $I=\int_{0}^{100}\left[\tan^{-1}x\right]dx$ ($\tan^{-1}x$ is shown below)





$\therefore \int_{0}^{100}\left[\tan^{-1}x\right]dx$ is shown as

$= 1\times(100 - \tan 1)$

149. The value of $\int_{0}^{2} f(x)dx$, where $f(x)=\begin{cases}0, & \text{when } x=\dfrac{n}{n+1}, n=1,2,3,\ldots \\ 1, & \text{otherwise}\end{cases}$ is equal to

(1) 1 (2) 2 (3) 3 (4) 0

Sol. Answer (2)

$$\int_{0}^{2} f(x)dx=\int_{0}^{1/2}1.dx+\int_{1/2}^{2/3}1.dx+\int_{2/3}^{3/4}1.dx+\ldots\ldots+\int_{\frac{n-1}{n}}^{\frac{n}{n+1}}1.dx+\ldots\ldots+\int_{1}^{2}1.dx$$

$$=\frac{1}{2}+\left(\frac{2}{3}-\frac{1}{2}\right)+\left(\frac{3}{4}-\frac{2}{3}\right)+\ldots\ldots+\left(\frac{n}{n+1}-\frac{n-1}{n}\right)+\ldots\ldots+1$$

$$=\frac{n}{n+1}+\ldots\ldots+1 \qquad \text{as } n\to\infty$$

We take, Limit $n\to\infty$

$$\therefore \int_{0}^{2} f(x)dx = 1+1=2$$

150. Let $g(x)$ be a continuous and differentiable function such that $\int_0^2 \left\{ \int_{\sqrt{2}}^{\sqrt{5/2}} \left[2x^2-3\right]dx \right\} . g(x)dx = 0$, then $g(x) = 0$ when $x \in (0, 2)$ has (where [.] denote greatest integer function)

(1) Exactly one real root (2) Atleast one real root (3) No real root (4) Two real roots

Sol. Answer (2)

As $1 < 2x^2 - 3 < 2 \quad \forall\ x \in \left(\sqrt{2}, \sqrt{\frac{5}{2}}\right)$

$\therefore \quad \int_{\sqrt{2}}^{\sqrt{5/2}} \left[2x^2-3\right]dx > 0 \quad \forall\ x \in (0, 2)$

$\Rightarrow$ $g(x)$ = 0 should have at least one real root in (0, 2) $\{\because\ g'(x) \neq 0\}$

151. Let $\frac{d}{dx}(F(x)) = \frac{e^{\sin x}}{x}, x > 0$. If $\int_1^4 \frac{2\, e^{\sin x^2}}{x} dx = F(K) - F(1)$, then the possible value of 'K' is ;

(1) 10 (2) 14 (3) 16 (4) 18

Sol. Answer (3)

Let $\frac{d}{dx}(F(x)) = \frac{e^{\sin x}}{x}$

$\Rightarrow \int \frac{e^{\sin x}}{x} dx = F(x)$

Now $\int_1^4 \frac{2e^{\sin x^2}}{x} dx = \int_1^4 \frac{e^{\sin x^2}}{x^2} d(x^2) = \int_1^{16} \frac{e^{\sin t}}{t} dt \quad \{\text{put} \quad x^2 = t\}$

$= \lfloor F(t) \rfloor_1^{16} = F(16) - F(1) \Rightarrow K = 16$

152. Let $f : (0, \infty) \to R$ and $F(x) = \int_0^x t.f(t)dt$. If $F(x^2) = x^4 + x^5$, then $\sum_{r=1}^{12} f(r^2)$ equals,

(1) 216 (2) 219 (3) 221 (4) 225

Sol. Answer (2)

$F(x^2) = x^4 + x^5 = \int_0^{x^2} t.f(t)dt$

Differentiating w.r.t x,

$x^2 f(x^2).2x = 4x^3 + 5x^4$

$\therefore \quad f(x^2) = 2 + \frac{5}{2}x$

$\Rightarrow \quad f(r^2) = 2 + \frac{5}{2}r$

$\therefore \quad \sum_{r=1}^{n} f(r^2) = 2n + \frac{5}{2}\left[\frac{n(n+1)}{2}\right] = n\left[\frac{5n+13}{4}\right]$

$\therefore \quad \sum_{r=1}^{12} f(r^2) = 12\frac{(60+13)}{4} = 219$

153. $\int_{-\pi/4}^{\pi/4} x^3 \sin^4 x\,dx$ equals

(1) 0 (2) –1 (3) $\frac{\pi^2}{16}$ (4) $\frac{-\pi^2}{16}$

Sol. Answer (1)

$f(x) = x^3 \sin^4 x$

Then $f(-x) = (-x)^3 \sin^4(-x) = -x^3 \sin^4 x = -f(x)$

$\Rightarrow$ $f(x)$ is an odd function

$\Rightarrow \quad \int_{-\pi/4}^{\pi/4} x^3 \sin^4 x\; dx = 0$

154. The value of $\int_{-10}^{10}\left[\{f(f(x)\}\times\left\{f\left(f\left(\frac{1}{x}\right)\right)\right\}\right]dx$, where $f(x) = \frac{1-x}{1+x}$

(1) 0 (2) 20 (3) 30 (4) 25

Sol. Answer (2)

Given $f(x) = \frac{1-x}{1+x}$

$\Rightarrow \quad f(f(x)) = f\left(\frac{1-x}{1+x}\right)$

$= f(a),\ a = \frac{1-n}{1+x}$

$= \frac{1-a}{1+a},$

$= \frac{1-\frac{1-x}{1+x}}{1+\frac{1-x}{1+x}} = x$

Similarly $f\left(f\left(\frac{1}{x}\right)\right)=\frac{1}{x}$

$\Rightarrow \ = \int_{-10}^{10}\left(x\times\frac{1}{x}\right)dx$

$= \int_{-10}^{10} dx$

= (10 + 10)

= 20

155. The value of $\int_{-2}^{2}\left(ax^{2011}+bx^{2001}+c\right)dx$, where a, b, c are constants, depends on the value of

(1) a (2) c (3) b (4) a & b

Sol. Answer (2)

Now, $\int_{-2}^{2}\left(ax^{2011}+bx^{2001}+c\right)dx$

$= c(2 + 2)$ $\left(\because\ x^{2011}, x^{2001} \text{ are odd functions}\right)$

156. If for non-zero x, $3f(x)+4f\left(\frac{1}{x}\right)=\frac{1}{x}-10$, then the value of $\int_{2}^{3}f(x)dx$ is

(1) $\frac{4}{7}\log\frac{2}{3}$ (2) $\frac{3}{4}\log\frac{3}{2}$ (3) $\frac{3}{7}\log\frac{2}{3}$ (4) 10

Sol. Answer (3)

Given $3f(x)+4f\left(\frac{1}{x}\right)=\frac{1}{x}-10$ ….(1)

Replacing x by $\frac{1}{x}$, we have

$3f\left(\frac{1}{x}\right)+4f(x)=x-10$ ….(2)

$\therefore$ (1) × (3) – (2) × 4, we have

$9f(x) - 16f(x) = \frac{3}{x}-30-4x+40$

$$\Rightarrow \quad -7f(x) = 10 + \frac{3}{x} - 4x$$

$$\Rightarrow \quad f(x) = \frac{4x}{7} - \frac{3}{7x} - \frac{10}{7}$$

$$\Rightarrow \quad \int_2^3 f(x)\,dx = \left(\frac{2x^2}{7} - \frac{3}{7}\log|x| - \frac{10}{7}x\right)\Bigg|_2^3$$

$$= \frac{1\cancel{8}}{\cancel{7}} - \frac{3}{7}\log 3 - \frac{3\cancel{0}}{\cancel{7}} - \frac{\cancel{8}}{\cancel{7}} + \frac{3}{7}\log 2 + \frac{2\cancel{0}}{\cancel{7}}$$

$$= \frac{3}{7}\log 2 - \frac{3}{7}\log 3$$

$$= \frac{3}{7}\log\left(\frac{2}{3}\right)$$

157. The value of $\displaystyle\int_{2010\pi+\frac{\pi}{6}}^{2010\pi+\frac{\pi}{3}} (\sin x + \cos x)dx$ is

(1) $\sqrt{2}-1$ (2) $\sqrt{3}-1$ (3) $\sqrt{3}+1$ (4) $\sqrt{2}+1$

Sol. Answer (2)

$$\int_{2010\pi+\frac{\pi}{6}}^{2010\pi+\frac{\pi}{3}} (\sin x + \cos x)dx$$

$$= \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} (\sin x + \cos x)dx$$

$$= (\sin x - \cos x)\Big|_{\pi/6}$$

$$= \frac{\sqrt{3}}{2} - \frac{1}{2} - \frac{1}{2} + \frac{\sqrt{3}}{2}$$

$$= \sqrt{3} - 1$$

158. Let $\displaystyle\int_a^b f(x)dx = \int_{a+c}^{b+c} f(x-c)dx$. Then the value of $\displaystyle\int_0^{\pi} \sin^{2010} x.\cos^{2009} x\,dx$ is

(1) 2010 (2) 2009 (3) 0 (4) 2011

Sol. Answer (3)

$$\int_0^{\pi} \sin^{2010} x.\cos^{2009} x\,dx$$

$$= \int_{-\pi/2}^{\pi/2} \sin^{2010}\left(x+\frac{\pi}{2}\right).\cos^{2009}\left(x+\frac{\pi}{2}\right)dx$$

$$= -\int_{-\pi/2}^{\pi/2} \cos^{2010} x.\sin^{2009} x\,dx$$

$= 0$

159. If $x^{f(x)} = e^{x-f(x)}$, then the value of integral

$$\int_1^2 f'(x)\left\{(1+\log x)^2\right\}dx+1 \text{ is}$$

(1) 3 log 3 (2) 2 log 2 (3) log 2 (4) log 3

Sol. Answer (2)

Given, $x^{f(x)} = e^{x-f(x)}$

$\Rightarrow f(x)\log x = x - f(x)$

$\Rightarrow f(x)(1 + \log x) = x$

$$\Rightarrow f(x) = \frac{x}{1+\log x}$$

$$\Rightarrow f'(x) = \frac{\log x}{(1+\log x)^2}$$

$\Rightarrow f'(x)(1+\log x)^2 = \log x$

$\Rightarrow \int f'(x)(1+\log x)^2\,dx = \int \log x\,dx$

$= x \log x - x$

$$\therefore \int_1^2 f'(x)\left((1+\log x)^2\right)dx+1$$

$$= x\log x - x\Big|_1^2 + 1$$

$$= 2\log 2 - 2 + 1 + 1$$

$= 2\log 2$

160. If $\int_0^x tf(t)dt = \sin x - x\cos x - \frac{x^2}{2}, \forall\, x \in R - \{0\}$. Then the value of $f\left(\frac{\pi}{6}\right)$ is

(1) 0 (2) 1 (3) $-\frac{1}{2}$ (4) 10

Sol. Answer (3)

Given, $\int_0^x t\,f(t)dt = \sin x - x\cos x - \frac{x^2}{2}$

$\Rightarrow$ $x\,f(x) = \cos x + x\sin x - \cos x - x$

$\Rightarrow$ $f(x) = \sin x - 1$

$\Rightarrow$ $f\left(\frac{\pi}{6}\right) = \sin\frac{\pi}{6} - 1 = \frac{1}{2} - 1 = -\frac{1}{2}$

161. The value of the integral $\int_{-2012\pi}^{2012\pi} \left\{\frac{d}{dx}\int_0^{x^2}(\cos t^2)dt\right\}dx$ is

(1) 2012π (2) 4024π (3) 0 (4) -2012π

Sol. Answer (3)

Now, $\frac{d}{dx}\int_0^{x^2}(\cos t^2)dt$

$= 2x\cos\left(x^4\right)$

$\Rightarrow$ $\int_{-2012\pi}^{2012\pi}\left\{\frac{d}{dx}\int_0^{x^2}(\cos t^2)dt\right\} = \int_{-2012\pi}^{2012\pi} 2x\cos\left(x^4\right)dx = 0$

162. The value of $\int_{-2010\pi}^{2010\pi}\left\{x^{2011}\left(1+\tan\left(\frac{\pi}{3}-x\right)\right)\left(1+\tan\left(x-\frac{\pi}{12}\right)\right)\right\}dx$ is

(1) $\frac{\pi}{2012}$ (2) $\frac{\pi}{2010}$ (3) 0 (4) $\frac{(2012)^2}{2}\theta$

Sol. Answer (3)

As we know that, $(1 + \tan A)(1 + \tan B) = 2$

where $A + B = \frac{\pi}{4}$

Here $A = \frac{\pi}{3} - x, B = x - \frac{\pi}{12}$

$$\Rightarrow A + B = \frac{\pi}{3} - \cancel{x} + \cancel{x} - \frac{\pi}{12} = \frac{4\pi - \pi}{12} = \frac{3\pi}{12} = \frac{\pi}{4}$$

$$\therefore \int_{-2010\pi}^{2010\pi} \left\{ x^{2011}\left(1+\tan\left(\frac{\pi}{3}-x\right)\right)\left(1+\tan\left(x-\frac{\pi}{12}\right)\right) dx\right.$$

$$= \int_{-2010\pi}^{2010\pi} \left(x^{2011}x^2\right) dx$$

$$= 0$$

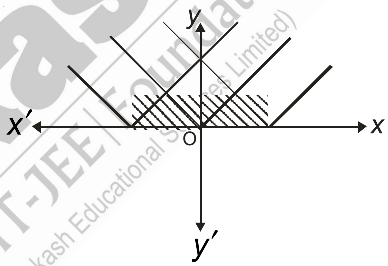
163. Let $f(x)=\min\{|x+2|,|x|,|x-2|\}$, then the value of the integral $\int_{-2}^{2} f(x)dx$ is

(1) 4 sq. units (2) 10 sq. units (3) 2 sq. units (4) 7 sq. units

Sol. Answer (3)

Required Area = $2\times\frac{1}{2}\times 2\times 1$

$= 2$ sq. units



164. If $\int \frac{dx}{x(x^2+1)^3} = \frac{1}{4(f(x))^2} + \frac{1}{2f(x)} + \frac{1}{2}\ln\left(\frac{x^2}{f(x)}\right) + c$, then $\lim\limits_{x\to 0}(f(x))^{1/x}$ is

(1) 3 (2) e (3) e^2 (4) 1

Sol. Answer (4)

Consider $I = \int \frac{dx}{x(x^2+1)^3} = \int \frac{x\,dx}{x^2(1+x^2)^3}$

Put $1 + x^2 = t$ so that $2x \cdot dx = dt$

$$I = \frac{1}{2}\int \frac{dt}{(t-1)t^3} = \frac{1}{2}\int \frac{1}{t^3}\left[\frac{t^3}{t-1} - 1 - t - t^2\right] dt$$

$$= \frac{1}{2}\int \left(\frac{1}{t-1} - \frac{1}{t^3} - \frac{1}{t^3} - \frac{1}{t}\right) dt$$

$$= \frac{1}{2}\left|\ln(t-1) + \frac{1}{2t^2} + \frac{1}{t} - \ln t\right| + c$$

$$= \frac{1}{4}t^2 + \frac{1}{2t} + \frac{1}{2}\ln\left(\frac{t-1}{t}\right) + c$$

Put the value of t, we get

$$I = \frac{1}{4}\frac{1}{(x^2+1)^2} + \frac{1}{2(x^2+1)} + \frac{1}{2}\ln\left(\frac{x^2}{x^2+1}\right) + c$$

Clearly $f(x) = 1 + x^2$

$$\lim_{x \to 0} f(x)^{1/x} = 1$$

165. $\int_{1/2}^{7/2}\left[\frac{1}{2}\left(|x-3|+|1-x|-4\right)\right]dx$ equals

(1) $-\frac{3}{2}$ (2) $\frac{9}{8}$ (3) $\frac{1}{4}$ (4) $-\frac{11}{4}$

Sol. Answer (4)

$$\int_{1/2}^{7/2}\left[\frac{1}{2}\left(|x-3|+|1-x|-4\right)\right]dx$$

$$= \frac{1}{2}\int_{1/2}^{7/2}|x-3|\,dx + \frac{1}{2}\int_{1/2}^{7/2}|1-x|\,dx - \frac{4}{2}\int_{1/2}^{7/2}dx$$

$|x-3| = x-3$ if $x \ge 3$ and $|1-x| = 1-x$ if $\begin{matrix}1-x \ge 0\\ x \le 1\end{matrix}$

$$= \frac{1}{2}\left[-\int_{1/2}^{3}(x-3)dx + \int_{3}^{7/2}(x-3)dx + \int_{1/2}^{1}(1-x)dx - \int_{1}^{7/2}(1-x)dx - 4\int_{1/2}^{7/2}dx\right] = -11$$

166. $\lim_{n\to\infty}\sum_{r=2n+1}^{3n}\frac{n}{r^2-n^2}$ is equal to

(1) $\ln\sqrt{\frac{2}{3}}$ (2) $\ln\sqrt{\frac{3}{2}}$ (3) $\ln\frac{2}{3}$ (4) $\ln\frac{3}{2}$

Sol. Answer (2)

$$\lim_{n\to\infty}\sum_{r=2n+1}^{3n}\frac{n}{r^2-n^2} = \lim_{n\to\infty}\frac{1}{n}\sum_{r=2n+1}^{3n}\frac{1}{\left(\frac{r^2}{n^2}-1\right)} = \int_2^3\frac{dx}{x^2-1} = \frac{1}{2}\left[\log\left|\frac{x-1}{x+1}\right|\right]_2^3 = \log\sqrt{\frac{3}{2}}$$

167. If $\left[\int_0^1 \frac{dt}{t^2+2t\cos\alpha+1}\right]x^2 - \left[\int_{-3}^{3} \frac{t^2 \sin 2t}{t^2+1}dt\right]x - 2 = 0$ $(0 < \alpha < \pi)$ then the value of x is

(1) $\pm\sqrt{\frac{\alpha}{2\sin\alpha}}$ (2) $\pm\sqrt{\frac{2\sin\alpha}{\alpha}}$ (3) $\pm\sqrt{\frac{\alpha}{\sin\alpha}}$ (4) $\pm 2\sqrt{\frac{\sin\alpha}{\alpha}}$

Sol. Answer (4)

$$\left[\int_0^1 \frac{dt}{t^2+2t\cos\alpha+1}\right]x^2 - \left[\int_{-3}^{3} \frac{t^2 \sin 2t}{t^2+1}dt\right]x - 2 = 0 \qquad (0 < \alpha < \pi)$$

$\int_{-a}^{a} f(x)dx = 0$ if $f(x)+f(-x)=0$

Let $I = \int_0^1 \frac{dt}{(t+\cos\alpha)^2+\sin^2\alpha} = \frac{1}{\sin\alpha}\left[\tan^{-1}\left(\frac{t+\cos\alpha}{\sin\alpha}\right)\right]_0^1$

$$= \frac{1}{\sin\alpha}\left[\tan^{-1}\frac{2\cos^2\frac{\alpha}{2}}{2\sin\frac{\alpha}{2}\cos\frac{\alpha}{2}} - \tan^{-1}\cot\alpha\right] = \frac{1}{\sin\alpha}\left[\tan^{-1}\cot\frac{\alpha}{2} - \tan^{-1}\cot\alpha\right]$$

$$= \frac{1}{\sin\alpha}\left[\tan^{-1}\tan\left(\frac{\pi}{2}-\frac{\alpha}{2}\right) - \tan^{-1}\tan\left(\frac{\pi}{2}-\alpha\right)\right] = \frac{1}{\sin\alpha}\left[\left(\frac{\pi}{2}-\frac{\alpha}{2}\right)-\left(\frac{\pi}{2}-\alpha\right)\right] = \frac{\alpha/2}{\sin\alpha} = \frac{\alpha}{2\sin\alpha}$$

$\therefore$ From the question

$$x^2 = \frac{2}{\int_0^1 \frac{dt}{t^2+2t\cos\alpha+1}} = \frac{2\sin\alpha}{\alpha/2} = 4\frac{\sin\alpha}{\alpha}$$

$$x = \pm 2\sqrt{\frac{\sin\alpha}{\alpha}}$$

168. Let a, b, c be non-zero real numbers such that $\int_0^1 (1+\cos^8 x)(ax^2+bx+c)dx = \int_0^2 (1+\cos^8 x)\ (ax^2+bx+c)dx$, then the equation $ax^2+bx+c = 0$ has

(1) No root in (0, 2) (2) At least one root in (0, 2)

(3) Two roots in (0, 2) (4) None of these

Sol. Answer (2)

$\Rightarrow \int_1^2 (1+\cos^8 x)(ax^2+bx+c)dx = 0$

$\therefore$ $(1+\cos^8 x)(ax^2+bx+c) = 0$ always its sign in (1, 2)

i.e., $ax^2+bx+c=0$ has at least one root in (0, 2).

169. $\int_0^1 \left(\log\frac{1}{x}\right)^{n-1} .dx$, equals

(1) $\overline{|n}$ (2) $\overline{|n+1}$ (3) $\overline{|(n-1)}$ (4) $\overline{\left|\frac{(n-1)}{2}\right.}$

Sol. Answer (1)

$$I = \int_0^1 \left(\log\frac{1}{x}\right)^{n-1} dx, \text{ put } \log\frac{1}{x} = t \Rightarrow dx = -e^{-t}dt$$

$$\therefore \quad I = \int_\infty^0 t^{n-1}(-e^{-t})dt = \int_0^\infty e^{-t}t^{n-1}dt$$

$$\therefore \quad \int_0^1 \left(\log\frac{1}{x}\right)^{n-1} dx = \overline{|n}$$

170. The value of $\int_2^{10} \{(x-1)(x-2)(x-3)\text{.........}(x-11)\}dx$ is

(1) 45 (2) 55 (3) 0 (4) 110

Sol. Answer (3)

$$\int_2^{10} \{(x-1)(x-2)(x-3)(x-4)(x-5)\text{.........}(x-11)\}\,dx$$

Put $x - 6 = t$

$$= \int_{-4}^{4} \{(t+5)(t+4)(t+3)(t+2)(t+1)(t-1)\text{.......}(t-5)\}\,dt$$

= 0. ($\because$ the given function is an odd function).

171. The value of $\int_{-2}^{2} \tan\left(\sin\left(\log\left(\sqrt{x^2+1}+x\right)\right)\right)dx$ is

(1) cot 4 (2) tan 4 (3) 0 (4) sin 4

Sol. Answer (3)

$$\int_{-2}^{2} \underset{\downarrow \atop 0}{\tan}\left(\underset{\downarrow \atop 0}{\sin}\left(\log\left(\underset{\downarrow \atop 0}{\sqrt{x^2}}+1+x\right)\right)\right)dx$$

= 0 ($\because$ Compositions of all odd functions is an odd function)

172. Let $x^2-(f(x))^2=t-\frac{1}{t}$ and $x^4+(f(x))^4=t^2+\frac{1}{t^2}$. Then the value of $\int_{-2012}^{2012}\left(x^3\cdot f(x)\cdot f'(x)+x\right)dx$ is

(1) $\frac{x^4}{4}+c$ (2) $\frac{\{f(x)\}^4}{4}+c$ (3) –4024 (4) $\frac{f'(x)}{3f(x)}+c$

Sol. Answer (3)

Given, $x^2-(f(x))^2=t-\frac{1}{t}$

$\Rightarrow\ x^4-2x^2\{f(x)\}^2+(f(x))^4=t^2+\frac{1}{t^2}-2$

$\Rightarrow\ t^2+\frac{1}{t^2}-2x^2\{f(x)\}^2=t^2+\frac{1}{t^2}-2$

$\Rightarrow\ x^2\{f(x)\}^2=1$

$\Rightarrow\ f(x)=\pm\frac{1}{x}$

Now, $x^3f(x).f'(x)=x^3\times\frac{1}{x}\times-\frac{1}{x^2}=-1$

$\Rightarrow\ x^3\cdot f(x)\cdot f'(x)+x=x-1$

$\therefore\ \int_{-2012}^{2012}\left(x^3.f(x).f'(x)+x\right)dx$

$\Rightarrow\ \int_{-2012}^{2012}(x-1)dx=-4024$

173. Let $f(x)=\left[\frac{x}{(1+x)}+\frac{x}{(1+x)(1+2x)}+........+\frac{x}{(1+9x)(1+10x)}\right]$ then the value of the integral $\int_{-5}^{5}\left(f(x)+\frac{1}{(1+10x)}\right)dx$ is

(1) 20 (2) 9 (3) 10 (4) 40

Sol. Answer (3)

Given,

$$f(x)=\left[\frac{x}{(1+x)}+\frac{x}{(1+x)1+2x}+.......+\frac{x}{(1+9x)(1+10x)}\right]=\left[1-\frac{1}{1+10x}\right]$$

$$f(x)=\frac{10x}{1+10x}$$

$$\because \int_{-5}^{5}\left(f(x)+\frac{1}{1+10x}\right)dx$$

$$=\int_{-5}^{5}\left(\frac{10x}{10x+1}+\frac{1}{10x+1}\right)dx$$

$$=\int_{-5}^{5}dx=[x]_{-5}^{5}=10$$

174. Let $f(x) = \cos(\log x)$. Then the value of the integral $\int_{-2010}^{2010}\left(f(x).f(f(x))-\frac{1}{2}\left\{f\left(\frac{x}{f(x)}\right)+f\left(xf(x)\right)\right\}\right)dx$ is

(1) 2010 (2) 4020 (3) 0 (4) 1005

Sol. Answer (3)

Let $f(x) = y$

Now, $f(x).f(y)-\frac{1}{2}\left(f\left(\frac{x}{y}\right)+f(xy)\right)$

$$=\frac{1}{2}\left[2\cos(\log x)\cos(\log y)-\left(\cos\left(\log\frac{x}{y}\right)+\cos\log(xy)\right)\right]$$

$$=\frac{1}{2}\left[\cos\log xy+\cos\log\left(\frac{x}{y}\right)-\cos\left(\log\left(\frac{x}{y}\right)-\cos\log(xy)\right)\right]$$

$= 0$

$$\Rightarrow \int_{-2010}^{2010}\left\{f(x).f(f(x))-\frac{1}{2}f\left(\frac{x}{f(x)}\right)+f\left(xf(x)\right)\right\}dx$$

$= 0$

175. Let $f(x)=\int_{0}^{\infty}e^{-x}x^{n-1}dx,\ n\in N, x\in Q^{+}$. Then the value of $f(5)$ is

(1) 20 (2) 5 (3) 24 (4) 120

Sol. Answer (3)

Given, $f(x)=\int_0^{\infty}e^{-x}x^{n-1}dx$

$\therefore \quad f(x+1) = \int_0^\infty e^{-x}x^n dx$

$$= x^n \left.\frac{e^{-x}}{-1}\right|_0^\infty - \int_0^\infty nx^{n-1}x\frac{e^{-x}}{-1}dx$$

$$= 0 + x\int_0^\infty e^{-x}x^{n-1}dx = n\,f(x)$$

$\Rightarrow f(x+1) = n.(n-1)(n-2)\ldots\ldots 3.2\, f(1)$

$= n!$

$\Rightarrow f(x+1) = n!$

$\Rightarrow f(5) = 4! = 24$

176. The value of the integral $\int_{-2}^{2}\left([x]+\log\left(\frac{1+x}{1-x}\right)+\sin\left(\log\left(\frac{2+x}{2-x}\right)\right)+x^{2011}\right)dx$ is

(1) 4 (2) 2 (3) –2 (4) –4

Sol. Answer (3)

We have,

$$\int_{-2}^{2}\left([x]+\log\left(\frac{1+x}{1-x}\right)+\sin\left(\log\left(\frac{2+x}{2-x}\right)\right)+x^{2011}\right)dx$$

$$= \int_{-2}^{2}([x]dx+0+0+0)$$

$$= \left(\int_{-2}^{-1}[x]+\int_{-1}^{0}[x]+\int_{0}^{1}[x]+\int_{0}^{2}[x]\right)dx$$

$$= \int_{-2}^{-1}(-2)dx+\int_{-1}^{0}(-1)dx+\int_{0}^{1}0.dx+\int_{1}^{2}1\,dx$$

= –2(–1 + 2) + –1(0 + 1) + 0 + 1(2 – 1)

= –2 –1 + 1

= –2

177. The value of the integral $\int_{-4}^{4}\left(x^3-6x^2+12x-8\right)dx$ is

(1) –64 (2) 72 (3) 50 (4) –320

Sol. Answer (4)

$$\int_{+4}^{0}(x^3-6x^2+12x-8)dx$$

$$=\int_{-4}^{4}(x-2)^3\,dx=\int_{-6}^{2}x^3 dx$$

$$=\frac{1}{4}[x^4]_{-6}^{2}=\frac{1}{4}[2^4-6^4]$$

$$=4-2^2\cdot 3^4$$

$$= -320$$

178. If $\int_0^{\pi/2}\log\sin x\,dx=k,$ then the value of the definite integral $\int_0^{\pi/4}\log(1+\tan\theta)d\theta$ is

(1) $-\frac{K}{8}$ (2) $-\frac{K}{4}$ (3) $\frac{K}{8}$ (4) $\frac{K}{4}$

Sol. Answer (2)

Given, $\int_0^{\pi/2}\log\sin x\,dx=k$

$\therefore$ Let $I=\int_0^{\pi/4}\log(1+\tan\theta)d\theta$

$$=\int_0^{\pi/4}\log\left(1+\tan\left(\frac{\pi}{4}-\theta\right)\right)d\theta$$

$$=\int_0^{\pi/4}\log\left(1+\frac{1-\tan\theta}{1+\tan\theta}\right)d\theta$$

$$=\int_0^{\pi/4}\log\left(\frac{2}{1+\tan\theta}\right)d\theta$$

$$=\int_0^{\pi/4}\log 2d\theta-\int_0^{\pi/4}\log(1+\tan\theta)d\theta$$

$$=\frac{\pi}{4}\log 2-I$$

$$\Rightarrow\ 2I=\frac{\pi}{4}\log 2$$

$$\Rightarrow \quad I = \frac{\pi}{8}\log 2 = -\frac{\pi}{2}\times\log\left(\frac{1}{2}\right)\times\frac{1}{4}$$

$$= -\frac{K}{4}\left(\because \int_0^{\pi/2}\log\sin x\,dx = -\frac{\pi}{2}\log\left(\frac{1}{2}\right)\right)$$

179. $\frac{36}{\pi}\int_{\frac{\pi}{6}}^{\frac{\pi}{3}}\frac{dx}{1+\sqrt{\cot x}}$ equals to ____.

(1) 3 (2) 6 (3) 9 (4) 1

Sol. Answer (1)

$$I = \frac{36}{\pi}\int_{\pi/6}^{\pi/3}\frac{dx}{1+\sqrt{\cot x}} = \frac{36}{\pi}\int_{\pi/6}^{\pi/3}\frac{\sqrt{\sin x}}{\sqrt{\sin x}+\sqrt{\cos x}}dx = \frac{36}{\pi}\int_{\pi/6}^{\pi/3}\frac{\sqrt{\sin\left(\frac{\pi}{2}-x\right)}}{\sqrt{\sin\left(\frac{\pi}{2}-x\right)}+\sqrt{\cos\left(\frac{\pi}{2}-x\right)}}dx$$

$$\Rightarrow \quad 2I = \frac{36}{\pi}\int_{\pi/6}^{\pi/3}dx = 6 \qquad \left(\int_a^b f(x)dx = \int_a^b f(a+b-x)dx\right)$$

$\Rightarrow \quad I = 3.$

180. The following integral $\int_{\frac{\pi}{4}}^{\frac{\pi}{2}}(2\operatorname{cosec} x)^{17}dx$ is equal to **[JEE(Advanced)-2014]**

(1) $\int_0^{\log(1+\sqrt{2})}2(e^u+e^{-u})^{16}du$ (2) $\int_0^{\log(1+\sqrt{2})}(e^u+e^{-u})^{17}du$

(3) $\int_0^{\log(1+\sqrt{2})}(e^u-e^{-u})^{17}du$ (4) $\int_0^{\log(1+\sqrt{2})}2(e^u-e^{-u})^{16}du$

Sol. Answer (1)

$$I = \int_{\frac{\pi}{4}}^{\frac{\pi}{2}}(2\operatorname{cosec} x)^{17}dx = \int_{\frac{\pi}{4}}^{\frac{\pi}{2}}\left(\operatorname{cosec} x+\cot x+\frac{1}{\operatorname{cosec} x+\cot x}\right)^{16}\cdot 2\operatorname{cosec} x dx$$

Let cosec$x$ + cot$x$ = e^u

$\Rightarrow$ –cosecxdx = du

$$\Rightarrow \quad I = -2\int_{\ln(\sqrt{2}+1)}^{0}(e^u+e^{-u})^{16}du = \int_0^{\ln(\sqrt{2}+1)}2(e^u+e^{-u})^{16}du$$

181. Let $f : [0, 2] \to \mathbb{R}$ be a function which is continuous on [0, 2] and is differentiable on (0, 2) with $f(0) = 1$.

Let $F(x) = \int_0^{x^2} f(\sqrt{t})dt$ for $x \in [0, 2]$. If $F'(x) = f'(x)$ for all $x \in (0, 2)$, then $F(2)$ equals

[JEE(Advanced)-2014]

(1) $e^2 - 1$ (2) $e^4 - 1$ (3) $e - 1$ (4) e^4

Sol. Answer (2)

$F'(x) = f(x).2x = f'(x)$

$\Rightarrow \dfrac{f'(x)}{f(x)} = 2x$

$\Rightarrow f(x) = ke^{x^2}$

Given, $f(0) = 1 \Rightarrow f(x) = e^{x^2}$

So, $F(x) = \int_0^{x^2} e^x dx = e^{x^2} - 1$

So, $F(2) = e^4 - 1$

SECTION - B

Objective Type Questions (More than one options are correct)

1. $\int \dfrac{dx}{\sqrt{x^2 + 2x + 1}} = A\log|x+1| + C, x \neq 1$ then

(1) $A = \pm 1$ (2) $A = 1$ if $x > -1$ (3) $C \in R$ (4) $A = -1$ if $x < -1$

Sol. Answer (2, 3, 4)

$\int \dfrac{dx}{\sqrt{x^2 + 2x + 1}} = A \log|x+1| + C,\ x \neq 1$

$= \int \dfrac{dx}{|x+1|}$ $|x+1| = x+1$ if $x+1 \geq 0,\ x \geq -1$

$= \int \dfrac{dx}{x+1} = \log|x+1| + C$

Also, $|x+1| = -(x+1)$ if $x + 1 < 0$

$-\int \dfrac{dx}{x+1} = -\log|x+1| + C$

Hence, $\int \dfrac{dx}{x+1} = \pm\log|x+1| + C,\ x \neq 1$

$A = 1$, if $x > -1$

$A = -1$, if $x < -1$

$C \in R$

2. $\int \frac{x^2+\cos^2 x}{1+x^2}\operatorname{cosec}^2 x dx$ is equal to

(1) $\cot x + \cot^{-1} x + C$

(2) $C - \cot x + \cot^{-1} x$

(3) $-\tan^{-1} x - \frac{\cos x}{\sin x} + C$

(4) $-e^{\ln \tan^{-1} x} - \cot x - C$

Sol. Answer (2, 3, 4)

$$\int \frac{x^2+\cos^2 x}{1+x^2}\operatorname{cosec}^2 x\,dx$$

$$= \int \frac{(x^2+1)-(1-\cos^2 x)}{1+x^2}\operatorname{cosec}^2 x\,dx$$

$$= \int\left(\operatorname{cosec}^2 x - \frac{1}{1+x^2}\right)dx$$

$$I = -\cot x - \tan^{-1} x + C$$

$$= C - \cot x - e^{\ln \tan^{-1} x}$$

3. If $I = \int(\sqrt{\tan x}+\sqrt{\cot x})dx = f(x)+C$ then $f(x)$ is equal to

(1) $\sqrt{2}\sin^{-1}(\sin x - \cos x)$

(2) $\frac{\pi}{\sqrt{2}} - \sqrt{2}\cos^{-1}(\sin x - \cos x)$

(3) $\sqrt{2}\tan^{-1}\left(\frac{\tan x - 1}{\sqrt{2\tan x}}\right)$

(4) $\sin 2x$

Sol. Answer (1, 2, 3)

$$I = \int\left(\sqrt{\tan}\,x + \sqrt{\cot}\,x\right)dx = f(x)+C$$

$$= \int \frac{\sin x + \cos x}{\sqrt{\sin x \cos x}}dx = \sqrt{2}\int \frac{\sin x + \cos x}{\sqrt{2\sin x \cos x}}dx$$

$$= \sqrt{2}\int \frac{(\sin x + \cos x)}{\sqrt{1-(\sin x - \cos x)^2}}dx$$

Let $\sin x - \cos x = t$

$(\cos x + \sin x)\,dx = dt$

$$= \sqrt{2}\int \frac{dt}{\sqrt{1-t^2}} = \sqrt{2}\sin^{-1} t = \sqrt{2}\sin^{-1}(\sin x - \cos x) + C$$

$$f(x) = \sqrt{2}\,\sin^{-1}(\sin x - \cos x)$$

$$= \sqrt{2}\left(\frac{\pi}{2} - \cos^{-1}(\sin x - \cos x)\right) = \frac{\pi}{\sqrt{2}} - \sqrt{2}\,\cos^{-1}(\sin x - \cos x)$$

$$= \sqrt{2}\,\tan^{-1}\left(\frac{\sin x - \cos x}{\sqrt{2\sin x\cos x}}\right) = \sqrt{2}\,\tan^{-1}\left(\frac{\tan x - 1}{\sqrt{2\tan x}}\right)$$

4. $\int \frac{dx}{5+4\cos x} = \lambda \tan^{-1}\left(m\tan\frac{x}{2}\right) + C$ then

(1) $\lambda = \frac{2}{3}$ (2) $m = \frac{1}{3}$ (3) $\lambda = \frac{1}{3}$ (4) $m = \frac{2}{3}$

Sol. Answer (1, 2)

$$\int \frac{dx}{5+4\cos x} = \lambda \tan^{-1}\left(m\tan\frac{x}{2}\right) + C, \text{ then}$$

$$\int \frac{dx}{5+4\left(\frac{1-\tan^2 x/2}{1+\tan^2 x/2}\right)} = \int \frac{\sec^2 x/2}{3^2+\tan^2 x/2}\,dx$$

Let $\tan x/2 = t$

$$\frac{1}{2}\sec^2 x/2\,dx = dt$$

$$\sec^2 x/2\,dx = 2dt$$

$$\int \frac{2dt}{t^2+3^2} = \frac{2}{3}\tan^{-1}\left(\frac{t}{3}\right) = \frac{2}{3}\tan^{-1}\left(\frac{\tan x/2}{3}\right) + C$$

$$\lambda = \frac{2}{3},\ m = \frac{1}{3}$$

5. $\int e^{\sin^2 x}(\cos x + \cos^3 x)\sin x\,dx$ equals

(1) $\frac{1}{2}e^{\sin^2 x}(3-\sin^2 x) + C$ (2) $e^{\sin^2 x}\left(1+\frac{1}{2}\cos^2 x\right) + C$

(3) $e^{\sin^2 x}(3\cos^2 x + 2\sin^2 x) + C$ (4) $e^{\sin^2 x}(2\cos^2 x + 3\sin^2 x) + C$

Sol. Answer (1, 2)

$$I = \int e^{\sin^2 x}(\cos x + \cos^3 x)\sin x\, dx$$

$$= \int e^{\sin^2 x}\sin x \cos x\, dx + \int e^{\sin^2 x}\cdot \cos^2 x \cdot \cos x \cdot \sin x\, dx$$

$$= I_1 + I_2 \text{ (say)}$$

$$I_1 = \int e^{\sin^2 x}\cdot \sin x \cos x\, dx \text{ Let } \sin^2 x = t \Rightarrow 2\sin x \cos x\, dx = dt$$

$$= \frac{1}{2}\int e^t\, dt = \frac{e^t}{2} + C_1 = \frac{e^{\sin^2 x}}{2} + C_1$$

$$I_2 = \int e^{\sin^2 x}\cdot \cos^2 x \cdot \cos x \sin x\, dx = \int e^{\sin^2 x}(1 - \sin^2 x)\sin x \cos x\, dx$$

Let $\sin^2 x = t$

$2\sin x \cos x\, dx = dt$

$$= \frac{1}{2}\int e^t (1-t)\, dt$$

$$= \frac{1}{2}\left[e^t - (te^t - e^t)\right] = C_2 = \frac{1}{2}[2e^t - te^t] + C_2 = \frac{e^t}{2}(2-t) + C_2$$

$$= e^{\sin^2 x} - \frac{1}{2}\sin^2 x \cdot e^{\sin^2 x} + C_2$$

$$I = I_1 + I_2 = \frac{e^{\sin^2 x}}{2} + e^{\sin^2 x} - \frac{\sin^2 x}{2}e^{\sin^2 x} + C_1 + C_2$$

$$= \frac{e^{\sin^2 x}}{2} + e^{\sin^2 x} - \frac{\sin^2 x}{2}e^{\sin^2 x} + C$$

$$= \frac{1}{2}e^{\sin^2 x}(3 - \sin^2 x) + C$$

$$= e^{\sin^2 x}\left(1 + \frac{\cos^2 x}{2}\right) + C$$

6. If $\int (3x^3 - 17)e^{2x}dx = \{Ax^3 + Bx^2 + Cx + D\}e^{2x} + k$ then

(1) $3A + B = 0$ (2) $B + C = 0$ (3) $B - D = 0$ (4) $A + B + C + D = -\frac{71}{8}$

Sol. Answer (1, 2, 4)

We have,

$\int(3x^3 - 17)e^{2x}dx = \{Ax^3 + Bx^2 + Cx + D\}e^{2x} + K$

Differentiate both sides w.r.t x, we get

$(3x^3 - 17)e^{2x} = 2\{Ax^3 + Bx^2 + Cx + D\}e^{2x} + e^{2x}\{3Ax^2 + 2Bx + C\}$

$\therefore\ 3x^3 - 17 = 2Ax^3 + (2B + 3A)x^2 + (2C + 2B)x + (2D + C)$

Equating, coefficients of equal powers of x, we get

$2A = 3,\ 2B + 3A = 0,\ 2C + 2B = 0,\ 2D + C = -17$

Solving the system, we get

$A = \frac{3}{2},\ B = \frac{-9}{4},\ C = \frac{9}{4},\ D = \frac{-77}{8}$

Hence option (1), (2), (4) correct

7. The relation $\int(\sin 2x + \cos 2x)dx = \frac{1}{\sqrt{2}}\sin(2x - a) + c$ is true for

(1) $a = \frac{\pi}{4}$ (2) $a = -\frac{\pi}{4}$ (3) $c \in R$ (4) c is arbitrary

Sol. Answer (1, 4)

$I = \int(\sin 2x + \cos 2x)\,dx = -\frac{1}{\sqrt{2}}[\cos 2x - \sin 2x] +$

$= \frac{1}{\sqrt{2}}\sin(2x - \pi/4) + C$

Hence option (1), (4) correct

8. The value of λ for which $\int\frac{4x^3 + \lambda 4^x}{4^x + x^4}dx = \log(4^x + x^4)$ is

(1) 1 (2) $\log_e 4$ (3) $\log_4 e$ (4) $\log_{e^2} 16$

Sol. Answer (2, 4)

We have, $\int\frac{4x^3 + \lambda 4^x}{4^x + x^4}dx = \log(4^x + x^4)$

Differentiate both sides,

$$\frac{4x^3 + \lambda\cdot 4^x}{4^x + x^4} = \frac{4^x\cdot\log e^4 + 4x^3}{4^x + x^4}$$

On comparison $\lambda = \log e^4$

9. If primitive of $\sqrt{(1 + \sec x)}$ is $2\,fog(x) + c$, then

(1) $g(x) = \sec x - 1$ (2) $g(x) = \sqrt{(\sec x - 1)}$ (3) $f(x) = 2\tan^{-1} x$ (4) $f(x) = \tan^{-1} x$

Sol. Answer (2, 4)

$I = \int\sqrt{1+\sec x}$

$= \int\frac{\sqrt{\sec^2 x - 1}\cdot \sec x}{\sec x\cdot\sqrt{\sec x - 1}}\,dx$

$\therefore\ I = \int\frac{\sec x \tan x}{\sec x\sqrt{\sec x - 1}}\,dx$

Put $\sec x - 1 = y^2$

$\Rightarrow \sec x \tan x\, dx = 2y\, dy$

$\therefore\ I = 2\int\frac{y\,dy}{(y^2+1)y} = 2\tan^{-1}y + c$

$\therefore\ I = 2\tan^{-1}(\sqrt{\sec x - 1}) + c$

Hence option (2), (4) correct

10. Let $f(0) = f'(0) = 0$ and $f''(x) = \sec^4 x + 4$, then

(1) $f\left(\frac{\pi}{4}\right) = \frac{1}{3}\log 2 + \frac{1}{6} + \frac{\pi^2}{8}$

(2) $f\left(-\frac{\pi}{4}\right) = \frac{1}{3}\log 2 + \frac{1}{6} + \frac{\pi^2}{8}$

(3) $f(x) = \frac{2}{3}\log|\cos x| + \frac{1}{6}\tan^2 x + 2x^2$

(4) $f(x) = \frac{2}{3}\log|\cos x| - \frac{1}{6}\tan^2 x + 2x^2$

Sol. Answer (1, 2)

$f''(x) = \sec^4 x + 4$

$\Rightarrow f'(x) = 4x + \int(1+\tan^2 x)\sec^2 x\, dx + C_1$

$= 4x + \tan x + \frac{1}{3}\tan^3 x + C_1$

$\therefore\ f'(0) = 0 \Rightarrow a = 0$

Now, $f(x) = 2x^2 + \log|\sec x| + \frac{1}{3}\int(\sec^2 x - 1)\tan x\, dx + C_2$

$\therefore\ f(x) = 2x^2 + \log|\sec x| + \frac{1}{3}\frac{\tan^2 x}{2} - \frac{1}{3}\log|\sec x| + C_2$

$\because\ f(0) = 0 \Rightarrow C_2 = 0$

$\therefore\ f(x) = 2x^2 + \frac{2}{3}\log|\sec x| + \frac{1}{6}\tan^2 x$

Hence option (1), (2) correct

11. If $\int \sec 2x\, dx = f\{g(x)\} + c$, then

(1) Domain of $f(x) = R - \{0\}$

(2) Range of $g(x) = R$

(3) $f'(x) = \frac{1}{2x} \forall x \in R^{+}$

(4) $g'(x) = -\text{cosec}^2\left(\frac{\pi}{4} - x\right)$

Sol. Answer (1, 2, 3)

$$\int \sec 2x\, dx = \frac{1}{2}\log|\sec 2x + \tan 2x| + c$$

$$= \frac{1}{2}\log\left|\frac{1+\sin 2x}{\cos 2x}\right| + c$$

$$= \frac{1}{2}\log\left|\frac{1+\cos\left(\frac{\pi}{2} - 2x\right)}{\sin\left(\frac{\pi}{2} - 2x\right)}\right| + c$$

$$= \frac{1}{2}\log\left|\cot\left(\frac{\pi}{4} - x\right)\right| + c$$

$\therefore$ $f(x) = \frac{1}{2}\log|x|$ and $g(x) = \cot\left(\frac{\pi}{4} - x\right)$

$\therefore$ Dom of $f(x) = R - \{0\}$, Range of $g(x) = R$

Also, $f'(x) = \frac{1}{2x} \ \forall\ x \in R^{+}$

Also, $g'(x) = -\text{cosec}^2\left(\frac{\pi}{4} - x\right)(-1)$

$= \text{cosec}^2\left(\frac{\pi}{4} - x\right)$; option (1), (2), (3) correct

12. If $\int \tan^4 x\, dx = A\tan^3 x + B\tan x + \phi(x)$, then

(1) $A = \frac{1}{3}$

(2) $B = 1$

(3) $\phi(x) = x + c$

(4) $B = -1$

Sol. Answer (1, 3, 4)

$$\int \tan^4 x\, dx = \int \tan^2 x(\sec^2 x - 1)dx = \int \tan^2 x \sec^2 x\, dx - \int \tan^2 x\, dx$$

$$= \frac{\tan^3 x}{3} - \int (\sec^2 x - 1)\, dx = \frac{\tan^2 x}{3} - \tan x + x + c$$

$\therefore$ $A = \frac{1}{3}$, $B = -1$, $\phi(x) = x + c$

13. If primitive of sin (log x) is $f(x)\cdot\{\sin g(x) - \cos h(x)\}+c$, ($c$ being constant of integration), then

(1) $\lim_{x\to 2} f(x) = 1$ (2) $\lim_{x\to 1} \frac{g(x)}{h(x)} = 1$ (3) $g(e^3) = 3$ (4) $h(e^5) = 5$

Sol. Answer (1, 2, 3, 4)

$I = \int \sin(\log x)dx,$ Put $\log x = t \Rightarrow x = e^t$

$\therefore\ dx = e^t\, dt$

$\therefore\ I = \int e^t \sin t\, dt$

$I = \frac{e^t}{2}\{\sin t - \cos t\} + c$

$= \frac{x}{2}\{\sin(\log x) - \cos(\log x)\} + c$

On comparing, we get

$(x) = \frac{x}{2}$, $g(x) = \log x$, $h(x) = \log x$

Hence option (1), (2), (3), (4) correct.

14. If $\int\left\{\cos^{-1} x + \cos^{-1}\sqrt{1-x^2}\right\} dx = Ax + f(x)\sin^{-1} x - 2\sqrt{1-x^2} + c\ \forall\ x \in [-1, 0)$, then

(1) $f(x) = x$ (2) $f(x) = -2x$ (3) $A = \frac{\pi}{4}$ (4) $A = \frac{\pi}{2}$

Sol. Answer (2, 4)

$\because\ x \in [-1, 0)\ \cos^{-1}\sqrt{1-x^2} = -\sin^{-1}x$

$\therefore\ \int\{\cos^{-1} x + \cos^{-1}\sqrt{1-x^2}\}\, dx = \int(\cos^{-1} x - \sin^{-1} x)dx = \int\left(\frac{\pi}{2} - 2\sin^{-1} x\right) dx$

$= \frac{\pi}{2} x - 2x \sin^{-1}x + 2\{-\sqrt{1-x^2}\} + C$

On comparison, we get

$A = \frac{\pi}{2}$, $f(x) = -2x$

15. If $\int \frac{3x+4}{x^3 - 2x - 4} dx = \log|x-2| + k\log f(x) + c$, then

(1) $f(x) = |x^2 + 2x + 2|$ (2) $f(x) = x^2 + 2x + 2$ (3) $k = \frac{1}{4}$ (4) $k = -\frac{1}{2}$

Sol. Answer (1, 2, 4)

$$\int \frac{3x+4}{x^3-2x-4}\,dx = \int \frac{dx}{x-2} - \frac{1}{2}\int \frac{2x+2}{x^2+2x+2}\,dx \text{ \{Using partial fraction\}}$$

$$= \log_e|x-2| - \frac{1}{2}\log|x^2+2x+2| + C$$

$\Rightarrow K = \frac{-1}{2}$, $f(x) = |x^2 + 2x + 2|$

16. A primitive of $\sin 6x$ is

(1) $\frac{1}{3}(\sin^6 x - \sin^3 x) + c$

(2) $-\frac{1}{3}\cos^2 3x + c$

(3) $\frac{1}{3}\sin^2 3x + c$

(4) $\frac{1}{3}\sin\left(3x + \frac{\pi}{7}\right)\sin\left(3x - \frac{\pi}{7}\right) + c$

Sol. Answer (2, 3, 4)

$$\int \sin 6x\, dx = -\frac{1}{6}\cos 6x + c$$

$$= \frac{-1}{6}\{1 - 2\sin^2 3x\} + c = \frac{1}{3}\sin^2 3x + d$$

$$= \frac{-1}{6}\cos 6x + c = \frac{-1}{6}\{2\cos^2 3x - 1\} + c$$

$$= \frac{-1}{3}\cos^2 3x + c$$

Also, derivative of $\frac{1}{3}\sin\left(3x + \frac{\pi}{7}\right)\sin\left(3x + \frac{\pi}{7}\right)$ is sin6x

17. If $\int \frac{\sin x}{\sin(x-a)}\,dx = f(x)\sin a + g(x)\cos a + c$, then

(1) $f(x) = \log|\sin(x-a)|$ (2) $g(x) = x - a$ (3) $g(x) = \sin(x-a)$ (4) $f(x) = \sin x$

Sol. Answer (1, 2)

Let $x - a = t$ $\therefore$ $I = \int \frac{\sin(a+t)}{\sin t}\,dt$

$\Rightarrow dx = dt$ $\quad I = \int \frac{\sin a \cos t + \cos a \sin t}{\sin t}\,dt \Rightarrow I = \sin a \int \cot t\, dt + \cos a \int dt$

$\therefore I = \sin a \log|\sin(x-a)| + (x-a)\cos a + c$

18. If $f(x)=\int_1^x \frac{\ln t}{1+t}dt$ where $x > 0$ then the values of x satisfying the equation $f(x)+f\left(\frac{1}{x}\right)=2$ are

(1) 2 (2) e (3) e^{-2} (4) e^{2}

Sol. Answer (3, 4)

$$f(x)=\int_1^x \frac{\log t}{1+t}dt \text{ where } x>0$$

$$f\left(\frac{1}{x}\right)=\int_1^{1/x}\frac{\log t}{1+t}dt$$

Put $t=\frac{1}{u}$ so that $dt=-\frac{1}{u^2}du$

$$f\left(\frac{1}{x}\right)=\int_1^x \frac{\log\left(\frac{1}{u}\right)\left(-\frac{1}{u^2}\right)du}{1+\frac{1}{u}} = \int_1^x \frac{\log u}{u(u+1)}du = \int_1^x \frac{\log t}{t(1+t)}dt$$

$$f(x)+f\left(\frac{1}{x}\right)=\int_1^x \frac{\log t}{(1+t)}dt+\int_1^x \frac{\log t}{t(1+t)}dt = \int_1^x \frac{\log t}{t}dt = \frac{1}{2}(\log x)^2$$

$$f(x)+f\left(\frac{1}{x}\right)=2 \text{ (given)}$$

$$2=\frac{1}{2}(\log x)^2$$

$$4=(\log x)^2$$

$$\log_e x = 2$$

$$x=e^2$$

If $\log_e x=-2$

$$x=e^{-2}$$

Then the value of x satisfying the equation $f(x)+f\left(\frac{1}{x}\right)=2$ are e^2 and e^{-2}.

19. $\int_0^\infty \frac{x\,dx}{(1+x)(1+x^2)}$

(1) $\frac{\pi}{4}$ (2) $\frac{\pi}{2}$

(3) Is same as $\int_0^\infty \frac{dx}{(1+x)(1+x^2)}$ (4) None of these

Sol. Answer (1, 3)

$$\int_0^{\infty} \frac{xdx}{(1+x)(1+x^2)}$$

Let $x = \tan\theta$

$dx = \sec^2\theta d\theta$

$$I = \int_0^{\pi/2} \frac{\tan\theta.\sec^2\theta d\theta}{(1+\tan\theta).\sec^2\theta} = \int_0^{\pi/2} \frac{\sin\theta}{\sin\theta+\cos\theta} d\theta = \frac{\pi}{4}\frac{1}{2}$$

20. Let $f(x) = \int_0^x \frac{\sin t}{t} dt$ $(x > 0)$ then $f(x)$ has $(n \in N)$

(1) Maximum if $x = (2n - 1)\pi$

(2) Minimum if $x = 2n\pi$

(3) Maximum if $x = 2n\pi$

(4) The function is monotonic

Sol. Answer (1, 2)

$$f(x) = \int_0^x \frac{\sin t}{t} dt \quad (x > 0)$$

$\Rightarrow$ $f'(x) = \frac{\sin x}{x}$

and $f''(x) = \frac{x\cos x - \sin x}{x^2}$

Now $f'(x) = 0$

$\Rightarrow$ $\sin x = 0$ $\Rightarrow x = n\pi,\ n \in N.$

When $x = 2n\pi$

$\Rightarrow$ $f'(x)$ is positive.

$\Rightarrow$ $f(x)$ has minimum at $x = 2n\pi$

When $x = (2n-1)\pi$

$\Rightarrow$ $f''(x) < 0$

$\Rightarrow$ $f(x)$ has maximum if $x = (2n-1)\pi$.

21. $\int_0^1 \frac{(2x^2+3x+3)dx}{(x+1)(x^2+2x+2)}$ equals

(1) $\frac{\pi}{4}+2\ln 2-\tan^{-1}2$

(2) $\frac{\pi}{4}+2\ln 2-\tan^{-1}\frac{1}{3}$

(3) $2\ln 2-\cot^{-1}3$

(4) $-\frac{\pi}{4}+\ln 4+\cot^{-1}2$

Sol. Answer (1, 3, 4)

$$\int_0^1 \frac{(2x^2+3x+3)dx}{(x+1)(x^2+2x+2)} = \int_0^1 \frac{(2x^2+4x+4)-(x+1)}{(x+1)(x^2+2x+2)}dx$$

$$= 2\int_0^1 \frac{dx}{x+1} - \int_0^1 \frac{dx}{(x+1)^2+1^2}$$

$$= 2[\log(x+1)]_0^1 - [\tan^{-1}(x+1)]_0^1$$

$$= 2\log 2 - \tan^{-1}2 + \frac{\pi}{4}$$

$$= 2\log 2 - \left(\frac{\pi}{2} - \cot^{-1}2\right) + \frac{\pi}{4}$$

$$= \log 4 - \frac{\pi}{4} + \cot^{-1}2$$

$$= 2\log 2 - [\tan^{-1}2 - \tan^{-1}1]$$

$$= 2\log 2 - \tan^{-1}\frac{1}{3} = 2\log 2 - \cot^{-1}3$$

22. $\int_a^b \text{sgn}(x)dx$ equals $(a, b \in R)$

(1) $|b| - |a|$

(2) $(b - a)\ \text{sgn}\ (b - a)$

(3) $b\ \text{sgn}(b) - a\ \text{sgn}(a)$

(4) $|a| - |b|$

Sol. Answer (1, 3)

$$\int_a^b \text{sgn}(x)dx \qquad (a, b \in R)$$

Note that sgn $(x) = -1$ if $x < 0$

$= 0$ if $x = 0$

$= 1$ if $x > 0$

If $0 \le a < b$ then

$$I = \int_a^b dx = b - a = |b| - |a|$$

If $a < 0 \le b$

$$\int_a^0 (-1)dx + \int_0^b 1dx = a + b = b - (-a) = |b| - |a|$$

If $a < b < 0$

$$\int_a^b (-1)dx = -b + a = |b| - |a|$$

$$b\,\text{sgn}(b) - a\,\text{sgn}(a) = b.\frac{|b|}{b} - a.\frac{|a|}{a} = |b| - |a|.$$

23. If $A_n = \int_0^{\frac{\pi}{2}} \frac{\sin(2n-1)x}{\sin x}dx$; $B_n = \int_0^{\frac{\pi}{2}} \left(\frac{\sin nx}{\sin x}\right)^2 dx$; for $n \in N$, then

(1) $A_{n-1} = A_n$ (2) $B_{n+1} = B_n$ (3) $A_{n+1} - A_n = B_{n+1}$ (4) $B_{n+1} - B_n = A_{n+1}$

Sol. Answer (1, 4)

We observe that

$$A_n - A_{n-1} = \int_0^{\pi/2}\left(\frac{\sin(2n-1)x}{\sin x} - \frac{\sin(2n-3)x}{\sin x}\right)dx$$

$$= \int_0^{\pi/2}\frac{\sin(2n-1)x - \sin(2n-3)x}{\sin x}dx$$

$$= \int_0^{\pi/2}\frac{2\cos(2n-2)x\sin x}{\sin x}dx$$

$$= \int_0^{\pi/2} 2\cos 2(n-1)x dx = 2\left[\frac{\sin 2(n-1)x}{2(n-1)}\right]_0^{\pi/2} = \frac{1}{n-1}[\sin 2(n-1)x]_0^{\pi/2} = 0$$

$$A_n - A_{n-1} = 0$$

$$A_n = A_{n-1}$$

Also, $B_{n+1} - B_n = \int_0^{\pi/2}\left[\left(\frac{\sin(n+1)x}{\sin x}\right)^2 - \left(\frac{\sin nx}{\sin x}\right)^2\right]dx$

$$= \int_0^{\pi/2}\frac{(\sin(n+1)x + \sin x)(\sin(n+1)x - \sin nx)dx}{\sin^2 x}$$

$$= \int_0^{\pi/2}\frac{2\sin\left(\frac{2n+1}{2}\right)x\cos\frac{x}{2}.2\cos\left(\frac{2n+1}{2}\right)x\sin\frac{x}{2}.dx}{\sin^2 x}$$

$$= \int_0^{\pi/2}\frac{\left(2\sin\frac{x}{2}\cos\frac{x}{2}\right)\left(2\sin\left(\frac{2n+1}{2}\right)x\cos\left(\frac{2n+1}{2}\right)x\right)dx}{\sin^2 x}$$

$$= \int_0^{\pi/2}\frac{\sin(2n+1)x\,dx}{\sin x} = A_{n+1}$$

$B_{n+1} - B_n = A_{n+1}$.

24. $\int_0^1 \sin(x^2+2x+1)dx - \int_1^2 \sin x^2\, dx$ is equal to

(1) 0
(2) 1
(3) 2
(4) $\int_{-4}^{4} \cos x^2 dx - 8\int_0^1 \cos 16(2x-1)^2 dx$

Sol. Answer (1, 4)

Using the transformation

$$\int_a^b f(x)dx = (b-a)\int_0^1 f((b-a)x+a)dx$$

$$\int_0^1 \sin(x^2+2x+1)dx - \int_1^2 \sin\, x^2 dx$$

$$= \int_0^1 \sin(x+1)^2 - \int_0^1 \sin(x+1)^2 = 0$$

Also $\int_{-4}^{4} \cos x^2 dx - 8\int_0^1 \cos 16(2x-1)^2 dx$

$$8\int_0^1 \cos(8x-4)^2 dx - 8\int_0^1 \cos 16(2x-1)^2 dx$$

$$= 8\int_0^1 \cos 16(2x-1)^2 dx - 8\int_0^1 \cos 16(2x-1)^2 dx$$

25. If $x = \int_0^y \frac{dt}{\sqrt{1+9t^2}}$ and $\frac{d^2y}{dx^2} = a^2 y$ then a is equal to

(1) 0
(2) 3
(3) –3
(4) 1

Sol. Answer (2, 3)

$$x = \int_0^y \frac{dt}{\sqrt{1+9t^2}} \Rightarrow \frac{dx}{dy} = \frac{1}{\sqrt{1+9y^2}}$$

$$\Rightarrow \frac{dy}{dx} = \sqrt{1+9y^2} \Rightarrow \frac{d^2y}{dx^2} = \frac{18y}{2\sqrt{1+9y^2}} \times \frac{dy}{dx} \Rightarrow 9y$$

$\Rightarrow$ $a^2 = 9$ or $a = \pm 3$

26. The integral $\int_0^{\pi} x\, f(\sin x)dx$ is equal to

(1) $\frac{\pi}{2}\int_0^{\pi} f(\sin x)dx$ (2) $\frac{\pi}{4}\int_0^{\pi} f(\sin x)dx$ (3) $\pi\int_0^{\pi/2} f(\sin x)dx$ (4) $\pi\int_0^{\pi/2} f(\cos x)\, dx$

Sol. Answer (1, 3, 4)

Let $I = \int_0^{\pi} x\, f(\sin x)dx$ ....(i)

$I = \int_0^{\pi} (\pi - x)(\sin x)dx$ ....(ii)

From (i) and (ii)

$2I = \pi\int_0^{\pi} f(\sin x)dx \Rightarrow \therefore I = \frac{\pi}{2}\int_0^{\pi} f(\sin x)dx$ [Option 1 is correct]

$= \frac{\pi}{2}\times 2\int_0^{\pi/2} f(\sin x)dx = \pi\int_0^{\pi/2} f(\sin x)dx$ [Option 3 is correct]

$= \pi\int_0^{\pi/2} f(\cos x)dx$ [Option 4 is correct]

27. Given *f* an odd function periodic with period 2 continuous $\forall x \in R$ and $g(x) = \int_0^{x} f(t)dt$ then

(1) $g(x)$ is odd function (2) $g(x + 2) = 1$ (3) $g(2) = 0$ (4) $g(x)$ is even function

Sol. Answer (3, 4)

$g(x+2) = \int_0^{x+2} f(t)dt$

$\Rightarrow \int_0^{2} f(t)dt + \int_2^{x+2} f(t)dt = g(2) + \int_0^{x} f(t)dt$

$\therefore$ $g(x + 2) = g(2) + g(x)$ $\Rightarrow$ $g(x)$ is periodic with period 2

Also, $g(2) = \int_0^{2} f(t)dt = \int_0^{1} f(t)dt + \int_1^{2} f(t)dt$

$$= \int_0^1 f(t)dt + \int_{-1}^0 f(t)dt \text{ [putting } t = z+2]$$

$$= \int_{-1}^1 f(t)dt = 0 \qquad (\because f(x) \text{ is odd function})$$

Also, $g(-x) = \int_0^{-x} f(t)dt = \int_0^x f(-z)(-dz) \qquad t = -z,\ dt = -dz$

$$= \int_0^x f(z)dz = g(x)$$

Hence $g(x)$ is even function.

28 Let $f:(0,\infty)\to R$ and $F(x)=\int_0^x t\,f(t)dt$. If $F(x^2)=x^4+x^5,\ x>0$ then

(1) $F(4) = 7$

(2) $f(x)$ is continuous $\forall\ x > 0$

(3) $f(x)$ increases for all $x > 0$

(4) $f(x)$ is onto

Sol. Answer (2, 3)

$$F(x^2) = \int_0^{x^2} t\, f(t)dt = x^4 + x^5$$

Differentiating both sides, we get

$$f'(x^2)\cdot 2x = x^2 f(x^2)\cdot 2x = 4x^3 + 5x^4$$

$$f(x^2) = \frac{1}{2}(4+5x)$$

$$x \to \sqrt{x}$$

$$f(x) = \frac{1}{2}\left(4+5\sqrt{x}\right) x > 0$$

Clearly $f(x)$ is continuous and increasing and $f(4) = 2 + \frac{5}{2}\times 2 = 7$

Range of the function $(2,\infty) \subset R$, so $f(x)$ is onto.

$\because$ $f(x)$ is continuous and increasing $f(4) = 7$

Range = $[2, \infty)$

$\because$ Range $\neq$ co-domain

$\therefore$ $f(x)$ is not onto.

29. Let $I_1 = \int_0^{10^4} \frac{\{\sqrt{x}\}}{\sqrt{x}} dx$ and $I_2 = \int_0^{10} x\{x^2\}\, dx$, where {x} denotes fractional part of x then

(1) $I_1 = I_2$ (2) $I_1 < I_2$ (3) $I_1 = 4I_2$ (4) $I_1 = 100$

Sol. Answer (3, 4)

$$I_1 = 2\int_0^{10^2} \{t\}\, dt \qquad \left[\text{Putting } \sqrt{x} = t \Rightarrow \frac{dx}{\sqrt{x}} = 2dt\right.$$

$$I_2 = \int_0^{10^2} \frac{\{t\}}{2} dt \qquad \left[\text{Putting } x^2 = t \Rightarrow 2x\, dx = dt\right.$$

$$\therefore \quad \frac{I_1}{I_2} = 4 \quad \text{Also, } I_1 = 2\int_0^{10^2} \{t\}\, dt = 2\times 10^2 \int_0^1 \{x\}\, dx = 100$$

30. If $f(x) = x^2 \int_0^1 f(t)dt + 2$, then

(1) $f(x) = 3x^2 + 2$ (2) $f(x) = 12x^2 + 2$

(3) Slope of $y = f(x)$ at $x = 1$ is 8 (4) $\int_0^2 f(x)dx = 12$

Sol. Answer (1, 3, 4)

$f(x) = x^2\lambda + 2$

Where, $\lambda = \int_0^1 f(t)dt = \int_0^1 (\lambda t^2 + 2)dt$

$$\lambda = \left[\frac{\lambda t^3}{3} + 2t\right]_0^1 = \frac{\lambda}{3} + 2$$

$$\lambda - \frac{\lambda}{3} = 2 \Rightarrow \frac{2\lambda}{3} = 2 \Rightarrow \lambda = 3$$

$$\because \quad f(x) = 3x^2 + 2$$

$$f'(x) = 6x + 2$$

$$f'(1) = 8$$

$$\because \quad \int_0^2 f(x)dx = \int_0^2 (3x^2 + 2)\, dx = [x^3 + 2x] = 12$$

31. If $f(x)=\int_0^x tf(t)dt+2$, then

(1) $f(x)$ is strictly increasing function for $x \in R$

(2) $f(x)$ is strictly increasing function for $x \in (0, \infty)$

(3) $f(x)$ has a point of minima at $x = 0$

(4) $2<\int_0^1 f(x)dx<2\sqrt{e}$

Sol. Answer (2, 3, 4)

Differentiating w.r.t. 'x'

$f'(x)=xf(x)+0$

$\Rightarrow \dfrac{f'(x)}{f(x)}=x \Rightarrow \ln f(x)=\dfrac{x^2}{2}+\ln c$ ....(1)

$\Rightarrow f(x)=c\,e^{x^2/2}$

Putting $x = 0$ in given equation

$f(0) = 0 + 2 = 2$

Putting $f(0) = 2$ in (1)

$2=c e^0=c$

$\therefore f(x)=2e^{x^2/2}$

$f'(x)=2.e^{x^2/2}.\dfrac{2x}{2}=2xe^{x^2/2}=0$

$f'(0^-)<0$

$f'(0^+)>0$

$\Rightarrow$ $x = 0$ is *kt* of minima of $f(x)$

$f(x)$ is increasing if all $x > 0$ and decreasing if all $x < 0$

$\therefore 1.f(0)<\int_0^1 2e^{x^2/2}dx<1.f(1)$

$2<\int_0^1 2e^{x^2/2}dx<2\sqrt{e}$

32. $\int\sqrt{1+\sec 2x}\,dx$ equals

(1) $\sin^{-1}(\sqrt{2}\sin x)+C$

(2) $\sqrt{2}\cos^{-1}\sqrt{\cos x}+C$

(3) $C - 2\sin^{-1}(1 - 2\sin x)$

(4) $\cos^{-1}\sqrt{1-2\sin^2 x}+C$

Sol. Answer (1, 4)

Let $I = \int \sqrt{1+\sec x}\,dx$

$$= \int \sqrt{1+\frac{1}{\cos x}}\,dx = \int \frac{\sqrt{1+\cos x}}{\sqrt{\cos x}}\,dx$$

$$= \int \frac{\sqrt{1+2\cos^2\frac{x}{2}-1}}{\sqrt{1-2\sin^2\frac{x}{2}}}\,dx = \frac{\sqrt{2}}{\sqrt{2}}\int \frac{\cos\frac{x}{2}\,dx}{\sqrt{\left(\frac{1}{\sqrt{2}}\right)^2-\sin^2\frac{x}{2}}}$$

Put $\sin\frac{x}{2} = t$

$\Rightarrow \quad \cos\frac{x}{2}dx = 2dt$

$$I = \int \frac{2dt}{\sqrt{\left(\frac{1}{\sqrt{3}}\right)^2 - t^2}}$$

$$I = 2\sin^{-1}(\sqrt{2}t) + C$$

$$= 2\sin^{-1}\left(\sqrt{2}\sin\frac{x}{2}\right) + C = \sin^{-1}\left(\sin\frac{x}{2}\sqrt{\cos x}\right) + C$$

33. If $I = \int \frac{x+\sqrt[3]{x^2}+\sqrt[6]{x}}{x(1+\sqrt[3]{x})}dx = Ax^B + C\,arc\ \tan x^D + k$, then

(1) $AB = 1$ (2) $CD = 1$ (3) $A - B = 0$ (4) $C + D = 0$

Sol. Answer (1, 2)

The least common multiple of 3 and 6 is 6, so

Let $x = t^6 \Rightarrow dx = 6t^5\ dt$

$$\therefore\ I = 6\int\frac{(t^6+t^4+t)t^5}{t^6(1+t^2)}dt = 6\int\frac{t^5+t^3+1}{1+t^2}dt = 6\int t^3dt + 6\int\frac{dt}{t^2+1} = \frac{3}{2}t^4 + 6 \text{ are } \tan t + K$$

$$\therefore\ I = \frac{3}{2}x^{2/3} + 6 \text{ are } \tan\sqrt[6]{x} + K$$

Hence option (1), (2) correct

34. If $I = \int x^{-11}(1+x^4)^{-1/2}dx = A\left[\dfrac{\left(1+\frac{1}{x^4}\right)^{5/2}}{5} + \dfrac{\left(1+\frac{1}{x^4}\right)^{3/2}}{B} + \left(1+\frac{1}{x^4}\right)^{1/2}\right] + c$ then

(1) $A = -\dfrac{1}{2}$ (2) $A = \dfrac{1}{2}$ (3) $A + B = -2$ (4) $A + B = 2$

Sol. Answer (1, 3)

Put $1 + x^4 = t^2x^4$

$$\Rightarrow \frac{-4}{x^5}dx = 2t\,dt$$

$$\therefore I = \int \frac{dx}{x^{11}\{1+x^4\}^{1/2}} = \int \frac{dx}{x^{11}\cdot x^2\left\{1+\frac{1}{x^4}\right\}^{1/2}}$$

$$= \int \frac{dx}{x^{13}\left\{1+\frac{1}{x^4}\right\}^{1/2}} = \frac{-1}{4}\int \frac{2t\,dt}{x^8\,t}$$

$$= \frac{-1}{2}\int (t^2-1)^2 dt = \frac{-1}{2}\int (t^4 - 2t^2 + 1)\,dt$$

$$= \frac{-1}{2}\left\{\frac{t^5}{5} - \frac{2t^3}{3} + t\right\} + c$$

Hence, option (1), (3) correct

35. If $f(x) = \lim\limits_{n\to\infty}\{2x + 4x^3 + \ldots\ldots + 2n\,x^{2n-1}\}$, $0 < x < 1$ and $\int f(x)\,dx = F(x)$ & $F\left(\dfrac{1}{\sqrt{2}}\right) = 2$, then

(1) $F(0) = 1$ (2) $F\left(\dfrac{1}{2}\right) = \dfrac{4}{3}$ (3) $F(0) = -1$ (4) $F\left(-\dfrac{1}{2}\right) = \dfrac{4}{3}$

Sol. Answer (1, 2, 4)

Let $S_n = 2x + 4x^3 + 6x^5 + \ldots + 2n\,x^{2n-1}$

$\therefore$ $x^2S_n = 2x^3 + 4x^5 + 6x^7 + \ldots + 2n\,x^{2n+1}$

$\therefore$ $S_n\{1 - x^2\} = 2x + 2x^3 + 2x^5 + 2x^7 + \ldots + 2n\,x^{2n+1}$

$$\therefore S_n = \frac{2x\{1-x^{2n}\}}{\{1-x^2\}^2} - \frac{2n\,x^{2n+1}}{1-x^2} \qquad \because 0 < x < 1$$

$$\therefore \lim_{n\to\infty} S_n = \frac{2x}{(1-x^2)^2}$$

$$f(x)=\frac{2x}{(1-x^2)^2}$$

$$F(x)=\int f(x)dx=\int\frac{2x}{(1-x^2)^2}dx$$

$$F(x)=\frac{1}{1-x^2}+c$$

$$\because\quad f\left(\frac{1}{\sqrt{2}}\right)=\frac{1}{1-\frac{1}{2}}+c$$

$$2=2+c\ \Rightarrow\ c=0$$

$$\therefore\quad F(x)=\frac{1}{1-x^2}$$

36. If $f(x)=\lim\limits_{n\to\infty} e^{x\tan\left(\frac{1}{n}\right)\log\left(\frac{1}{n}\right)}$ and $\int\frac{f(x)}{\sqrt[3]{(\sin^{11}x\cos x)}}dx=g(x)+c$, then

(1) $g\left(\frac{\pi}{4}\right)=\frac{3}{2}$

(2) $g(x)$ is continuous $\forall\, x$

(3) $g\left(\frac{\pi}{4}\right)=\frac{-15}{8}$

(4) $g(x)$ is non-differentiable at infinitely many points

Sol. Answer (3, 4)

$$\lim_{n\to\infty}\tan\left(\frac{1}{n}\right)\ \log\left(\frac{1}{n}\right)=\lim_{n\to\infty}\frac{\tan\left(\frac{1}{n}\right)}{\left(\frac{1}{n}\right)}\cdot\frac{\log(n)}{(n)}=-1\lim_{n\to\infty}\frac{\frac{1}{n}}{1}=0$$

Then, $f(x) = e^0 = 1$

$$\therefore\quad \int\frac{1}{3\sqrt{\sin^{11}x\cos x}}dx=\int\frac{dx}{\sin^{11/3}x\cos^{1/3}x}=\int\sin^{-11/3}x\cdot\cos^{-1/3}x\,dx$$

$$=\int(\tan x)^{-11/3}\cdot\sec^4 x\,dx=\int(\tan x)^{-11/3}\{1+\tan^2 x\}\sec^2 x\,dx$$

$$=\frac{-3}{8}\{\tan x)^{-8/3}-\frac{3}{2}\{\tan x\}^{-2/3}+c$$

$$\therefore\ g(x)=\frac{-3}{8}\{\tan x\}^{-8/3}-\frac{3}{2}\{\tan x\}^{-2/3}$$

$$g(\pi/4)=\frac{-3}{8}-\frac{3}{2}=\frac{-15}{8}$$

and $g(x)$ is non-differentiable at $\tan x = 0$

or $x = n\pi,\ n \in I$

37. If $\int \frac{x + x^{2/3} + 2x^{1/6}}{x(1+x^{1/3})}dx = A\cdot x^{2/3} + B\tan^{-1}(x^{1/6}) + c$, then

(1) $A = \frac{3}{2}$ (2) $B = 12$ (3) $A = \frac{2}{3}$ (4) $B = \frac{1}{6}$

Sol. Answer (1, 2)

Let $x = y^6 \Rightarrow dx = 6y^5dy$

$$\therefore\ I = \int \frac{y^6 + y^4 + 2y}{y^6(1+y^2)}\cdot 6y^5dy$$

$$= 6\int \frac{y^3(y^2+1)+2}{(1+y^2)}dy$$

$$= 6\int \left\{y^3 + \frac{2}{1+y^2}\right\}dy$$

$$= 6\left\{\frac{y^4}{4} + 2\tan^{-1}y\right\} + c$$

$= \frac{3}{2}\cdot x^{2/3} + 12\tan^{-1}(x^{1/6}) + C$; option (1), (2) correct.

38. Which of the following are true?

(1) $\int_a^{\pi-a} xf(\sin x)dx = \frac{\pi}{2}\int_a^{\pi-a} f(\sin x)dx$ (2) $\int_{-a}^{a} f(x^2)dx = 2\int_0^a f(x^2)dx$

(3) $\int_0^{n\pi} f(\cos^2 x)dx = n\int_0^{\pi} f(\cos^2 x)dx, n\in N$ (4) $\int_0^{b-c} f(x+c)dx = \int_c^b f(x)dx$

Sol. Answer (1, 2, 3, 4)

(1) $\int_a^{\pi-a} xf(\sin x)dx = \frac{\pi}{2}\int_a^{\pi-a} f(\sin x)dx$ $\int_a^b f(x)dx = \int_a^b f(a+b-x)dx$

Let $I = \int_a^{\pi-a} xf(\sin x)dx$ … (i)

$\Rightarrow$ $I = \int_a^{\pi-a} (\pi - x)f(\sin x)dx$ … (ii) $\sin(\pi - x) = \sin x$

Adding equations (i) and (ii), we get

$$2I = \pi\int_a^{\pi-a} f(\sin x)dx$$

$$I = \frac{\pi}{2}\int_a^{\pi-a} f(\sin x)dx$$

(2) $\int_{-a}^{a} f^2(x)dx = 2\int_{-a}^{0} f^2(x)dx + \int_{0}^{a} f^2(x)dx$

$= 2\int_{0}^{a} f^2(x)dx$

(3) $\int_{0}^{n\pi} f(\cos^2 x)dx = n\int_{0}^{\pi} f(\cos^2 x)dx \quad \int_{0}^{nT} f(x)dx == n\int_{0}^{T} f(x)dx$

where T is a period of the function. i.e. $f(x+T) = f(x)$

(4) $\int_{0}^{b-c} f(x+c)dx = \int_{c}^{b} f(x)dx$

Let $x + c = t$

$dx = dt$

$\int_{c}^{b} f(t)dt = \int_{c}^{b} f(x)dx$.

39. Let $f(a) > 0$ and let $f(x)$ be a non-increasing continuous function in [a, b], then $\frac{1}{b-a}\int_{a}^{b} f(x)\,dx$ has the

(1) Minimum value $f(b)$ (2) Maximum value $f(a)$

(3) Maximum value $bf(b)$ (4) Minimum value $\frac{f(a)}{b-a}$

Sol. Answer (1, 2)

Here $f'(x) \le 0$ in [a, b], so, f(x) is monotonically decreasing.

Hence $f(a) \ge f(x) \ge f(b)$

$$\therefore \int_{a}^{b} f(a)dx \ge \int_{a}^{b} f(x)dx \ge \int_{a}^{b} f(b)dx$$

$$(b-a)\,f(a) \ge \int_{a}^{b} f(x)dx \ge (b-a)\,f(b)$$

$$f(a) \ge \frac{1}{b-a}\int_{a}^{b} f(x)dx \ge f(b)$$

40. The value of α, which satisfy $\int_{\pi/2}^{\alpha} \sin x\,dx = \sin 2\alpha(\alpha \in [0, 2\pi]$ are equal to

(1) $\frac{\pi}{2}$ (2) $\frac{3\pi}{2}$ (3) $\frac{7\pi}{6}$ (4) $\frac{11\pi}{6}$

Sol. Answer (1, 2, 3, 4)

$$\int_{\pi/2}^{\alpha} \sin x\, dx = \sin 2\alpha \;\Rightarrow\; -(\cos\alpha - 0) = \sin 2\alpha$$

$$\Rightarrow\; \cos\alpha(2\sin\alpha + 1) = 0 \quad \because \cos\alpha = 0 \text{ and } \sin\alpha = -\frac{1}{2}$$

$$\therefore\quad \alpha = \frac{\pi}{2}, \frac{3\pi}{2} \text{ and } \alpha = \pi + \frac{\pi}{6}, 2\pi - \frac{\pi}{6}$$

$$\alpha = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}$$

41. If $\int_0^{\pi/2}\left[a^2\left(\frac{\cos 3x}{4} + \frac{3}{4}\cos x\right) + a\sin x - 20\cos x\right]dx \le -\frac{a^2}{3}$, then a can be equal to

(1) 1 (2) 2 (3) 3 (4) 4

Sol. Answer (1, 2, 3, 4)

The L.H.S. of given inequality is equal to

$$\left[a^2\left(\frac{\sin 3x}{3} + \frac{3}{4}\sin x\right) - a\cos x - 20\cos x\right]_0^{\pi/2}$$

$$\Rightarrow\; a^2\left(-\frac{1}{12} + \frac{3}{4}\right) - a(0-1) - 20 = \frac{2a^2}{3} + a - 20$$

Thus the given inequality is

$$\frac{2a^2}{3} + a - 20 \le -\frac{a^2}{3} \quad \textit{i.e.,} \quad a^2 + a - 20 \le 0$$

$$-5 \le a \le 4$$

$$a = 1, 2, 3, 4$$

42. The integral $\int_{\tan^{-1}\alpha}^{\cot^{-1}\alpha} \frac{\tan x}{\tan x + \cot x}dx\ \alpha \in R$ cannot take the value

(1) π (2) $\frac{\pi}{2}$ (3) $\frac{\pi}{4}$ (4) $-\frac{\pi}{4}$

Sol. Answer (1, 4)

Let $I = \int_{\tan^{-1}\alpha}^{\cot^{-1}\alpha} \frac{\tan x}{\tan x + \cot x}dx = \int_{\tan^{-1}\alpha}^{\cot^{-1}\alpha} \frac{\cot x}{\tan x + \cot x}dx$

$\because \quad \left(\tan^{-1}\alpha + \cot^{-1}\alpha = \frac{\pi}{2}\right)$

$\therefore \quad 2I = \int_{\tan^{-1}\alpha}^{\cot^{-1}\alpha} dx = \cot^{-1}\alpha - \tan^{-1}\alpha$

$\Rightarrow \quad I = \frac{1}{2}\left[\frac{\pi}{2} - \tan^{-1}\alpha\right] \;\therefore\; -\frac{\pi}{4} < I < \frac{3\pi}{4}\,\forall\alpha$

43. Let $f(x)$ be a continuous function for all x, which is not identically zero such that $\{f(x)\}^2 = \int_0^x f(t)\frac{2\sec^2 t}{4+\tan t}dt$ and $f(0) = \ln 4$, then

(1) $f\left(\frac{\pi}{4}\right) = \log 5$ (2) $f\left(\frac{\pi}{2}\right) = \log 4$

(3) $f\left(\frac{3\pi}{4}\right) = \log 3$ (4) $f(x) = 0$ has infinite roots

Sol. Answer (1, 3, 4)

Differentiating $2f(x)f'(x) = f(x)\frac{2\sec^2 x}{4+\tan x}$

$\because \quad f(x) \neq 0$ so $f'(x) = \frac{\sec^2 x}{4+\tan x} \Rightarrow f(x) = \log(4+\tan x) + c$

Given that $f(0) = \log 4$, so $f(x) = \log(4 + \tan x)$

44. The value of $\lim_{x\to\infty} \frac{\left(\int_0^{x+y} e^{t^2}dt\right)^2}{\int_0^{x+y} e^{2t^2}dt}$ is equal to

(1) 0 (2) 1 (3) $\lim_{x\to 0}\frac{\int_0^x \cos t^2 dt}{x} - 1$ (4) –1

Sol. Answer (1, 3)

$$\lim_{x\to\infty} \frac{\left(\int_0^{x+y} e^{t^2}dt\right)^2}{\int_0^{x+y} e^{2t^2}dt} = \lim_{x\to\infty} \frac{2\left(\int_0^{x+y} e^{t^2}dt\right)e^{(x+y)^2}}{e^{2(x+y)^2}}$$

$$= \lim_{x\to\infty} \frac{2\int_0^{x+y} e^{t^2} dt}{e^{(x+y)^2}} = \lim_{x\to\infty} \frac{2e^{(x+y)^2}}{e^{(x+y)^2}.2(x+y)}$$

$$\Rightarrow \lim_{x\to\infty} \frac{1}{x+y} = 0$$

$$\text{And } \lim_{x\to 0} \frac{\int_0^x \cos t^2 dt}{x} = \lim_{x\to 0} \frac{\cos x^2}{1} = \cos 0 = 1$$

$$\therefore \lim_{x\to 0} \frac{\int_0^x \cos t^2 dt}{x} - 1 = 1 - 1 = 0$$

45. $\int_{-\frac{1}{2}}^{\frac{1}{2}} \sqrt{\left(\frac{x+1}{x-1}\right)^2 + \left(\frac{x-1}{x+1}\right)^2 - 2}\, dx$ is equal to

(1) $4\log\frac{3}{4}$ (2) $4\log\frac{4}{3}$ (3) $2\log\frac{16}{9}$ (4) $-\log\frac{81}{256}$

Sol. Answer (2, 4)

$$= \int_{-\frac{1}{2}}^{\frac{1}{2}} \sqrt{\left(\frac{x+1}{x-1} - \frac{x-1}{x+1}\right)^2} dx = \int_{-\frac{1}{2}}^{\frac{1}{2}} \sqrt{\left(\frac{(x+1)^2-(x-1)^2}{x^2-1}\right)^2} dx$$

$$= \int_{-\frac{1}{2}}^{\frac{1}{2}} \sqrt{\left(\frac{4x}{x^2-1}\right)^2} = \int_{-\frac{1}{2}}^{\frac{1}{2}} \left|\frac{4x}{x^2-1}\right| dx = 4.2\int_0^{\frac{1}{2}} \left|\frac{x}{x^2-1}\right| dx$$

$$= -8\int_0^{\frac{1}{2}} \frac{xdx}{x^2-1} = -4\left|\log(x^2-1)\right|_0^{\frac{1}{2}}$$

$$= -4\left[\log\frac{3}{4} - 0\right] = 4\log\frac{4}{3} = -\log\frac{81}{256}$$

46. Let $y(x,t)=\begin{cases} x(t-1); & \text{when} \quad x\le t \\ t(x-1); & \text{when} \quad t<x \end{cases}$ and t is continuous function of x in [0, 1].

If $g(x)=\int_0^1 f(t)y(x,t)dt$ then

(1) $g(0) + g(1) = 0$ (2) $g(0) = 0$ (3) $g(1) = 1$ (4) $g''(x) = f(x)$

Sol. Answer (1, 2, 4)

$g(0,t)=0$ for $t\ge 0$ so $g(0)=\int_0^1 f(t).0\ dt=0$

$y(1,t)=t.(1-t)=0$ for $t<1$

Hence $g(1)=\int_0^1 f(t).0\,dt=0$

Also $g(x)=\int_0^x f(t)t(x-1)dt+\int_x^1 f(t)x(t-1)dt$

$$=(x-1)\int_0^x t\,f(t)dt+x\int_x^1 f(t)(t-1)dt$$

Hence $g'(x)=(x-1)\,x\,f(x)+\int_0^x t\,f(t)dt+\int_x^1 f(t)(t-1)-xf(x)(x-1)$

$$=\int_0^1 t\,f(t)dt-\int_x^1 f(t)dt \qquad \text{Thus} \quad g''(x)=f(x)$$

47. The value of $\frac{\pi}{2}-\int_0^{\pi/4}\tan^{-1}\left(\frac{2\cos^2\theta}{2-\sin 2\theta}\right)\sec^2\theta\,d\theta$ is equal to

(1) 3 (2) ln2 (3) $\int_0^{\infty}\left[2e^{-x}\right]dx$ (4) 0

Sol. Answer (2, 3)

$$I=\int_0^{\pi/4}\tan^{-1}\left(\frac{2\cos^2\theta}{2-\sin 2\theta}\right)\sec^2\theta d\theta$$

$$=\int_0^{\pi/4}\tan^{-1}\left(\frac{1}{\sec^2\theta-\tan\theta}\right)\sec^2\theta d\theta$$

$$= \int_0^{\pi/4} \tan^{-1}\left(\frac{1}{1+\tan^2\theta - \tan\theta}\right)\sec^2\theta - d\theta$$

Putting $\tan\theta = t$ then $\sec^2\theta d\theta = dt$

The given integral reduces to

$$\int_0^1 \tan^{-1}\left(\frac{1}{1+t^2-t}\right)dt = \int_0^1 \tan^{-1}\frac{1-(t-1)}{1+t(t-1)} = \int_0^1 \tan^{-1} t\,dt - \int_0^1 \tan^{-1}(t-1)\,dt$$

$$\Rightarrow \int_0^1 \tan^{-1} t + \int_0^1 \tan^{-1}(1-t)dt = 2\int_0^1 \tan^{-1} t - dt$$

Now $\displaystyle\int_0^1 \tan^{-1} t\,dt = \left|t \tan^{-1} t\right|_0^1 - \int_0^1 \frac{t}{1+t^2}dt = \frac{\pi}{4} - \left|\frac{1}{2}\log(1+t^2)\right|_0^1 = \frac{\pi}{4} - \frac{1}{2}\log 2$

$$I = \frac{\pi}{2} - \log 2 \Rightarrow \frac{\pi}{2} - I = \log 2$$

Also for $0 < x < \log 2 \Rightarrow \left[2\,e^{-x}\right] = 1$ and for $x > \log 2$ $\left[2\,e^{-x}\right] = 0$

$$\therefore \int_0^{\infty}\left[2\,e^{-x}\right]dx = \int_0^{\log 2} 1.\,dx = \log 2$$

48. $I_1 = \int_0^1 3^{x^2}dx, I_2 = \int_0^1 3^{x^3}dx, I_3 = \int_1^2 3^{x^2}dx, I_4 = \int_1^2 3^{x^3}dx$, then

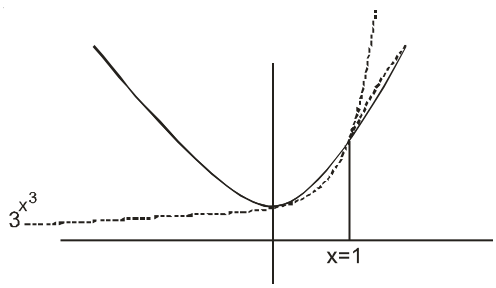
(1) $I_1 > I_2$ (2) $I_2 > I_1$ (3) $I_3 > I_4$ (4) $I_4 > I_3$

Sol. Answer (1, 4)

Clearly in $x \in (0, 1)$

Area under the curve

$3^{x^2} > e^{x^3}$



And in (1, 2), area under the curve $3^{x^3} > 3^{x^2}$

49. Let $f(x)$ be a cubic polynomial such that $x = 1$ is a repeated root of order 2 and $x=\frac{5}{3}$ is a point of local Minima. If $\int_0^1 f(x)dx=\frac{-7}{12}$, then

(1) $f(x)=(x-1)^2(x-2)$

(2) $f(x)=-2(x-1)^2(x-2)$

(3) Area enclosed by the curve and x-axis lies in 1st quadrant

(4) Area enclosed by the curve and x-axis lies in the fourth quadrant

Sol. Answer (1, 4)

Let $f(x)=(x-1)^2(ax+b)$

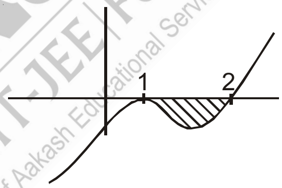
$f'(x)=2(x-1)(ax+b)+a.(x-1)^2=0$ at $x=\frac{5}{3}$

$\Rightarrow\ 2\times\frac{2}{3}\times\left(\frac{5a}{3}+b\right)+\frac{4}{9}a=0$

$5a + 3b + a = 0 \Rightarrow 2a + b = 0$ ….(1)

$\therefore\ f(x)=(x-1)^2.(ax-2a)$

$=a(x-1)^2(x-2)$



$\because\ \int_0^1 f(x)dx=\frac{-7}{12}$

$\therefore\ a\int_0^1 (x-1)^2(x-2)dx=\frac{-7}{12}$

$\Rightarrow\ a\int_0^1 (x^3-4x^2+5x-2)dx=\frac{-7}{12}$

$\Rightarrow\ a\left(\frac{x^4}{4}-\frac{4}{3}x^3+\frac{5}{2}x^2-2x\right)_0^1=\frac{-7}{12}$

$\Rightarrow\ a\left(\frac{1}{4}-\frac{4}{3}+\frac{5}{2}-2\right)=\frac{-7}{12}$

$\Rightarrow\ a\left(\frac{3-16+30-24}{12}\right)=\frac{-7}{12}$

$a = 1$

Area lies in the 4^{th} quadrant.

50. Let $\lim_{n\to\infty}\left(\frac{(kn)!}{n^{kn}}\right)^{\frac{1}{n}} = \lambda(k)$, where $k \in N$

(1) $\lambda(1) = \frac{1}{e}$ (2) $\lambda(2) = \frac{4}{e^2}$ (3) $\lambda(1) = \frac{2}{e}$ (4) $\lambda(2) = \frac{1}{e}$

Sol. Answer (1, 2)

$$\ln\lambda = \lim_{n+\infty}\frac{1}{n}\ln\left\{\frac{1.2.........(kn)}{n.n........n}\right\}$$

$$= \lim_{n\to\infty}\frac{1}{n}\sum_{r=1}^{kn}\ln\frac{r}{n}$$

$$= \int_0^k \ln x\,dx$$

$$= \left(x\ln x - x\right)_0^k$$

$$= \left(k\ln k - k\right) - \lim_{x\to 0}(x\ln x) + 0$$

$$= k(\ln k - \ln e) - \lim_{x\to 0}\left(\frac{\ln x}{1/x}\right)$$

$$= \ln\left(\frac{k}{e}\right)^k - \lim_{x\to 0}\frac{\frac{1}{x}}{\frac{-1}{x^2}} = \ln\left(\frac{k}{e}\right)^k$$

$$\therefore \quad \lambda = \left(\frac{k}{e}\right)^k$$

$$\therefore \quad \lambda(1) = \frac{1}{e}, \lambda(2) = \frac{4}{e^2}$$

51. $\int_0^{2\pi}\frac{x}{1+\tan^{24}x}dx,\ n>0$

(1) π^2

(2) $\frac{\pi^2}{4}$

(3) $\int_0^a f(x)dx = \int_0^a f(a-x)dx$ can be used to evaluate it

(4) Integrand is odd function

Sol. Answer (1, 3, 4)

$$I = \int_0^{2\pi} \frac{x}{1+\tan^{2n} x} dx$$

$$= \int_0^{2\pi} \frac{x\cos^{2n} x}{\sin^{2n} x + \cos^{2n} x} dx \qquad \ldots(1)$$

$$= \int_0^{2\pi} \frac{(2\pi - x)\cos^{2n} x}{\sin^{2n} x + \cos^{2n} x} dx \qquad \ldots(2) \quad \Rightarrow \text{(Option (3) is correct)}$$

$$2I = 2\pi \int_0^{2\pi} \frac{\cos^{2n} x}{\sin^{2n} x + \cos^{2n} x} dx$$

$$\Rightarrow \quad I = \pi \,.\, 2 \,.\, 2 \int_0^{\pi/2} \frac{\cos^{2n} x}{\sin^{2n} x + \cos^{2n} x} dx \quad \left(\text{Applying } \int_0^{2a} f(x)dx = 2\int_0^{a} f(x)dx \text{ twice}\right)$$

$$= 4\pi \times \frac{\pi}{4} = \pi^2$$

52. Let $I = \int_0^n [x]\, dx, n > 0$, where [ ] is G.I.F.,

(1) If n is an integer then $I = \dfrac{n(n-1)}{2}$

(2) If $n = x$ then $I = x[x] - \dfrac{[x]}{2}([x]+1)$

(3) If n = an integer then $I = \dfrac{n(n+1)}{2}$

(4) If $n = x$ then $I = x[x] + 1\dfrac{[x]}{2}([x]+1)$

Sol. Answer (1, 2)

n is not an integer. Let $k < n < k + 1$

$$\Rightarrow \int_0^1 0\, dx + \int_1^2 1 dx + \ldots\ldots\ldots + \int_{k-1}^{k} (k-1)dx + \int_k^n k\, dx$$

$$\equiv 0 + 1 + 2 + \ldots\ldots + (k-1) + k(n-k)$$

$$= \frac{(k-1)k}{2} + k(n-k)$$

$$= \frac{k^2}{2} - \frac{k}{2} + kn - k^2$$

$$= kn - \frac{k^2}{2} - \frac{k}{2}$$

$$= kn - \frac{k}{2}(k+1)$$

$$= n[n] - \frac{[n]}{2}([n]+1)$$

If n = integer

$\Rightarrow$ $[n] = n$

$$\therefore \quad f = n^2 - \frac{n^2}{2} - \frac{n}{2} = \frac{n^2}{2} - \frac{n}{2} = \frac{n(n-1)}{2}$$

If $n = x \Rightarrow I = x[x] - \frac{[x]}{2}2([x]+1)$

53. If $f(x) = \int_0^{\cos^2 x} \sec x \frac{\sqrt{t}}{1+t^3} dt$, then

(1) $f'(\pi) = 0$ (2) $f'(\pi) = 1$ (3) $f'(2\pi) = 0$ (4) $f'(2\pi) = -1$

Sol. Answer (1, 3)

$$\frac{d}{dx}(f(x)) = \frac{d}{dx} \int_0^{\cos^2 x} \sec x \frac{\sqrt{t}}{1+t^3} dt$$

$$= \sec x \cdot \tan x \int_0^{\cos^2 x} \frac{\sqrt{t}}{1+t^3} dt - 2\cos x \cdot \sin x \cdot \sec x \frac{\cos x}{1+\cos^6 x}$$

$$\Rightarrow \sin x \int_0^{\cos^2 x} \frac{\sqrt{t}}{1+t^3} dt - 2\sin x \frac{\cos x}{1+\cos^6 x}$$

$f'(\pi) = 0$, $f'(2\pi) = 0$

54. Let $f : (0, \infty) \to \mathbb{R}$ be given by

$f(x) = \int_{\frac{1}{x}}^{x} e^{-\left(t+\frac{1}{t}\right)} \frac{dt}{t}$. Then

[JEE(Advanced)-2014]

(1) $f(x)$ is monotonically increasing on $[1, \infty)$ (2) $f(x)$ is monotonically decreasing on $(0, 1)$

(3) $f(x) + f\left(\frac{1}{x}\right) = 0$, for all $x \in (0, \infty)$ (4) $f(2^x)$ is an odd function of x on $\mathbb{R}$

Sol. Answer (1, 3, 4)

$$f(x) = \int_{\frac{1}{x}}^{x} e^{-\left(t+\frac{1}{t}\right)} \frac{dt}{t}$$

$$f'(x) = \frac{2e^{-\left(x+\frac{1}{x}\right)}}{x} > 0$$

Also, $f(x) + f\left(\frac{1}{x}\right) = \int_{\frac{1}{x}}^{x} e^{-\left(t+\frac{1}{t}\right)} \frac{dt}{t} + \int_{x}^{\frac{1}{x}} e^{-\left(t+\frac{1}{t}\right)} \frac{dt}{t}$

and so, $f(2^x) + f(2^{-x}) = 0$

i.e., (1, 3, 4) is correct answer.

55. Let $f(x) = 7\tan^8 x + 7\tan^6 x - 3\tan^4 x - 3\tan^2 x$ for all $x \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$. Then the correct expression(s) is(are)

[JEE(Advanced)-2015]

(1) $\int_0^{\pi/4} xf(x)\,dx = \frac{1}{12}$ (2) $\int_0^{\pi/4} f(x)\,dx = 0$ (3) $\int_0^{\pi/4} xf(x)\,dx = \frac{1}{6}$ (4) $\int_0^{\pi/4} f(x)\,dx = 1$

Sol. Answer (1, 2)

$f(x) = 7\tan^8 x + 7\tan^6 x - 3\tan^4 x - 3\tan^2 x$

$f(x) = \sec^2 x(7\tan^6 x - 3\tan^2 x)$

Now,

$$\int_0^{\pi/4} \sec^2 x\left(7\tan^6 x - 3\tan^2 x\right)dx$$

$$= \int_0^1 \left(7t^6 - 3t^2\right)dt$$

$$= \frac{7t^7}{7} - \frac{3t^3}{3}$$

$= 1^7 - 1^3 = 0$

Also

$$\int_0^{\pi/4} x\sec^2 x(7\tan^6 x - 3\tan^2 x)dx$$

$\tan x = t$

$$\int_0^1 \tan^{-1} t(7t^6 - 3t^2)dt$$

$$\tan^{-1} t\left(t^7 - t^3\right)\Big|_0^1 - \int_0^1 \frac{1}{1+t^2}(t^7 - t^3)dt$$

$$= -\int_0^1 \frac{1}{1+t^2} t^3(t^4 - 1)dt$$

$$= +\int_0^1 \frac{t^3\left(1+t^2\right)\left(1-t^2\right)dt}{1+t^2} = \int_0^1 \left(t^3 - t^5\right)dt$$

$$= \frac{1}{4} - \frac{1}{6} = \frac{1}{12}$$

56. The option(s) with the values of a and L that satisfy the following equation is(are)

$$\frac{\int_0^{4\pi} e^t\left(\sin^6 at + \cos^4 at\right)dt}{\int_0^{\pi} e^t\left(\sin^6 at + \cos^4 at\right)dt} = L?$$

[JEE(Advanced)-2015]

(1) $a = 2,\ L = \dfrac{e^{4\pi} - 1}{e^{\pi} - 1}$ (2) $a = 2,\ L = \dfrac{e^{4\pi} + 1}{e^{\pi} + 1}$ (3) $a = 4,\ L = \dfrac{e^{4\pi} - 1}{e^{\pi} - 1}$ (4) $a = 4,\ L = \dfrac{e^{4\pi} + 1}{e^{\pi} + 1}$

Sol. Answer (1, 3)

$$I = \int_0^{4\pi} e^t(\sin^6 2t + \cos^4 2t)dt$$

$$= \underbrace{\int_0^{\pi} e^t(\sin^6 2t + \cos^4 2t)dt}_{I_1} + \underbrace{\int_{\pi}^{2\pi} e^t(\sin^6 2t + \cos^4 2t)dt}_{I_2} + \underbrace{\int_{2\pi}^{3\pi} e^t(\sin^6 2t + \cos^4 2t)dt}_{I_3} + \underbrace{\int_{3\pi}^{4\pi} e^t(\sin^6 2t + \cos^4 2t)dt}_{I_4}$$

$t = x + \pi$

$dt = dx$

$$I_2 = \int_0^{\pi} e^{x+\pi}(\sin^6 2x + \cos^4 2x)dx = e^{\pi} I_1$$

$t = 2\pi + x$

$I_3 = e^{2\pi} I_1$

$I_4 = e^{3\pi} I_1$

$I = I_1(1 + e^{\pi} + e^{2\pi} + e^{3\pi})$

$$\frac{I}{I_1} = \frac{e^{4\pi} - 1}{e^{\pi} - 1}$$

The same for $a = 4$

57. Let $f'(x)=\dfrac{192x^3}{2+\sin^4 \pi x}$ for all $x \in \mathbb{R}$ with $f\left(\dfrac{1}{2}\right)=0$. If $m \le \displaystyle\int_{1/2}^{1} f(x)dx \le M$, then the possible values of m and M are **[JEE(Advanced)-2015]**

(1) $m = 13$, $M = 24$ (2) $m=\dfrac{1}{4}, M=\dfrac{1}{2}$ (3) $m = -11$, $M = 0$ (4) $m = 1$, $M = 12$

Sol. Answer (4)

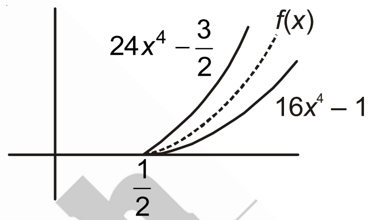
Assuming $f'(x)$ to more increasing or less increasing

$$\frac{192x^3}{3} < f'(x) < \frac{192x^3}{2}$$

$$64x^3 < f'(x) < 96x^3$$

$$\int 64x^3\,dx < \int f'(x)dx < \int 96x^3\,dx$$

$$16x^4 - 1 < f(x) < 24x^4 - \frac{3}{2}$$



$$\int_{\frac{1}{2}}^{1}(16x^4 - 1)\,dx < \int_{\frac{1}{2}}^{1} f(x)\,dx < \int_{\frac{1}{2}}^{1}\left(24x^4 - \frac{3}{2}\right)dx$$

$$\frac{26}{10} < \int_{\frac{1}{2}}^{1} f(x)\,dx < \frac{78}{20}$$

Hence option (4) is correct.

SECTION - C

Linked Comprehension Type Questions

Comprehension-I

Reduction formulas can be used to compute integrals of higher power of sin$x$, cos$x$, tan$x$ etc.

1. $\displaystyle\int \sin^5 x\,dx = -\frac{1}{5}\sin^4 x\cos x + A\sin^2 x\cos x - \frac{8}{15}\cos x + C$ then A equals

(1) $-\dfrac{2}{15}$ (2) $-\dfrac{3}{5}$ (3) $-\dfrac{4}{15}$ (4) $-\dfrac{1}{15}$

Sol. Answer (3)

$$\int \sin^n x\,dx = \frac{-\sin^{n-1} x\cdot\cos x}{n} + \left(\frac{n-1}{n}\right)\int \sin^{n-2} x\,dx$$

$$I_n = \frac{-\sin^{n-1} x\cos x}{n} + \left(\frac{n-1}{n}\right)I_{n-2}$$

$$\int \sin^5 x\, dx = \frac{-\sin^4 x \cos x}{5} + \frac{4}{5} I_3$$

$$= \frac{-\sin^4 x \cdot \cos x}{5} + \frac{4}{5}\left(\frac{-\sin^2 x \cos x}{3} + \frac{2}{3} I_1\right)$$

$$= \frac{-\sin^4 x \cos x}{5} - \frac{4}{15}\sin^2 x \cos x - \frac{8}{15}\cos x + C$$

$$\Rightarrow \quad \int \sin^n x\, dx = \frac{-1}{5}\sin^4 x \cos x + A \sin^2 x \cos x - \frac{8}{15}\cos x + C$$

$$\Rightarrow \; A = \frac{-4}{15}$$

2. $\int \tan^6 x\, dx = \frac{1}{5}\tan^5 x + A\tan^3 x + \tan x - x + C$ then

(1) $A = \frac{1}{3}$ (2) $A = \frac{2}{3}$ (3) $A = -\frac{2}{3}$ (4) $A = -\frac{1}{3}$

Sol. Answer (4)

$$\int \tan^6 x\, dx = \frac{1}{5}\tan^5 x + A\tan^3 x + \tan x - x + C$$

According to reduction formula

$$I_n = \frac{\tan^{n-1} x}{n-1} - I_{n-2}$$

$$I_n = \frac{\tan^5 x}{5} - I_4 = \frac{\tan^5 x}{5} - \left(\frac{\tan^3 x}{3} - I_2\right) = \frac{\tan^5 x}{5} - \frac{\tan^3 x}{3} + \tan x - x + C$$

$$\Rightarrow \; A = \frac{-1}{3}$$

3. $\int \sec^6 x\, dx = \frac{1}{5}\tan^5 x + A\tan^3 x + \tan x + C$ then

(1) $A = \frac{1}{3}$ (2) $A = \frac{2}{3}$ (3) $A = -\frac{1}{3}$ (4) $A = -\frac{2}{3}$

Sol. Answer (2)

$$\int \sec^6 x\, dx = \frac{\sec^4 x \tan x}{5} + \frac{4}{15}\sec^2 x \tan x + \frac{4}{5} A \tan x + C$$

$$\int \sec^n x\, dx = \frac{\sec^{n-2} x \tan x}{(n-1)} + \left(\frac{n-2}{n-1}\right) I_{n-2}$$

$$\int \sec^6 x\, dx = \frac{\sec^4 x \tan x}{5} + \frac{4}{5} I_4$$

$$= \frac{\sec^4 x \tan x}{5} + \frac{4}{5}\left(\frac{\sec^2 x \tan x}{3} + \frac{2}{3} I_2\right)$$

$$= \frac{\sec^4 x \tan x}{5} + \frac{4}{15}\sec^2 x \tan x + \frac{8}{15}\tan x + C \quad \text{... (i)}$$

Put $\sec^2 x = 1 + \tan^2 x$ in equation (i), we get

$$\int \sec^6 x\, dx = \frac{1}{5}\tan^5 x + \frac{2}{3}\tan^3 x + \tan x + C, \text{ then } A = \frac{2}{3}$$

Comprehension-II

If the integrand is a rational function of x and fractional power of a linear fractional function of the form $\frac{ax+b}{cx+d}$, then rationalization of the integral is at substitution by $\frac{ax+b}{cx+d} = t^m$, where m is L.C.M. of fractional powers of $\frac{ax+b}{cx+d}$.

1. If $\int \frac{dx}{(x-1)^{3/4}(x+2)^{5/4}} = A\left(\frac{x-1}{x+2}\right)^{1/4} + C$ then

(1) $A = \frac{1}{3}$ (2) $A = \frac{2}{3}$ (3) $A = \frac{3}{4}$ (4) $A = \frac{4}{3}$

Sol. Answer Answer (4)

$$\int \frac{dx}{(x-1)^{3/4}(x+2)^{5/4}} = A\left(\frac{x-1}{x+2}\right)^{1/4} + C$$

$$\int \frac{dx}{\left(\frac{x-1}{x+2}\right)^{3/4}(x+2)^{\frac{5}{4}+\frac{3}{4}}} = \int \frac{dx}{\left(\frac{x-1}{x+2}\right)^{3/4}(x+2)^2}$$

Let $\frac{x-1}{x+2} = t$, then $\frac{3}{(x+2)^2}dx = dt$

$$= \frac{1}{3}\int \frac{dt}{t^{3/4}} = \frac{4}{3}t^{1/4} = \frac{4}{3}\left(\frac{x-1}{x+2}\right)^{1/4} + C$$

$\Rightarrow A = \frac{4}{3}$

2. $\int \frac{dx}{(x+1)^{2/3}(x-1)^{4/3}} = k\left[\frac{1+x}{1-x}\right]^{1/3} + C$ then

(1) $k = \frac{2}{3}$ (2) $k = \frac{3}{2}$ (3) $k = \frac{1}{3}$ (4) $k = \frac{1}{2}$

Sol. Answer (2)

$$\int \frac{dx}{(x+1)^{2/3}(x-1)^{4/3}} = k\left[\frac{1+x}{1-x}\right]^{1/3} + C$$

$$\int \frac{dx}{\left(\frac{1+x}{1-x}\right)^{2/3}(1-x)^2}$$

Let $\frac{1+x}{1-x} = t,$ then $\frac{2}{(1-x)^2}dx = dt$

$$\frac{1}{2}\int \frac{dt}{t^{2/3}} = \frac{3}{2}t^{1/3} = \frac{3}{2}\left[\frac{1+x}{1-x}\right]^{1/3} + C$$

$\Rightarrow k = \frac{3}{2}$

3. If $\int \frac{dx}{(1-x)\sqrt{1-x^2}} = k\sqrt{\frac{x+1}{1-x}} + C$ then

(1) $k = \frac{1}{2}$ (2) $k = 1$ (3) $k = \frac{1}{3}$ (4) $k = \frac{2}{3}$

Sol. Answer (2)

$$\int \frac{dx}{(1-x)\sqrt{1-x^2}} = \int \frac{dx}{(1-x)\sqrt{1-x}\,\sqrt{1+x}}$$

$$= \int \frac{dx}{(1-x)^{3/2}(1+x)^{1/2}} = \int \frac{dx}{\left(\frac{1+x}{1-x}\right)^{1/2}\cdot(1-x)^2}$$

Let $\frac{1+x}{1-x} = t,$ then $\frac{2}{(1-x)^2}dx = dt$

$$= \frac{1}{2}\int \frac{dt}{t^{1/2}} = t^{1/2} = \sqrt{\frac{1+x}{1-x}}$$

$\Rightarrow k = 1$

Comprehension-III

Let $f(x)=\dfrac{9^x}{9^x+3}$

1. If $g(x)=\int\left(f\left(\frac{1}{11}\right)+f\left(\frac{2}{11}\right)+...+f\left(\frac{10}{11}\right)\right)$, then

(1) $g(x) = 2x + c$ (2) $g(x) = 10x + c$ (3) $g(x) = 5x + c$ (4) $g(x)=\dfrac{10}{11}x+c$

Sol. Answer (3)

$\because f(x)=\dfrac{9^x}{9^x+3}$ ... (1)

$\therefore\ f(1-x)=\dfrac{9^{1-x}}{9^{1-x}+3}=\dfrac{9}{9+3.9^x}=\dfrac{3}{3+9^x}$ ... (2)

Adding (1) and (2),

$$f(x)+f(1-x)=\frac{9^x}{9^x+3}+\frac{3}{9^x+3}=1$$

$$g(x)=\int\left\{f\left(\frac{1}{11}\right)+f\left(\frac{2}{11}\right)+..............+f\left(\frac{10}{11}\right)\right\}dx$$

$$=\int\left[\left\{f\left(\frac{1}{11}\right)+f\left(\frac{10}{11}\right)\right\}+\left\{f\left(\frac{2}{11}\right)+f\left(\frac{9}{11}\right)\right\}+...........+\left\{f\left(\frac{5}{11}\right)+f\left(\frac{6}{11}\right)\right\}\right]dx$$

$$=\int\left\{f\left(\frac{1}{11}\right)+f\left(1-\frac{1}{11}\right)\right\}+\left\{f\left(\frac{2}{11}\right)+f\left(1-\frac{2}{11}\right)\right\}+.............+\left\{f\left(\frac{5}{11}\right)+f\left(1-\frac{5}{11}\right)\right\}dx$$

$$=5\int dx$$

$g(x) = 5x + c$

2. If $g(x)$ passes through origin, then area bounded by $g(x)$, x-axis and the line $x = 4$ is

(1) 20 sq. units (2) 40 sq. units (3) 10 sq. units (4) 80 sq. units

Sol. Answer (2)

$\because f(x)=\dfrac{9^x}{9^x+3}$ ... (1)

$\therefore\ f(1-x)=\dfrac{9^{1-x}}{9^{1-x}+3}=\dfrac{9}{9+3.9^x}=\dfrac{3}{3+9^x}$ ... (2)

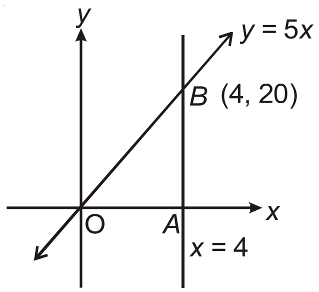
Adding (1) and (2),

$$f(x)+f(1-x)=\frac{9^x}{9^x+3}+\frac{3}{9^x+3}$$

$$= 1$$

$\because$ $g(x)$ passes through origin

$\therefore$ $0 = 0 + c \Rightarrow c = 0$



$g(x) = 5x$

Required area = Area of ΔOAB

$$= \frac{1}{2}OA \times AB$$

$$= \frac{1}{2}\times 4\times 20$$

$= 40$ sq.units.

3. Let $h(x) = \int f(x)dx$. If $h(\log_9 6) = 1$, then $h(x) =$

(1) $\ln(9^x + 3)$ (2) $\log_6(9^x + 3)$ (3) $\frac{\log_6(9^x+3)}{\ln 9}$ (4) $\log_9(9^x+3)$

Sol. Answer (4)

$$\because f(x)=\frac{9^x}{9^x+3} \quad \text{... (1)}$$

$$\therefore\ f(1-x)=\frac{9^{1-x}}{9^{1-x}+3}=\frac{9}{9+3.9^x}=\frac{3}{3+9^x} \quad \text{... (2)}$$

Adding (1) and (2),

$$f(x)+f(1-x)=\frac{9^x}{9^x+3}+\frac{3}{9^x+3} = 1$$

$h(x) = \int \frac{9^x}{9^x + 3} dx$ Let $9^x + 3 = t$

$= \frac{1}{\ln 9} \int \frac{dt}{t}$

$= \frac{1}{\ln 9} \cdot \ln|t| + c$

$= \frac{1}{\ln 9} \cdot \ln(9^x + 3) + c$

$\because \; h(\log_9 6) = 1$

$\therefore \; 1 = \frac{1}{\ln 9} \cdot \ln\left(9^{\log_9 6} + 3\right) + c$

$\Rightarrow \; 1 = \frac{1}{\ln 9} . \ln 9 + C \quad C = 0$

$\therefore \; h(x) = \frac{1}{\ln 9} . \ln(9^x + 3)$

$= \log_9(9^x + 3)$

Comprehension-IV

The average value of a function $f(x)$ over the interval $[a, b]$ is the number $\mu = \frac{1}{b-a}\int_a^b f(x)dx$. The square root $\left\{\frac{1}{b-a}\int_a^b f^2(x)dx\right\}^{\frac{1}{2}}$ is called the root mean square of f on $[a, b]$. The average value μ is attained if f is continuous on $[a, b]$.

1. The average ordinate of $y = \sin x$ over the interval $[0, \pi]$ is

(1) $\frac{1}{\pi}$ (2) $\frac{2}{\pi}$ (3) $\frac{4}{\pi^2}$ (4) $\frac{2}{\pi^2}$

Sol. Answer (2)

We have, $\mu = \frac{1}{b-a}\int_a^b f(x)dx = \frac{1}{\pi - 0}\int_0^{\pi} \sin x dx = \frac{1}{\pi}[-\cos x]_0^{\pi}$

$= \frac{1}{\pi}[-\cos\pi + \cos 0] = \frac{2}{\pi}$

2. The average value of pressure varying from 2 to 10 atm if the pressure P and the volume are related by $PV^{\frac{3}{2}} = 160$ is

(1) $\dfrac{(20)^{\frac{2}{3}}}{(10)^{\frac{1}{3}}+2^{\frac{1}{3}}}$ (2) $\dfrac{10}{(10)^{\frac{1}{3}}+2^{\frac{1}{3}}}$ (3) $\dfrac{2(20)^{\frac{2}{3}}}{(10)^{\frac{1}{3}}+2^{\frac{1}{3}}}$ (4) $\dfrac{8(20)^{\frac{2}{3}}}{(10)^{\frac{1}{3}}+(2)^{\frac{1}{3}}}$

Sol. Answer (3)

As P varies from 2 to 10 atm, V transverses the interval $[4\sqrt[3]{4}, 4\sqrt[3]{100}]$, average value = $\dfrac{1}{4(\sqrt[3]{100}-\sqrt[3]{4})}$

$$\int_{4\sqrt[3]{4}}^{4\sqrt[3]{100}} 160V^{-3/2}dV$$

$$= \frac{40}{\sqrt[3]{20}(\sqrt[3]{10}+\sqrt[3]{2})} = \frac{2(20)^{2/3}}{(10)^{1/3}+(2)^{1/3}}.$$

3. The average value of $f(x) = \dfrac{\cos^2 x}{\sin^2 x + 4\cos^2 x}$ on $\left[0, \dfrac{\pi}{2}\right]$ is

(1) $\dfrac{\pi}{6}$ (2) $\dfrac{4}{\pi}$ (3) $\dfrac{6}{\pi}$ (4) $\dfrac{1}{6}$

Sol. Answer (4)

Average value = $\dfrac{1}{\pi/2}\displaystyle\int_0^{\pi/2} \frac{\cos^2 x}{\sin^2 x + 4\cos^2 x}dx = \frac{2}{\pi}\int_0^{\infty} \frac{dt}{(t^2+4)(t^2+1)}$

Put $t = \tan x$

$$= \frac{1}{6}$$

Comprehension-V

In some cases of integration we take recourse to the method of successive reduction of the integrand which mostly depends on the repeated application of integration by parts. This is specially the case when the integrands are complicated in nature and depend on certain parameters. These parameters may be positive, negative or fractional indices, as for *e.g.*, $x^n e^{ax}$, $\tan^n x$, $(x^2 + a^2)^{n/2}$, $\cos^n x$ etc. The formula in which a certain integral involving some parameter is connected with some integral of lower order is called a reduction formula. *e.g.* obtaining the reduction formula for $\int x^n e^{ax} dx$ we write

$$I_n = \int x^n e^{ax} dx = x^n.\frac{e^{ax}}{a} - \frac{n}{a}\int x^{n-1}e^{ax}dx$$

$$I_n = x^n.\frac{e^{ax}}{a} - \frac{n}{a}I_{n-1} \quad (x \in I)$$

1. The reduction formula for $\int \sec^n x\, dx$ is

(1) $I_n = \dfrac{\sec^{n-2} x \tan x}{n+1} + \dfrac{n-2}{n-1} I_{n-2}$

(2) $I_n = \dfrac{\sec^{n-2} x \tan x}{n-1} + \dfrac{n-2}{n-1} I_{n-2}$

(3) $I_n = \dfrac{\sec^{n-2} x \tan x}{n+1} + \dfrac{n-1}{n-2} I_{n-2}$

(4) None of these

Sol. Answer (2)

The reduction formula for $\int \sec^n x\, dx$

Let $I_n = \int \sec^n x\, dx = \int \sec^{n-2} x \cdot \sec^2 x\, dx$

$$= \sec^{n-2} x \cdot \tan x - \int (n-2) \sec^{n-3} x \cdot \sec x \tan x \cdot \tan x\, dx$$

$$= \sec^{n-2} x \cdot \tan x - (n-2) \int \sec^{n-2} x (\sec^2 x - 1)\, dx$$

$$I_n = \sec^{n-2} x \cdot \tan x - (n-2) \int \sec^n x\, dx + (n-2) \int \sec^{n-2} x\, dx$$

$$(1 + n - 2) I_n = \sec^{n-2} x \tan x + (n-2) I_{n-2}$$

$$I_n = \frac{\sec^{n-2} x \cdot \tan x}{(n-1)} + \left(\frac{n-2}{n-1}\right) I_{n-2}$$

2. If $I_n = \int \cot^n x dx$ and $I_0 + I_1 + 2(I_2 + ... + I_8) + I_9 + I_{10} = A\left(\mu + \dfrac{\mu^2}{2} + ... + \dfrac{\mu^9}{9}\right)$ + constant, where $\mu = \cot x$ then

(1) $A = 2$ (2) $A = -1$ (3) $A = 1$ (4) $A = -2$

Sol. Answer (2)

$$I_n = \int \cot^n x\, dx = -\frac{\cot^{n-1} x}{(n-1)} - I_{n-2}$$

$$I_2 = -\cot x - I_0$$

$$I_3 = \frac{-\cot^2 x}{2} - I_1$$

$$I_4 = \frac{-\cot^3 x}{3} - I_2$$

$$I_5 = \frac{-\cot^4 x}{4} - I_3$$

$$I_6 = \frac{-\cot^5 x}{5} - I_4$$

$$I_7 = \frac{-\cot^6 x}{6} - I_5$$

$$I_8 = \frac{-\cot^7 x}{7} - I_6$$

$$I_9 = \frac{-\cot^8 x}{8} - I_7$$

$$I_{10} = \frac{-\cot^9 x}{9} - I_8$$

$$I_0 + 2(I_2 + I_3 + \ldots\ldots + I_8) + I_9 + I_{10}$$

$$= -\left[\cot x + \frac{\cot^2 x}{2} + \frac{\cot^3 x}{3} + \ldots\ldots + \frac{\cot^9 x}{9}\right]$$

$\Rightarrow A = -1$

3. If $I_n = \int x^n \sqrt{a-x}\,dx$, then $(2n+3)I_n$ equals

(1) $2anI_{n-1} - 2x^n\sqrt{a-x}$

(2) $-2anI_n + 2x^n(a-x)^{3/2}$

(3) $2anI_{n-1} - 2x^n(a-x)^{3/2}$

(4) None of these

Sol. Answer (3)

$$I_n = \int x^n \sqrt{a-x}\,dx$$

$$= x^n \cdot \frac{(a-x)^{3/2}}{3/2}(-1) + \frac{2}{3}\int nx^{n-1}(a-x)^{3/2}\,dx$$

$$= -\frac{2}{3}x^n(a-x)^{3/2} + \frac{2n}{3}\int x^{n-1}(a-x)^{3/2}\,dx$$

$$= -\frac{2}{3}x^n(a-x)^{3/2} + \frac{2na}{3}\int x^{n-1}\sqrt{a-x}\,dx - \frac{2n}{3}\int x^n\sqrt{a-x}\,dx$$

$$\left(1+\frac{2n}{3}\right)I_n = \frac{-2}{3}x^n(a-x)^{3/2} + \frac{2na}{3}I_{n-1}$$

$$(3+2n)I_n = -2x^n(a-x)^{3/2} + 2an\,I_{n-1}$$

Comprehension-VI

$$I = \int f\left\{x^m (ax^n + b)^{p/q}\right\} dx , \qquad \begin{matrix} p,\ q \in I \\ m, n \in Q \end{matrix}$$

If $\frac{m+1}{n}$ = integer then we can solve by substituting $ax^n + b = t^q$ and if $\frac{m+1}{n} \neq$ integer but $\frac{m+1}{n} + \frac{p}{q}$ = Integer, we can substitute $a + bx^{-n} = t^q$ it converts function in integrable form.

1. $\int \frac{dx}{\sqrt{a^{5x}} \cdot \sqrt[4]{(a^{2x} + a^{-2x})^3}} =$

(1) $\frac{1}{\ln a}(1+a^{-4x})^{1/4} + c$

(2) $-\frac{1}{\ln a}(1+a^{-4x})^{1/4} + c$

(3) $\ln a(1+a^{-4x})^{1/4} + c$

(4) $-\ln a(1+a^{-4x})^{1/4} + c$

Sol. Answer (2)

$$\int \frac{dx}{\sqrt{a^{5x}} \cdot \sqrt[4]{(a^{2x} + a^{-2x})^3}}$$

$$= \frac{1}{\ln a}\int \frac{dt}{t.t^{5/2}\left\{t^2 + t^{-2}\right\}^{3/4}} \qquad \text{Let } a^x = t \Rightarrow a^x.\ln a.dx = dt$$

$$= \frac{1}{\ln a}\int \frac{dt}{t^{7/2} \cdot \frac{(1+t^4)^{3/4}}{t^{3/2}}} \qquad \left(\because \frac{m+1}{n} - \frac{p}{q} = \text{intger } \frac{-2+1}{4} - \frac{3}{4} = -1\right)$$

$$= \frac{1}{\ln a}\int \frac{dt}{t^2 \cdot (1+t^4)^{3/4}}$$

$$= \frac{1}{\ln a}\int \frac{dt}{t^5 \cdot \left(\frac{1}{t^4} + 1\right)^{3/4}} \qquad \frac{1}{t^4} + 1 = u^4 \Rightarrow -\frac{4}{t^5}dt = 4u^3 du$$

$$= \frac{1}{\ln a}\int \frac{-u^3 du}{u^3}$$

$$= -\frac{1}{\ln a}\left\{\frac{1}{t^4} + 1\right\}^{1/4} + c$$

$$= -\frac{1}{\ln a}\{a^{-4x} + 1\}^{1/4} + c$$

2. $\int\sqrt{\text{cosec}\,x}\cdot\left(2\cdot\sqrt[3]{\text{cosec}\,x}+1\right)^{3/2}\cdot\cos x\,dx=$

(1) $2t^3+12t+6\sqrt{2}\ln\left|\dfrac{t-\sqrt{2}}{t+\sqrt{2}}\right|+c$, where $t=\sqrt{2+\sin^{1/3}x}$

(2) $2t^3-12t-6\sqrt{2}\ln\left|\dfrac{t-\sqrt{2}}{t+\sqrt{2}}\right|+c$, where $t=\sqrt{2+\sin^{1/3}x}$

(3) $2t^3+12t-6\sqrt{2}\ln\left|\dfrac{t-\sqrt{2}}{t+\sqrt{2}}\right|+c$, where $t=\sqrt{2+\sin^{1/3}x}$

(4) $-2t^3+t+3\sqrt{2}\ln\left|\dfrac{t-\sqrt{2}}{t+\sqrt{2}}\right|+c$, where $t=\sqrt{2+\sin^{1/3}x}$

Sol. Answer (1)

$$\int\sqrt{\text{cosec}x}\left(2.\sqrt[3]{\text{cosec}x}+1\right)^{3/2}\cdot\cos x\ dx$$

$$\int\frac{1}{\sqrt{\sin x}}\cdot\frac{\left(2+\sqrt[3]{\sin x}\right)^{3/2}}{\sin^{1/2}x}\cos x\ dx$$

Let $2+\sin^{1/3}x=t^2$

$$\frac{1}{3}\frac{\cos x}{\sin^{2/3}x}dx=2t\ dt$$

$$=\int\frac{\left(2+\sqrt[3]{\sin x}\right)^{3/2}}{\sin^{1/3}x}\cdot\frac{\cos x}{\sin^{2/3}x}dx$$

$$=\int\frac{t^3.6t}{t^2-2}dt$$

$$=6\int\frac{t^4-4+4}{t^2-2}dt$$

$$=6\int\left[t^2+2+\frac{4}{t^2-2}\right]dt$$

$$=6\left[\frac{t^3}{3}+2t+\frac{4}{2\sqrt{2}}\ln\left|\frac{t-\sqrt{2}}{t+\sqrt{2}}\right|\right]+c$$

$$=2t^3+12t+6\sqrt{2}\ln\left|\frac{t-\sqrt{2}}{t+\sqrt{2}}\right|+c\quad\text{where }t=\sqrt{2+\sin^{1/3}x}$$

3. $\int x^{1/5}(1+x^{3/5})^{2/5}dx=$

(1) $\dfrac{25}{3}\left[\dfrac{(1+x^{3/5})^{12/5}}{12}-\dfrac{(1+x^{3/5})^{7/5}}{7}\right]+c$

(2) $\dfrac{25}{3}\left[\dfrac{(1+x^{3/5})^{12/5}}{12}\right]+c$

(3) $\dfrac{25}{12}\left[(1+x^{3/5})^{12/5}+(1+x^{3/5})^{7/5}\right]+c$

(4) None of these

Sol. Answer (1)

$$\int x^{1/5}(1+x^{3/5})^{2/5}dx$$

$$= \int x^{3/5}\cdot(1+x^{3/5})^{2/5}\cdot x^{-2/5}dx \qquad \because \frac{\frac{1}{5}+1}{\frac{3}{5}} = 2 = \text{integer}$$

$$= \int (t^5-1)\cdot t^2\cdot\frac{25}{3}t^4dt \qquad \therefore \text{ Putting } 1+x^{3/5} = t^5$$

$$= \frac{25}{3}\left[\frac{t^{12}}{12}-\frac{t^7}{7}\right]+C \qquad \frac{3}{5}x^{-2/5}dx = 5t^4dt$$

$$= \frac{25}{3}\left[\frac{(1+x^{3/5})^{12/5}}{12}-\frac{(1+x^{3/5})^{7/5}}{7}\right]+c$$

Comprehension-VII

If a function f defined for all real numbers R, takes both positive and negative values in R, then the equation $f(x) = 0$ possesses a root in R.

Let $g(x) = \cos x - x\sin x,\ \forall\ x \in R$

1. Let $h(x) = \int g(x)\,dx$. If h(x) = 0 has one root $x = 0$, then $h(x) = 0$ has

(1) No other root in R

(2) Infinitely many roots at $x = n\pi, n \in \mathbb{Z}$

(3) Infinitely many roots at $x = (2n+1)\frac{\pi}{2},\ n \in \mathbb{Z}$

(4) None of these

Sol. Answer (3)

$h(x) = \int g(x)dx = \int (\cos x - x\sin x)dx$

= sin x – {x(-cos x) – 1.(-sin x)}+ C

= x cos x + C

$\because$ $h(0) = 0 \Rightarrow C = 0 \quad \therefore h(x) = x\cos x$

$h(x) = 0 \quad x\cos x = 0$

$x = 0,\ x = (2n+1)\frac{\pi}{2},\ n \in Z$

2. Let $\phi(x) = \int h(x)\,dx$ where $h(x)$ is the particular solution obtained in question 1. If $\phi(0) = 1$, then $\phi(x) = 0$ has

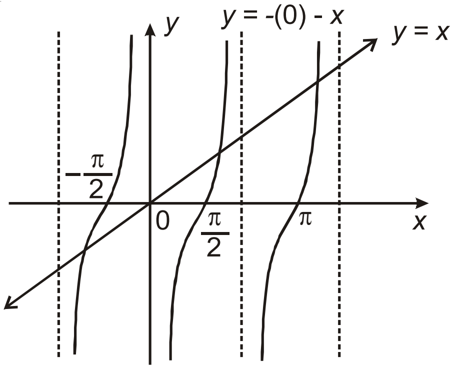
(1) No root in R

(2) Infinitely many roots at $x = n\pi, n \in \mathbb{Z}$

(3) Infinitely many roots at $x = (2n+1)\frac{\pi}{2},\ n \in \mathbb{Z}$

(4) Infinitely many roots in R

Sol. Answer (4)



$\phi(x) = \int h(x)dx$

$= \int (x\cos x)dx$

$= x(\sin x) - 1.(-\cos x) + C_1$

$\because \quad \phi(0) = 1$

$\therefore \quad 1 = 0 + 1 + C_1 \qquad C_1 = 0$

$\therefore \quad \phi(x) = x\sin x + \cos x$

$\phi(x) = 0 \quad x\sin x + \cos x = 0$

$x = -\cot x$

Clearly there are infinitely many solutions in R.

3. Let $F(x) = \int \{g(x) + \phi(x)\}\, dx$, where $\phi(x)$ is the particular solution obtained in question 2. If $F(0) = 0$, then $F(x) = 4$ has

(1) No root in R

(2) One root in R

(3) Infinitely many roots in R

(4) None of these

Sol. Answer (1)

$F(x) = \int \{g(x) + \phi(x)\}dx$

$g(x) = \cos x - x\sin x$

$\phi(x) = x\sin x + \cos x$

$\Rightarrow \; g(x) + \phi(x) = 2\cos x$

$\therefore \; F(x) = \int 2\cos x \; dx$

$= 2\sin x + C$

$\because \; F(0) = 0$

$\therefore \; 0 = 0 + C \;\; \Rightarrow \;\; C = 0$

$F(x) = 2\sin x$

Equation

$2\sin x = 4 \quad \sin x = 2$ has no solution

Comprehension-VIII

$$|x| = \begin{bmatrix} x & ; & x \geq 0 \\ -x & ; & x < 0 \end{bmatrix}$$

$[x] = n$ where $n \in I$ and

$n = x \,;\, x \in I$

$n = x - 1 \,;\, x \notin I$

1. $\int_0^4 (|x-1|+|x-3|)dx$ equals

(1) 5 (2) 8 (3) 10 (4) 9

Sol. Answer (3)

We have, $\int_0^4 (|x-1|+|x-3|)dx$

$= -\int_0^1 (x-1)dx + \int_1^4 (x-1)dx - \int_0^3 (x-3)dx + \int_3^4 (x-3)dx$

= 10

2. $\int_{0.5}^{4.5} [x]dx + \int_{-1}^{1} |x|dx$ equal to

(1) 6 (2) 7 (3) 8 (4) 9

Sol. Answer (4)

$\int_{0.5}^{4.5} [x]dx + \int_{-1}^{1} |x|\,dx$

$= \int_{0.5}^{1} 0dx + \int_1^2 1dx + \int_2^3 2dx + \int_3^4 3dx + \int_4^{4.5} 4dx - \int_{-1}^{0} xdx + \int_0^1 xdx$

= 9

3. $\int_0^{\frac{\pi}{3}} \left[\sqrt{3}\tan x\right]dx$ is equal to

(1) $\frac{\pi}{2} - \tan^{-1}\frac{2}{\sqrt{3}}$ (2) $\frac{5\pi}{6} - \tan^{-1}\frac{2}{\sqrt{3}}$ (3) $\frac{5\pi}{6}$ (4) $\frac{\pi}{6}$

Sol. Answer (1)

$\int_0^{\pi/3} [\sqrt{3}\tan x]dx = \int_0^{\pi/6} 0dx + \int_{\pi/6}^{\tan^{-1}\frac{2}{\sqrt{3}}} 1.dx + \int_{\tan^{-1}\frac{2}{\sqrt{3}}}^{\pi/3} 2.dx$

$= \frac{\pi}{2} - \tan^{-1}\frac{2}{\sqrt{3}}.$

Comprehension-IX

If function $f(x)$ and $g(x)$ are continuous on the interval $[a, b]$ and $g(x)$ retain the same sign on $[a, b]$ then there is $C \in (a, b)$ such that $\int_a^b g(x).f(x)dx = f(c)\int_a^b g(x)dx$. This is known as Mean Value Theorem. This result can be used to estimate some definite integrations. Other results which can be used for estimation are

(i) If f increases and has a concave graph in the interval $[a, b]$ then

$$(b-a)f(a) < \int_a^b f(x)dx < (b-a)\frac{f(a)+f(b)}{2}$$

(ii) If f increases and has a convex graph in the interval $[a, b]$ then

$$(b-a)\frac{f(a)+f(b)}{2} < \int_a^b f(x)dx < (b-a)f(b)$$

(iii) $$\left|\int_a^b f(x)g(x)dx\right|^2 \le \left(\int_a^b f^2(x)dx\right)\left(\int_a^b g^2(x)dx\right)$$

1. Using Mean-Value Theorem, the best upper bound of $\int_0^1 \frac{\sin x}{1+x^2}dx$ is

(1) $\frac{\pi}{4}\sin 1$ (2) $\pi \sin 1$ (3) $\frac{\pi}{2}\sin 1$ (4) $\frac{\pi}{4}\sin\frac{1}{2}$

Sol. Answer (1)

$$\int_0^1 \frac{\sin x}{1+x^2}dx = \sin c\int_0^1 \frac{dx}{1+x^2} = \sin c \,|\tan^{-1}x|_0^1 = \sin c(\tan^{-1}1-\tan^{-1}0)$$

$$= \sin c\left(\frac{\pi}{4}-0\right) = \frac{\pi}{4}\sin c \quad (0 < c < 1)$$

Since the function $\sin x$ increases on $[0, 1]$ so $\sin c < \sin 1$.

Hence $\int_0^1 \frac{\sin x}{1+x^2}dx < \frac{\pi}{4}\sin 1$.

2. Using (i) and (ii) above the best upper bound of $\int_0^1 \sqrt{1+x^4}\,dx$ is

(1) $1+\sqrt{2}$ (2) $\frac{1+\sqrt{2}}{2}$ (3) $\frac{\sqrt{2}-1}{2}$ (4) $2(\sqrt{2}-1)$

Sol. Answer (2)

$f(x) = \sqrt{1+x^4}$ is concave on $[0, 1]$

$$f'(x) = \frac{4x^3}{2\sqrt{1+x^4}} = \frac{2x^3}{\sqrt{1+x^4}}$$

$$f''(x) = \frac{2x^2(x^4+3)}{(1+x^4)^{3/2}} > 0,\ 0 \le x \le 1$$

So, $1 < \int_0^1 \sqrt{1+x^4}\,dx < \frac{1.f(0)+f(1)}{2}$

$$1 < \int_0^1 \sqrt{1+x^4}\,dx < \frac{1+\sqrt{2}}{2}$$

Best upper bound of $\int_0^1 \sqrt{1+x^4}\,dx = \frac{1+\sqrt{2}}{2}$.

3. Using (iii) above the best upper bound of $\int_0^1 \sqrt{1+x^4}\, dx$ is

(1) 1.2 (2) $\sqrt{1.22}$ (3) $\sqrt{1.2}$ (4) $\sqrt{1.4}$

Sol. Answer (3)

$$\left|\int_0^1 \sqrt{1+x^4}\,dx\right| = \int_0^1 \sqrt{1+x^4}\,dx < \sqrt{\int_0^1 (1+x^4)dx \int_0^1 1^2 dx}$$

$$= \sqrt{\left|x+\frac{x^5}{5}\right|_0^1} = \sqrt{1.2}\,.$$

Comprehension-X

If f is a continuous even function then,

$$\int_{-a}^{a} f(x)dx = 2\int_0^a f(x)dx$$

If f is a continuous odd function $\int_{-a}^{a} f(x)dx = 0$

It f is a periodic function with period T then

$$\int_a^{a+nT} f(x)dx = n\int_0^T f(x)dx$$

1. $\int_{-1/2}^{1/2} \cos x \ln\left(\frac{1+x}{1-x}\right)dx$ is equal to

(1) 0 (2) 1 (3) 2 (4) $\frac{1}{2\sqrt{3}}$

Sol. Answer (1)

Let $f(x) = \cos x \ln\left(\frac{1+x}{1-x}\right)$

$$f(-x) = \cos x \ln\left(\frac{1-x}{1+x}\right)$$

$$f(x) + f(-x) = 0$$

So f is an odd function hence the value of 0.

2. $\int_2^4 \frac{\sqrt{x}}{\sqrt{6-x}+\sqrt{x}}\,dx$ is equal to

(1) 1 (2) 2 (3) 3 (4) 6

Sol. Answer (1)

$$I = \int_2^4 \frac{\sqrt{x}}{\sqrt{6-x}+\sqrt{x}}\,dx$$

$$I = \int_2^4 \frac{\sqrt{6-x}}{\sqrt{x}+\sqrt{6-x}}\,dx$$

$$2I = \int_2^4 dx = 2$$

$$I = 1$$

Comprehension-XI

Given that for each $a \in (0, 1)$,

$$\lim_{h\to 0^+} \int_h^{1-h} t^{-a}(1-t)^{a-1}\,dt$$

exists. Let this limit be $g(a)$. In addition, it is given that the function $g(a)$ is differentiable on (0, 1).

[JEE(Advanced)-2014]

1. The value of $g\left(\frac{1}{2}\right)$ is

(1) π (2) 2π (3) $\frac{\pi}{2}$ (4) $\frac{\pi}{4}$

Sol. Answer (1)

$$g(a) = \lim_{h\to 0^+} \int_h^{(1-h)} t^{-a}(1-t)^{a-1}\,dt$$

$$g = \lim_{h\to 0^+} \int_h^{(1-h)} t^{-1/2}(1-t)^{-1/2}\,dt$$

$$= \lim_{h\to 0^+} \int_h^{(1-h)} \frac{1}{\sqrt{t(1-t)}}\,dt$$

$$= \lim_{h\to 0^+} \sin^{-1}\left(\frac{t-\frac{1}{2}}{\frac{1}{2}}\right)\Bigg|_h^{1-h}$$

$$= \lim_{h\to 0^+} [\sin^{-1}(2(1-h)-1)) - \sin^{-1}(2h-1)]$$

$$= \frac{\pi}{2} + \frac{\pi}{2} = \pi$$

2. The value of $g'\left(\frac{1}{2}\right)$ is

(1) $\frac{\pi}{2}$ (2) π (3) $-\frac{\pi}{2}$ (4) 0

Sol. Answer (4)

$$g(a) = \lim_{h\to 0^+} \int_h^{1-h} t^{-a}(1-t)^{a-1}dt$$

$$g'(a) = \lim_{h\to 0^+} \int_h^{1-h} t^{-a}(1-t)^{a-1} \ln\left(\frac{1-t}{t}\right) dt$$

$$g'\left(\frac{1}{2}\right) = \lim_{h\to 0^+} \int_h^{1-h} t^{-\frac{1}{2}}(1-t)^{-\frac{1}{2}} \ln\left(\frac{1-t}{t}\right) dt$$

$$\Rightarrow \quad g'\left(\frac{1}{2}\right) = 0$$

SECTION - D

Assertion-Reason Type Questions

1. STATEMENT-1 : $\int \frac{xe^x dx}{(1+x)^2} = \frac{e^x}{x+1} + C$

and

STATEMENT-2 : $\int e^x[f(x) + f'(x)]dx = e^x f(x) + C$

Sol. Answer (1)

Statement-1 : $\int \frac{x\,e^x dx}{(1+x)^2} = \frac{e^x}{x+1} + C$

$\Rightarrow \quad \int \frac{(x+1-1)e^x}{(1+x)^2}\,dx = \int e^x\left(\frac{1}{1+x}-\frac{1}{(1+x)^2}\right)dx = \frac{e^x}{x+1}+C$

$\therefore \quad \int e^x (f(x)+f'(x)\,dx = e^x f(x)+C$

Statement-1 is true.

Statement-2 : $\int e^x [f(x)+f'(x)]\,dx = e^x f(x)+C$

Statement-2 is true and statement-2 is a correct explanation for statement-1.

2. STATEMENT-1 : $\int x^x(1+\log x)dx = x^x + C$

and

STATEMENT-2 : $\frac{d}{dx}x^x = x^x(1+\log x)$

Sol. Answer (1)

Statement-1 : $\int x^x(1+\log x)\,dx = x^x + C$

We know that $\frac{d}{dx}(x^x) = x^x(1+\log x)$

Statement-1 is true.

Statement-2 : $\frac{d}{dx}x^x = x^x(1+\log x)$

Statement-2 is true and is a correct explanation for statement-1.

3. STATEMENT-1 : $\int \frac{x^2-1}{x^2} e^{\left(\frac{x^2+1}{x}\right)}dx = e^{\frac{x^2+1}{x}} + c$

and

STATEMENT-2 : $\int f(x)e^{f(x)}dx = e^{f(x)} + c$

Sol. Answer (3)

Let $\frac{x^2+1}{x} = x+\frac{1}{x} = t$

$1-\frac{1}{x^2}dx = dt$

$\int e^t dt = e^t + c = e^{\frac{x^2+1}{x}} + c$

Statement-2 is not true.

4. STATEMENT-1 : If $I_n = \int \tan^n x\, dx$ then $9(I_8 + I_{10}) = \tan^9 x$

and

STATEMENT-2 : $I_n = \int \tan^n x\, dx$ then $I_n = \dfrac{\tan^{n+1} x}{n+1} + c,\ n \neq -1$.

Sol. Answer (3)

$$I_n = \int \tan^n x\, dx$$

$$= \int \tan^{n-2} x \sec^2 x - \int \tan^{n-2} x\, dx$$

$$= \frac{\tan^{n-1} x}{n-1} - I_{n-2}$$

Putting $n = 10$ $9(I_{10} + I_8) = \tan^9 x$

Statement-1 is true and statement-2 is false

5. STATEMENT-1 : $\int \tan 5x \tan 3x \tan 2x\, dx$ is equal to $\dfrac{\log|\sec 5x|}{5} - \dfrac{\log|\sec 3x|}{3} - \dfrac{\log|\sec 2x|}{2} + c$

and

STATEMENT-2 : $\tan 5x - \tan 3x - \tan 2x = \tan 5x \tan 3x \tan 2x$.

Sol. Answer (1)

$5x = 3x + 2x \Rightarrow \tan 5x = \tan(3x + 2x)$

$$\tan 5x = \frac{\tan 3x + \tan 2x}{1 - \tan 3x \cdot \tan 2x}$$

$\tan 5x - \tan 3x - \tan 2x = \tan 5x + \tan 3x + \tan 2x$

$$\int \tan 5x \tan 3x \tan 2x\, dx$$

$$= \int (\tan 5x - \tan 3x - \tan 2x)\, dx$$

$$= \frac{\log|\sec 5x|}{5} - \frac{\log|\sec 3x|}{3} - \frac{\log|\sec 2x|}{2} + c$$

6. STATEMENT-1 : $\displaystyle\int \frac{dx}{e^x + e^{-x} + 2} = -\frac{1}{e^x + 1} + c$

and

STATEMENT-2 : $\displaystyle\int \frac{d(f(x))}{(f(x))^2} = -\frac{1}{f(x)} + c$

Sol. Answer (1)

Statement-2 is true

Now, $\int \frac{dx}{e^x + e^{-x} + 2} = \int \frac{e^x dx}{(e^x+1)^2}$

$= \int \frac{d(e^x+1)}{(e^x+1)^2}$

$= -\frac{1}{e^x+1} + c$

7. STATEMENT-1 : If $f(x) = \int \frac{dx}{\sin^{1/2} x \cos^{7/2} x}$; then the value of $f\left(\frac{\pi}{4}\right) - f(0)$ is equal to $\frac{12}{5}$.

and

STATEMENT-2 : To find the $\int \sin^m x \cos^n x \, dx$ if $m + n$ = –ve even; then we can substitute tan $x = t$.

Sol. Answer (1)

Hence $-\frac{1}{2} - \frac{7}{2} = -4$ = negative even number

But tan $x = t \Rightarrow \sec^2 x \, dx = dt$

$f(x) \int \frac{dx}{\frac{\sin^{1/2} x}{\cos^{1/2} x} \cos^4 x} = \int \frac{(1+\tan^2 x)\sec^2 x}{\sqrt{\tan x}} dx$

$= \int \left(\frac{1+t^2}{\sqrt{t}}\right) dt$

$= \int \left(t^{-\frac{1}{2}} + t^{\frac{3}{2}}\right) dt$

$= 2t^{\frac{1}{2}} + \frac{2}{5} t^{\frac{5}{2}} + C$

$= 2\sqrt{\tan x} + \frac{2}{5}(\tan x)^{\frac{5}{2}} + C$

$\therefore \quad f\left(\frac{\pi}{4}\right) - f(0) = 2 + \frac{2}{5} = \frac{12}{5}$

8. STATEMENT-1 : If $f\left(\frac{3x-4}{3x+4}\right) = x+2$, then $\int f(x)\,dx$ is equal to $\frac{2}{3}x - \frac{8}{3}\log|x-1| + c$.

and

STATEMENT-2 : If $f\left(\frac{3x-4}{3x+4}\right) = x+2$, then $f(x) = \frac{2}{3} - \frac{8}{3(x-1)}$.

Sol. Answer (1)

$$f\left(\frac{3x-4}{3x+4}\right) = x+2$$

Let $\frac{3x-4}{3x+4} = t \Rightarrow 3x - 4 = 3xt + 4t$

$$x = \frac{4t+4}{3(1-t)}$$

$$\therefore\ f(t) = \frac{4t+4}{3(1-t)} + 2 = \frac{4(x-t-1)}{3(1-t)} + 2$$

$$= 2 - \frac{4}{3} - \frac{8}{3(t-1)} = \frac{2}{3} - \frac{8}{3(t-1)}$$

$$\therefore\ f(x) = \frac{2}{3} - \frac{8}{3(x-1)}$$

$$\Rightarrow\ \int f(x)dx = \frac{2}{3}x - \frac{8}{3}\log|x-1| + C$$

9. STATEMENT-1 : $\int \frac{e^{\log\left(1+\frac{1}{x^2}\right)}}{x^2+\frac{1}{x^2}}dx = \frac{1}{\sqrt{2}}\tan^{-1}\frac{x^2-1}{\sqrt{2}x} + c$

and

STATEMENT-2 : $e^{\log x}$ is equal to x if x > 0.

Sol. Answer (2)

$\int \frac{1+\frac{1}{x^2}}{x^2+\frac{1}{x^2}}dx$ putting $x - \frac{1}{x} = t \Rightarrow 1 + \frac{1}{x^2}dx = dt$

$$\int \frac{dt}{t^2 + (\sqrt{2})^2} = \frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{x^2-1}{\sqrt{2}x}\right) + C$$

10. STATEMENT-1 : $\int_0^{\frac{\pi}{2}} \frac{x\sin x\cos x}{\sin^4 x+\cos^4 x}dx = \frac{\pi^2}{32}$

and

STATEMENT-2 : $\int_0^a f(x)dx = \int_0^a f(a-x)dx$

Sol. Answer (4)

Statement-1 : $\int_0^{\pi/2} \frac{x\sin x\cos x}{\sin^4 x+\cos^4 x} = \frac{\pi^2}{32}$

Let $I = \int_0^{\pi/2} \frac{x\sin x\cos x}{\sin^4 x+\cos^4 x}dx$ ... (1) $\left(\int_0^a f(x)dx = \int_0^a f(a-x)dx\right)$

$$I = \int_0^{\pi/2} \frac{\left(\frac{\pi}{2}-x\right)\sin\left(\frac{\pi}{2}-x\right)\cos\left(\frac{\pi}{2}-x\right)dx}{\sin^4\left(\frac{\pi}{2}-x\right)+\cos^4\left(\frac{\pi}{2}-x\right)}$$

$$I = \int_0^{\pi/2} \frac{\left(\frac{\pi}{2}-x\right)\cos x\sin x dx}{\cos^4 x+\sin^4 x} \quad \dots (2)$$

Adding equations (1) and (2), we get

$$2I = \frac{\pi}{2}\int_0^{\pi/2} \frac{\sin x\cos x}{\sin^4 x+\cos^4 x}dx$$

$$= \frac{\pi}{2}\int_0^{\pi/2} \frac{\sin x\cos x dx}{1-2\sin^2 x\cos^2 x}$$

$$= \frac{\pi}{4}\int_0^{\pi/2} \frac{\sin 2x}{1-\frac{1}{2}\sin^2 2x}dx$$

$$= \frac{\pi}{4}\int_0^{\pi/2} \frac{\sin 2x dx}{1-\frac{1}{2}(1-\cos^2 2x)}$$

$$= \frac{\pi}{2}\int_0^{\pi/2} \frac{\sin 2x}{1+\cos^2 2x}dx$$

Let $\cos 2x = t$

$-2\sin 2x dx = dt$

$= \frac{\pi}{2}\left(-\frac{1}{2}\right)\int \frac{dt}{1+t^2} = \frac{-\pi}{4}\tan^{-1} t$

$= -\frac{\pi}{4}|\tan^{-1}\cos 2x|_0^{\pi/2}$

$2I = \frac{\pi^2}{8} \Rightarrow I = \frac{\pi^2}{16}$.

Statement-1 is false.

Statement-2 : $\int_0^a f(x)dx = \int_0^a f(a-x)dx$

Statement-2 is true.

11. STATEMENT-1 : $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin\left(\log(x+\sqrt{1+x^2})\right)dx = 0$

and

STATEMENT-2 : $\int_{-a}^{a} f(x) = 0$ if $f(x)$ is an even function

Sol. Answer (3)

Statement-1 : $\int_{-\pi/2}^{\pi/2} \sin(\log(x+\sqrt{1+x^2})dx = 0$ $\left(\int_{-a}^{a} f(x)dx = 0, \text{if } f(-x) = -f(x)\right)$

$\sin(\log(x+\sqrt{1+x^2}))$ is an odd function

because $\sin(\log(-x+\sqrt{1+x^2})) = -\sin(\log(x+\sqrt{1+x^2}))$

Statement-1 is true.

Statement-2 : $\int_{-a}^{a} f(x)dx = 0$ if $f(x)$ is an even function.

Statement –2 is false.

12. STATEMENT-1 : If $|a|<1$, then $\int_0^{\frac{\pi}{2}} \frac{\log(1+a\cos x)}{\cos x}dx = \sin^{-1} a$

and

STATEMENT-2 : $\int_0^a f(x)d(x) = \int_0^a f(a-x)dx$.

Sol. Answer (4)

$$f(a)=\int_0^{\pi}\log\frac{(1+a\cos x)}{\cos x}dx$$

$$f'(a)=\int_0^{\pi}\frac{\cos x}{\cos x.(1+\cos x)}dx \qquad \text{\{Differentiating w.r.t. "}a\text{"\}}$$

$$=\int_0^{\pi}\frac{dx}{1+a\cos x}$$

Put $\tan\frac{x}{2}=t,\frac{dx}{dt}=\frac{2}{1+t^2}$ where $x = 0 \Rightarrow t = 0$

as $x=\pi \Rightarrow t\to\infty$

$$f'(a)=\int_0^{\infty}\frac{\frac{2dt}{1+t^2}}{1+a\left(\frac{1-t^2}{1+t^2}\right)}=\int_0^{\infty}\frac{2dt}{(1-a)t^2+(1+a)}=\frac{2}{\sqrt{1-a^2}}.\frac{\pi}{2}=\frac{\pi}{\sqrt{1-a^2}}$$

$$f(a)=\pi\sin^{-1}a+c$$

13. STATEMENT-1 : $\int_{-2\pi}^{5\pi}\cot^{-1}(\tan x)dx=7\frac{\pi^2}{2}$

and

STATEMENT-2 : $\int_a^b f(x)dx=\int_a^c f(x)dx+\int_c^b f(x)dx\,;a<c<d$

Sol. Answer (1)

$$\cot^{-1}(\cot x)=\begin{cases}x, & 0<x<\pi/2\\ \pi+x, & \pi/2<x<\pi\end{cases}$$

$$I=7\left\{\int_0^{\pi/2}\left(\frac{\pi}{2}-x\right)dx+\int_{\pi/2}^{\pi}\left(\pi+\frac{\pi}{2}-x\right)dx\right\}$$

$$=7\left\{\left(\frac{\pi}{2}x-\frac{x^2}{2}\right)_0^{\pi/2}+\left(\frac{3}{2}\pi x-\frac{x^2}{2}\right)_{\pi/2}^{\pi}\right\}$$

$$=\frac{7}{2}\pi^2$$

14. STATEMENT-1 : $\int_0^{\frac{\pi}{2}} \sin 2kx.\cot x.dx = \frac{\pi}{2}$; where $k \in I^+$

and

STATEMENT-2 : $\frac{\sin 2kx}{\sin x} = 2[\cos x + \cos 3x + ... + \cos(2k-1)x]$

Sol. Answer (1)

Since $\left(\frac{\sin 2kx}{\sin x}\right) = 2\cos x\left[\cos x + \cos 3x + ... + \cos(2k-1)^{\pi}\right]dx$

$$\int_0^{\pi/2}\left(\frac{\sin 2kx}{\sin x}\right)dx = \int_0^{\pi/2}(1+\cos 2x)dx + \int_0^{\pi/2}(\cos 4x + \cos 2x)dx + ... + \int_0^{\pi/2}(\cos 2kx + \cos(2k-2)x)dx$$

But we know that $\int_0^{\pi/2}(\cos 2nx)dx = 0 \forall n \in I \quad n \neq 0$

$$\int_0^{\pi/2}\sin 2kx.\cot x\, dx = \int_0^{\pi/2} 1dx = \frac{\pi}{2}$$

15. STATEMENT-1 : If $n \in N \int_{-n}^{n}(-1)^{[x]}dx = 2n$

and

STATEMENT-2 : $(-1)^{[x]}$ is odd if $x \notin I$

Sol. Answer (4)

Consider

$f(x) = (-1)^{[x]}$

$f(-x) = (-1)^{[-x]}$

$f(x) + f(-x) = 0$

$$\int_{-n}^{n}(-1)^{[x]} = 0$$

16. STATEMENT-1 : $\int_0^2 [x + [x + [x]]]\, dx = 3$

and

STATEMENT-2 : $\int_a^b f(x)dx = \int_a^b f(a+b-x)\, dx$

Sol. Answer (2)

$$I = \int_0^2 [x + [x + [x]]]\,dx$$

$$I = \int_0^2 [x + 2 + [x]]\,dx$$

$$= \int_0^2 3[x]\,dx = 3\left\{\int_0^1 [x]dx + \int_1^2 [x]dx\right\}$$

$= 3\ (2 - 1) = 3$

17. STATEMENT-1 : The function $f(x) = \int \cos^2 x\, dx$ satisfies $f(\pi + x) = f(x)\ \forall\ x \in R$.

and

STATEMENT-2 : $\cos^2 x = \cos^2(\pi + x)$

Sol. Answer (4)

$$f(x) = \int \cos^2 x\, dx = \frac{1}{2}\int(1 + \cos 2x)dx$$

$$= \frac{1}{2}\left(x + \frac{\sin 2x}{2}\right)$$

$$f(x+\pi) - f(x) = \frac{1}{2}\left((x+\pi) + \frac{\sin 2(x+\pi)}{2}\right) - \frac{1}{2}\left(x + \frac{\sin 2x}{2}\right)$$

$$= \frac{\pi}{2} \neq 0$$

$\therefore\ f(x+\pi) \neq f(x)$

18. STATEMENT-1 : $\displaystyle\int \frac{5-2x}{\sqrt{2+2x-x^2}}dx = 2\sqrt{2+2x-x^2} + 3\sin^{-1}\frac{x-1}{\sqrt{3}} + c$

and

STATEMENT-2 : $\displaystyle\int \frac{dx}{\sqrt{a^2-x^2}} = \frac{x}{2}\sqrt{a^2-x^2} + \frac{a^2}{2}\sin^{-1}\frac{x}{a}$.

Sol. Answer (3)

$$\int \frac{(2-2x)+3}{\sqrt{3-(x^2-2x+1)}}dx = \int \frac{2-2x}{\sqrt{2+2x-x^2}}dx + \int \frac{3dx}{\sqrt{3-(x-1)^2}}$$

$$= 2\sqrt{2+2x-x^2} + 3\sin^{-1}\frac{x-1}{3}$$

19. STATEMENT-1 : $\int \frac{3+4\cos x}{(4+3\cos x)^2}dx = \left(\frac{\sin x}{4+3\cos x}\right)+c$

and

STATEMENT-2 : $\int (f(x))^n f'(x)\,dx = \frac{(f(x))^{n+1}}{n+1}+c$.

Sol. Answer (3)

$I = \int \frac{3+4\cos x}{(4+3\cos x)^2}dx$; multiplying and dividing by cosec^2x

$$\int \frac{3\text{cosec}^2 x + 4\cot x \,\text{cos}\,ecx}{(4\text{cosec}x + 3\cot x)^2}dx$$

$\therefore\ I = -\int (f(x))^{-2} f'(x)dx$

where $f(x)$ = 4 cosec x + 3 cot x

$$= \frac{1}{f(x)} = \frac{1}{4\text{cosec}x + 3\cot x} = \frac{\sin x}{4+3\cos x}+C$$

Statement-2 is not true of $n = -1$

20. STATEMENT-1 : If $0 < x < \frac{\pi}{2}$, then $\int \sqrt{1+2\tan x(\tan x+\sec x)}\,dx$ equal to $\log(\sec x+\tan x)+\log\sec x+c$

and

STATEMENT-2 : $\int x^3 d(\tan^{-1}x)$ is equal to $\frac{x^2}{2}-\frac{1}{2}\log(1+x^2)+c$.

Sol. Answer (2)

$I = \int \sqrt{1+2\tan^2 x+2\tan x\sec x}\ dx = \sqrt{(\sec^2 x+\tan^2 x+2\tan x\sin x)}\ dx$

$= \int (\sec x+\tan x)dx = \log(\sec x+\tan x)+\log\ \sec x + C$

$I = \int x^2 d(\tan^{-1}x) = \int \frac{x^3}{1+x^2}dx = \int \left(x - \frac{x}{1+x^2}\right)dx$

$= \frac{x^2}{2} - \frac{1}{2}\log(1+x^2)+C$

21. STATEMENT-1 : The value of $\int \frac{dx}{x^2(x^n+1)^{n-1/n}}$ is equal to $-(1+x^{-n})^{1/n}+c$.

and

STATEMENT-2 : The value of $\int \frac{dx}{x^n(1+x^n)^{1/n}}$ is equal to $\frac{1}{1-n}(1+x^{-n})^{\frac{n-1}{n}}$.

Sol. Answer (1)

$$\int \frac{dx}{x^2(x^n+1)^{\frac{n-1}{n}}} = \int \frac{dx}{x^2 x^{n-1}\left(1+\frac{1}{x^n}\right)^{\frac{n-1}{n}}}$$

$$= \int \frac{dx}{x^{n+1}(1-x^{-n})^{\frac{n-1}{n}}} \quad \text{putting } 1+x^{-n}=t \ \therefore -nx^{-n-1}dx = dt \ \text{or} \ \frac{dx}{x^{n+1}} = -\frac{dt}{n}$$

$$= \int \frac{dx}{x^2(x^n+1)^{\frac{n-1}{n}}} = -\frac{1}{n}\int \frac{dt}{t^{\frac{n-1}{n}}} = -\frac{1}{n}\int t^{\frac{1}{n}-1}dt$$

$$= -\frac{1}{n}\frac{t^{\frac{1}{n}-1+1}}{\frac{1}{n}-1+1} + C = -t^{\frac{1}{n}} + C \ \Rightarrow \ -(1+x^{-n})^{\frac{1}{n}} + C$$

St-2 $\displaystyle\int \frac{dx}{x^n(1+x^n)^{\frac{1}{n}}} = \int \frac{dx}{x^{n+1}\left(1+\frac{1}{x^n}\right)^{\frac{1}{n}}}$

$$1+x^{-n} = t \ \Rightarrow -nx^{-n-1}dx = dt \ \Rightarrow \frac{1}{x^{n+1}}dx = -\frac{dt}{n}$$

$$= -\frac{1}{n}\int \frac{dt}{t^{\frac{1}{n}}} = -\frac{1}{n}\frac{t^{-\frac{1}{n}+1}}{-\frac{1}{n}+1} = -\frac{1}{n}\frac{t^{\frac{-1+n}{n}}}{\frac{-1+n}{n}}$$

$$= \frac{1}{1-n}t^{\frac{n-1}{n}} \ \Rightarrow \ \frac{1}{1-n}(1+x^{-n})^{\frac{n-1}{n}}$$

22. STATEMENT-1 : $\displaystyle\int_0^{100} e^{x-[x]}\,dx = 100(e-1)$

and

STATEMENT-2 : If $f(x)$ is a periodic function with period T then $\displaystyle\int_0^{nT} f(x)dx = n\int_0^T f(x)dx \quad (n\in N)$

Sol. Answer (1)

Statement-1 : $\displaystyle\int_0^{100} e^{x-[x]}dx = 100\ (e-1)$

x – [x] is a function of fundamental period 1.

$$\int_0^{100} e^{x-[x]}dx = 100\int_0^1 e^{x-[x]}dx \ = \ 100\int_0^1 e^x dx = 100[e^x]_0^1 = 100(e-1)$$

Statement-1 is true.

Statement-2 : If $f(x)$ is a periodic function with period T, then $\int_0^{nT} f(x)dx = n\int_0^T f(x)dx \ \ (n \in N)$.

Statement-2 is true and it is a correct explanation for statement-1.

23. STATEMENT-1 : $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{dx}{e^{\sin x}+1} = \pi$

and

STATEMENT-2 : $\int_a^b f(x)dx = \int_a^b f(a+b-x)dx$

Sol. Answer (4)

Statement-1 : $\int_{-\pi/2}^{\pi/2} \frac{dx}{e^{\sin x}+1} = \pi$

$$I = \int_{-\pi/2}^{\pi/2} \frac{dx}{e^{\sin x}+1} \quad \dots (1) \qquad \left(\int_a^b f(x)dx = \int_a^b f(a+b-x)dx\right)$$

$$I = \int_{-\pi/2}^{\pi/2} \frac{dx}{e^{-\sin x}+1}dx$$

$$= \int_{-\pi/2}^{\pi/2} \frac{e^{\sin x}dx}{e^{\sin x}+1} \quad \dots (2)$$

Adding equations (1) and (2), we get

$$2I = \int_{-\pi/2}^{\pi/2} dx = [x]_{-\pi/2}^{\pi/2} = \pi$$

$I = \frac{\pi}{2}$, Statement-1 is false.

Statement-2 : $\int_a^b f(x)dx = \int_a^b f(a+b-x)dx$

Statement-1 is false and statement-2 is true.

24. STATEMENT-1 : $\int_{-\pi}^{\pi} \frac{\cos^2 x\,dx}{1+a^x}, (a>0) = \frac{\pi}{2}$

and

STATEMENT-2 : $\int_a^b f(x)dx = \int_a^b f(a+b-x)dx$

Sol. Answer (1)

Statement-1 : $\int_{-\pi}^{\pi}\frac{\cos^2 x}{1+a^x}dx$, $(a > 0)$ = $\frac{\pi}{2}$

Let $I = \int_{-\pi}^{\pi}\frac{\cos^2 x}{1+a^x}dx$ … (1) $\left(\int_a^b f(x)dx = \int_a^b f(a+b-x)dx\right)$

$I = \int_{-\pi}^{\pi}\frac{\cos^2 xdx}{1+a^{-x}}$ = $\int_{-\pi}^{\pi}\frac{a^x\cos^2 x}{1+a^x}dx$ … (2)

Adding equations (1) and (2) we get

$$2I = \int_{-\pi}^{\pi}\cos^2 xdx = 2\int_0^{\pi}\cos^2 xdx = 4\int_0^{\pi/2}\cos^2 xdx$$

$$2I = 4\left(\frac{\pi}{4}\right)$$

$$I = \frac{\pi}{2}.$$

Statement-1 is true.

Statement-2 : $\int_a^b f(x)dx = \int_a^b f(a+b-x)dx$

Statement-2 is true and it is a correct explanation for statement-1.

25. STATEMENT-1 : $\lim\limits_{n\to\infty}\frac{1}{n}\sum\limits_{r=1}^{n}\left(\frac{r}{n+r}\right) = 2-\log 2$

and

STATEMENT-2 : $\lim\limits_{n\to\infty}\frac{1}{n}\sum\limits_{r=1}^{n}\left(\frac{r}{n+r}\right) = \int\limits_0^1\frac{x\,dx}{(x+1)}$

Sol. Answer (4)

$$\lim_{n\to\infty}\frac{1}{n}\sum_{r=1}^{n}\left(\frac{r}{n+r}\right)$$

$$= \lim_{n\to\infty}\frac{1}{n}\sum_{r=1}^{n}\left(\frac{r/n}{1+r/n}\right)$$

$$= \int_0^1\frac{x\,dx}{(x+1)} = \int_0^1 dx - \int_0^1\frac{dx}{1+x} = 1-\log 2$$

26. STATEMENT-1 : $\int_0^{[x]} (x-[x])dx = \frac{[x]}{2}$

and

STATEMENT-2 : $\int_0^{nT} f(x)dx = n\int_0^{T} f(x)dx$; where T is the period of f.

Sol. Answer (1)

$$\int_0^{[x]} (x-[x]dx) = [x]\int_0^{1}(x-[x])dx$$

$$= [x]\int_0^{1}(x-0)dx = \frac{[x]}{2}$$

SECTION - E

Matrix-Match Type Questions

1. Match $\int f(x)dx$ from column I to column II

Column-I	Column-II
(A) $f(x) = e^x \sin x$	(p) $\frac{-e^x}{5}(\sin 2x + 3\cos 2x) + C$
(B) $f(x) = \cos(\log x)$	(q) $\frac{e^x(\sin x - \cos x)}{2} + C$
(C) $f(x) = \sin(\log x)$	(r) $\frac{x}{2}[\cos(\log x) + \sin(\log x)] + C$
(D) $f(x) = e^x(\sin 2x - \cos 2x)$	(s) $\frac{x}{2}[-\cos(\log x) + \sin(\log x)] + C$

Sol. Answer A(q), B(r), C(s), D(p)

(A) $I = \int e^x \sin x\, dx = e^x(-\cos x) + \int e^x \cos x\, dx$

$$= -e^x \cos x + e^x \sin x - \int e^x \sin x\, dx$$

$2I = e^x (\sin x - \cos x)$

$$I = \frac{e^x(\sin x - \cos x)}{2} + C$$

(B) $I = \int \cos(\log x)\, dx$ (Let $\log x = t \Rightarrow x = e^t \Rightarrow dx = e^t dt$)

$= \int \cos t \cdot e^t\, dt$

$= e^t \sin t - \int e^t \sin t\, dt$

$I = e^t \sin t + e^t \cos t - \int e^t \cos t\, dt$

$2I = e^t (\sin t + \cos t)$

$I = \dfrac{e^t(\sin t + \cos t)}{2} = \dfrac{x}{2}\left[\sin(\log x) + \cos(\log x)\right] + C$

(C) $I = \int \sin(\log x)\, dx$ Let $\log x = t$, $dx = e^t\, dt$

$= \int (\sin t) e^t\, dt = \dfrac{e^t(\sin t - \cos t)}{2} + C$

$= \dfrac{x}{2}\left[\sin(\log x) - \cos(\log x)\right] + C$

(D) $I = \int e^x(\sin 2x - \cos 2x)\, dx$

$= (\sin 2x - \cos 2x)e^x - \int (2\cos 2x + 2\sin 2x)\, e^x\, dx$

$= (\sin 2x - \cos 2x)\, e^x - 2(\cos 2x + \sin 2x)\, e^x + 2\int(-2\sin 2x + 2\cos 2x)e^x\, dx$

$= e^x(\sin 2x - \cos 2x - 2\cos 2x - 2\sin 2x) - 4\int e^x(\sin 2x - \cos 2x)\, dx$

$5I = e^x(-\sin 2x - 3\cos 2x)$

$I = \dfrac{-e^x(\sin 2x + 3\cos 2x)}{5} + C$

2. Match the followings

Column-I		**Column-II**
(A) $\int \dfrac{(\sqrt{3}+1)x - 1}{2x(x-1)}dx = A\ln\lvert x\rvert + B\ln\lvert x-1\rvert + C$	(p)	$A = \tan^2\dfrac{\pi}{6}$
(B) $\int (\tan x)^{1/3}\, dx = A\ln\dfrac{t^4 - t^2 + 1}{(t^2+1)^2} + B\tan^{-1}\dfrac{(2t^2-1)}{\sqrt{3}} + C$, where $t = \tan^{1/3} x$	(q)	$A = \cos^2\dfrac{\pi}{3}$

(C) $\int\left(\frac{\sin x+\sin^3 x}{\cos 2x}\right)dx = A\cos x + B\ln\left(\frac{\sqrt{2}\cos x+1}{\sqrt{2}\cos x-1}\right)+C$ (r) $A=\sin\frac{\pi}{6}$

(D) $\int\frac{dx}{(x^2+1)(x^2+4)} = A\tan^{-1}x + B\tan^{-1}\left(\frac{x}{2}\right)+C$ (s) $B=\cos\frac{\pi}{6}$

(t) $B=\cos^2\frac{\pi}{6}.\sin\frac{\pi}{4}$

Sol. Answer A(r, s), B(q, s), C(r, t), D(p)

(A) $\int\frac{(\sqrt{3}+1)x-1}{2x(x-1)}dx = \frac{1}{2}\ln|x| + \frac{\sqrt{3}}{2}\ln|x-1|$

$\therefore A=\frac{1}{2}=\sin\frac{\pi}{6},\ B=\frac{\sqrt{3}}{2}=\cos\frac{\pi}{6}$

(B) Putting $\tan x = t^3$ we get

$$\int(\tan x)^{1/3}dx = \frac{1}{4}\ln\frac{t^4-t^2+1}{(t^2+1)^2}+\frac{\sqrt{3}}{2}\tan^{-1}\left(\frac{2t^2-1}{\sqrt{3}}\right)+C$$

$\therefore A=\frac{1}{4}=\cos^2\frac{\pi}{3},\qquad B=\frac{\sqrt{3}}{2}=\cos\frac{\pi}{6}$

(C) $\int(\tan x)^{1/3}dx = \frac{1}{4}\ln\frac{t^4-t^2+1}{(t^2+1)^2}+\frac{\sqrt{3}}{2}\tan^{-1}\left(\frac{2t^2-1}{\sqrt{3}}\right)+C$

$A=\frac{1}{2}=\sin\frac{\pi}{6},\quad B=\frac{3}{4\sqrt{2}}=\cos^2\frac{\pi}{6}.\sin\frac{\pi}{4}$

(D) $\int(\tan x)^{1/3}dx = \frac{1}{4}\ln\frac{t^4-t^2+1}{(t^2+1)^2}+\frac{\sqrt{3}}{2}\tan^{-1}\left(\frac{2t^2-1}{\sqrt{3}}\right)+C$

$A=\frac{1}{3}=\tan^2\frac{\pi}{6}$

3. Match the followings

Column-I **Column-II**

(A) $\int\frac{\sin x-\cos x}{|\sin x-\cos x|}dx = f(x)\left(\frac{\pi}{4}<x<\frac{\pi}{2}\right)$, then $f(x)$ is (p) A transcendental function

(B) $\int\frac{x^2}{(x^3+1)(x^3+2)}dx = \frac{1}{3}f\left(\frac{x^3+1}{x^3+2}\right)+C$, then $f(x)$ is (q) $\sin x$

(C) $\int \sin^{-1} x.\cos^{-1} x\, dx =$ (r) $x + C$

$f^{-1}(x)\left[\frac{\pi}{2}x - xf^{-1}(x) - 2\sqrt{1-x^2}\right] + 2x + \frac{\pi}{2}\sqrt{1-x^2} + C$,

then $f(x)$ is

(D) $\int \frac{dx}{x f(x)} = f(f(x)) + C$, then $f(x)$ is (s) $\ln |x|$

(t) An algebraic function

Sol. Answer A(r, t), B(p, s), C(p, q), D(p, s)

(A) $\frac{\pi}{4} < x < \frac{\pi}{2} \quad \Rightarrow \sin x - \cos x > 0$

$\Rightarrow \frac{\sin x - \cos x}{|\sin x - \cos x|} = 1$

$\int 1.dx = x + c \quad \therefore f(x) = x + C$

(B) $\int \frac{x^2 dx}{(x^3+1)(x^3+2)} = \frac{1}{3}\ln\left|\frac{x^3+1}{x^3+2}\right| + C$

$\therefore f(x) = \ln |x|$

(C) $\int \sin^{-1} x.\cos^{-1} x\, dx$

$= \sin^{-1} x\left[\frac{\pi}{2}x - x\sin^{-1} x - 2\sqrt{1-x^2}\right] + \frac{\pi}{2}\sqrt{1-x^2} + 2x + C$

$\therefore f^{-1}(x) = \sin^{-1} x \Rightarrow f(x) = \sin x$

(D) Clearly $f(x) = \ln |x|$

4. Match the followings

Column-I **Column-II**

(A) $\int \frac{x\, dx}{1+x^4}$ (p) $\frac{2}{7}\ln\left(x^{7/2} + \sqrt{1+x^7}\right) + c$

(B) $\int \frac{(1+x)^2}{x+x^3}$ (q) $\frac{5}{2}\sqrt{1+x^{4/5}} + c$

(C) $\int \frac{x^{5/2}}{\sqrt{1+x^7}}dx$ (r) $\ln x + 2\tan^{-1} x + c$

(D) $\int \frac{dx}{x^{1/5}(1+x^{4/5})^{1/2}}$ (s) $\frac{1}{2}\tan^{-1}(x^2) + c$

Sol. Answer A(s), B(r), C(p), D(q)

(A) $I = \int \frac{x\,dx}{x^4+1} = \frac{1}{2}\int \frac{dt}{1+t^2}$ $\left[\begin{array}{l}\text{Put } x^2 = t \text{ so that} \\ 2xdx = dt\end{array}\right]$

$= \frac{1}{2} \cdot \tan^{-1} t + c$

$= \frac{1}{2}\tan^{-1}(x^2) + c$

(B) $I = \int \frac{(1+x)^2}{x+x^3}dx = \int \frac{1+x^2+2x}{x+x^3}dx$

$= \int\left(\frac{1}{x} + \frac{2}{1+x^2}\right)dx$

$= \ln x + 2\tan^{-1} x + c$

(C) $I = \int \frac{x^{5/2}}{\sqrt{1+x^7}}dx$

Put $x^{7/2} = t$, so that $\frac{7}{2} \cdot x^{5/2}dx = dt$

$I = \frac{2}{7}\int \frac{dt}{\sqrt{1+t^2}} = \frac{2}{7} \cdot \ln(t + \sqrt{1+t^2}) + c$

$= \frac{2}{7}\ln\left[x^{7/2} + \sqrt{1+x^7}\right] + c$

5. If f is even or odd function then

Column-I	**Column-II**
(A) $\int_{-a}^{a} f^2(x)dx$	(p) 0
(B) $\int_{-a}^{a} xf^2(x)dx$	(q) $\int_0^a \left(f^4(x)+f^4(a-x)\right)dx$
(C) $\int_{-a}^{a} (x^2+x^3)f^2(x)dx$	(r) $2\int_0^a f^2(x)dx$
(D) $\int_{-a}^{a} f^4(x)dx$	(s) $2\int_0^a x^2f^2(x)dx$

Sol. Answer A(r), B(p), C(s), D(q)

If f is even or odd function then

(A) $\int_{-a}^{a} f^2(x)dx = 2\int_0^a f^2(x)dx$ $\left\{\int_{-a}^{a} f(x)dx = 2\int_0^a f(x)dx \text{ if } f(-x) = f(x)\right.$

$\therefore$ $f^2(x)$ is an even function. $\left\{\int_{-a}^{a} f(x)dx = 0 \text{ if } f(-x) = -f(x)\right.$

(B) $\int_{-a}^{a} xf^2(x) = 0$ $xf^2(x)$ is an odd function.

(C) $\int_{-a}^{a} (x^2 + x^3)f^2(x)dx = \int_{-a}^{a} x^2f^2(x)dx + \int_{-a}^{a} x^3f^2(x)dx$

$\therefore$ $x^2f^2(x)$ is an even function

$x^3f^2(x)$ is an odd function

$\int_{-a}^{a} (x^2 + x^3)f^2(x)dx = 2\int_0^a x^2f^2(x)dx$

(D) $\int_{-a}^{a} f^4(x)dx = 2\int_0^a f^4(x)dx = \int_0^a f^4(x)dx + \int_0^a f^4(a-x)dx$

$= \int_0^a (f^4(x) + f^4(a-x))dx$

6. Match the following

Column-I	**Column-II**
(A) $\int_{-1}^{1} \frac{3x^2}{1+4^{\tan x}} dx = \phi$	(p) $\phi = 5$
(B) $\int_{6}^{10} \frac{\sin x^2 dx}{\sin(x-16)^2 + \sin x^2} dx = \phi$	(q) $\phi = \frac{1}{2}$
(C) $\frac{1}{66}\int_{1}^{12} [x] dx = \phi$ [[.] is integral part function]	(r) $\phi = 1$
(D) $\frac{1}{\pi \ln 2}\int_{\pi/2}^{0} \ln \sin x \, dx = \phi$	(s) $\{\phi\} = 0$ [{.} is fractional part function)]
	(t) $\phi = 2$

Sol. Answer A(r, s), B(s, t), C(r, s), D(q)

(A) $\phi = \int_{-1}^{1} \frac{3x^2}{1+4^{\tan x}} dx = \int_{-1}^{0} \frac{3x^2}{1+4^{\tan x}} + \int_{0}^{1} \frac{3x^2}{1+4^{\tan x}} dx = \int_{0}^{1} \frac{4^{\tan x}.3x^2}{1+4^{\tan x}} dx + \int_{0}^{1} \frac{3x^2}{1+4^{\tan x}} dx$

$= \int_0^1 3x^2 dx = \left[x^3\right]_0^1 = 1$

$\phi = 1 \quad \{\phi\} = 0$

(B) $\phi = \int_6^{10} \frac{\sin x^2}{\sin(x-16)^2 + \sin x^2} dx = \frac{1}{2}[10-6] = 2$

(C) $\phi = \frac{1}{66}\int_0^{12} [x] dx = \frac{1}{66}[0+1+2+....+11] = \frac{1}{66} x \frac{11\times 12}{2} = \frac{1}{2}\times 2 = 1, \{\phi\} = 0$

(D) $\int_0^{\pi/2} \ln\sin x\, dx = -\frac{\pi}{2} \ln 2$

$\Rightarrow \int_{\pi/2}^{0} \ln\sin x\, dx = \frac{\pi}{2} \ln 2$

$\Rightarrow \frac{1}{\pi \ln 2}.\int_{\pi/2}^{0} \ln\sin x\, dx = \frac{1}{2} \qquad \Rightarrow \phi = \frac{1}{2}$

7. $\int f(x) dx$ when

Column-I	Column-II
(A) $f(x) = \frac{1}{e^x+1}$	(p) $2\log(e^{x/2} + e^{-x/2}) + C$
(B) $f(x) = e^{e^x + x}$	(q) $\log \frac{e^x}{e^x+1} + C$
(C) $f(x) = \frac{e^x - 1}{e^x + 1}$	(r) $\frac{1}{4} e^{2x^2} + C$
(D) $f(x) = e^{2x^2 + \log x}$	(s) $e^{e^x} + C$

Sol. Answer A(q), B(s), C(p), D(r)

(A) $\int \frac{1}{e^x+1}dx = \int \frac{e^{-x}}{1+e^{-x}}dx = -\log(1+e^{-x}) = -\log\left(\frac{e^x+1}{e^x}\right)$

$= \log\left(\frac{e^x}{e^x+1}\right) + C$

(B) $\int e^{e^x+x}dx = \int e^{e^x}\cdot e^x\,dx$

Let $e^{e^x} = t$

$e^{e^x}\cdot e^x \cdot dx = dt$

$\int dt = t + C = e^{e^x} + C$

(C) $\int \frac{e^x-1}{e^x+1}dx = \int \frac{e^{x/2}-e^{-x/2}}{e^{x/2}+e^{-x/2}}dx$

Let $e^{x/2}+e^{-x/2} = t$

$\frac{1}{2}(e^{x/2}-e^{-x/2})\,dx = dt$

$\int \frac{2dt}{t} = 2\log|t| + C = 2\log(e^{x/2}+e^{-x/2}) + C$

(D) $I = \int e^{2x^2+\log x}dx$

$= \int e^{2x^2}.xdx$, Let $2x^2 = t$, $xdx = \frac{1}{4}dt$

$= \frac{1}{4}\int e^t dt$

$= \frac{1}{4}e^t + C = \frac{1}{4}e^{2x^2} + C$

8. $\int f(x)dx$ when

Column-I	**Column-II**
(A) $f(x) = \frac{1}{(a^2+x^2)^{3/2}}$	(p) $\frac{1}{a}\sec^{-1}\frac{x}{a} + C$

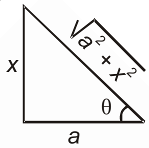
(B) $f(x)=\dfrac{x^2}{\sqrt{a^2-x^2}}$ (q) $\dfrac{a^2}{2}\sin^{-1}\dfrac{x}{a}-\dfrac{x}{2}\sqrt{a^2-x^2}+C$

(C) $f(x)=\dfrac{1}{(x^2-a^2)^{3/2}}$ (r) $C-\dfrac{x}{a^2\sqrt{x^2-a^2}}$

(D) $f(x)=\dfrac{1}{|x|\sqrt{x^2-a^2}}$ (s) $\dfrac{x}{a^2\sqrt{x^2+a^2}}+C$

Sol. Answer A(s), B(q), C(r), D(p)

(A) $\displaystyle\int\frac{dx}{(a^2+x^2)^{3/2}}$



Let $x = a\tan\theta$

$dx = a\sec^2\theta\, d\theta$

$$\int\frac{a\sec^2\theta\, d\theta}{(a^2+a^2\tan^2\theta)^{3/2}}=\frac{a}{a^3}\int\frac{\sec^2\theta\, d\theta}{\sec^3\theta\, d\theta}$$

$$=\frac{1}{a^2}\int\cos\theta\, d\theta=\frac{1}{a^2}\sin\theta+C \quad =\frac{1}{a^2}\left(\frac{x}{\sqrt{a^2+x^2}}\right)+C$$

(B) $$\int\frac{x^2}{\sqrt{a^2-x^2}}\,dx=-\int\frac{a^2-x^2-a^2}{\sqrt{a^2-x^2}}\,dx=-\int\sqrt{a^2-x^2}\,dx+a^2\int\frac{dx}{\sqrt{a^2-x^2}}$$

$$=-\left[\frac{x\sqrt{a^2-x^2}}{2}+\frac{a^2}{2}\sin^{-1}\frac{x}{a}\right]+a^2\sin^{-1}\frac{x}{a}$$

$$=\frac{a^2}{2}\sin^{-1}\frac{x}{a}-\frac{x\sqrt{a^2-x^2}}{2}+C$$

(C) $\displaystyle\int\frac{dx}{(x^2-a^2)^{3/2}}$

Let $x=a\sec\theta$

$dx=a\sec\theta\tan\theta\, d\theta$

$$\frac{a}{a^3}\int\frac{\sec\theta\tan\theta\, d\theta}{\tan^3\theta}=\frac{1}{a^2}\int\frac{\sec\theta}{\tan^2\theta}\,d\theta=\frac{1}{a^2}\int\frac{\cos\theta}{\sin^2\theta}\,d\theta$$

Let $\sin\theta=t,\ \cos\theta\, d\theta=dt$

$$\Rightarrow \frac{1}{a^2}\int\frac{dt}{t^2} = \frac{-1}{a^2t} = \frac{-1}{a^2\sin\theta}$$

$$\cos\theta = \frac{a}{x} \quad \Rightarrow \quad \sin\theta = \sqrt{1-\frac{a^2}{x^2}} = \frac{\sqrt{x^2-a^2}}{x}$$

$$\Rightarrow \int\frac{dx}{(x^2-a^2)^{3/2}} = \frac{-x}{a^2\sqrt{x^2-a^2}} + C$$

(D) $\displaystyle\int\frac{dx}{|x|\sqrt{x^2-a^2}} = \frac{-1}{a}\operatorname{cosec}^{-1}\frac{|x|}{a} = \frac{-1}{a}\sin^{-1}\frac{a}{|x|} + C$

9. $\int f(x)dx$, when

Column-I	Column-II
(A) $f(x) = \dfrac{\sin^3 x}{\cos x}$	(p) $\tan x + \dfrac{1}{3}\tan^3 x + C$
(B) $f(x) = \dfrac{\cos^3 x}{\sin^4 x}$	(q) $2\sqrt{\cos x}\left(\dfrac{\cos^2 x}{5} - 1\right) + C$
(C) $f(x) = \dfrac{\sin^3 x}{\sqrt{\cos x}}$	(r) $\operatorname{cosec} x - \dfrac{1}{3}\operatorname{cosec}^3 x + C$
(D) $f(x) = \dfrac{1}{\cos^4 x}$	(s) $\dfrac{\cos^2 x}{2} - \log|\cos x| + C$

Sol. Answer A(s), B(r), C(q), D(p)

(A) $\displaystyle\int\frac{\sin^3 x}{\cos x}\,dx = \int\frac{(1-\cos^2 x)\sin x}{\cos x}\,dx$

Let cos x = t

–sin$x$ dx = dt

$$= \int\frac{-dt\,(1-t^2)}{t} = \int\left(t-\frac{1}{t}\right)dt$$

$$= -\log t + \frac{t^2}{2} + C = -\log|\cos x| + \frac{\cos^2 x}{2} + C$$

(B) $\displaystyle\int\frac{\cos^3 x}{\sin^4 x}\,dx = \int\frac{\cos^2 x\cdot\cos x\,dx}{\sin^4 x}$

Let sin x = t

$\cos x\, dx = dt$

$$= \int \left(\frac{1-t^2}{t^4}\right) dt = \int (t^{-4} - t^{-2})\, dt = \frac{1}{t} - \frac{1}{3t^3} + C$$

$$= \operatorname{cosec} x - \frac{\operatorname{cosec}^3 x}{3} + C$$

(C) $\displaystyle\int \frac{\sin^3 x}{\sqrt{\cos x}}\, dx$

Let $\cos x = t^2$

$-\sin x\, dx = 2t\, dt$

$$-\int \frac{2t\, dt\,(1-t^4)}{t} = 2\int (t^4 - 1)\, dt = 2\left(\frac{t^5}{5} - t\right) + C$$

$$= 2\sqrt{\cos x}\left(\frac{\cos^2 x}{5} - 1\right) + C$$

(D) $\displaystyle\int \frac{1}{\cos^4 x}\, dx = \int \sec^4 x\, dx = \int (1+\tan^2 x)\cdot \sec^2 x\, dx$

Let $\tan x = t$

$\sec^2 x\, dx = dt$

$$\int (1+t^2)\, dt = t + \frac{t^3}{3} + C = \tan x + \frac{\tan^3 x}{3} + C$$

10. Match the following

Column-I	**Column-II**
(A) $\displaystyle\int_0^{\pi} x \sin^4 x\, dx$	(p) $\dfrac{\pi}{2}(\pi - 2)$
(B) $\displaystyle\int_0^{\pi} x \sin^4 x \cos^6 x\, dx$	(q) $\dfrac{\pi}{3\sqrt{3}}$
(C) $\displaystyle\int_0^{\frac{\pi}{2}} \frac{\cos^2 x\, dx}{1+\cos x \sin x}$	(r) $\dfrac{3\pi^2}{512}$
(D) $\displaystyle\int_0^{\pi} \frac{x \tan x\, dx}{\sec x + \tan x}$	(s) $\dfrac{3\pi^2}{16}$

Sol. Answer A(s), B(r), C(q), D(p)

(A) $I = \int_0^{\pi} x\sin^4 x dx$ ... (1)

$= \int_0^{\pi} (\pi - x)\sin^4(\pi - x)dx$ ... (2)

Adding equations (1) and (2), we get,

$2I = \pi\int_0^{\pi} \sin^4 x dx$ $\quad \int_0^{2a} f(x)dx = 2\int_0^{a} f(x)dx$ if $f(2a-x) = f(x)$

$2I = 2\pi\int_0^{\pi/2} \sin^4 x dx$

$I = \pi\int_0^{\pi/2} \sin^4 x dx = \pi\frac{3.1}{4.2}\frac{\pi}{2} = \frac{3\pi^2}{16}$

(B) $I = \int_0^{\pi} x\sin^4 x\cos^6 x dx$ ... (1)

$I = \int_0^{\pi} (\pi - x)\sin^4(\pi - x)\cos^6(\pi - x)dx$ ... (2)

Adding equations (1) and (2), we get,

$2I = \pi\int_0^{\pi} \sin^4 x.\cos^6 x dx = 2\pi\int_0^{\pi/2} \sin^4 x.\cos^6 x dx$

$I = \pi\int_0^{\pi/2} \sin^4 x.\cos^6 x\, dx = \frac{3.1.5.3.1}{10.8.6.4.2}.\frac{\pi^2}{2}$

$= \frac{3\pi^2}{512}$ (Using Gamma Functions)

(C) $I = \int_0^{\pi/2} \frac{\cos^2 x dx}{1+\cos x\sin x}$ ... (1) $\quad \int_0^{a} f(x)dx = \int_0^{a} f(a-x)dx$

$I = \int_0^{\pi/2} \frac{\sin^2 x}{1+\sin x\cos x}dx$ ... (2)

Adding equations (1) and (2) we get

$2I = \int_0^{\pi/2} \frac{1}{1+\cos x\sin x}dx = \int_0^{\pi/2} \frac{\sec^2 x dx}{1+\tan^2 x+\tan x}$

Put $\tan x = t$

$\sec^2 x\, dx = dt$

$\Rightarrow 2I = \int_0^{\infty} \frac{dt}{t^2+t+1} = \int_0^{\infty} \frac{dt}{\left(t+\frac{1}{2}\right)^2+\left(\frac{\sqrt{3}}{2}\right)^2} = \frac{2}{\sqrt{3}}\left|\tan^{-1}\frac{2t+1}{\sqrt{3}}\right|_0^{\infty} = \frac{2\pi}{3\sqrt{3}}$

$\Rightarrow\ 2I = \dfrac{2\pi}{3\sqrt{3}}$

$\Rightarrow\ I = \dfrac{\pi}{3\sqrt{3}}$

(D) $I = \int_0^{\pi} \dfrac{x\tan x dx}{\sec x + \tan x}$ ... (1) $\int_0^a f(x)dx = \int_0^a f(a-x)dx$

$I = \int_0^{\pi} \dfrac{(\pi - x)\tan(\pi - x)dx}{\sec(\pi - x) + \tan(\pi - x)}$ ... (2)

Adding equations (1) and (2), we get

$$2I = \pi\int_0^{\pi} \frac{\tan x dx}{\sec x + \tan x}dx \ = \ 2\pi\int_0^{\pi/2} \frac{\sin x}{1+\sin x}dx$$

$$I = \pi\int_0^{\pi/2}\left(\frac{1+\sin x - 1}{1+\sin x}\right)dx \ = \ \pi\int_0^{\pi/2}dx - \pi\int_0^{\pi/2}\frac{1}{1+\sin x}dx$$

$$= \pi\left(\frac{\pi}{2}\right) - \frac{2}{2}\int_0^{\pi/2}\frac{1}{1+\cos\left(\frac{\pi}{2} - x\right)}dx = \frac{\pi^2}{2} - \frac{1}{2}\int_0^{\pi/2}\sec^2\left(\frac{\pi}{4} - \frac{x}{2}\right)dx$$

$$= \frac{\pi^2}{2} - \frac{1}{2}\left[\frac{\tan\left(\frac{\pi}{4} - \frac{x}{2}\right)}{\left(-\frac{1}{2}\right)}\right]_0^{\pi/2} \ = \ \frac{\pi^2}{2} + \left[\tan\left(\frac{\pi}{4} - \frac{x}{2}\right)\right]_0^{\pi/2}$$

$$= \frac{\pi^2}{2} - 1 = \frac{\pi}{2}(\pi - 2)$$

11. $I_a = \int_0^{\frac{\pi}{2}} \dfrac{dx}{2\cos x + \sin x + a}$ then the value of I_a for

Column-I		**Column-II**
(A) $a = 1$	(p)	$\log\left(\dfrac{3}{2}\right)$
(B) $a = 3$	(q)	$\dfrac{1}{2}\log(3)$
(C) $a = 2$	(r)	$\dfrac{2}{\sqrt{11}}\left(\tan^{-1}\dfrac{3}{\sqrt{11}} - \tan^{-1}\dfrac{1}{\sqrt{11}}\right)$
(D) $a = 4$	(s)	$\tan^{-1}\left(\dfrac{1}{3}\right)$

Sol. Answer A(q), B(s), C(p), D(r).

$$I_a = \int_0^{\pi/2} \frac{dx}{2\cos x + \sin x + a}$$

$$I_a = \int_0^{\pi/2} \frac{dx}{2\left(\frac{1-\tan^2(x/2)}{1+\tan^2(x/2)}\right) + \frac{2\tan(x/2)}{1+\tan^2(x/2)} + a}$$

$$= \int_0^{\pi/2} \frac{dx.\sec^2(x/2)}{2 - 2\tan^2(x/2) + 2\tan x/2 + a + a\tan^2(x/2)}$$

$$I_a = \int_0^{\pi/2} \frac{\sec^2(x/2)dx}{(a-2)\tan^2(x/2) + 2\tan(x/2) + (a+2)}$$

Let $\tan\frac{x}{2} = t$

$\Rightarrow \quad \frac{1}{2}\sec^2\frac{x}{2}dx = dt$

When $x = 0,\ t = 0$

$x = \frac{\pi}{2},\ \ t = \tan\frac{\pi}{4} = 1$

Thus, $I_a = \int_0^1 \frac{2dt}{(a-2)t^2 + 2t + (a+2)}$

(A) If $a = 1$

$$I_a = \int_0^1 \frac{2dt}{-t^2 + 2t + 3} = -\int_0^1 \frac{2dt}{t^2 - 2t - 3}$$

$$= -2\int_0^1 \frac{dt}{(t-3)(t+1)} = -\frac{2}{4}\int_0^1 \left(\frac{1}{t-3} - \frac{1}{t+1}\right)dt$$

$$= -\frac{1}{2}\left[\log\left|\frac{t-3}{t+1}\right|\right]_0^1$$

$$= -\frac{1}{2}\left[\log\frac{2}{2} - \log 3\right] = -\frac{1}{2}[\log 1 - \log 3]$$

$$= \frac{1}{2}\log 3 .$$

$I_a = \frac{1}{2}\log 3$ if $a = 1$.

(B) If $a = 3$

$$I_a = \int_0^1 \frac{2dt}{t^2+2t+5} = 2\int_0^1 \frac{dt}{(t+1)^2+2^2}$$

$$= 2.\frac{1}{2}\left[\tan^{-1}\left(\frac{t+1}{2}\right)\right]_0^1 = \tan^{-1}1 - \tan^{-1}\frac{1}{2}$$

$$= \tan^{-1}\left(\frac{1-\frac{1}{2}}{1+1.\frac{1}{2}}\right) \quad \left(\tan^{-1}x - \tan^{-1}y = \tan^{-1}\left(\frac{x-y}{1+xy}\right)\right)$$

$$= \tan^{-1}\frac{\frac{1}{2}}{\frac{3}{2}} = \tan^{-1}\left(\frac{1}{3}\right)$$

$I_a = \tan^{-1}\left(\frac{1}{3}\right)$, if $a = 3$.

(C) If $a = 2$

$$I_a = \int_0^1 \frac{2dt}{2t+4} = \int_0^1 \frac{dt}{t+2} = [\log|t+2|]_0^1$$

$$= \log 3 - \log 2 = \log\left(\frac{3}{2}\right)$$

$I_a = \log\left(\frac{3}{2}\right)$ if $a = 2$.

(D) If $a = 4$

$$I_a = \int_0^1 \frac{2dt}{2t^2+2t+6} = \int_0^1 \frac{dt}{t^2+t+3}$$

$$= \int_0^1 \frac{dt}{\left(t+\frac{1}{2}\right)^2 + \left(\sqrt{\frac{11}{4}}\right)^2} = \frac{1}{\left(\frac{\sqrt{11}}{2}\right)}\left[\tan^{-1}\left(\frac{t+\frac{1}{2}}{\frac{\sqrt{11}}{2}}\right)\right]_0^1$$

$$= \frac{2}{\sqrt{11}}\left[\tan^{-1}\left(\frac{2t+1}{\sqrt{11}}\right)\right]_0^1 = \frac{2}{\sqrt{11}}\left[\tan^{-1}\frac{3}{\sqrt{11}} - \tan^{-1}\frac{1}{\sqrt{11}}\right]$$

$I_a = \frac{2}{\sqrt{11}}\left[\tan^{-1}\frac{3}{\sqrt{11}} - \tan^{-1}\frac{1}{\sqrt{11}}\right]$ if $a= 4$.

12. Match the following

Column I		Column II
(A) $\int_0^2 [x^2]\,dx = u$	(p)	$u = 3$
(B) $\int_0^2 \{x^2\}\,dx = u$	(q)	$u = 2$
(C) $\int_0^3 (\text{sgn}(x))^2\,dx = u$	(r)	$\{u\} = 0$
(D) $2\int_{1/2}^{7/2} \{x\}\,dx = u$	(s)	$[u] = 0$
	(t)	$[u] = 1$

In this question [.] represents greatest integer function and {.} represents fractional part function.

Sol. Answer A(t), B(s), C(p, r), D(p, r)

(A) $$u = \int_0^2 [x^2]dx = \int_0^1 [x^2]dx + \int_1^{\sqrt{2}} [x^2]dx + \int_{\sqrt{2}}^{\sqrt{3}} [x^2]dx + \int_{\sqrt{3}}^2 [x^2]dx$$

$$= 0 + \int_1^{\sqrt{2}} 1.dx + \int_{\sqrt{2}}^{\sqrt{3}} 2.dx + \int_{\sqrt{3}}^2 3.dx$$

$$= 0 + \sqrt{2} - 1 + 2(\sqrt{3} - \sqrt{2}) + 3(2 - \sqrt{3}) = 5 - \sqrt{3} - \sqrt{2}$$

$$1 < u = 5 - \sqrt{3} - \sqrt{2} < 2 \quad \therefore [u] = 1$$

(B) $$u = \int_0^2 \{x^2\}dx$$

$$= \int_0^1 x^2dx + \int_1^{\sqrt{2}} (x^2 - 1)dx + \int_{\sqrt{2}}^{\sqrt{3}} (x^2 - 2)dx + \int_{\sqrt{3}}^2 (x^2 - 3)dx$$

$$= \int_0^2 x^2dx - \left[\int_1^{\sqrt{2}} 1.dx + \int_{\sqrt{2}}^{\sqrt{3}} 2.dx + \int_{\sqrt{3}}^2 3.dx\right]$$

$$= \frac{8}{3} - [5 - \sqrt{3} - \sqrt{2}]$$

$$= \sqrt{3} + \sqrt{2} - \frac{7}{3}$$

Now $0 < u < 1$ $\quad [u] = 0$

(C) $u = \int_0^3 \{\text{sgn}(x)\}^2 dx = \int_0^3 1.dx = 3 \quad \therefore \{u\} = 0$

(D) $x = 2.\int_{1/2}^{7/2} \{x\}\, dx = 2\int_0^3 \{x\}\, dx = 2.3\int_0^1 \{x\}\, dx = 6.\frac{1}{2} = 3, \{u\} = 0$

13. Match the following **[JEE(Advanced)-2014]**

Column I		**Column II**
(A) The number of polynomials $f(x)$ with non-negative integer coefficients of degree ≤ 2, satisfying $f(0) = 0$ and $\int_0^1 f(x)dx = 1$, is	(p)	8
(B) The number of points in the interval $[-\sqrt{13}, \sqrt{13}]$ at which $f(x) = \sin(x^2) + \cos(x^2)$ attains its maximum value, is	(q)	2
(C) $\int_{-2}^{2} \frac{3x^2}{(1+e^x)} dx$ equals	(r)	4
(D) $\dfrac{\left(\int_{-\frac{1}{2}}^{\frac{1}{2}} \cos 2x \log\left(\frac{1+x}{1-x}\right) dx\right)}{\left(\int_0^{\frac{1}{2}} \cos 2x \log\left(\frac{1+x}{1-x}\right) dx\right)}$ equals	(s)	0

Sol. Answer A(q), B(r), C(p), D(s)

(A) Let $f(x) = a_0x^2 + a_1x + a_2$; $a_0, a_1, a_2 > 0$

Now, $a_2 = 0$ and $\int_0^1 f(x)dx = 1$

$$\Rightarrow \left\{\frac{a_0x^3}{3} + \frac{a_1x^2}{2}\right\}_0^1 = 1$$

$$\Rightarrow \frac{a_0}{3} + \frac{a_1}{2} = 1$$

$\Rightarrow 2a_0 + 3a_1 = 6$

i.e., for integral solution

$$\left.\begin{array}{ll} \text{either} & a_0 = 3, \quad a_1 = 0 \\ \text{or} & a_0 = 0, \quad a_1 = 2 \end{array}\right\}$$

i.e., $f(x) = 3x^2$ or $f(x) = 2x$

A $\rightarrow$ (q)

(B) $f(x)_{\max} = \sqrt{2}$ at $x^2 = \frac{\pi}{4}, \frac{9\pi}{4}$

$\Rightarrow$ $x = \pm\sqrt{\frac{\pi}{4}}, \pm\sqrt{\frac{9\pi}{4}}$

So, number of points = 4

B $\rightarrow$ (r)

(C) $I = \int_{-2}^{2} \frac{3x^2}{1+e^x} dx$ again, $I = \int_{-2}^{2} \frac{3x^2}{1+e^{-x}} dx$

Adding, $2I = \int_{-2}^{2} 3x^2 dx = \{x^3\}_{-2}^{2} = 16$

$\Rightarrow$ $I = 8$

C $\rightarrow$ (p)

(D) As $\cos 2x \cdot \log\left(\frac{1+x}{1-x}\right)$ is an odd function

So, $\int_{-\frac{1}{2}}^{\frac{1}{2}} \cos 2x \cdot \log\left(\frac{1+x}{1-x}\right) dx = 0$

D $\rightarrow$ (s)

SECTION - F

Integer Answer Type Questions

1. If the graph of the anti derivative $g(x)$ of $f(x) = \frac{1}{x \ln x}$ passes through $\left(e^{e^2}, 4\right)$, then the term independent of x in $g(x)$ is

Sol. Answer (2)

$g(x) = \int \frac{1}{x \ln x} dx = \ln(\ln x) + c$

$4 = g\left(e^{e^2}\right) = \ln\left(\ln e^{e^2}\right) + c = 2 + c$

$\Rightarrow c = 4 - 2 = 2$

2. If $\int \frac{\sin x + 3\cos x}{\sin x + \cos x} dx = \frac{kx}{2} + \ln(\sin x + \cos x)$, then k is

Sol. Answer (4)

$$\int \frac{\sin x + 3\cos x}{\sin x + \cos x} dx = 2x + \ln c(\sin x + \cos x)$$

Hence $\frac{k}{2} = 2 \Rightarrow k = 4$

3. If the graph of anti-derivative $g(x)$ of $\frac{3x}{(x-1)(x+2)}$ intersect the line $x = 2$ at a point with ordinate 5 + ln 16, then the term independent of x in $g(x)$ is

Sol. Answer (5)

$$g(x) = \int \frac{3x}{(x-1)(x+2)} dx = \ln|x-1| + 2\ln|x+2| + c$$

$$5 + \ln 16 = \ln 1 + 2\ln 4 + c = \ln 16 + c \Rightarrow c = 5$$

4. If $\int \frac{\ln\left(\frac{x-1}{x+1}\right)}{x^2-1} dx = \frac{1}{a} \cdot \left(\ln\left|\frac{x-1}{x+1}\right|\right)^b + c$, then $a^2 - b^2 - ab$ is equal to

Sol. Answer (4)

Let $I = \int \frac{\ln\left(\frac{x-1}{x+1}\right)}{x^2-1} dx$

Put $\ln\left(\frac{x-1}{x+1}\right) = t$ so that $\frac{dx}{x^2-1} = \frac{dt}{2}$

$\therefore \quad I = \int \frac{t\,dt}{2} = \frac{t^2}{4} + c$

$$= \frac{1}{4}\left[\ln\left|\frac{x-1}{x+1}\right|\right]^2 + c$$

Clearly $a = 4$, $b = 2$

$\therefore\ a^2 - b^2 - ab = 16 - 4 - 8 = 4$

5. If $\int \frac{x \cdot \ln(x+\sqrt{1+x^2})}{\sqrt{1+x^2}}\,dx = A\cdot\sqrt{1+x^2}\ln(x+\sqrt{1+x^2})+Bx+c$, then $\frac{3A^2+5B^2}{A-B}$ equal to ______.

Sol. Answer (4)

$$I = \int \frac{x\ln(x+\sqrt{1+x^2})}{\sqrt{1+x^2}}\,dx$$

Put $\sqrt{1+x^2} = t$ so that $\frac{x}{\sqrt{1+x^2}}dt$

$$I = \int \ln(\sqrt{(t^2-1}+t)\,dt$$

$$= \ln(\sqrt{t^2-1}+t)\cdot t - \int\left(\frac{1+\frac{2t}{2\sqrt{t^2-1}}}{\sqrt{t^2-1}+t}\right)$$

$$= t\cdot\ln\left(\sqrt{t^2-1}+t\right) - \int\frac{t}{\sqrt{t^2-1}}dt$$

$$= t\ln(\sqrt{t^2-1}+t) - \sqrt{t^2-1}+c$$

$$= \sqrt{x^2+1}\ln(x+\sqrt{1+x^2}) - x + c$$

Clearly $A = 1$, $B = -1$

$$\therefore \quad \frac{3A^2+5B^2}{A-B} = \frac{8}{2} = 4$$

6. If $\int_0^{18\pi}[\sin x+\cos x]\,dx = -k\pi$ and [.] denotes the greatest integer function, then k is __________.

Sol. Answer (9)

$$\int_0^{2n\pi}[\sin x+\cos x]\,dx = -n\pi$$

$$-k\pi = \int_0^{18\pi}(\sin x+\cos x)\,dx = -9\pi \quad \therefore \quad k = 9$$

7. If $\int_{\pi/6}^{\pi/3} \sin x\, dx = k \int_{\pi/6}^{\pi/3} \cos x\, dx$, then k is ________.

Sol. Answer (1)

$$\int_{\pi/6}^{\pi/3} \sin x\, dx = \int_{\pi/6}^{\pi/3} \sin\left(\frac{\pi}{3}+\frac{\pi}{6}-x\right) dx = \int_{\pi/6}^{\pi/3} \sin\left(\frac{\pi}{2}-x\right) dx = \int_{\pi/6}^{\pi/3} \cos x\, dx$$

Hence $k = 1$

8. If $I = \int_1^2 \frac{dx}{\sqrt{x}-\sqrt{x-1}}$, then value of $\frac{9}{32}(I)^2$, is__________ .

Sol. Answer (1)

$$I = \int_1^2 \frac{dx}{\sqrt{x}-\sqrt{x-1}} = \int_1^2 \frac{\sqrt{x}+\sqrt{x-1}}{x-(x-1)} dx$$

$$\therefore \quad I = \frac{2}{3}\left\lfloor x^{3/2} + (x-1)^{3/2} \right\rfloor_1^2 = \frac{4\sqrt{2}}{3}$$

$$\Rightarrow \quad \frac{9}{32}(I)^2 = \frac{9}{32} \times \frac{16\times 2}{9} = 1$$

9. If $I = \int_0^{\pi/2} \frac{\cos x}{1+\sin^2 x} dx$, then $\frac{4}{\pi} I$ is _________.

Sol. Answer (1)

Let $\sin x = t \Rightarrow \cos x\, dx = dt$

$$\therefore \quad I = \int_0^1 \frac{dt}{1+t^2} = \left\lfloor \tan^{-1}(t) \right\rfloor_0^1$$

$$\therefore \quad I = \tan^{-1} 1 - \tan 0 = \frac{\pi}{4}$$

$$\Rightarrow \quad \frac{4}{\pi} I = 1$$

10. If $\int_0^{\pi} \frac{x^2 \sin 2x \sin\left\{\frac{\pi}{2}\cos x\right\}}{2x-\pi} dx = \frac{A}{\pi}$, then A is __________ .

Sol. Answer (8)

$$I=\int_0^{\pi}\frac{x^2\sin 2x\sin\left(\frac{\pi}{2}\cos x\right)}{2x-\pi}dx$$

$$\therefore\quad I=\int_0^{\pi}\frac{(\pi-x)^2\sin 2(\pi-x)\sin\left(\frac{\pi}{2}\cos(\pi-x)\right)}{2(\pi-x)-\pi}dx$$

$$\therefore\quad I=\int_0^{\pi}\frac{-(\pi-x)^2\sin 2x\sin\left(-\frac{\pi}{2}\cos x\right)}{\pi-2x}dx$$

$$\therefore\quad I=-\int_0^{\pi}\frac{(\pi-x)^2\sin 2x\sin\left(\frac{\pi}{2}\cos x\right)}{2x-\pi}dx$$

$$\therefore\quad 2I=\int_0^{\pi}\frac{\left\{x^2-(\pi-x)^2\right\}\sin 2x\sin\left(\frac{\pi}{2}\cos x\right)}{2x-\pi}dx$$

$$\therefore\quad 2I=\int_0^{\pi}\pi 2\sin x\cos x\sin\left(\frac{\pi}{2}\cos x\right)dx$$

$$\therefore\quad \text{Let } \frac{\pi}{2}\cos x=t\Rightarrow-\frac{\pi}{2}\sin x dx=dt$$

$$\therefore\quad 2I=\int_{\pi/2}^{-\pi/2}8\frac{t}{\pi}\sin t\,dt$$

$$2I=\frac{16}{\pi}[-t\cos t+\sin t]_0^{\pi/2}=\frac{16}{\pi}$$

$$\Rightarrow\quad I=\frac{8}{\pi}\Rightarrow A=8$$

11. If $I=\int_0^{1/2}\frac{x\sin^{-1}x}{\sqrt{1-x^2}}dx=A-\frac{\pi}{12},\sqrt{3}$ then $2A$ is ________ .

Sol. Answer (1)

Let $\sin^{-1}x=t\ \Rightarrow\ \frac{1}{\sqrt{1-x^2}}dx=dt$

$$\therefore\quad I=\int_0^{\pi/6}t\,\sin t\,dt$$

$\therefore \quad I = \lfloor t(-\cos t \rfloor_0^{\pi/6} - \int\limits_0^{\pi/6} (-\cos t)dt$

$I = \frac{-\pi}{6}\cos\frac{\pi}{6} + \lfloor \sin t \rfloor_0^{\pi/6}]$

$\therefore \quad I = \frac{1}{2} - \frac{\pi\sqrt{3}}{12} \Rightarrow 2A = 1$

12. The value of $\int_0^1 4x^3 \left\{\frac{d^2}{dx^2}(1-x^2)^5\right\} dx$ is **[JEE(Advanced)-2014]**

Sol. Answer (2)

Using by parts

$4x^2.\frac{d(1-x^2)^5}{dx}\Bigg]_0^1 - \int\limits_0^1 d(1-x^2)^5.12x^2dx$

$= -12x^2.(1-x^2)^5\Big]_0^1 + \int\limits_0^1 24x(1-x^2)^5 dx$

$1 - x^2 = t$

$-2xdx = dt$

$= -\int\limits_1^0 12t^5 dt$

$= 2$

13. Let $f : \mathbb{R} \to \mathbb{R}$ be a function defined by $f(x) = \begin{cases} [x], & x \le 2 \\ 0, & x > 2 \end{cases}$, where $[x]$ is the greatest integer less than or equal to x. If $I = \int\limits_{-1}^{2} \frac{xf(x^2)}{2+f(x+1)}dx,$ then the value of $(4I - 1)$ is **[JEE(Advanced)-2015]**

So. Answer (0)

$$f(x+1) = \begin{cases} [x+1], & x+1 \le 2 \\ 0, & x+1 > 2 \end{cases} = \begin{cases} [x]+1, & x \le 1 \\ 0, & x > 1 \end{cases} = \begin{cases} 0, & -1 \le x < 0 \\ 1, & 0 \le x < 1 \\ 0, & 1 < x < 2 \end{cases}$$

$$f(x^2) = \begin{cases} [x^2], & x^2 \le 2 \\ 0, & x^2 > 2 \end{cases} = \begin{cases} [x^2], & |x| \le \sqrt{2} \\ 0, & |x| > \sqrt{2} \end{cases} = \begin{cases} 0, & -1 \le x < 0 \\ 0, & 0 \le x < 1 \\ 1, & 1 \le x < \sqrt{2} \\ 0, & \sqrt{2} < x < 2 \end{cases}$$

$$I = \int_{-1}^{2} \frac{x.f(x^2)}{2+f(x+1)}dx$$

$$= \int_{-1}^{0} 0dx + \int_{0}^{1} 0dx + \int_{1}^{\sqrt{2}} \frac{xdx}{2+0} + \int_{\sqrt{2}}^{2} 0\,dx$$

$$= \frac{1}{4}[x^2]_1^{\sqrt{2}} = \frac{1}{4}[2-1] = \frac{1}{4}$$

$4I = 1$

$4I - 1 = 0$

14. If $\alpha = \int_0^1 (e^{9x+3\tan^{-1}x})\left(\frac{12+9x^2}{1+x^2}\right)dx$ where $\tan^{-1}x$ takes only principal values, then the value of $\left(\log_e|1+\alpha|-\frac{3\pi}{4}\right)$ is **[JEE(Advanced)-2015]**

Sol. Answer (9)

$$\alpha = \int_0^1 (e^{9x+3\tan^{-1}x})\left(\frac{12+9x^2}{1+x^2}\right)dx$$

$$= \int_0^1 (e^{9x+3\tan^{-1}x})\left(9+\frac{3}{1+x^2}\right)dx$$

$$= (e^{9x+3\tan^{-1}x})\Big|_0^1$$

$$= e^{9+\frac{3\pi}{4}} - 1$$

$$\Rightarrow \quad \ln|(1+\alpha)| - \frac{3\pi}{4} = 9$$

15. Let $f:\mathbb{R}\to\mathbb{R}$ be a differentiable function such that $f(0) = 0$, $f\left(\frac{\pi}{2}\right) = 3$ and $f'(0) = 1$.

If $g(x) = \int_x^{\frac{\pi}{2}} [f'(t)\operatorname{cosec} t - \cot t \operatorname{cosec} t\, f(t)]dt$ for $x \in \left(0, \frac{\pi}{2}\right]$, then $\lim_{x\to 0} g(x) =$ **[JEE(Advanced)-2017]**

Sol. Answer (2)

$$g(x) = \int_x^{\pi/2} d((\operatorname{cosec} t)(f(t))) = \{(\operatorname{cosec} t).f(t)\}_x^{\pi/2}$$

So, $\lim_{x\to 0} g(x) = \lim_{x\to 0}\left(f\left(\frac{\pi}{2}\right) - f(x).\operatorname{cosec} x\right) = 3 - \lim_{x\to 0}\frac{f(x)}{\sin x} = 3 - \lim_{x\to 0}\frac{f'(x)}{\cos x} = 2$

16. The value of the integral $\displaystyle\int_0^{\frac{1}{2}} \frac{1+\sqrt{3}}{\left((x+1)^2(1-x)^6\right)^{\frac{1}{4}}}dx$ is ____. **[JEE(Advanced)-2018]**

Sol. Answer (2.00)

$$I = \int_0^{1/2} \frac{1+\sqrt{3}}{\left((x+1)^2(1-x)^6\right)^{1/4}}dx = \int_0^{1/2} \frac{1+\sqrt{3}}{\left((x+1)^8\left(\frac{1-x}{1+x}\right)^6\right)^{1/4}}dx = \int_0^{1/2} \frac{1+\sqrt{3}}{(1+x)^2\left(\frac{1-x}{1+x}\right)^{3/2}}dx$$

Put $\dfrac{1-x}{1+x} = t \Rightarrow \dfrac{-2dx}{(1+x)^2} = dt$

$$\Rightarrow \quad I = -\frac{1}{2}\int_1^{1/3} \frac{1+\sqrt{3}}{t^{3/2}} = -\frac{1}{2}\times(1+\sqrt{3})\left[\frac{t^{-1/2}}{\frac{-1}{2}}\right]_1^{1/3} = (1+\sqrt{3})(\sqrt{3}-1) = 2$$

SECTION - G

Multiple True-False Type Questions

1. STATEMENT-1 : If $\displaystyle\int \frac{2^x}{\sqrt{1-4^x}} = k\sin^{-1}(2^x)$, then k equals $\dfrac{1}{\log 2}$.

STATEMENT-2 : If $\int f(x)\,dx = -f(x)+c$, then $f(\log_e 2) = \dfrac{1}{2}$

STATEMENT-3 : $\displaystyle\int \frac{e^x}{\sqrt{1+e^x}}dx = -2\sqrt{1+e^x}+c$

(1) T T T (2) T T F (3) T F T (4) F F T

Sol. Answer (2)

STATEMENT-1 : True,

Put $2^x = z \quad \therefore 2^x \log 2\, dx = dz$

$$\therefore \text{LHS} = \frac{1}{\log 2}\int \frac{dz}{\sqrt{1-z^2}} = \frac{1}{\log 2}\sin^{-1} z$$

STATEMENT-2 : True,

Clearly, $f(x) = e^{-x}$

$$\therefore f(\log_e 2) = e^{-\log_e^2} = \frac{1}{2}$$

STATEMENT-3 : False,

$$\int \left(1+e^x\right)^{-1/2}.e^x dx = \frac{\left(1+e^x\right)^{1/2}}{1/2}+c$$

2. STATEMENT-1 : If $\int \frac{1}{f(x)} dx = 2\log|f(x)| + c$, then $f(x) = \frac{x}{2}$.

STATEMENT-2 : When $f(x) = \frac{x}{2}$, then $\int \frac{1}{f(x)} dx = \int \frac{2}{x} dx = 2\log|x| + c$

STATEMENT-3 : $\int e^{x^2} dx = e^{x^2} + c$

(1) T T T (2) T T F (3) F T T (4) F F F

Sol. Answer (2)

STATEMENT-1 and STATEMENT-2 True

$$\int \frac{1}{f(x)} dx = 2 \log|f(x)| + C$$

Differentiate w.r.t. x,

$$\frac{1}{f(x)} = \frac{2}{f(x)} f'(x)$$

$$\therefore f'(x) = \frac{1}{2} \qquad \Rightarrow f(x) = \frac{x}{2} + C$$

If $f(0) = 0$, then $f(x) = \frac{x}{2}$

STATEMENT-3 : False,

$\int e^{x^2} dx$ cannot be expressed in terms of elementary function, then integral is known as inexpressible as that is "cannot be found".

3. STATEMENT-1 : $\int_{-3}^{3} |x| dx = 9$

STATEMENT-2 : $\int_{0}^{1} \tan^{-1} x \, dx = \frac{\pi}{4} - \ln\sqrt{2}$

STATEMENT-3 : $\int_{0}^{\pi/2} \frac{\sqrt{\cos x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx = \frac{\pi}{4}$

(1) T F T (2) T T T (3) F F F (4) F F T

Sol. Answer (2)

STATEMENT-1 : True, $\int_{-3}^{3}|x|dx = 2\int_{0}^{3}|x|dx = 2\int_{0}^{3}x\,dx = 2\left[\frac{x^2}{2}\right]_0^3 = 9$

STATEMENT-2 : True, $\int_{0}^{1}\tan^{-1}x = \left[x\tan^{-1}x - \ln\sqrt{1+x^2}\right]_0^1 = \frac{\pi}{4} - \ln 2$

STATEMENT-3 : True, Obvious (using property)

4. STATEMENT-1 : $\int_{0}^{\infty}\frac{dx}{1+e^x} = \ln 2 - 1$

STATEMENT-2 : $\int_{0}^{\infty}\frac{\sin\left(\tan^{-1}x\right)}{1+x^2}dx = \pi$

STATEMENT-3 : $\int_{0}^{\pi^2/4}\frac{\sin\sqrt{x}}{\sqrt{x}}dx = 1$

(1) T F T (2) T T T (3) F F F (4) F F T

Sol. Answer (3)

STATEMENT-1 : False, $\int_{0}^{\infty}\frac{dx}{1+e^x} = \int_{1}^{\infty}\frac{dt}{t(t+1)} = \ln 2$

STATEMENT-2 : False, $\int_{0}^{\infty}\frac{\sin\left(\tan^{-1}x\right)}{1+x^2}dx = \int_{0}^{\pi/2}\sin\theta\,d\theta = 1$

STATEMENT-3 : False, $\int_{0}^{\pi^2/4}\frac{\sin\sqrt{x}}{\sqrt{x}}dx = 2\int_{0}^{\pi/2}\sin\theta\,d\theta = 2$

5. STATEMENT-1 : If $g(x)=\int_{0}^{x}\cos^4 t\,dt$ then $g(x+\pi)$ is equal to $g(x)+g(\pi)$

STATEMENT-2 : If $\{x\}$ represents the fractional part of x then $\int_{0}^{100}\left\{\sqrt{x}\right\}$ is equal to $\frac{2000}{3}$

STATEMENT-3 : The value of $\int_{-\frac{1}{2}}^{\frac{1}{2}}\left(\alpha\log\left(\frac{1+x}{1-x}\right)+\beta\right)dx$ depends on the value of β.

(1) T F T (2) T T T (3) F F F (4) F F T

Sol. Answer (1)

STATEMENT-1 : $g(x+\pi)=\int_0^{x+\pi}\cos^4 t\,dt=\int_0^{\pi}\cos^4 t\,dt+\int_{\pi}^{x+\pi}\cos^4 t\,dt=d(\pi)+I$

$\cos^4 t$ is periodic function with period π

Hence, $\int_{\pi}^{x+\pi}\cos^4 t\,dt=\int_0^{x}\cos^4 t\,dt$

Hence, $g(x+\pi)=g(x)+g(\pi)$

STATEMENT-2 : $I=\int_0^{100}\left\{\sqrt{x}\right\}dx=\int_0^{100}\left(\sqrt{x}-\left[\sqrt{x}\right]\right)dx$

$I_1=\int_0^{100}\left[\sqrt{x}\right]dx$ If $0\le x<1$ $\left[\sqrt{x}\right]=0$

$1\le x<4$ $\left[\sqrt{x}\right]=1$

$4\le x<0$ $\left[\sqrt{x}\right]=2$

$=\int_0^1 0\,dx+\int_1^4 1.dx+\int_4^9 2dx..........+\int_{81}^{100} I\,dx$

$81\le x<100\left[\sqrt{x}\right]=9$

$=0+1(4-1)+2(9-4)+.........9(100-81)$

$=3+10+21+36+55+78+105+136+171=165$

$\therefore\ I=\frac{2000}{3}-615=\frac{155}{3}$

STATEMENT-3 : Since $\log\frac{1+x}{1-x}$ is an odd function $\therefore\int_{-1/2}^{1/2}\log\frac{1+x}{1-x}dx=0$

$\therefore\int_{-1/2}^{1/2}\left(\alpha\log\frac{1+x}{1-x}+\beta\right)dx=\beta\int_{-1/2}^{1/2}dx=\beta$ Hence depend on the value of β

6. STATEMENT-1 : $\int \frac{dx}{(x-1)(x-2)} = \ln\left|\frac{x-2}{x-1}\right| + c$.

STATEMENT-2 : $\int \frac{2x}{x^2+3x+2} dx = 2\ln\frac{(x+2)^2}{|x+1|} + c$.

STATEMENT-3 : $\int \frac{x^2+1}{x^4-x^2+1} dx = \tan^{-1}\left(\frac{x^2-1}{x}\right) + c$.

(1) T F T (2) T T T (3) F F F (4) F F T

Sol. Answer (2)

STATEMENT-1 : True,

$$\int \frac{1}{(x-2)(x-1)} dx = \int \frac{(x-1)-(x-2)}{(x-1)(x-2)} dx$$

$$= \int \frac{1}{x-2} dx - \int \frac{1}{x-1} dx$$

$$= \log(x-2) - \log(x-1) + c$$

$$= \log\frac{|x-2|}{|x-1|} + c$$

STATEMENT-2 : True,

$$\int \frac{2x}{(x+2)(x+1)} dx = \int \frac{+4\,dx}{x+2} + \int \frac{-2\,dx}{x+1}$$

$$= 4\log|x+2| - 2\log|x+1|$$

$$= 2\ln\frac{(x+2)^2}{|x+1|} + c$$

STATEMENT-3 : True,

$$\int \frac{x^2+1}{x^4-x^2+1} dx = \int \frac{1+\frac{1}{x^2}\,dx}{\left(x^2+\frac{1}{x^2}\right)-1}$$

$$= \int \frac{1+\frac{1}{x^2}\,dx}{\left(x-\frac{1}{x}\right)^2+1} \qquad \text{let, } x-\frac{1}{x} = t$$

$$= \int \frac{dt}{1+t^2}$$

$$= \tan^{-1} t$$

$$= \tan^{-1}\left(\frac{x^2-1}{x}\right) + c$$

7. STATEMENT-1 : $\int \frac{\sin x}{1+\cos^2 x} dx = -\tan^{-1}(\cos x) + c$.

STATEMENT-2 : $\int x \ln x \, dx = \frac{x^2}{2} \log\left(\frac{x^2}{e}\right) + c$.

STATEMENT-3 : $\int \frac{1}{x(x^n+1)} dx = \frac{1}{n} \log\left(\frac{x^n}{x^n+1}\right) + c$.

(1) T F T (2) T T T (3) F F F (4) F F T

Sol. Answer (1)

STATEMENT-1 : True,

$$I = \int \frac{\sin x}{1+\cos^2 x} dx$$

$$\cos x = t$$

$$-\sin x \, dx = dt$$

$$I = -\int \frac{dt}{|t|^2}$$

$$I = \tan^{-1}(\cos x)$$

STATEMENT-2 : False,

$$\int x \ln x dx = \frac{x^2}{2} \ln\left(\frac{x}{\sqrt{e}}\right) + c$$

STATEMENT-3 : True,

$$I = \int \frac{1}{x(x^n+1)} dx$$

$$= \int \frac{x^{n-1}}{x^n(x^n+1)} dx$$

$$= \frac{1}{n} \int \frac{nx^{n-1} dx}{x^n(x^n+1)} \quad x^n = t$$

$$= \frac{1}{n} \int \frac{dt}{t(t+1)}$$

$$= \frac{-1}{n}\left[\int \frac{1}{t+1} dt - \frac{1}{t}\int dt\right]$$

$$= \frac{-1}{n}\left[\log\left(\frac{1+t}{t}\right)\right]$$

$$= \frac{-1}{n}\left[\log\left(\frac{1+x^n}{x^n}\right)\right]$$

$$= \frac{1}{n}\log\left(\frac{x^n}{1+x^n}\right) + c$$

8. STATEMENT-1 : $\int e^x\left[\frac{1+nx^{n-1}-x^{2n}}{(1-x^n)\sqrt{1-x^{2n}}}\right]dx = e^x\left[\frac{\sqrt{1-x^{2n}}}{1-x^n}\right]+c$.

STATEMENT-2 : If $\int \log(\sqrt{1-x}+\sqrt{1+x})\,dx = xf(x)+Ax+B\sin^{-1}x+c$, then $f(x)=\log(\sqrt{1-x}+\sqrt{1+x})$.

STATEMENT-3 : If $\int\frac{dx}{(x^2+1)(x^2+4)} = A\tan^{-1}x+B\tan^{-1}\frac{x}{2}+c$, then $A + 2B = 0$

(1) T T T (2) T T F (3) T F T (4) F F F

Sol. Answer (1)

STATEMENT-1 : True,

$$\int\frac{e^x(1-x^{2n})+nx^{n-1}}{(1-x^n)\sqrt{1-x^{2n}}}\,dx = \int e^x\left[\sqrt{\frac{1+x^n}{1-x^n}}+\frac{n.x^{n-1}}{(1-x^n)^2}.\sqrt{\frac{1-x^n}{1+x^n}}\right]dx$$

$$= e^x\sqrt{\frac{1+x^n}{1-x^n}}+C = e^x\frac{\sqrt{1-x^{2n}}}{1-x^n}+C$$

STATEMENT-2 : True,

$$\text{Let } I = \int \underset{\text{II}}{1}.\log\left(\underset{\text{I}}{\sqrt{1-x}}+\sqrt{1+x}\right)dx$$

$$= x\log\left(\sqrt{1-x}+\sqrt{1+x}\right)-\int\frac{1}{\sqrt{1-x}+\sqrt{1+x}}\left[\frac{-1}{2\sqrt{1-x}}+\frac{1}{2\sqrt{1+x}}\right]x\,dx$$

$$\therefore\quad I = x\log\left(\sqrt{1-x}+\sqrt{1+x}\right)+\frac{1}{2}\int\left(\frac{1}{\sqrt{1-x^2}}-1\right)dx$$

$$\therefore\quad I = x\,\underbrace{\log\left\{\sqrt{1-x}+\sqrt{1+x}\right\}}_{f(x)}+\frac{1}{2}\sin^{-1}x-\frac{x}{2}+C$$

STATEMENT-3 : True,

using partial fraction

$$I = \frac{1}{3}\int\frac{dx}{x^2+1}-\frac{1}{3}\int\frac{dx}{x^2+2^2}$$

$$\therefore\quad I=\frac{1}{3}\tan^{-1}x-\frac{1}{6}\tan^{-1}\left(\frac{x}{2}\right)\Rightarrow A+2B=0$$

9. STATEMENT-1 : The value of the integral $\int_0^{\infty}\frac{dx}{1+x^2}$ is $\frac{\pi}{2}$.

STATEMENT-2 : The value of the integral $\int_0^{1}\frac{dx}{\sqrt{1-x^2}}$ is $\frac{\pi}{2}$.

(1) T F (2) T T (3) F F (4) F T

Sol. Answer (2)

STATEMENT-1 : We have $\int_0^{\infty}\frac{dx}{1+x^2}=\frac{\lim}{t\to\infty}\int_0^{t}\frac{dx}{1+x^2}$

$= \lim\limits_{t\to\infty} \tan^{-1}t - \tan^{-1}0 = \frac{\pi}{2}$

STATEMENT-2 : $\int_0^{1}\frac{dx}{\sqrt{1-x^2}} = \lim\limits_{h\to 0}\int_0^{1-h}\frac{1}{\sqrt{1-x}},dx = \lim\limits_{h\to 0}\left|\sin^{-1}x\right|_0^{1-h}$

$= \lim\limits_{h\to 0}\sin^{-1}(1-h) = \frac{\pi}{2}$

10. STATEMENT-1 : The value of $\int_{-10}^{10}\frac{3^x}{3^{[x]}}dx$ is equal to $\frac{60}{\log 3}$

STATEMENT-2 : The value of $\int_0^{\infty}\left[\frac{3}{x^2+1}\right]dx$ is equal to $\sqrt{2}$

STATEMENT-3 : The value of $\int_{-1}^{1}[x[1+\sin\pi x]+1]dx$ is equal to 2 (where [.] denotes the, G.I.F.)

(1) T F T (2) T T T (3) F F F (4) F F T

Sol. Answer (4)

STATEMENT-1 : $\int_{-10}^{10}\frac{3^x}{3^{[x]}}dx = 20\int_0^{1}3^{x-[x]}dx = 20\int_0^{1}3^x dx = 20\left|\frac{3^x}{\log 3}\right|_0^1$

$=\frac{40}{\log 3}$

STATEMENT-2 : $0 < \dfrac{3}{x^2+1} \le 3$

$\dfrac{3}{x^2+1} = 2 \Rightarrow x = \dfrac{1}{\sqrt{2}}$ and $\dfrac{3}{x^2+1} = 1 \Rightarrow x = \sqrt{2}$

$$\therefore \quad I = \int_0^{1/\sqrt{2}} 2dx + \int_{\frac{1}{\sqrt{2}}}^{\sqrt{2}} 1.dx + \int_{\sqrt{2}}^{\infty} 0.dx = \sqrt{2} + \sqrt{2} - \frac{1}{\sqrt{2}} = \frac{3}{\sqrt{2}}$$

STATEMENT-3 : $\int_{-1}^{1} \left[x[1+\sin \pi x]+1\right] dx = \int_{-1}^{0} \left[x[1+\sin \pi x]+1\right] dx + \int_0^1 \left[x[1+\sin \pi]+1\right] dx$

Now $-1 < x < 0 \Rightarrow \left[1+\sin \pi x\right] = 0$

And $0 < x < 1 \Rightarrow \left[1+\sin \pi x\right] = 1 \Rightarrow \left[x[1+\sin \pi x]+1\right] = 1$

So $\int_{-1}^{1} \left[x[1+\sin \pi x]+1\right] dx = 2$

11. STATEMENT-1 : If $a+b=\dfrac{\pi}{2}$ then $\int_a^b \dfrac{f(\cos x)}{f(\cos x)+f(\sin x)} dx$ is equal to $\dfrac{b-a}{2}$

STATEMENT-2 : $\int_0^{1/4} \dfrac{dx}{1+(\tan(2\pi x))^{\sqrt{3}}}$ is equal to $\dfrac{1}{4}$

STATEMENT-3 : $\int_2^8 \dfrac{[x^2]dx}{[x^2-20x+100]+[x^2]} = 3$; where [.] be G.I.F.

(1) T F T (2) T T T (3) F F F (4) F F T

Sol. Answer (1)

STATEMENT-1 : $I = \int_a^b \dfrac{f(\cos x)}{f(\cos x)+f(\sin x)} dx \Rightarrow I = \int_a^b \dfrac{f(\sin x)}{f(\sin x)+f(\cos x)} dx$

$$\because \int_a^b f(x)dx = \int_a^b f(a+b-x)dx$$

$$\Rightarrow \quad 2I = \int_a^b \frac{f(\cos x)+f(\sin x)}{f(\cos x)+f(\sin x)} dx = \int_a^b dx = b-a$$

Hence $I = \dfrac{b-a}{2}$

STATEMENT-2 : $I = \int_0^{1/4} \frac{(\cos 2\pi x)^{\sqrt{3}}\,dx}{(\cos 2\pi x)^{\sqrt{3}} + (\sin 2\pi x)^{\sqrt{3}}}$

$$I = \int_0^{1/4} \frac{(\sin 2\pi x)^{\sqrt{3}}}{(\sin 2\pi x)^{\sqrt{3}} + (\cos 2\pi x)^{\sqrt{3}}}\,dx \Rightarrow 2I = \int_0^{1/4} dx = \frac{1}{4}$$

$$\therefore \quad I = \frac{1}{8}$$

STATEMENT-3 : $I = \int_2^8 \frac{[x^2]\,dx}{[(10-x)^2] + [x^2]} \Rightarrow I = \int_2^8 \frac{[(10-x)^2]\,dx}{[x^2] + [(10-x)^2]}$

$$2I = \int_2^8 dx \Rightarrow 2I = 6 \Rightarrow I = 3$$

12. STATEMENT-1 : $\lim_{n\to\infty}\left[\frac{1}{n+1} + \frac{1}{n+2} + \frac{1}{n+3} .... \frac{1}{6n}\right]$ is equal to $\int_0^1 \frac{dx}{1+x}$

STATEMENT-2 : The value of $\lim_{n\to\infty}\left(1+\frac{1}{n^2}\right)^{\frac{2}{n^2}}\left(1+\frac{2^2}{n^2}\right)^{\frac{4}{n^2}}\left(1+\frac{3^2}{n^2}\right)^{\frac{6}{n^2}} .... \left(1+\frac{n^2}{n^2}\right)^{\frac{2n}{n^2}}$ is equal to $\frac{4}{e}$

STATEMENT-3 : $\lim_{n\to\infty}\left\{\frac{n+1}{n^2+1^2} + \frac{n+2}{n^2+n^2} + \frac{n+3}{n^2+3^2} .... + \frac{1}{n}\right\}$ is equal to $\frac{\pi}{4}\log 2$

(1) T F T (2) T T T (3) F T F (4) F F T

Sol. Answer (3)

STATEMENT-1 : $\lim_{n\to\infty}\sum_{r=1}^{5n}\frac{1}{n+r} \Rightarrow \sum_{r=1}^{5n}\frac{1}{n}\left(\frac{1}{1+r/n}\right) = \int_0^5 \frac{1}{1+x}dx$

STATEMENT-2 : $y = \lim_{n\to\infty}\left(1+\frac{1}{n^x}\right)^{2/n^2}\left(1+\frac{2^2}{n^2}\right)^{4/n^2}\left(1+\frac{3^2}{n^2}\right)^{6/n^2} .... \left(1+\frac{n^2}{n^2}\right)^{2x/n^2}$

$$\Rightarrow \log y = \lim_{x\to\infty}\sum_{r=1}^{n}\log\left(1+\frac{r^2}{n^2}\right).\frac{2r}{n}\times\frac{1}{n}$$

$$= \int_0^1 2x\left(\log(1+x^2)\right)dx = \int_1^2 \log t\, dt$$

$$= |t\log t - t|_1^2 = 2\log 2 - 1 = \log\frac{4}{e}$$

$$\Rightarrow\ y = \frac{4}{e}$$

STATEMENT-3 : $\lim\limits_{n\to\infty}\sum\limits_{r=1}^{n}\frac{n+r}{n^2+r^2} \Rightarrow \lim\limits_{n\to\infty}\sum\limits_{r=1}^{n}\frac{1}{n}\frac{1+\frac{r}{n}}{1+\frac{r^2}{n^2}}$

$$= \int_0^1 \frac{1+x}{1+x^2}dx = \left|\tan^{-1}x + \frac{1}{2}\log(1+x^2)\right|_0^1$$

$$= \frac{\pi}{4} + \frac{1}{2}\log 2$$

SECTION - H

Aakash Challengers Questions

1. If $\int \frac{f(x)}{x^3-1}dx = -\log\left|\frac{x^2+x+1}{x-1}\right| + \frac{A}{948\sqrt{3}}\left(\tan^{-1}\frac{2x+1}{\sqrt{3}}\right) + C$, where $f(x)$ is a polynomial of second degree in x such that $f(0) = f(1) = 3f(2) = 3$, then A equals.....

Sol. $\int \frac{f(x)}{x^3-1}dx = -\log\left|\frac{x^2+x+1}{x-1}\right| + \frac{A}{948\sqrt{3}}\cdot\tan^{-1}\left(\frac{2x+1}{\sqrt{3}}\right) + C$

Differentiate w.r.t. x

$$\frac{f(x)}{x^3-1} = -\left\{\frac{1}{x^2+x+1} - \frac{1}{x-1}\right\} + \frac{A}{948\sqrt{3}}\left\{\frac{1}{1+\left(\frac{2x+1}{\sqrt{3}}\right)^2}\times\frac{2}{\sqrt{3}}\right\}$$

Put $x = 0$

$$-3 = -\{1+1\} + \frac{A}{948\sqrt{3}}\left\{\frac{3}{4}\cdot\frac{2}{\sqrt{3}}\right\}$$

$$-1 = \frac{A}{948\times 2}$$

$A = -948 \times 2$

$= -1896$

2. For any $t \in R$ and f a continuous function, let $I_1 = \int_{\sin^2 t}^{1+\cos^2 t} x f[x(2-x)]dx$ and $I_2 = \int_{\sin^2 t}^{1+\cos^2 t} f[x(2-x)]dx$ then $\frac{I_1}{I_2}$ equals to

Sol. $I_1 = \int_{\sin^2 t}^{1+\cos^2 t} xf[x(2-x)]dx = \int_{\sin^2 t}^{1+\cos^2 t} (2-x)f[(2-x).x]dx \quad \left(\int_a^b f(x)dx = \int_a^b f(a+b-x)dx\right)$

$2I_1 = 2\int_{\sin^2 t}^{1+\cos^2 t} f[x(2-x)]dx$

and $I_2 = \int_{\sin^2 t}^{1+\cos^2 t} f[x(2-x)]dx$

$\Rightarrow \frac{I_1}{I_2} = 1$.

3. Evaluate $\lim_{n\to\infty}\left\{\tan\frac{\pi}{2n}\tan\frac{2\pi}{2n}\tan\frac{3\pi}{2n}......\tan\frac{n\pi}{2n}\right\}^{\frac{1}{n}}$

Sol. Let $P = \lim_{n\to\infty}\left\{\tan\frac{\pi}{2n}\tan\frac{2\pi}{2n}.......\tan\frac{n\pi}{2n}\right\}^{\frac{1}{n}}$

Taking logarithm,

$$\ln P = \lim_{n\to\infty}\frac{1}{n}\left[\ln\left(\tan\frac{\pi}{2n}\right)+\ln\left(\tan\frac{2\pi}{2n}\right)+......+\ln\left(\tan\frac{n\pi}{2n}\right)\right]$$

$$= \lim_{n\to\infty}\frac{1}{n}\sum_{r=1}^{n}\ln\left(\tan\frac{r\pi}{2n}\right)$$

$$= \int_0^1 \ln\left(\tan\frac{\pi}{2}x\right)dx \quad \left(\begin{array}{l}\text{using the definition of definite}\\ \text{integral as the limit of a sum}\end{array}\right)$$

Let $\frac{\pi x}{2} = \theta \Rightarrow dx = \frac{2d\theta}{\pi}$, limit become 0 to $\frac{\pi}{2}$

$$\therefore \ln P = \frac{2}{\pi}\int_0^{\frac{\pi}{2}}\ln(\tan\theta).d\theta$$

$$= \frac{2}{\pi}\int_0^{\frac{\pi}{2}} \ln\left(\frac{\sin\theta}{\cos\theta}\right)d\theta$$

$$= \frac{2}{\pi}\left[\int_0^{\frac{\pi}{2}} \ln(\sin\theta)d\theta - \int_0^{\frac{\pi}{2}} \ln(\cos\theta)d\theta\right]$$

$$= \frac{2}{\pi}\left[\int_0^{\frac{\pi}{2}} \ln(\sin\theta)d\theta - \int_0^{\frac{\pi}{2}} \ln(\sin\theta)d\theta\right]$$

$\ln P = \frac{2}{\pi}\times 0 = 0$

$\Rightarrow P = e^0 = 1$

4. Suppose $g(x)$ satisfies $g(x) = x + \int_0^1 (xy^2 + yx^2)g(y)dy$, then find $g(x)$.

Sol. $g(x) = x + \int_0^1 (xy^2 + yx^2)g(y)dy$

$$= x + \int_0^1 xy^2 g(y)dy + \int_0^1 yx^2 g(y)dy$$

$$= x + x\int_0^1 y^2 g(y)dy + x^2\int_0^1 yg(y)dy$$

$= x + \alpha x + \beta x^2$

Where $\alpha = \int_0^1 y^2 g(y)dy$ and $\beta = \int_0^1 y\,g(y)dy$

So $g(x) = x + \alpha x + \beta x^2$ .....(A)

Changing x to y, we have

$g(y) = y + \alpha y + \beta y^2$

$\therefore \quad \alpha = \int_0^1 y^2 g(y)dy = \int_0^1 y^2(y + \alpha y + \beta y^2)dy$

$$= \left[\frac{y^4}{4}+\frac{\alpha y^4}{4}+\frac{\beta y^5}{5}\right]_0^1 = \frac{1}{4}+\frac{\alpha}{4}+\frac{\beta}{5}$$

$$\Rightarrow \quad \frac{3\alpha}{4}-\frac{\beta}{5}=\frac{1}{4} \qquad \text{..... (i)}$$

Again $\beta = \int_0^1 y\, g(y)dy = \int_0^1 y(y+\alpha y+\beta y^2)dy$

$$= \left[\frac{y^3}{3}+\alpha\frac{y^3}{3}+\beta\frac{y^4}{4}\right]_0^1 = \frac{1}{3}+\frac{\alpha}{3}+\frac{\beta}{4}$$

$$\Rightarrow \quad \frac{3\beta}{4}-\frac{\alpha}{3}=\frac{1}{3} \qquad \text{.....(ii)}$$

Solving (i) and (ii) for α and β we have

$\alpha = \frac{61}{119}, \beta = \frac{80}{119}$

Hence from (A),

$g(x) = x + \alpha x + \beta x^2$

$$= x+\frac{61}{119}x+\frac{80}{119}x^2$$

$$= \frac{180}{119}x+\frac{80}{119}x^2$$

5. Find the integral $\int \frac{x^2-1}{x^2+1}\cdot\frac{dx}{\sqrt{1+x^4}}$.

Sol. $\int \frac{1-\frac{1}{x^2}}{x+\frac{1}{x}} \times \frac{dx}{\sqrt{x^2+\frac{1}{x^2}}}$

Put $x+\frac{1}{x}=t$

$\left(1-\frac{1}{x^2}\right)dx = dt$

$= \int \frac{dt}{t\sqrt{t^2-2}}$

Now let $t^2 - 2 = z^2$

$\Rightarrow$ $2tdt = 2zdz$

$$= \int \frac{tdt}{t^2\sqrt{t^2-2}}\, dt = \int \frac{dz}{\left(z^2+2\right)z}$$

Now resolving this into partial fraction and simplifying we get.

$$I = \frac{1}{\sqrt{2}}\cos^{-1}\frac{x\sqrt{2}}{x^2+1} + c$$

6. Evaluate $\int \left\{ \log\left(\frac{1+\sin 2\theta}{1-\sin 2\theta}\right)^{\cos^2\theta} + \log\left(\frac{\cos 2\theta}{1+\sin 2\theta}\right) \right\} d\theta$.

Sol. Here $I = \int \left\{ \log\left(\frac{1+\sin 2\theta}{1-\sin 2\theta}\right)^{\cos^2\theta} + \log\left(\frac{\cos 2\theta}{1+\sin 2\theta}\right) \right\} d\theta$

$$= \int \left(2\cos^2\theta \log\left(\frac{\cos\theta+\sin\theta}{\cos\theta-\sin\theta}\right) - \log\left(\frac{\cos\theta+\sin\theta}{\cos\theta-\sin\theta}\right) \right) d\theta$$

$$= \int \left(2\cos^2\theta - 1\right)\log\left(\frac{\cos\theta+\sin\theta}{\cos\theta-\sin\theta}\right) d\theta$$

Applying by parts $= \frac{\sin 2\theta}{2}\log\left(\frac{\cos\theta+\sin\theta}{\cos\theta-\sin\theta}\right) + \frac{1}{2}\log\left|\cos 2\theta\right| + c$

7. Evaluate $\int \frac{dx}{\tan x + \cot x + \sec x + \operatorname{cosec} x}$.

Sol. $= \int \frac{(\sin x \cos x)\,dx}{1+\sin x + \cos x}$

$$= \int \frac{\sin x}{\sec x + \tan x + 1}\, dx$$

$$= \int \frac{\sin x(1+\tan x - \sec x)}{(1+\tan x)^2 - \sec^2 x}\, dx$$

$$= \int \frac{\sin x(1+\tan x - \sec x)}{2\tan x}\, dx$$

$$= \frac{1}{2}\int \cos x(1+\tan x - \sec x)\, dx$$

$$= \frac{1}{2}\int (\cos x + \sin x - 1)\, dx$$

$$= \frac{1}{2}(\sin x - \cos x - x) + c$$

8. Evaluate the integral $\int \frac{dx}{\sqrt{e^{5x}}\sqrt[4]{(e^{2x}+e^{-2x})^3}}$.

Sol. $\int \frac{dx}{\sqrt{e^{5x}}\sqrt[4]{\frac{\left(e^{4x}+1\right)^3}{e^{6x}}}}$

$$= \int \frac{e^{-x}dx}{\sqrt[4]{\left(e^{4x}+1\right)^3}} = \int \frac{e^{-x}dx}{\sqrt[4]{e^{12x}\left(1+e^{-4x}\right)^3}}$$

$$= \int \frac{e^{-4x}dx}{\sqrt[4]{\left(1+e^{-4x}\right)^3}}$$

Let $1 + e^{-4x} = t$

$-4e^{-4x}\,dx = dt$

$$\Rightarrow \quad I = \frac{-1}{4}\int \frac{dt}{t^{3/4}}$$

$$= \frac{-1}{4}t^{1/4} \times 4 + c$$

$= -t^{1/4} + c$

$= -(1 + e^{-4x})^{1/4} + c$

9. Evaluate $\int \frac{\tan\left(\frac{\pi}{4}-x\right)}{\cos^2 x\sqrt{\tan^3 x+\tan^2 x+\tan x}}\,dx$.

Sol. $= \int \frac{\left(1-\tan^2 x\right)dx}{(1+\tan x)^2\cos^2 x\sqrt{\tan^3 x+\tan^2 x+\tan x}}$

$$= \int \frac{-\left(1-\frac{1}{\tan^2 x}\right)\sec^2 x\,dx}{\left(\tan x+2+\frac{1}{\tan x}\right)\sqrt{\tan x+1+\frac{1}{\tan x}}}$$

Let $y = \sqrt{\tan x+1+\frac{1}{\tan x}}$

$$\Rightarrow \quad 2y\,dy = \left(\sec^2 x-\frac{1}{\tan^2 x}.\sec^2 x\right)dx$$

$$I = \int \frac{-2y\ dy}{\left(y^2+1\right).y}$$

$$= -2\int \frac{dy}{y^2+1} = -2\tan^{-1} y + C$$

$$= -2\tan^{-1}\left(\sqrt{\tan x + 1 + \frac{1}{\tan x}}\right) + c$$

10. $\int_0^{\infty}\left[ne^{-x}\right]dx$, is equal to ; where n is a natural number. ([·] represents greatest integer function)

Sol. $$\int_0^{\infty}\left[ne^{-x}\right]dx = \int_0^{\ln\frac{n}{n-1}}(n-1)dx + \int_{\ln\frac{n}{n-1}}^{\ln\frac{n}{n-2}}(n-2)dx + \int_{\ln\frac{n}{n-2}}^{\ln\frac{n}{n-3}}(n-3)dx + \int_{\ln\frac{n}{n-3}}^{\ln\frac{n}{n-4}}(n-4)dx + \ldots\ldots$$

$$= n\left[\frac{n}{n-1} + \left(\frac{n}{n-2} - \frac{n}{n-1}\right) + \left(\frac{n}{n-3} - \frac{n}{n-2}\right) + \ldots\ldots\right]$$

$$-\left(\frac{n}{n-1} + 2\left(\frac{n}{n-2} - \frac{n}{n-1}\right) + 3\left(\frac{n}{n-3} - \frac{n}{n-2}\right) + 4\left(\frac{n}{n-4} - \frac{n}{n-3}\right) + \ldots\ldots\right)$$

$$= \ln\left(\frac{n^n}{n!}\right)$$

11. Let f be a real valued function satisfying $f(x) + f(x + 4) = f(x + 2) + f(x + 6)$ and let $g(x) = \int_x^{x+8} f(t)dt$ and $g(1) = 2011$; then $g(2011)$ is equal to

(1) 1 (2) 2011 (3) 2010 (4) 8

Sol. Answer (2)

Given $f(x)+f(x+4)=f(x+2)+f(x+6)$ ….(i)

$\Rightarrow\ f(x+2)+f(x+6)=f(x+4)+f(x+8)$ ….(ii)

Adding (i) and (ii) we get, $f(x)=f(x+8)$ ….(iii)

Now $g(x) = \int_x^{x+8} f(t)dt$

$\Rightarrow\ g'(x)=f(x+8)-f(x)=0$ {using (iii)}

As $g'(x)$ is zero, So g is a constant function. $\Rightarrow$ $g(2011) = 2011$

12. If $f(x)=\int_1^x \frac{\log t\,dt}{1+t+t^2}, \forall\, x \geq 1$, then $f(2)$ is equal to

(1) $f(0)$ (2) $f(1)$ (3) $f\left(\frac{1}{2}\right)$ (4) $f(4)$

Sol. Answer (3)

Given that, $f(x)=\int_1^x \frac{\log t}{1+t+t^2}, \forall x \geq 1$

Substitute $z=\frac{1}{t}$ in the integral

$$\int_1^x \frac{\log t}{1+t+t^2}dt=\int_1^x \frac{\log t}{t^2\left(1+\frac{1}{t}+\frac{1}{t^2}\right)}$$

$$\boxed{dz=-\frac{dt}{t^2}}$$

$$=\int_1^{\left(\frac{1}{x}\right)} \frac{\log z}{z^2+z+1}dz$$

$$=\int_1^{\left(\frac{1}{x}\right)} \frac{\log t}{t^2+t+1}dt$$

$\Rightarrow\ f(x)=f\left(\frac{1}{x}\right)$

$\Rightarrow\ f(x)=f\left(\frac{1}{2}\right)$

13. Given a real valued function $f(x)$ which is monotonic and differentiable, prove that for any real numbers a and

b : $\int_a^b\left\{f^2(x)-f^2(a)\right\}dx=\int_{f(a)}^{f(b)} 2x\left\{b-f^{-1}(x)\right\}dx$

Sol. As f is differentiable and monotonic

$f^{-1}(x)$ exists

Let $f^{-1}(x) = t \Rightarrow x = f(x)$ or $dx = f'(t)dt$

$\Rightarrow x = f(a) \Rightarrow f^{-1}\{f(a)\} = t \Rightarrow t = a$

$x = f(b) \Rightarrow f^{-1}(f(b)) = t \Rightarrow t = b$

$$\therefore \int_{f(a)}^{f(b)} 2x\left\{b - f^{-1}(x)\right\}dx$$

$$= \int_a^b 2f(t)(b-t)f'(t)\,dt$$

Now Integrating by parts we get

$$= -(b-a)\{f(a)\}^2 + \int_a^b \{f(t)\}^2\,dt$$

$$= \int_a^b \left(\{f(t)\}^2 - (f(a))^2\right)dt$$

14. If $\int_0^{\pi}\left(\frac{x}{1+\sin x}\right)^2 dx = A$, then $\int_0^{\pi}\frac{\left(2x^2\cos^2 x/2\right)}{(1+\sin x)^2}dx$ is equal to

(1) A (2) $A+2\pi$ (3) $A+2\pi-\pi^2$ (4) $A+2\pi-2\pi^2$

Sol. Answer (3)

Let $B = \int_0^{\pi}\frac{\left(2x^2\cos^2\frac{x}{2}\right)}{(1+\sin x)^2}dx$

$$B - A = \int_0^{\pi}\frac{x^2\left(2\cos^2\frac{x}{2}-1\right)}{(1+\sin x)^2}dx$$

$$= \int_0^{\pi}\frac{x^2\cos x}{(1+\sin x)^2}dx$$

Integrating by parts,

$$B - A = \left\{\frac{-x^2}{1+\sin x}\right\}_0^{\pi} + 2\int_0^{\pi}\frac{x}{1+\sin x}dx$$

$$B - A = -\pi^2 + 2k$$

Where, $k = \int_0^{\pi} \frac{x\,dx}{(1+\sin x)} = \int_0^{\pi} \frac{(\pi - x)}{1+\sin x} dx$

$$k = \pi \int_0^{\pi} \frac{dx}{1+\sin x} - k$$

Solving, we get, $k = \pi$

$\Rightarrow$ $B - A = -\pi^2 + 2\pi$

$\Rightarrow$ $B = A - \pi^2 + 2\pi$

15. Find $\int \left(\frac{1}{\sqrt[4]{(x-1)^3 (x+2)^5}} \right) dx$

Sol. $\int \frac{1}{\sqrt[4]{(x-1)^3 (x+2)^5}} \, dx$

$$\int \frac{1}{(x-1)(x+2)} \sqrt[4]{\frac{x-1}{x+2}} \, dx$$

Let $\frac{x-1}{x+2} = t^4$

$$\frac{(x+2)-(x-1)}{(x+2)^2} dx = 4t^3 \ dt$$

$$\frac{1}{(x+2)^2} dx = \frac{4}{3} t^3 \ dt$$

$$= \int \frac{(x+2)}{(x-1)} \times \sqrt[4]{\frac{x-1}{x+2}} . \frac{dx}{(x+2)^2}$$

$$= \int \frac{1}{t^4} \cdot t \frac{4}{3} t^3 dt = \frac{4}{3} t + C$$

$$= \frac{4}{3} \left(\frac{x-1}{x+2} \right)^{1/4} + C$$

16. Evaluate $\int \frac{x^2+n(n-1)}{(x\sin x+n\cos x)^2}\,dx$.

Sol. Multiplying and dividing by x^{2n-2}, we get

$$I=\int \frac{(x^2+n(n-1))x^{2n-2}}{(x^n\sin x+nx^{n-1}\cos x)^2}\,dx$$

Put $x^n \sin x + nx^{n-1}\cos x = t$

$\Rightarrow x^{n-2}\cos x\,(x^2+n(n-1))\,dx = dt$

$$\Rightarrow I=\int \frac{(x^2+n(n-1).x^{n-2}\cos x}{(x^n\sin x+nx^{n-1}\cos x)^2}x^n.\sec x\,dx$$

Integrating by parts we get

$$I=-\frac{x\sec x}{x\sin x+n\cos x}+\int \sec^2 x\,dx$$

$$=-\frac{(x\sec x)}{x\sin x+n\cos x}+\tan x+c$$

17. Evaluate the integral $\int \frac{\tan^{-1}x}{x^4}dx$

Sol. $I=\int \frac{\tan^{-1}x}{x^4}\,dx=\int \tan^{-1}x\,\frac{1}{x^4}\,dx$

Integrate by parts to get

$$=\left(\tan^{-1}x\right)\left(-\frac{1}{3x^3}\right)-\int \frac{1}{1+x^2}\cdot\frac{1}{\left(-3x^3\right)}\,dx$$

$$=-\frac{\tan^{-1}x}{3x^3}+\frac{1}{3}\int \frac{dx}{x^3(1+x^2)}$$

$$I=-\frac{\tan^{-1}x}{3x^3}+\frac{1}{6}\int \frac{dt}{(t-1)^2.t} \qquad \text{where } 1+x^2=t$$

Resolving into partial fraction we get

$$I=-\frac{\tan^{-1}x}{3x^3}-\frac{1}{6}\left|\frac{x^2+1}{x^2}\right|-\frac{1}{6x^2}+c$$

18. Find the integral $I = \int x^2 \log(1-x^2)dx$ and hence prove that $\frac{1}{1\cdot5}+\frac{1}{2\cdot7}+\frac{1}{3\cdot9}+.....=\frac{8}{9}-\frac{2}{3}\log 2$.

Sol. We know that

$$\log(1-x) = -\left\{x+\frac{x^2}{2}+\frac{x^3}{3}+\frac{x^4}{4}+...\infty\right\}$$

Replace x^2 by x we get

$$\log\left(1-x^2\right) = -\left\{x^2+\frac{x^4}{2}+\frac{x^6}{3}+\frac{x^8}{4}+...\infty\right\}$$

$$x^2\log\left(1-x^2\right) = -\left[x^4+\frac{x^6}{2}+\frac{x^8}{3}+\frac{x^{10}}{4}+\ldots\infty\right]$$

Now Integrating both sides we get

$$\int x^2\log\left(1-x^2\right)dx = -\left[\frac{x^5}{1.5}+\frac{x^7}{2.7}+\frac{x^9}{3.9}+\ldots\infty\right]+c$$

Now to find constant of integration, put $x = 0$

$\Rightarrow C = 0$

Apply integration by parts and take limit 0 to 1

$$\left(\frac{x^3}{3}\log\left(1-x^2\right)\right)_0^1 - \int_0^1\frac{x^3}{3}.\frac{1(-2x)}{1-x^2}\,dx$$

Taking $\log(1-x^2) = \log(1+x) + \log(1-x)$

$$\log\left(\frac{1+x}{1-x}\right) = \log(1+x)-\log(1-x)$$

$$\frac{1}{3}\log 2+\frac{1}{3}\log 2-\frac{2}{3}-\frac{2}{9}+\lim_{x\to 1}\left(\frac{x^3}{3}-\frac{1}{3}\right)\log(1-x)$$

$$\frac{2}{3}\log 2-\frac{8}{9} = \text{R.H.S}$$

19. If $I_n = \int z^n e^{1/z}dz$, then show that $(n+1)!\,I_n = I_0 + e^{1/z}(1!\,z^2+2!\,z^3+......+n!\,z^{n+1})$.

Sol. $I_n = \int z^n e^{1/z}dz$

Applying by parts, we get

$$\mathrm{I}_n = \frac{e^{1/z}.z^{n+1}}{(n+1)} - \int e^{1/z}\left(-\frac{1}{z^2}\right).\frac{z^{n+1}}{n+1}\,dz$$

$$= \frac{e^{1/z}.z^{n+1}}{(n+1)} + \frac{1}{(n+1)}\int e^{1/z}.z^{n-1}\,dz$$

$$I_n = \frac{e^{1/z}.z^{n+1}}{(n+1)} + \frac{I_{n-1}}{(n+1)}$$

$$= \frac{e^{1/z}.z^{n+1}}{(n+1)} + \frac{1}{(n+1)}\left[\frac{e^{1/z}.z^{n}}{n} + \frac{1}{n}I_{n-2}\right)$$

$$\frac{e^{1/z}(z)^{n+1}}{(n+1)} + \frac{e^{1/z}.(z)^{n}}{(n+1)^n} + \frac{1}{n(n+1)}I_{n-2}$$

$$= \frac{e^{1/z}(z)^{n+1}}{(n+1)} + \frac{e^{1/z}.(z)^{n}}{(n+1)^n} + \frac{e^{1/z}.(z)^{n-1}}{(n+1).n.(n-1)} + \frac{1}{(n+1)n(n-1)}I_{n-3}$$

Multiplying both sides by $(n + 1)!$ we get.

$(n+1)!\, I_n = (e^{1/z}.\, z^{n+1} - n! + e^{1/z}.z^n\,(n-1)! + \ldots + e^{1/z}.z^2.1!) + I_0$

$\Rightarrow I_n\,(n + 1)! = I_0 + e^{1/z}\,(1!z^2 + 2!\, z^3 + \ldots + n!\, z^{n+1})$

20. If $I_n = \int x^n\sqrt{a^2 - x^2}\,dx$, prove that $I_n = -x^{n-1}\frac{(a^2 - x^2)^{3/2}}{(n+2)} + \frac{(n-1)}{(n+2)}a^2 I_{n-2}$.

Sol. Applying by parts in the problems

$$I_n = \int x^2\sqrt{a^2 - x^2}\,dx$$

$$= \int x^{n-1}\left\{x\sqrt{a^2 - x^2}\right\}dx$$

$$= x^{n-1}\left\{-\frac{(a^2 - x^2)^{3/2}}{3}\right\} + \int (n-1)x^{n-1}\left\{-\frac{(a^2 - x^2)^{3/2}}{3}\right\}dx$$

$$= \frac{-x^{n-1}(a^2 - x^2)^{3/2}}{3} + \frac{(n-1)}{3}\int e^{n-2}.(a^2 - x^2)(a^2 - x^2)^{1/2}\,dx$$

$$I_n = -x^{n-1}\frac{(a^2 - x^2)^{3/2}}{3} + \frac{(n-1)}{3}I_{n-2} - \frac{(n-1)}{3}I_n$$

$$\Rightarrow \left(\frac{n+2}{3}\right)I_n = -\frac{x^{n-1}(a^2 - x^2)^{3/2}}{3} + \frac{(n-1)a^2}{(n+2)}\;I_{n-2}$$

$$\Rightarrow I_n = \frac{-x^{n-1}(a^2 - x^2)^{3/2}}{(n+2)} + \frac{(n-1)a^2}{(n+2)}\;I_{n-2}$$

21. $\left[\left(\int_0^1 \frac{dx}{\sqrt{x}+\sqrt{2-x}}\right)+\log(\sqrt{2}+1)\right]^2$ equals to ____.

Sol. $\left[\left(\int_0^1 \frac{dx}{\sqrt{x}+\sqrt{2-x}}\right)+\log(\sqrt{2}+1)\right]^2 = 2$

Let us put $\sqrt{x} = \sqrt{2}\sin\theta$

$$\frac{1}{2\sqrt{x}}dx = \sqrt{2}\cos\theta d\theta$$

$$dx = 2\sqrt{x}.\sqrt{2}\cos\theta d\theta$$

$$= 2(\sqrt{2}\sin\theta)(\sqrt{2}\cos\theta)d\theta$$

$$= 4\sin\theta\cos\theta d\theta$$

When $x = 0$, $\theta = 0°$

$x = 1$, $\frac{1}{\sqrt{2}} = \sin\theta$

$\theta = \frac{\pi}{4}$.

$$\int_0^1 \frac{dx}{\sqrt{x}+\sqrt{2-x}}$$

$$\int_0^{\pi/4} \frac{4\sin\theta\cos\theta d\theta}{\sqrt{2}\sin\theta+\sqrt{2}\cos\theta} = 2\sqrt{2}\int_0^{\pi/4}\frac{\sin\theta\cos\theta}{\sin\theta+\cos\theta}d\theta$$

$$= \frac{2\sqrt{2}}{\sqrt{2}}\int_0^{\pi/4}\frac{\frac{1}{2}\sin 2\theta}{\sin\left(\frac{\pi}{4}+\theta\right)}d\theta$$

$$= \int_0^{\pi/4}\frac{\sin 2\left(\frac{\pi}{4}-\theta\right)}{\sin\left(\frac{\pi}{4}+\frac{\pi}{4}-\theta\right)}d\theta = \int_0^{\pi/4}\frac{\cos 2\theta}{\cos\theta}d\theta$$

$$= \sqrt{2}-\log(\sqrt{2}+1)$$

Hence $\left(\int_0^1 \frac{1}{\sqrt{x}+\sqrt{2-x}}dx + l_n(1\sqrt{2}-1)\right)^2 = (\sqrt{2})^2 = 2$

22. Prove that $\int_0^{\infty}\frac{\ln(1+a^2x^2)}{1+b^2x^2}dx = \frac{\pi}{b}\ln\frac{a+b}{b}$

Sol. Let $I = \int_0^{\infty}\frac{\ln(1+a^2x^2)}{1+b^2x^2}dx$ .....(i)

Differentiating w.r.t. a,

$$\frac{dI}{da} = \int_0^{\infty}\frac{2ax^2}{(1+a^2x^2)(1+b^2x^2)}.dx$$

$$= 2a\int_0^{\infty}\left\{\frac{1}{(b^2-a^2)(1+a^2x^2)} - \frac{1}{(b^2-a^2)(1+b^2x^2)}\right\}dx$$

$$= \frac{2a}{b^2-a^2}\left[\int_0^{\infty}\frac{1}{1+a^2x^2}dx - \int_0^{\infty}\frac{1}{1+b^2x^2}dx\right]$$

$$= \frac{2a}{b^2-a^2}\left[\frac{1}{a}\tan^{-1}ax - \frac{1}{b}\tan^{-1}bx\right]_0^{\infty}$$

$$= \frac{2a}{b^2-a^2}\left[\frac{1}{a}.\frac{\pi}{2} - \frac{1}{b}.\frac{\pi}{2}\right]$$

$$= \frac{2a}{b^2-a^2}.\frac{(b-a)\pi}{2ab}$$

$$\frac{dI}{da} = \frac{\pi}{b(a+b)}$$

Integrating, we get $I = \frac{\pi}{b}.\log_e(a+b)+K$ .....(ii)

(Where K is constant of integration)

From (i) when a = 0, I = 0 .....(iii)

From (ii) when a = 0, $I = \frac{\pi}{b}\ln b + K = 0$ {using equation (3)}

$\Rightarrow$ $K = -\frac{\pi}{b}\ln b$

Hence $I = \frac{\pi}{b}\ln(a+b) - \frac{\pi}{b}\ln b$

$$I = \frac{\pi}{b}\ln\left(\frac{a+b}{b}\right)$$

23. If $V_n = \int_0^{\pi} \frac{1-\cos nx}{1-\cos x} dx$, where n is a positive integer or zero, then show that $V_{n+2} + V_n = 2V_{n+1}$

Sol. $V_n = \int_0^{\pi} \frac{1-\cos nx}{1-\cos x} dx$

Construct

$$V_{n+2} - V_{n+1} = \int_0^{\pi} \frac{\{1-\cos(n+2)x\} - \{1-\cos(n+1)x\}}{1-\cos x} dx$$

$$= \int_0^{\pi} \frac{\cos(n+1)x - \cos(n+2)}{1-\cos x} dx$$

$$= \int_0^{\pi} \frac{2\sin\left(\frac{2n+3}{2}\right)x.\sin\frac{x}{2}}{2\sin^2\frac{x}{2}} dx$$

$$= \int_0^{\pi} \frac{\sin\left(\frac{2n+3}{2}\right)x}{\sin\frac{x}{2}} dx$$

$$= \int_0^{\pi} \frac{\sin\left(n+\frac{3}{2}\right)x}{\sin\frac{x}{2}} dx \quad .....(i)$$

Change n to $n - 1$ in (i) to obtain

$$V_{n+1} - V_n = \int_0^{\pi} \frac{\sin\left(n+\frac{1}{2}\right)x}{\sin\frac{x}{2}} dx \quad .....(ii)$$

Subtract (ii) from (i) to get

$$(V_{n+2} - V_{n+1}) - (V_{n+1} - V_n) = \int_0^{\pi} \frac{\sin\left(n+\frac{3}{2}\right)x - \sin\left(n+\frac{1}{2}\right)x}{\sin\frac{x}{2}}.dx$$

$$= \int_0^{\pi} \frac{2.\cos(n+1)x.\sin\frac{x}{2}}{\sin\frac{x}{2}} dx = 2.\left[\frac{\sin(n+1)x}{n+1}\right]_0^{\pi} = 0$$

$\therefore \quad V_{n+2} - V_{n+1} = V_{n+1} - V_n$

$\Rightarrow \quad V_{n+2} + V_n = 2\, V_{n+1}$

24. If $|x| < 1$, then find the sum of the series by integral method $\frac{1}{1+x}+\frac{2x}{1+x^2}+\frac{4x^3}{1+x^4}+\frac{8x^7}{1+x^8}+.....to\ \infty$

Sol. Let $S = \frac{1}{1+x}+\frac{2x}{1+x^2}+\frac{4x^3}{1+x^4}+\frac{8x^7}{1+x^8}+.....+\frac{2^n.x^{2^n-1}}{1+x^{2^n}}$

Integration gives

$$\int S\,dx = \ln(1+x)+\ln(1+x^2)+.....+\ln(1+x^{2^n})+K$$, K being a constant of integration

$$= \ln\left\{(1+x)(1+x^2).....(1+x^{2^n})\right\}+K \qquad ...(A)$$

Now, let $P = (1+x)(1+x^2).....(1+x^{2^n})$

$$\Rightarrow (1-x)P = \left\{(1-x)(1+x)\right\}(1+x^2).....(1+x^{2^n})$$

$$= \left\{(1-x^2)(1+x^2)\right\}.....(1+x^{2^n})$$

$$= (1-x^4)(1+x^4).....(1+x^{2n})$$

$$= (1-x^{2^{n+1}})$$

$$P = \frac{1-x^{2^{n+1}}}{1-x} \qquad ...(B)$$

From (A) and (B),

$$\int S\,dx = \ln\left(\frac{1-x^{2^{n+1}}}{1-x}\right)+C$$

Taking limits, when $n \to \infty$

$$\lim_{n\to\infty}\int S\,dx = \ln\left(\frac{1-0}{1-x}\right)+C = -\ln(1-x)+K$$

Differentiating both sides w.r.t. x,

$$\frac{1}{1+x}+\frac{2x}{1+x^2}+\frac{4x^3}{1+x^4}+.....\text{to } \infty = \frac{1}{1-x}$$

25. $\int_0^2 \frac{dx}{(17+8x-4x^2)(e^{6(1-x)}+1)}$ is equal to

(1) $\frac{1}{\sqrt{21}}\log\left(\frac{\sqrt{21}+2}{\sqrt{21}-2}\right)$

(2) $\frac{1}{2\sqrt{21}}\log\left(\frac{\sqrt{21}+2}{\sqrt{21}-2}\right)$

(3) $\frac{1}{4\sqrt{21}}\log\left(\frac{\sqrt{21}+2}{\sqrt{21}-2}\right)$

(4) $\frac{1}{8\sqrt{21}}\log\left(\frac{\sqrt{21}+2}{\sqrt{21}-2}\right)$

Sol. Answer (3)

$$\int_0^2 \frac{dx}{(17+8x-4x^2)(e^{6(1-x)}+1)}$$

$$= \int_0^2 \frac{dx}{(21-4(1-x)^2)(e^{6(1-x)}+1)}$$ Put $1 - x = t$

$dx = -\,dt$

$$= -\int_1^{-1} \frac{dt}{(21-4t^2)(e^{6t}+1)}$$

$$I = \int_{-1}^{1} \frac{dt}{(21-4t^2)(e^{6t}+1)}$$

$$\Rightarrow \quad I = \int_{-1}^{1} \frac{dt}{(21-4t^2)(e^{-6t}+1)}$$

$$\Rightarrow \quad 2I = \int_{-1}^{1} \frac{dt}{(21-4t^2)}$$

$$= 2\int_0^1 \frac{dt}{21-4t^2}$$

$$\Rightarrow \quad I = \int_0^1 \frac{dt}{21-4t^2}$$

$$= \frac{1}{4\sqrt{21}}\log\left(\frac{\sqrt{21}+2}{\sqrt{21}-2}\right)$$

26. If $I_n = \int_0^1 x^n \tan^{-1} x\,dx$ then prove that $(n+1)I_n + (n-1)I_{n-2} = \frac{\pi}{2} - \frac{1}{n}$.

Sol. $I_n = \int_0^1 x^n \tan^{-1} x\,dx$

Let $U = \tan^{-1} x$; $dv = x^n\,dx$

$$dv = \frac{1}{1+x^2}dx;\ v = \frac{x^{n+1}}{n+1}$$

$$I_n = \frac{1}{n+1}\left[x^{n+1}\tan^{-1} x\right]_0^1 \quad \frac{-1}{n+1}\int_0^1 \frac{x^{n+1}}{1+x^2}dx$$

$$\Rightarrow\ (n+1)I_n = \left[x^{n+1}\tan^{-1} x\right]_0^1 - \int_0^1 \frac{x^{n+1}}{1+x^2}dx$$

$$\Rightarrow\ (n-1)I_{n-2} = \left[x^{n-1}\tan^{-1} x\right]_0^1 - \int_0^1 \frac{x^{n-1}}{1+x^2}dx$$

$$\Rightarrow\ (n+1)\,I_n + (n-1)I_{n-2} = \frac{\pi}{2} - \int_0^{-1} x^{n-1}\frac{(1+x^2)}{1+x^2}dx = \frac{\pi}{2} - \frac{1}{n}$$

27. If $2f(x) + f(-x) = \frac{1}{x}\sin\left(x - \frac{1}{x}\right)$ then $\int_{\frac{1}{e}}^{e} f(x)dx$ is equal to

(1) e (2) $\ln e$ (3) 1 (4) 0

Sol. Answer (4)

$$2\,f(x) + f(-x) = \frac{1}{x}\sin\left(x - \frac{1}{x}\right)$$

$$2f(-x) + f(x) = \frac{1}{x}\sin\left(x - \frac{1}{x}\right)$$

$$\Rightarrow\ 3f(x) = \frac{1}{x}\sin\left(x - \frac{1}{x}\right)$$

$$I=\int_{1/e}^{e} f(x)dx$$

$$=\frac{1}{3}\int_{\frac{1}{e}}^{e}\frac{1}{x}\sin\left(x-\frac{1}{x}\right)dx$$

Putting $x=\frac{1}{t}$

$$I=\frac{1}{3}\int_{e}^{1/e} t\sin\left(\frac{1}{t}-t\right).\left(\frac{-1}{t^2}\right)dt$$

$$=-\frac{1}{3}\int_{\frac{1}{e}}^{e}\frac{1}{t}\sin\left(t-\frac{1}{t}\right)dt$$

$$I=-I \Rightarrow 2I=0 \Rightarrow I=0$$

$$\int_{\frac{1}{e}}^{e} f(x)dx=0$$

28. Prove that $\int_0^x\left\{\int_0^u f(t)dt\right\}du=\int_0^x (x-u)f(u)du$.

Sol. Here apply integration by part

$$\int_0^x 1.\left\{\int_0^u f(t)dt\right\}du=\left\{\int_0^u f(t)dt\right\}_0^x\{u\}_0^x-\int_0^x f(u).u\,du$$

$$=\left(u\int_0^u f(t)dt\right)_0^x-\int_0^x u f(u)du$$

$$=x\int_0^x f(t)dt-\int_0^x u f(u)du$$

$$=\int_0^x (x-u)f(u)du$$

29. $\displaystyle\int_{-\frac{1}{\sqrt{3}}}^{\frac{1}{\sqrt{3}}} \frac{\cos^{-1}\left(\frac{2x}{1+x^2}\right)+\tan^{-1}\left(\frac{2x}{1-x^2}\right)}{e^x+1}dx$ is equal to

(1) $\frac{\pi}{2\sqrt{3}}$ (2) $\frac{\pi}{\sqrt{3}}$ (3) $\frac{\pi}{\sqrt{3}}$ (4) $\frac{\pi}{8\sqrt{3}}$

Sol. Answer (1)

$$\int_{-1/\sqrt{3}}^{1/\sqrt{3}} \frac{\cos^{-1}\left(\frac{2x}{1+x^2}\right)+\sin^{-1}\left(\frac{2x}{1+x^2}\right)}{e^x+1}.dx$$

$$I=\frac{\pi}{2}\int_{-1/\sqrt{3}}^{1/\sqrt{3}}\frac{1}{e^x+1}dx$$

$$I=\frac{\pi}{2}\int_{-1/\sqrt{3}}^{1/\sqrt{3}}\frac{1}{e^{-x}+1}dx=\frac{\pi}{2}\int_{\frac{-1}{\sqrt{3}}}^{\frac{1}{\sqrt{3}}}\frac{e^x}{e^x+1}dx$$

$$2I=\frac{\pi}{2}\int_{\frac{-1}{\sqrt{3}}}^{1/\sqrt{3}}dx$$

$$I=\frac{\pi}{2}\int_0^{\frac{1}{\sqrt{3}}}dx=\frac{\pi}{2\sqrt{3}}$$

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